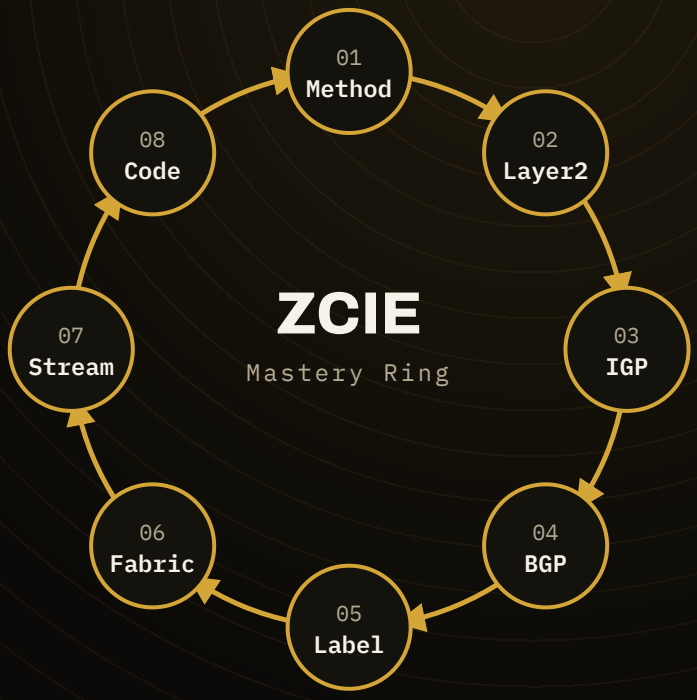
ZCIE

ZyeemZia™ Certified Internetwork Expert

Master the network. Prove it under pressure.



COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 39 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

An expert-level certification for senior engineers: methodical troubleshooting at speed, Layer 2 at scale, IGP and BGP mastery, MPLS and segment routing, EVPN fabrics and SD-WAN, multicast and QoS, and expert automation, rehearsed with full mock labs.

01
Learn
Work through 39 lessons. Each ends with a knowledge check.

02
Practise
Fill in the templates, answer the scenarios and case studies, review the glossary.

03
Certify
Take the 45-question official exam at zcie.zyeemzia.com, graded on the ZyeemZia server. Pass mark 85%.

04
Get certified
Download your certificate online. Anyone can verify its credential ID at zcie.zyeemzia.com/#verify. 92%+ and the capstone earn Distinction.

Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Month 1	m1 Method · m2 Layer2	Lab build, methodology drills	☐
Month 2	m3 IGP	IS-IS/OSPF deep labs, convergence	☐
Month 3	m4 BGP	Route reflection, policy, security	☐
Month 4	m5 Label	L3VPN, SR-MPLS labs	☐
Month 5	m6 Fabric	EVPN and SD-WAN labs	☐
Month 6	m7 Stream	Multicast and QoS labs	☐
Month 7	m8 Code	Automation, mock labs	☐
Month 8	Capstone	Full timed mock lab, certification exam	☐

Learner name .....

Start date ..... Target exam date .....

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3. Structured Troubleshooting at Speed · 35 min
4. Labs, Time and Documentation · 30 min

02 Layer2 — Layer 2 at Scale

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2. STP Tuning, Protection and Failure Modes · 40 min
3. First-Hop Security at Scale · 40 min
4. Multi-Chassis Aggregation and First-Hop Redundancy · 45 min
5. Provider Bridging (QinQ) and Layer 2 Loop Forensics · 45 min

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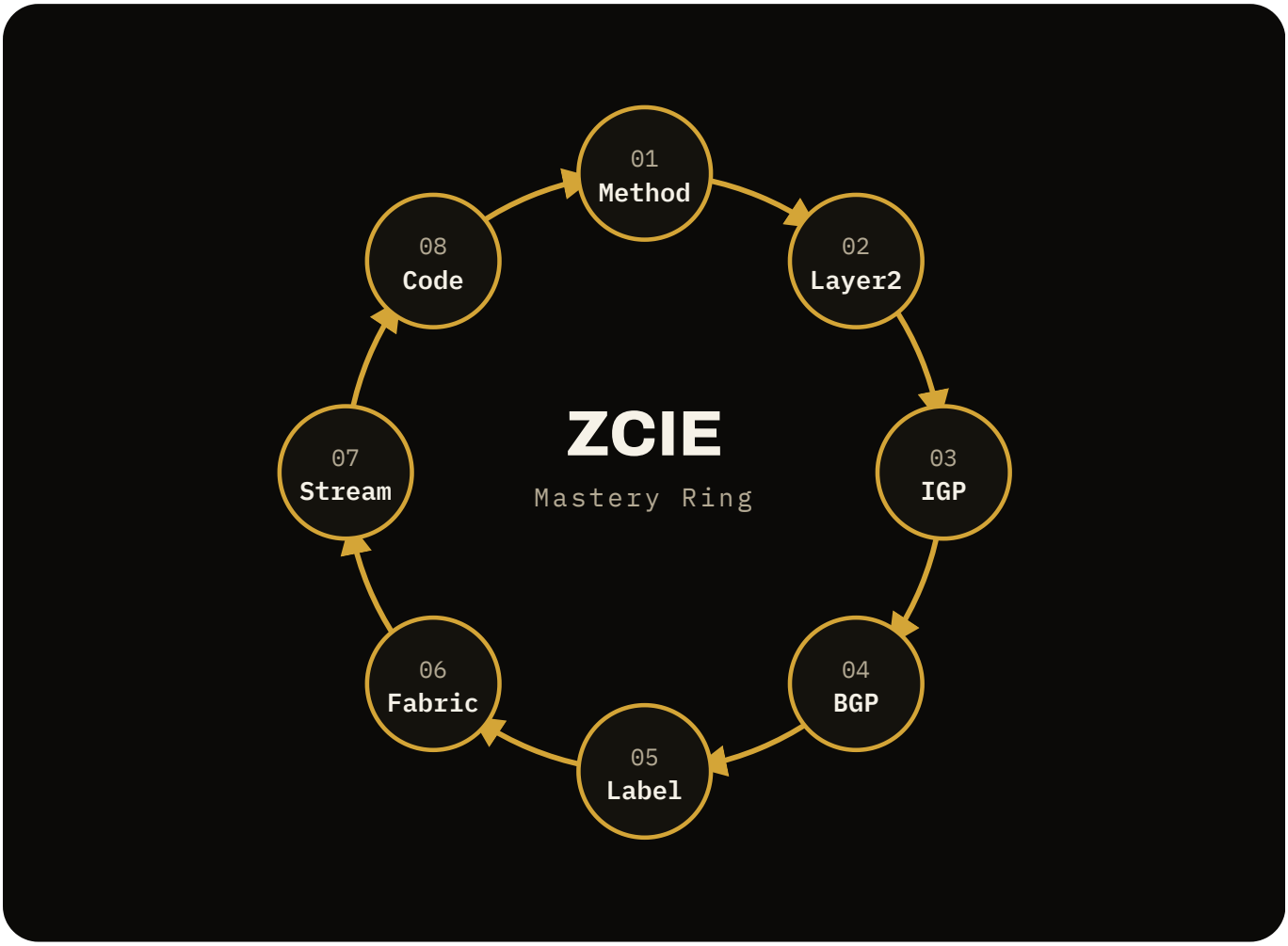
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THE FRAMEWORK

The ZCIE Mastery Ring

Eight disciplines an internetwork expert keeps sharp in a continuous loop: method, Layer 2, IGP, BGP, labels, fabrics, streams and code. Experts never finish the ring; they keep going round it.



01 Method

Requirement parsing, packet-walk reasoning, verification discipline and lab strategy

02 Layer2

MST, spanning tree protection, first-hop security, multi-chassis aggregation, gateway redundancy and loop forensics

03 IGP

OSPF and IS-IS databases, area and level design, fast convergence and safe redistribution

04 BGP

Reflection and confederations, community policy, best-path depth, advanced advertisement, convergence and security

05 Label

Build, scale, engineer and troubleshoot label-switched transport and the VPN services that ride on it.

06 Fabric

Design, build, scale and migrate overlay networks in the data centre and across the WAN.

07 Stream

Deliver one-to-many traffic reliably and give every class of traffic the treatment the business needs.

08 Code

Automate safely at scale, close the loop with telemetry, and bring every ring segment together

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01

MODULE 1 · METHOD

Expert Methodology

Requirement parsing, packet-walk reasoning, verification discipline and lab strategy

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Parse complex tasks into explicit requirements, implicit prerequisites and prohibitions
- Reason about any fault with a hop-by-hop packet walk across control and data planes
- Prove every change with the strongest available evidence
- Troubleshoot unfamiliar faults quickly with a structured, hypothesis-driven model
- Build repeatable labs and manage time and documentation in practical exams

LESSONS

1. Thinking Like an Expert: The Mastery Ring and Requirement Parsing · 30 min
2. Packet-Walk Reasoning and Verification Discipline · 35 min
3. Structured Troubleshooting at Speed · 35 min
4. Labs, Time and Documentation · 30 min

MODULE 1 · METHOD · LESSON 1 · 30 MIN

Thinking Like an Expert: The Mastery Ring and Requirement Parsing

Read every task as a set of constraints, predict the outcome, then build

- Explain the eight domains of the ZCIE Mastery Ring and how faults cross between them
- Decompose a lab task into explicit requirements, implicit requirements and prohibitions
- Define the verification evidence for a requirement before configuring anything

Expert-level practical work is not won by knowing more commands than everybody else. It is won by **deterministic reasoning**: you can predict what a network will do before you touch it, you change only what the requirement demands, and you can prove the result with evidence another engineer would accept. This module builds that method; every later module assumes it.

The ZCIE Mastery Ring

MTH
Method
Requirement parsing, packet-walk reasoning, verification discipline, structured troubleshooting, lab and time management.

L2
Layer2
Spanning tree at scale, first-hop security, multi-chassis aggregation, gateway redundancy, provider bridging and loop forensics.

IGP
IGP
Link-state databases, flooding, area and level design, fast convergence, summarisation and safe redistribution.

BGP
BGP
Scaling iBGP, policy with communities, best-path mastery, advanced advertisement, convergence and routing security.

LBL
Label
Label switching, layer 3 VPNs, segment routing and traffic engineering built on the IGP and BGP layers.

FAB
Fabric
Overlay fabrics with VXLAN and EVPN, and software-defined WAN edges.

STR
Stream
Multicast distribution and quality of service from edge to core.

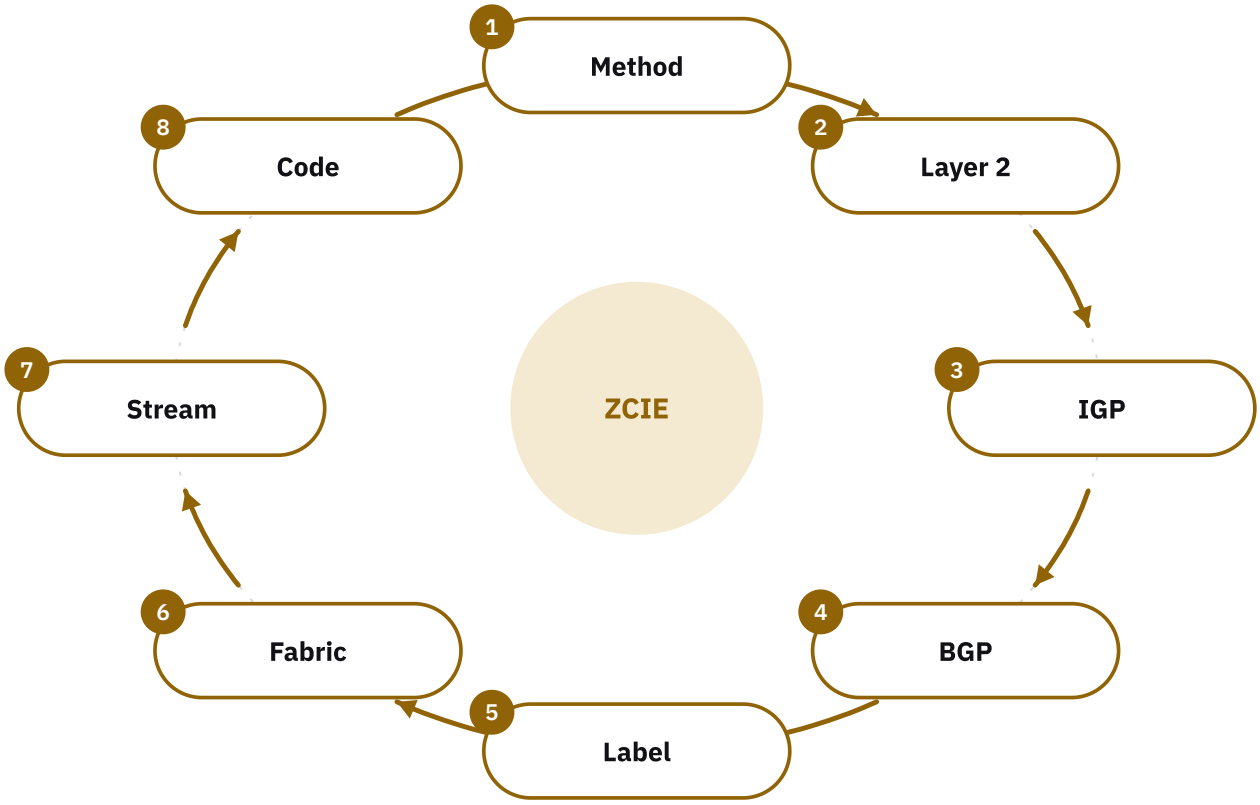
CODE
Code
Automation, programmatic verification, lab strategy and the capstone.

INSIGHT · THE RING IS A LOOP, NOT A LADDER

Real faults rarely respect domain boundaries. A BGP next hop becomes unreachable because an IGP summary hid the loopback; a VPN breaks because an MTU change in the underlay drops labelled frames; a multicast stream dies because a spanning-tree change moved the querier. Expert reasoning constantly asks **which neighbouring ring could be causing what I see in this one**.

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The Mastery Ring as a loop



1. **Method** — Parse, predict, prove
2. **Layer 2** — Loops, gateways, edge security
3. **IGP** — Loopback and next-hop reachability
4. **BGP** — Policy and scale on top of the IGP
5. **Label** — Transport and VPNs on IGP and BGP
6. **Fabric** — Overlays riding the underlay
7. **Stream** — Multicast and QoS across all of it
8. **Code** — Automate and verify every segment

Read it clockwise as a dependency chain, then notice the loop: a symptom in one segment is often caused by its neighbour, and Method wraps around everything.

Practical exam tasks and real change requests are written as **constraints**, not instructions. The words that matter are small: *only*, *without*, *must*, *prefer*, *must not*, *at all times*. An engineer who reads quickly and starts typing solves the problem they imagined, not the one that was asked. Requirement parsing is the deliberate act of turning prose into a precise list of outcomes, limits and proofs.

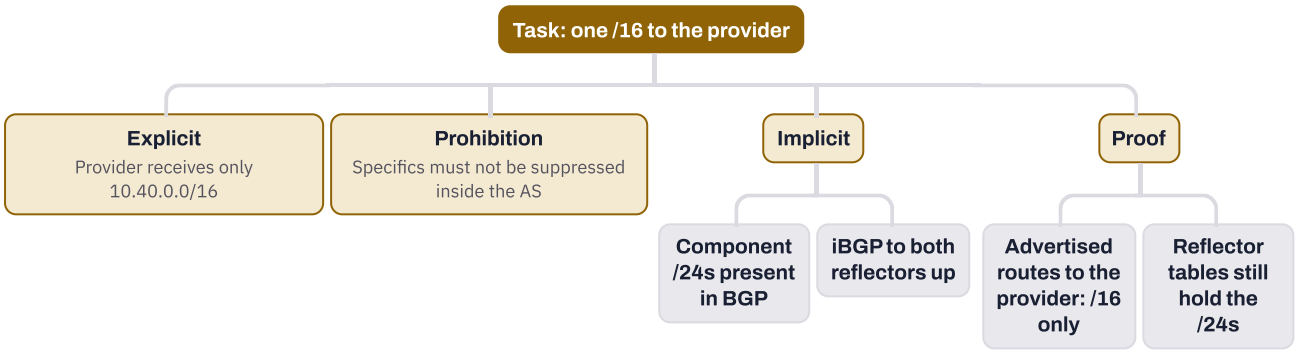
Requirement parsing in five passes

1. **Read for outcome** Read the whole task once without a keyboard. Write one sentence describing the end state in observable terms: who reaches what, over which path, under which failure.

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2. **Mark the constraints** Underline every restricting word: only, without, do not, minimum, must survive, prefer. Each becomes a line in your task notes.
3. **Find the implicit requirements** List what must already be true for the task to work: reachability of next hops, adjacencies, MTU, a route in the RIB for a network statement to match.
4. **Define the proof** Before configuring, write the exact command and the exact output that will prove success, including the failure case if one is mentioned.
5. **Map interactions** Note which other tasks touch the same devices or protocols. Changes in one task can silently break the proof of another.

Decomposing one task before typing



The aggregation task from the scenario below, split into the four lists an expert writes first. The proof is defined before any configuration exists.

Constraint language decoder

PHRASE IN THE TASK	WHAT IT REALLY CONSTRAINS	TYPICAL TRAP
"using a single command"	The solution must be one line, so it is usually a feature, not a combination	Building a route-map plus prefix-list when one aggregate or summary keyword was intended
"without modifying R3"	Every change happens on neighbours or via protocol signalling	Fixing it on R3 anyway because it is faster
"traffic must prefer"	Data-plane path, not just the RIB entry	Proving it with the routing table while CEF load-shares or policy-routes elsewhere
"must survive the failure of"	A backup path must exist and converge without manual action	Not testing the failure, or testing a different failure
"reachable from"	Both directions, from the stated source address	Pinging from the egress interface instead of the stated loopback
"do not use static routes"	Includes floating statics and static discard routes you add yourself	A summary that creates a protocol discard route is fine; a manual one is not

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Novice reading versus expert reading

Novice

- Starts configuring after the first sentence
- Assumes the default behaviour of the platform they know
- Proves success with a single ping
- Treats each task in isolation

Expert

- Reads every task before changing anything
- Lists implicit prerequisites and checks them first
- Writes the proof before the configuration
- Tracks how each change affects earlier tasks

Baseline snapshot before touching anything

R1 · IOS-style CLI

```
R1# show ip interface brief | exclude unassigned
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.1.12.1       YES manual up                    up
GigabitEthernet0/1     10.1.13.1       YES manual up                    down
Loopback0              10.0.0.1        YES manual up                    up
R1# show ip protocols | include Routing Protocol|Router ID
Routing Protocol is "ospf 1"
  Router ID 10.0.0.1
Routing Protocol is "bgp 65000"
R1# show ip route summary | include Total|ospf|bgp
ospf 1          4           9           1144        2808
bgp 65000       0           0           0           0
Total           12          11          1880        4108
```

Ninety seconds of baseline capture reveals two facts before any task is attempted: Gi0/1 is up/down (a layer 1 or 2 issue that will affect later tasks), and BGP is configured but has installed nothing. Record both in your notes.

SCENARIO · DECIDE · PARSING AN AGGREGATION TASK

Task text: "Advertise the branch networks 10.40.0.0/16 to the provider as a single prefix. Do not suppress any of the more specific routes inside your own AS." The branch /24s are in BGP on the edge router, which has iBGP sessions to two route reflectors and one eBGP session to the provider.

(A) Configure the aggregate with the keyword that suppresses all more-specific routes.

(B) Summarise 10.40.0.0/16 in the IGP at the area border so BGP only sees the summary.

(C) Aggregate without suppression, and filter the specifics outbound on the provider session only.

(D) Create a static route to a discard interface for the /16 and advertise it with a network statement.

Circle your choice. Answer and feedback: module answer key.

Before you configure any task

☐ I have read every task in the section, not just this one

☐ I have written the end state in observable terms

☐ Every restricting word is captured as a constraint

☐ Implicit prerequisites (adjacency, next hop, MTU, RIB entry) are listed and checked

☐ The proof command and expected output are written down

☐ I know which earlier tasks share devices or protocols with this one

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Key terms

Mastery Ring
The eight-domain ZCIE framework: Method, Layer2, IGP, BGP, Label, Fabric, Stream and Code, treated as interdependent rather than sequential.

Implicit requirement
A condition the task depends on without stating it, such as next-hop reachability or a matching RIB entry.

Verification evidence
The specific command output that proves a requirement is met, defined before configuring.

Blast radius
The set of devices, prefixes and tasks that a change can affect beyond its intended target.

Explicit requirement
An outcome stated directly in the task text, such as a preferred path or a single advertised prefix.

Prohibition constraint
A limit on how a task may be solved, for example no static routes or no changes on a named device.

Baseline snapshot
A short capture of interface, protocol and routing state taken before any change, used to separate pre-existing faults from new ones.

Task interaction map
A note of which tasks share devices or protocols, used to re-verify earlier work after later changes.

REFLECT

Think of the last change you made that had an unexpected side effect. Which implicit requirement or interaction did you miss, and which of the five parsing passes would have caught it?

Key takeaways

- Read every task before typing; constraints hide in small words such as only, without and prefer.
- Split each task into explicit requirements, implicit prerequisites, prohibitions and a written proof.
- Take a baseline snapshot first so pre-existing faults are not blamed on your changes.
- Faults cross ring segments: always ask which neighbouring segment could cause the symptom.

KNOWLEDGE CHECK

1. A task says traffic from the branch "must prefer" link A. Which evidence proves the requirement?

   (A) The routing table shows a route through link A with the lowest metric among the candidates

   (B) The configuration on the branch router contains a higher bandwidth value on link A

   (C) A ping across link A succeeds with zero loss from the branch router to the hub

   (D) A forwarding-plane check, such as an exact-route lookup or a trace from the source

2. Which item is an implicit requirement for an iBGP-learned prefix to be installed?

   (A) The prefix must be advertised with a community that the receiving router recognises

   (B) The receiving router must have a network statement that covers the prefix

   (C) The next hop of the prefix must be resolvable through the local routing table

   (D) The prefix must carry a local preference higher than the platform default of 100

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3. A task forbids static routes. Which approach still respects the constraint?

(A) A floating static route with a high distance that is only used on failure

(B) A protocol summary that installs its own automatic discard route

(C) A static route to a discard interface that anchors a BGP aggregate

(D) A static default route that is redistributed into the IGP with a tag

4. Why should you read every task in a section before starting the first one?

(A) Later tasks often depend on choices that you would otherwise make too early

(B) It allows you to memorise the device names and IP plan before typing anything

(C) Practical exams usually grade tasks in reverse order of their appearance

(D) It prevents the lab environment from timing out during long configurations

5. What is the main purpose of a baseline snapshot?

(A) To create a configuration backup that can be restored with a single command

(B) To measure device CPU before applying any debug commands to the router

(C) To document the topology for graders who review the lab after it ends

(D) To separate pre-existing faults from faults your changes introduce

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · METHOD · LESSON 2 · 35 MIN

Packet-Walk Reasoning and Verification Discipline

Follow one packet hop by hop and prove every claim with the strongest evidence available

- Walk a packet through ingress, lookup, resolution, rewrite, egress and return
- Distinguish control-plane evidence from data-plane evidence
- Apply a predict-change-observe loop to every configuration change

A packet walk is the expert's most reliable reasoning tool. Instead of asking "why is it broken?", you pick **one packet** with a precise source, destination, protocol and ingress point, then ask the same questions at every hop until the answer changes. The point where your prediction and reality diverge is where the fault lives.

The hop-by-hop packet walk

01 Ingress
Is the frame accepted? Physical state, VLAN membership, MAC destination, inbound ACL or filter.

→

02 Lookup
Which table is consulted (VRF, policy route, FIB) and which entry wins by longest match?

→

03 Resolve
Is the next hop resolved to an adjacency: ARP or neighbour entry, label, tunnel?

→

04 Rewrite
What changes: MAC addresses, TTL, labels, encapsulation, DSCP?

→

05 Egress
Outbound policy, queueing, MTU and fragmentation decisions.

→

06 Return
Does the reply have a valid path back to the original source address?

Where the evidence lives

Intent	`task requirements` `design documents` `policy statements`
Control plane	`neighbour state` `link-state database` `BGP table with attributes`
Routing table	`selected routes` `administrative distance` `recursive next hops`
Forwarding table	`FIB entries` `adjacencies` `load-sharing buckets`
Wire	`interface counters` `captures` `sourced traces and pings`

Verification discipline means every change has a predicted, observable effect, and you look for that effect at the layer the requirement names. If the requirement is about forwarding, a routing-table entry is only circumstantial evidence. If it is about a protocol attribute, the received and advertised attributes per neighbour are the proof, not the presence of a route-map in the configuration.

The evidence ladder

CLAIM	WEAK EVIDENCE	STRONG EVIDENCE
Adjacency is working	Configuration present on both sides	Neighbour state FULL or Up and the peer's database entries present

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CLAIM	WEAK EVIDENCE	STRONG EVIDENCE
Route is preferred	Route visible in the routing table	FIB exact-route lookup and a trace sourced from the stated address
Policy is applied	Route-map exists and is attached	Per-prefix attributes shown as received and as advertised to that neighbour
Failover works	A backup route exists in the table	Primary failed on purpose and traffic observed on the backup within the target time
Filter blocks traffic	ACL entry configured	Match counters increasing while the source sends test traffic

Proving the forwarding decision

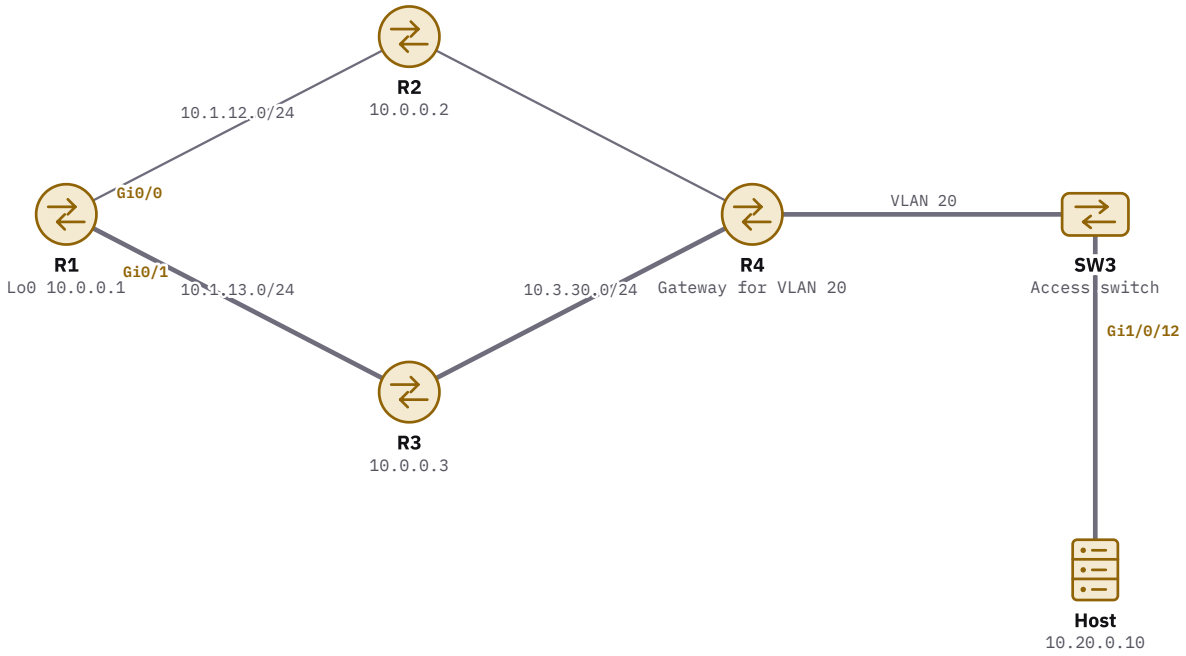
R1 · IOS-style CLI

```
R1# show ip route 10.20.0.10
Routing entry for 10.20.0.0/16
  Known via "ospf 1", distance 110, metric 30, type inter area
  * 10.1.12.2, from 10.0.0.2, via GigabitEthernet0/0
    10.1.13.3, from 10.0.0.3, via GigabitEthernet0/1
R1# show ip cef exact-route 10.0.0.1 10.20.0.10
10.0.0.1 -> 10.20.0.10 =>IP adj out of GigabitEthernet0/1, addr 10.1.13.3
R1# traceroute 10.20.0.10 source Loopback0 numeric
  1 10.1.13.3 2 msec 1 msec 1 msec
  2 10.3.30.4 3 msec 2 msec 3 msec
  3 10.20.0.10 4 msec 3 msec 4 msec
```

The routing table lists two equal-cost paths and marks Gi0/0 with an asterisk, which only shows the next path to be used for process-switched traffic. The FIB hash for this source and destination pair actually selects Gi0/1, confirmed by the sourced trace. A task asking for traffic to use R2 is not satisfied.

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The flow being walked



Both paths are equal cost in the RIB, but the FIB hash for 10.0.0.1 → 10.20.0.10 selects the bold path via R3. The last hop is resolved by ARP on R4 and by the MAC table on SW3.

WATCH OUT · A SUCCESSFUL PING IS NOT PROOF

A router ping uses the egress interface address as its source unless told otherwise, follows whatever hash bucket that source pair lands in, and proves only one direction of one flow. Always source tests from the address named in the task and confirm the path, not just the reply.

Walking the last hop at layer 2

R4 and SW3 · IOS-style CLI

```
R4# show ip arp 10.20.0.10
Internet  10.20.0.10     3   0050.56aa.0102  ARPA   Vlan20
SW3# show mac address-table address 0050.56aa.0102
Vlan    Mac Address       Type        Ports
  20    0050.56aa.0102    DYNAMIC     Gi1/0/12
SW3# show interfaces Gi1/0/12 | include line protocol|drops|output errors
GigabitEthernet1/0/12 is up, line protocol is up (connected)
  Input queue: 0/2000/0/0 (size/max/drops/flushes); Total output drops: 48211
     0 output errors, 0 collisions, 1 interface resets
```

The router has resolved the host, the switch knows exactly which port the MAC lives on, and the port is up. The walk continues to the next question: 48,211 output drops suggest congestion or a speed mismatch on the access port, not a routing fault.

The verification loop

1. **Predict** State what will change and where it will be visible: which table, which neighbour, which counter.
2. **Change** Apply the smallest change that can produce the predicted effect.
3. **Observe** Collect the strongest evidence available for the claim, not the most convenient.
4. **Compare** If observation and prediction differ, stop and explain the difference before changing anything else.

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5. **Record** Note the change, the proof and any side effect seen on other tasks.

SCENARIO · DECIDE · THE TRACE THAT STOPS AT HOP THREE

A sourced traceroute from R1 Loopback0 to 10.60.1.1 shows replies from hops 1 and 2, then asterisks from hop 3 onward. Every router on the path has a route to 10.60.0.0/16 and the destination host can ping its own gateway.

A. Check that the hop-3 router has a route back to R1's loopback, the trace source.

B. Clear the routing process on every router on the path so the tables are rebuilt cleanly.

C. Repeat the trace with a higher maximum TTL so the later hops have more time to respond.

D. Increase the trace timeout because hop 3 may be rate-limiting its control-plane replies.

Circle your choice. Answer and feedback: module answer key.

Verification discipline

☐ Every test is sourced from the address the requirement names

☐ Forwarding claims are proven with FIB lookups or traces, not only the RIB

☐ Policy claims are proven per neighbour, received and advertised

☐ Failover claims are proven by actually failing the primary

☐ Every unexpected observation is explained before the next change

Key terms

Packet walk
A hop-by-hop analysis of one specific flow that asks the same ingress, lookup, resolve, rewrite, egress and return questions at every device.

Adjacency
The resolved layer 2 rewrite information for a next hop, built from ARP, neighbour discovery or label bindings.

Return path
The route the reply takes back to the source address; it can differ from the forward path and fail independently.

Exact-route lookup
A query that shows which load-sharing path the FIB selects for a given source and destination pair.

FIB
Forwarding information base: the data-plane table derived from the routing table and used for actual packet forwarding.

Recursive lookup
Resolving a next hop that is not directly connected by looking it up again in the routing table until a connected interface is found.

uRPF
Unicast reverse path forwarding: a check that drops packets whose source address is not reachable back through the arrival interface.

Evidence ladder
A ranking of proof from weak (configuration present) to strong (forwarding state and observed traffic).

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REFLECT

Pick a recent incident. Write the packet walk for the failing flow and mark the first hop where your prediction would have differed from reality. How long would it have taken to reach that point with this method?

Key takeaways

- Walk one precisely defined packet hop by hop: ingress, lookup, resolve, rewrite, egress and return.
- Match the evidence to the claim: forwarding claims need FIB lookups or sourced traces, not only the RIB.
- Source every test from the address named in the requirement and check both directions.
- When observation and prediction differ, stop and explain the difference before changing anything else.

KNOWLEDGE CHECK

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6. A router shows two equal-cost routes for a prefix. How do you prove which one carries traffic for a specific flow?

   (A) Look for the asterisk that marks the active path in the routing table

   (B) Compare interface bandwidth values and assume the faster link wins

   (C) Use a FIB exact-route lookup for the flow's exact source and destination pair

   (D) Shut one interface and check whether the ping still succeeds afterwards

7. A trace from R1's loopback dies after hop 2 although forward routes exist everywhere. What is the strongest first hypothesis?

   (A) The third hop cannot route replies back to the loopback source

   (B) The destination host has a firewall that blocks all ICMP echo requests

   (C) The default maximum TTL is too small for a path of this length

   (D) Hop 2 is load-sharing the traffic onto a path with no route at all

8. Which evidence most strongly proves that an outbound route policy is working?

   (A) The route-map is attached to the neighbour in the running configuration

   (B) The prefix's attributes as they are advertised to that specific neighbour

   (C) The route-map match counters have increased since the last policy change

   (D) The local BGP table shows the attribute set on the best path for the prefix

9. In the verification loop, what should you do when the observation differs from your prediction?

   (A) Apply the next planned change because it may resolve the difference

   (B) Stop and explain the difference before making any further change

   (C) Revert the change immediately and move on to a different task

   (D) Clear the relevant protocol sessions to force a fresh calculation

10. Why is the return step part of every packet walk?

    (A) Routers always forward replies over the same interface the request used

    (B) Return traffic is the only direction that inbound access lists inspect

    (C) Replies follow their own lookup and can fail on their own

    (D) Traceroute only measures the return direction of a network path

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · METHOD · LESSON 3 · 35 MIN

Structured Troubleshooting at Speed

Hypothesis-driven fault isolation that stays fast under pressure

- Apply a six-stage troubleshooting model to unfamiliar faults
- Choose the fastest isolation method for a given symptom
- Use debugs safely and record evidence in a troubleshooting log

Speed in troubleshooting does not come from typing faster. It comes from **never running a command without a hypothesis** and from cutting the search space in half with every test. Engineers who look slow under pressure are usually re-reading output they did not need, or switching hypotheses before the current one is disproved.

The SCOPE-PATH troubleshooting model

S1
Scope
Define the failing flow precisely: source, destination, protocol, since when, who else is affected.

S2
Path
Draw the expected path and the control-plane dependencies that build it.

S3
Layer
Pick the layer most consistent with the symptom pattern; partial loss and total loss point to different layers.

S4
Hypothesis
State one falsifiable cause and the single test that would disprove it.

S5
Test
Run the test, read only the fields that decide the hypothesis, and record the result.

S6
Close
Fix, verify with the original failing flow, and check for side effects on other flows.

Choosing an isolation method

METHOD	FASTEST WHEN	RISK
Follow the path	The failing flow is known and the path is short	Slow on large meshes with many equal-cost branches
Divide and conquer	The path is long; test the middle first	Needs a good test point in the middle of the path
Compare working and broken	A similar flow, site or device still works	Differences that do not matter can distract you
Bottom-up	Physical or layer 2 symptoms: errors, flaps, partial loss	Wastes time when the fault is clearly in policy
Top-down	One application fails while others on the same path work	Misses lower-layer faults that affect only large packets

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Symptom to first suspects

SYMPTOM	FIRST SUSPECTS	DECIDING TEST
OSPF neighbour stuck in EXSTART or EXCHANGE	Interface MTU mismatch, duplicate router ID	Compare interface MTU and router IDs on both ends
OSPF neighbour stuck in INIT	One direction of hellos lost: filter, unidirectional link, multicast drop	Check whether our router ID appears in the peer's hello
BGP neighbour in Active	No route to the peer address, wrong update source, TCP 179 filtered	Route lookup for the peer address and a ping sourced from the update source
BGP route marked RIB-failure	A route with lower administrative distance already installed	Look up the prefix in the routing table and compare the source protocol
Prefix not advertised by network statement	No exact match (prefix and mask) in the routing table	Exact routing-table lookup for the prefix and mask
Small packets pass, large packets fail	MTU or fragmentation mismatch on the path	Sourced ping with don't-fragment set and increasing size

Debugging a stuck adjacency safely

R2 · IOS-style CLI

```
R2# show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.0.3          1   EXSTART/DR      00:00:36    10.1.23.3       GigabitEthernet0/2
R2# show logging | include buffer
    Buffer logging: level debugging, 1200000 messages logged
R2# debug condition interface GigabitEthernet0/2
R2# debug ip ospf adj
OSPF-1 ADJ   Gi0/2: Rcv DBD from 10.0.0.3 seq 0x1F4A opt 0x52 flag 0x7 len 32  mtu 9000 state
EXSTART
OSPF-1 ADJ   Gi0/2: Nbr 10.0.0.3 has larger interface MTU
R2# undebug all
R2# show interfaces GigabitEthernet0/2 | include MTU
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
```

The debug is limited to one interface and written to the buffer, not the console. One debug line decides the hypothesis: the neighbour advertises 9000 bytes in its database description packets while R2 uses 1500. Align the MTU; do not disable the MTU check unless the task explicitly requires it.

WATCH OUT · DEBUGS ARE A SCALPEL

On a busy router an unconditioned debug can saturate the control plane and cause the outage you were investigating. Condition debugs to an interface or neighbour, log to the buffer, turn them off immediately after the deciding line appears, and never leave them running when you move to the next task.

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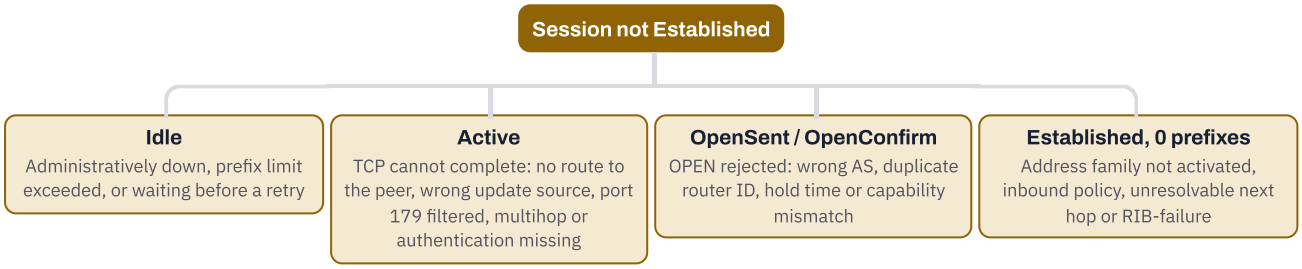
A BGP session that never leaves Active

R5 · IOS-style CLI

```
R5# show ip bgp summary | begin Neighbor
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.9        4 65000       0       0        1    0    0 never    Active
R5# show ip route 10.0.0.9
% Subnet not in table
R5# show ip ospf database router 10.0.0.9 | include Link ID|Advertising
R5#
```

The session cannot form because there is no route to the peer's loopback, and R9 has no router LSA in R5's database at all. The fault is in the IGP, not in BGP. Moving to the IGP ring immediately saves the time otherwise spent checking BGP configuration.

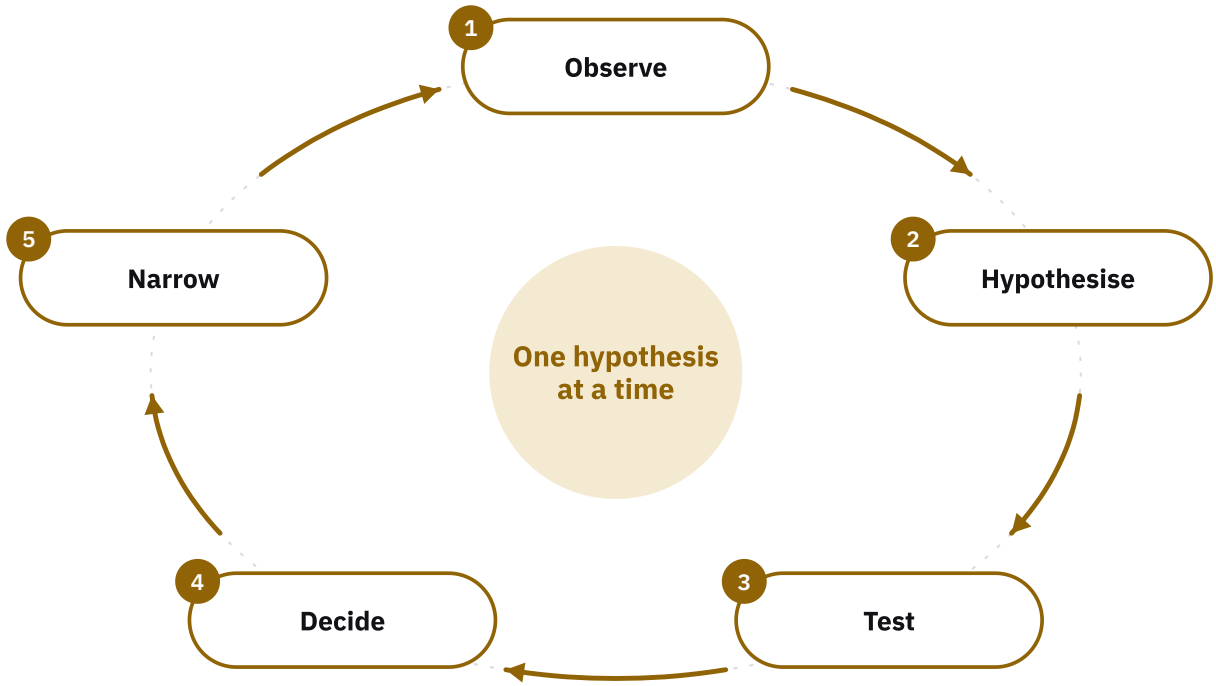
Triage a BGP session by its state



Read the state before opening the configuration: each state rules out whole layers. The R5 output above stops at Active, so the search moves straight to reachability.

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The hypothesis cycle



1. **Observe** — Collect the symptom precisely: source, destination, protocol, since when
2. **Hypothesise** — One falsifiable cause and the single test that decides it
3. **Test** — Run it and read only the deciding fields
4. **Decide** — Confirmed, refuted or inconclusive; write it in the log
5. **Narrow** — Cut the search space in half and go round again

Each turn of the loop tests exactly one hypothesis and should halve the search space; an inconclusive result means the test was badly chosen, not that you need more commands.

SCENARIO · DECIDE · TEN MINUTES AND NO ROOT CAUSE

During a timed practical lab you have spent ten minutes on a fault worth three points. You have checked the IGP and BGP configuration on two routers and found nothing. Six other tasks are still open.

- (A) Keep going on this task, because the root cause is probably close after ten minutes of searching.
- (B) Record the evidence so far, park the task with a note, and return after the independent tasks.
- (C) Remove the configuration on both routers and rebuild it from scratch to eliminate hidden errors.
- (D) Enable broad debugging on both routers and wait for an error message to identify the fault.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · THE FORTY-MINUTE OUTAGE THAT TOOK NINE HOURS

A regional payments processor added a new 10 Gb/s link between two data-centre routers during a change window. After the change, card authorisations from one data centre began timing out intermittently. The on-call

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team saw BGP sessions between the sites flapping every few minutes and spent six hours tuning BGP timers, replacing route reflectors in the design and rolling back unrelated firewall rules. A senior engineer joining late ran a sourced ping with the don't-fragment bit at 1500 bytes across the new link and saw it fail at 1476. The new link was a provider service with GRE-based transport, and the interface MTU had been left at 1500.

1. Which symptom should have pointed to MTU from the start?

2. Which step of the SCOPE-PATH model did the team skip?

3. Why did BGP timer tuning appear to help at times?

4. What evidence would you capture for the post-incident review?

Model answer: module answer key.

TEMPLATE **Troubleshooting log**

Record each hypothesis and test so you never repeat work, and can hand over or resume a parked task instantly.

TIME	OBSERVATION	HYPOTHESIS	DECIDING TEST	RESULT	NEXT ACTION
10:42	*R5 to R9 iBGP stuck Active*	*No route to 10.0.0.9*	*show ip route 10.0.0.9*	*Not in table; no R9 router LSA*	*Check R9 OSPF adjacency on Gi0/3*
10:46	*R9 Gi0/3 neighbour EXSTART*	*MTU mismatch*	*Compare interface MTU both ends*	*9000 vs 1500*	*Align MTU, re-verify BGP*

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Troubleshooting at speed

☐ The failing flow is defined with source, destination and protocol

☐ Every command I run is tied to a written hypothesis

☐ I read only the fields that decide the current hypothesis

☐ Debugs are conditioned, buffered and removed when done

☐ Parked tasks have a written evidence summary

☐ The fix is verified with the original failing flow

Key terms

Hypothesis-driven troubleshooting
An approach in which each test is chosen to confirm or refute one stated cause.

Conditional debug
A debug limited to a specific interface, neighbour or prefix to reduce load and noise.

Active state
A BGP state in which the router is trying to open a TCP connection to a peer that it cannot reach or that refuses the connection.

Time-boxing
Allocating a fixed time budget to a task and parking it with notes when the budget runs out.

Divide and conquer
Testing at the midpoint of a path or layer stack to halve the search space with each test.

EXSTART
An OSPF neighbour state in which the routers negotiate the master and slave roles for database exchange; being stuck here often indicates MTU mismatch.

RIB-failure
A BGP path that is valid and best but not installed because the routing table holds a route from a source with lower administrative distance.

Sunk-cost trap
Continuing an unproductive effort because of the time already invested rather than the expected return.

REFLECT

Which isolation method do you default to? Describe a fault where a different method would have been faster, and what signal should have triggered the switch.

Key takeaways

- Never run a command without a written hypothesis and the single test that would disprove it.
- Choose the isolation method from the symptom: follow the path, divide and conquer, compare, bottom-up or top-down.
- Condition debugs to one interface or neighbour, log to the buffer and stop them as soon as they answer.
- Time-box every task and park it with written evidence when no live hypothesis remains.

KNOWLEDGE CHECK

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11. An OSPF neighbour is stuck in INIT. Which test decides the most likely hypothesis?

(A) Compare the configured interface MTU values on both ends of the link

(B) Compare router IDs across the area to find any duplicate values

(C) Check whether the neighbour is listed in the database summary list

(D) Check whether our router ID appears in the peer's hellos

12. A BGP prefix is shown as RIB-failure. What is the most likely reason?

(A) A lower-distance route from another source is already installed

(B) The next hop of the prefix is not resolvable through the routing table

(C) The prefix was rejected by the inbound route-map applied to the peer

(D) The peer has exceeded the configured maximum number of prefixes

13. Which practice makes debugging on a production router safer?

(A) Enabling debugs on all interfaces so no relevant message is missed

(B) Conditioning the debug to one interface and logging to the buffer

(C) Sending debug output to the console where it is displayed immediately

(D) Leaving debugs running until the next maintenance window closes

14. Small pings succeed across a new link but file transfers stall. Which first test is most decisive?

(A) A sourced ping with don't-fragment set and stepwise larger sizes

(B) A traceroute between the two hosts to check the hop count and latency

(C) A comparison of the routing metrics on both sides of the new link

(D) A check of the interface duplex settings and speed on both ends

15. When is comparing a working and a broken system the fastest isolation method?

(A) When the failing path is short and every hop is easy to inspect directly

(B) When the symptoms point clearly to a physical layer fault on one link

(C) When the fault affects every flow across the entire network at once

(D) When a similar flow, site or device is still working under similar load

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · METHOD · LESSON 4 · 30 MIN

Labs, Time and Documentation

Build a lab you can trust, budget exam time, and document like an expert

- Design a reusable virtual lab with topology-as-code and snapshots
- Plan time in a practical exam using point budgets and dependency order
- Keep topology notes and change records that make verification fast

A lab is an instrument. If it is slow to rebuild, inconsistent between runs or poorly documented, every experiment costs more than it should and you will avoid running the experiments that build expert intuition. Treat your lab like production: version-controlled topology, repeatable startup configurations, snapshots and notes.

Lab platform options

Container-based network OS

- Fast boot and low memory per node
- Topology defined in a text file
- Some data-plane features may be simplified

VM-based network OS

- Closest to the real control plane
- Heavier on CPU and memory
- Good for platform-specific behaviour

Physical or remote rack

- Real hardware forwarding and timers
- Slow to rewire and reset
- Best for hardware-dependent topics such as QoS queues

A reusable lab stack

Scenarios	`task files` `expected outcomes` `fault-injection scripts`
Topology as code	`node list` `link list` `management addressing`
Node configuration	`startup configs` `baseline templates` `per-scenario overlays`
Automation	`deploy and destroy` `config snapshot and restore` `verification scripts`
Host	`CPU and memory budget` `hypervisor or container runtime` `storage for images`

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Topology as code for a practice pod

Linux host

```
# topology.yml - ZCIE core practice pod
name: zcie-core
mgmt: { network: 172.30.0.0/24 }
nodes:
  R1: { kind: router, startup: cfg/R1.cfg }
  R2: { kind: router, startup: cfg/R2.cfg }
  R3: { kind: router, startup: cfg/R3.cfg }
  SW1: { kind: switch, startup: cfg/SW1.cfg }
links:
  - [R1:eth1, R2:eth1]
  - [R1:eth2, R3:eth1]
  - [R2:eth2, SW1:eth1]
  - [R3:eth2, SW1:eth2]
# keep this file and cfg/ under version control
```

The exact syntax depends on the lab tool you use, but the principle is constant: nodes, links and startup configurations live in text files under version control, so any scenario can be rebuilt identically in minutes.

> **PRO TIP · TOPOLOGY NOTES ON ONE SHEET**
> Before any practical lab, draw one sheet with every device, link address, router ID, AS number, area or level boundary, VRF and redistribution point. Mark tasks on the drawing as you complete them. Under pressure, a glance at the sheet replaces dozens of show commands.

Time management for a practical lab

1. **Read everything first** Spend the first 15 to 20 minutes reading every task and drawing the dependency map. No configuration yet.
2. **Foundations before features** Layer 2, addressing and the IGP come first, because most later tasks depend on them.
3. **Budget by points** Give each task a time budget proportional to its points, and note the budget next to the task.
4. **Park, do not sink** When a task reaches twice its budget without a live hypothesis, park it with written evidence.
5. **Save regularly** Save configurations after every completed task so a reload never costs completed work.
6. **Reserve the final sweep** Keep the last 30 to 45 minutes for re-verifying every task, because later changes break earlier proofs.

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An eight-hour practical lab, planned

0:00
Read and map
Every task read, dependency map drawn, no configuration

0:20
Foundations
Layer 2, addressing and the IGP, each verified

1:45
Core services
BGP, labels and VPNs, proven one by one

4:00
Checkpoint
Points earned against plan; re-prioritise and park

4:15
Features
Overlays, multicast, QoS and automation tasks

7:20
Final sweep
Re-verify everything, remove debugs, save

8:00
End

An example for a 480-minute lab with a 20-minute read and a 40-minute sweep, matching the budget arithmetic below; scale the marks to the published duration.

Time budget arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Minutes per point	`(total minutes − reading time − final sweep) ÷ total points`	The working budget for each point; a 4-point task gets four times this value.
Park threshold	`time spent > 2 × task budget and no live hypothesis`	The moment to stop and move on, recording evidence first.
Dependency weight	`own points + points of tasks that depend on it`	Use this, not raw points, to order foundational work.

TEMPLATE

Lab and exam task planner

Plan the order, budget and proof of every task before configuring, and track status during the session.

TASK	POINTS	DEPENDS ON	DEVICES	BUDGET (MIN)	PROOF COMMAND	STATUS
2.3 OSPF area 10 NSSA	*3*	*2.1 adjacencies*	*R3, R6, R7*	*12*	*show ip ospf database nssa-external*	*Done 11:20*
4.1 BGP inbound policy	*4*	*2.x IGP, 3.1 iBGP*	*R1, R2*	*16*	*show ip bgp neighbors x received-routes*	*Parked: R2 policy order*

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TASK	POINTS	DEPENDS ON	DEVICES	BUDGET (MIN)	PROOF COMMAND	STATUS

SCENARIO · DECIDE · NINETY MINUTES LEFT

With 90 minutes remaining you have three unfinished tasks: a 4-point IGP redistribution task that currently leaves two remote loopbacks unreachable, a 2-point syslog formatting task, and a 3-point BGP policy task on routers whose loopbacks are among the unreachable ones.

(A) Do the syslog task first to bank easy points, then attempt the larger tasks in point order.

(B) Start with the BGP policy task because it is configuration-heavy and benefits from the most time.

(C) Skip the final verification sweep so that all three tasks can be attempted with more time.

(D) Fix the redistribution task first, then BGP, then syslog, keeping 30 minutes for the sweep.

Circle your choice. Answer and feedback: module answer key.

Final verification sweep

☐ Every task re-verified with its recorded proof command

☐ No interfaces left administratively down from testing

☐ All debugs and temporary filters removed

☐ Configurations saved on every device

☐ Earlier tasks re-checked after later redistribution or policy changes

☐ Notes reconcile with the final state of the lab

Key terms

Topology as code
Describing nodes, links and startup configuration in version-controlled text files so a lab can be rebuilt identically.

Snapshot
A saved state of all node configurations that can be restored to reset a scenario quickly.

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Fault injection
Deliberately introducing a failure or misconfiguration in a lab to practise detection and recovery.

Point budget
The time allocated to a task in proportion to its score.

Topology sheet
A single annotated drawing of devices, addressing, protocol boundaries and redistribution points.

Dependency map
A sketch of which tasks depend on others, used to order work in a practical lab.

Final sweep
The reserved period at the end of a lab for re-verifying every task in order.

Change record
A short written entry for each change: what, where, why, proof and side effects.

REFLECT

Design the lab you will use for the rest of this course. Which topologies do you need, how will you reset them, and how will you record what you learned from each experiment?

Key takeaways

- Treat the lab as an instrument: topology as code, versioned startup configurations and snapshots.
- Order work by dependency weight, not by raw points.
- Budget minutes per point after subtracting reading and sweep time; park a task at about twice its budget.
- Reserve the last 30 to 45 minutes to re-verify everything, because late changes break early proofs.

KNOWLEDGE CHECK

16. Why should foundational tasks such as the IGP usually be completed first in a practical lab?

(A) Foundational tasks are always worth more points than the feature tasks

(B) Graders review the configuration in the order the tasks are printed

(C) Many later tasks cannot be configured or proven without reachability

(D) Foundational tasks cannot be changed again once others are configured

17. A lab has 480 minutes, 100 points, 20 minutes of reading and a 40-minute final sweep. What is the budget for a 3-point task?

(A) About 14.4 minutes

(B) About 9.8 minutes

(C) About 12.6 minutes

(D) About 16.0 minutes

18. What is the main advantage of topology-as-code for expert practice?

(A) Scenarios can be rebuilt identically, so experiments are repeatable

(B) It makes container-based nodes forward traffic exactly like hardware does

(C) It removes the need for any startup configuration on the lab nodes

(D) It allows the lab to run without a management network for access

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19. When should a task be parked?

(A) As soon as the first test result disagrees with your initial prediction

(B) Only when every other task in the lab has been fully completed

(C) Whenever the task involves more than two devices or protocols

(D) When it exceeds about twice its budget and no live hypothesis remains

20. Why is a final verification sweep valuable even when every task was proven during the lab?

(A) Graders only accept proofs that were captured in the last thirty minutes

(B) Later changes, such as redistribution or policy, can silently break earlier proofs

(C) Configurations are not saved until the sweep commands are executed

(D) Protocol timers reset during the lab and must be checked again

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · METHOD

Answer key

Knowledge checks

1. D Prefer describes the data plane. The RIB can show one path while load sharing or policy forwarding sends packets elsewhere, so the proof must come from forwarding state or a sourced trace.

2. C An unreachable next hop makes the path invalid. The other items are either unrelated to installation or describe origination rather than reception.

3. B Automatic discard routes created by a summarisation feature are not static routes you configured. Every other option is a static route, however it is used.

4. A Dependencies and conflicts between tasks are the main reason experts read ahead. Early choices such as area design or tagging can make later tasks easy or impossible.

5. D Knowing the starting state stops you from chasing problems you did not cause, and reveals pre-seeded faults early.

6. C Per-flow load sharing is decided by the forwarding table hash. The routing-table marker and bandwidth do not reveal it, and shutting links changes the system under test.

7. A Silence from hop 3 onward with complete forward routing points to the return path for the source address.

8. B Advertised-route output per neighbour shows the result of outbound policy. Local tables show pre-policy state and counters prove a match, not the resulting attributes.

9. B Unexplained differences are the most valuable evidence you have. Stacking more changes on top makes the system harder to reason about.

10. C Forward and reverse paths are computed independently. Asymmetry, uRPF and stateful devices make return-path failures common.

11. D INIT means hellos are received but the peer does not list us, so hellos are lost in one direction. MTU and duplicate IDs cause later-state problems.

12. A RIB-failure means the path was selected but could not be installed. An unresolvable next hop makes the path invalid, and policy or prefix limits stop it from appearing at all.

13. B Conditioned, buffered and short-lived debugs limit control-plane load and noise.

14. A Size-dependent failure points to MTU. A DF-bit size sweep finds the exact maximum that passes.

15. D Comparison needs a working reference. Without one, following the path or divide and conquer is faster.

16. C Dependency, not points or print order, drives the sequence. Reachability underpins BGP, VPN and policy proofs.

17. C (480 − 20 − 40) ÷ 100 = 4.2 minutes per point, so 3 points gets about 12.6 minutes.

18. A Repeatability is the value. Data-plane fidelity and configuration needs are unchanged.

19. D Parking is triggered by time and the absence of a hypothesis, not by the first surprise or by task size.

20. B Interactions between tasks are the reason for the sweep; nothing about the grading clock requires it.

Scenarios & case studies

> **Scenario · Parsing an aggregation task**
> **C** is best. Correct. Both constraints are met: the provider sees one prefix, and the internal AS keeps the specifics because suppression happens per neighbour, not globally.

> **Scenario · The trace that stops at hop three**
> **A** is best. Correct. Time-exceeded messages and the final reply must be routed back to 10.0.0.1. A missing return route, filtered loopback range or uRPF drop at hop 3 produces exactly this pattern even though forward routing is complete.

> **Scenario · Ten minutes and no root cause**
> **B** is best. Correct. Time-boxing protects the overall score. The written notes mean you restart from evidence, not from scratch, and later tasks often reveal the fault indirectly.

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Case study · The forty-minute outage that took nine hours

BGP sessions establishing and then failing during large update transfers, combined with intermittent failures of larger application packets, is a classic MTU signature: small keepalives pass, full-size segments are dropped. The team skipped Scope and Layer: they never defined which packets failed and jumped straight to the protocol they could see flapping. Timer tuning appeared to help only because longer hold times delayed the reset while retransmissions occasionally got smaller segments through. For the review: the DF-bit ping series showing the exact maximum size, interface MTU on both ends, TCP retransmission counters for the BGP sessions, and the provider's documented service MTU. The fix was to set the interface and IP MTU correctly, then verify with the original failing flow.

Scenario · Ninety minutes left

D is best. Correct. The redistribution task has a dependency weight of 7 points. Solving it first also unlocks the BGP task, and the cheap syslog task fits in remaining time.

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02

MODULE 2 · LAYER2

Layer 2 at Scale

MST, spanning tree protection, first-hop security, multi-chassis aggregation, gateway redundancy and loop forensics

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Design and verify MST regions without pruning blackholes or accidental boundaries
- Harden spanning tree with the right guard at every trust boundary
- Deploy DHCP snooping, ARP inspection, source guard and storm control as one system
- Engineer multi-chassis aggregation and gateway redundancy that survive failures cleanly
- Deploy QinQ services and run a structured layer 2 loop investigation

LESSONS

1. MST Deep Dive: Regions, CIST and Boundaries · 40 min
2. STP Tuning, Protection and Failure Modes · 40 min
3. First-Hop Security at Scale · 40 min
4. Multi-Chassis Aggregation and First-Hop Redundancy · 45 min
5. Provider Bridging (QinQ) and Layer 2 Loop Forensics · 45 min

MODULE 2 · LAYER2 · LESSON 1 · 40 MIN

MST Deep Dive: Regions, CIST and Boundaries

How multiple spanning tree really builds its topologies and where it breaks

- Explain how an MST region is defined and why a single mismatch splits it
- Identify the CIST root, CIST regional root and boundary ports from show output
- Design VLAN-to-instance mappings and trunk pruning that cannot blackhole traffic

Per-VLAN spanning tree gives one topology per VLAN, which is flexible but expensive: one BPDU stream and one state machine per VLAN per port. Multiple Spanning Tree (IEEE 802.1s, now part of 802.1Q) maps many VLANs to a small number of **instances**, so a switch runs a handful of topologies regardless of how many VLANs exist. The price is a stricter model that punishes inconsistent configuration.

> DEFINITION · MST REGION
> A set of interconnected bridges that share exactly the same **region name**, **revision number** and **VLAN-to-instance mapping**. The mapping is compared as a digest carried in every MST BPDU. Any difference in any of the three places a bridge in a different region.

The MST hierarchy

CIST	`spans the whole bridged network` `each region appears as one virtual bridge` `interoperates with RSTP and legacy STP`
IST (MSTI 0)	`the CIST inside a region` `carries the region's link to the outside` `all unmapped VLANs belong here`
MSTI 1..n	`exist only inside the region` `carried as records inside the single MST BPDU` `independent root and port roles per instance`
Ports	`role and state per instance` `boundary ports follow the CIST role for all instances` `hop count limits instance diameter`

Roles you must recognise

ROLE OR BRIDGE	SCOPE	HOW IT IS DETERMINED
CIST root	Entire network	Lowest bridge ID among all bridges in all regions and any RSTP or legacy STP bridges
CIST regional root	One region	The bridge in the region with the lowest external path cost to the CIST root; the CIST root itself if it is inside the region
MSTI regional root	One instance in one region	Lowest bridge ID for that instance, using the per-instance priority
Boundary port	Region edge	A port connected to a bridge in a different region or to an RSTP or legacy STP bridge

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ROLE OR BRIDGE	SCOPE	HOW IT IS DETERMINED
Master port	Regional root of the CIST	The boundary port toward the CIST root; for every MSTI its role is master and it follows the CIST state

Building a consistent region

SW1 · IOS-style CLI

```
spanning-tree mode mst
spanning-tree mst configuration
 name DC-CORE
 revision 7
 instance 1 vlan 100-199
 instance 2 vlan 200-299
 exit
spanning-tree mst 0-1 priority 4096
spanning-tree mst 2 priority 8192
spanning-tree mst max-hops 20
!
! on SW2: same name, revision and mapping;
! priorities reversed for instance 2 so each core switch roots one instance
spanning-tree mst 0-1 priority 8192
spanning-tree mst 2 priority 4096
```

Changes in the MST configuration sub-mode only take effect when you leave it. Generate the region block from one source of truth and apply it identically everywhere, because a single typo creates a new region.

Spotting a region split

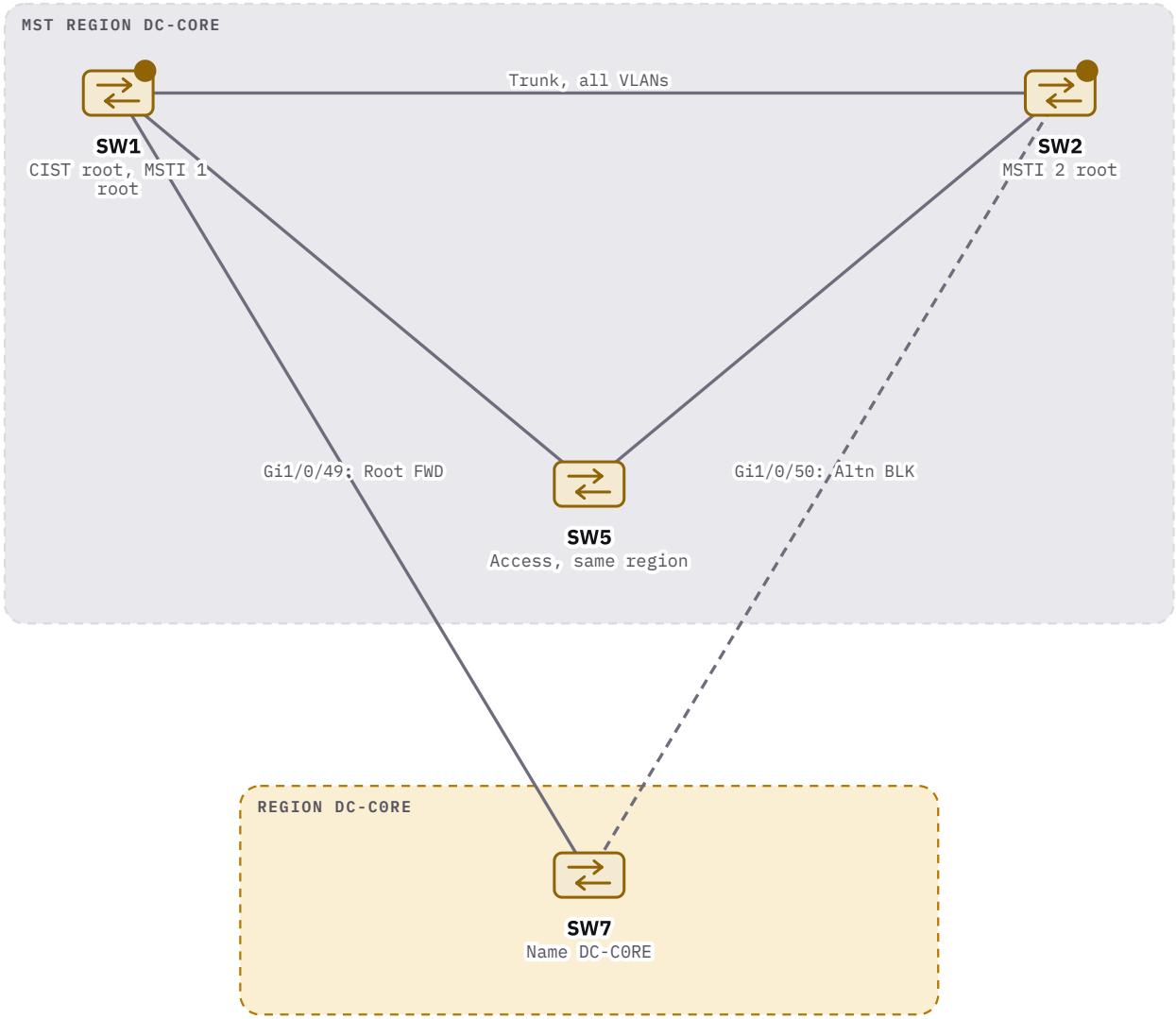
SW7 · IOS-style CLI

```
SW7# show spanning-tree mst configuration
Name      [DC-C0RE]
Revision  7     Instances configured 3
Instance  Vlans mapped
--------  ----------------------------------------
0         1-99,300-4094
1         100-199
2         200-299
SW7# show spanning-tree mst 0 | begin Interface
Interface        Role Sts Cost      Prio.Nbr Type
Gi1/0/49         Root FWD 20000     128.49   P2p Bound
Gi1/0/50         Altn BLK 20000     128.50   P2p Bound
```

The region name uses a zero instead of the letter O. SW7 is its own region, both uplinks are boundary ports, and at a boundary every instance follows the CIST role. Gi1/0/50 therefore blocks all VLANs, including those that the design intended to forward on it for instance 2.

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A one-character region split



SW7's region name has a zero instead of the letter O, so both of its uplinks are boundary ports. Every instance, including MSTI 2, follows the CIST role there, and Gi1/0/50 blocks all VLANs.

WATCH OUT · THE PRUNING BLACKHOLE

MST computes each instance topology without looking at which VLANs a trunk actually carries. If VLAN 210 is mapped to instance 2 but is allowed only on a link that instance 2 blocks, VLAN 210 is cut off even though spanning tree reports a healthy, loop-free topology. Inside a region, allow every VLAN of an instance on every trunk that participates in that instance, or prune only in ways that match the instance topology exactly.

Choosing a spanning tree model

Per-VLAN rapid STP

- One topology per VLAN
- Simple to load-share per VLAN
- Control-plane cost grows with VLAN count

MST

- Few instances, many VLANs
- Strict region consistency required
- Scales to thousands of VLANs

Single RSTP instance

- One topology for everything
- Simplest to operate
- No per-VLAN load sharing

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Migrating safely to MST

1. **Design the mapping once** Group VLANs by intended forwarding path and leave spare instances for growth; avoid remapping later because every change splits the region during rollout.
2. **Place the roots** Make the core switches the CIST root and the regional roots for every instance, so boundary behaviour stays predictable during migration.
3. **Pre-stage the region block** Configure the identical name, revision and mapping on every switch before changing the spanning tree mode.
4. **Convert from the core outward** Switch the core first, then distribution, then access, verifying root and port roles at each stage.
5. **Audit trunks** Compare allowed VLAN lists against instance topologies to remove pruning blackholes.

MST arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Remaining hops	`max_hops − bridges traversed inside the region`	BPDUs inside a region are discarded when the hop count reaches zero; the default limit is 20.
BPDU load, per-VLAN STP	`VLANs × trunk ports × 1 BPDU per hello`	Grows linearly with VLAN count.
BPDU load, MST	`trunk ports × 1 BPDU per hello (all instances inside)`	Independent of VLAN count; instance records ride in the same BPDU.

SCENARIO · DECIDE · BUILDING B LOSES VLAN 210

After a new access switch is installed in building B, users in VLAN 210 there lose connectivity while VLAN 110 users on the same switch work normally. On the new switch both uplinks show the Bound type in the instance 0 output and the second uplink is blocking. VLAN 210 is allowed only on the second uplink by design.

- (A) Fix the new switch's region name, revision or mapping, then audit trunk allowed lists.
- (B) Lower the instance 2 priority on the new switch so it becomes the regional root for instance 2.
- (C) Disable spanning tree on the second uplink so it forwards all VLANs regardless of role.
- (D) Configure the second uplink as an edge port so it moves to forwarding immediately.

Circle your choice. Answer and feedback: module answer key.

MST health check

☐ Region name, revision and digest identical on every switch in the region
☐ CIST root and every MSTI regional root on intended core switches
☐ No unexpected Bound ports inside the region
☐ Trunk allowed lists consistent with each instance topology
☐ Spare instances reserved to avoid remapping later
☐ Max hops sufficient for the region diameter

Key terms

MST region
Bridges with identical region name, revision number and VLAN-to-instance mapping.

CIST
Common and internal spanning tree: the single tree that spans all regions and non-MST bridges.

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IST
Internal spanning tree: instance 0, the portion of the CIST inside a region.

CIST regional root
The bridge in a region with the lowest external path cost to the CIST root.

Master port
The boundary port on the CIST regional root that leads toward the CIST root.

Max hops
The hop count limit for BPDUs inside an MST region, replacing message age within the region.

MSTI
A multiple spanning tree instance that exists only inside its region and carries a group of VLANs.

Boundary port
A port that connects to another region or to an RSTP or legacy STP bridge; all instances follow the CIST role on it.

Configuration digest
A hash of the VLAN-to-instance table carried in MST BPDUs to detect mapping mismatches.

REFLECT

Draw your current or a familiar campus as MST regions. Where are the boundaries, where is the CIST root, and which VLANs would be at risk from a pruning blackhole?

Key takeaways

- An MST region is defined by identical name, revision and VLAN-to-instance mapping; one typo creates a new region.
- At a region boundary every instance follows the CIST port role.
- MST ignores trunk allowed lists, so pruning that disagrees with the instance topology blackholes VLANs.
- Design the mapping once, root every instance on the core and pre-stage the region block before migrating.

KNOWLEDGE CHECK

1. Two switches have the same region name and revision but different VLAN-to-instance mappings. What happens?

   (A) They form different regions because the configuration digests differ

   (B) They form one region, and the switch with the lower bridge ID wins the mapping

   (C) They form one region and only the VLANs mapped identically are protected

   (D) They stop exchanging BPDUs and both treat the link as an edge port

2. On a boundary port, how are MST instances handled?

   (A) Each instance keeps its own role, computed from the external neighbour

   (B) Only instance 0 is blocked while other instances forward normally

   (C) Every instance follows the port role and the forwarding state of the CIST

   (D) All instances forward because boundary ports cannot create loops

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3. VLAN 210 is mapped to instance 2 and allowed only on trunk T2, which instance 2 blocks. What is the effect?

(A) Spanning tree unblocks T2 for VLAN 210 because it is the only path

(B) VLAN 210 is isolated even though spanning tree reports no problem

(C) VLAN 210 automatically moves to instance 0 for the affected trunk

(D) Instance 2 recalculates a new topology based on the allowed list

4. Which bridge is the CIST regional root when the CIST root is outside the region?

(A) The bridge in the region with the lowest instance 0 priority value

(B) The bridge in the region with the highest number of boundary ports

(C) The bridge that holds the regional root role for most of the instances

(D) The bridge with the lowest external path cost to the CIST root

5. What replaces message age as the diameter control inside an MST region?

(A) A per-instance timer that doubles every time a BPDU is relayed

(B) The forward delay timer applied separately to each instance

(C) A sequence number carried in each of the instance records

(D) A hop count that is decremented by each bridge in the region

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · LAYER2 · LESSON 2 · 40 MIN

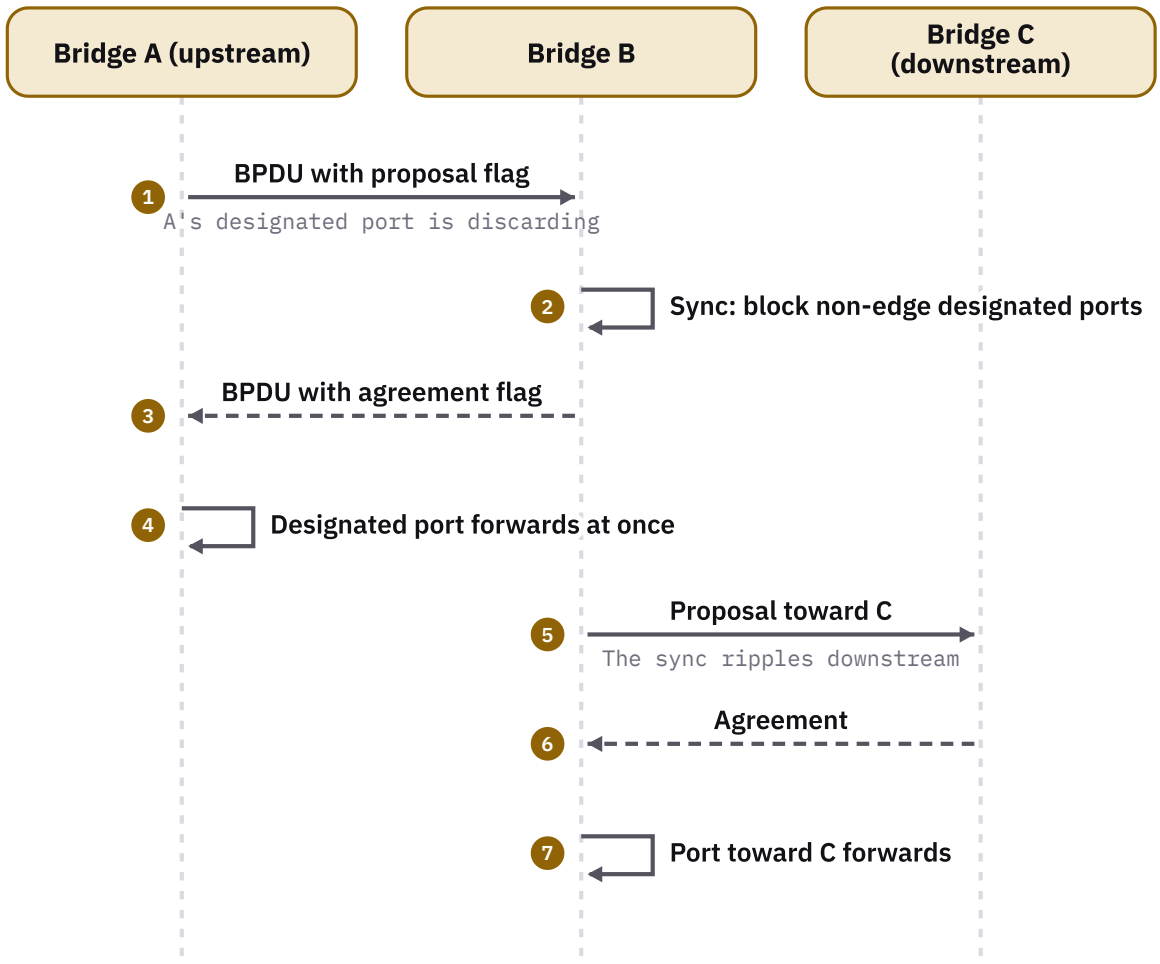
STP Tuning, Protection and Failure Modes

Rapid convergence mechanics, guard features and how spanning tree really fails

- Explain rapid spanning tree proposal and agreement and topology change handling
- Place BPDU guard, root guard, loop guard and unidirectional link detection correctly
- Diagnose the classic spanning tree failure modes from logs and counters

Rapid spanning tree converges in milliseconds on point-to-point links because it does not wait for timers. A designated port sends a **proposal**; the downstream bridge **syncs** by blocking its non-edge designated ports, then returns an **agreement**, and the designated port forwards immediately. The sync then ripples downstream. Timers matter only on shared segments, toward legacy bridges, and to detect lost BPDUs.

Proposal and agreement on one link



No timer appears anywhere: each link converges as soon as the downstream bridge has synced, and the handshake then ripples one link further.

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INSIGHT · TOPOLOGY CHANGES ARE A FLUSH, NOT A RECALCULATION

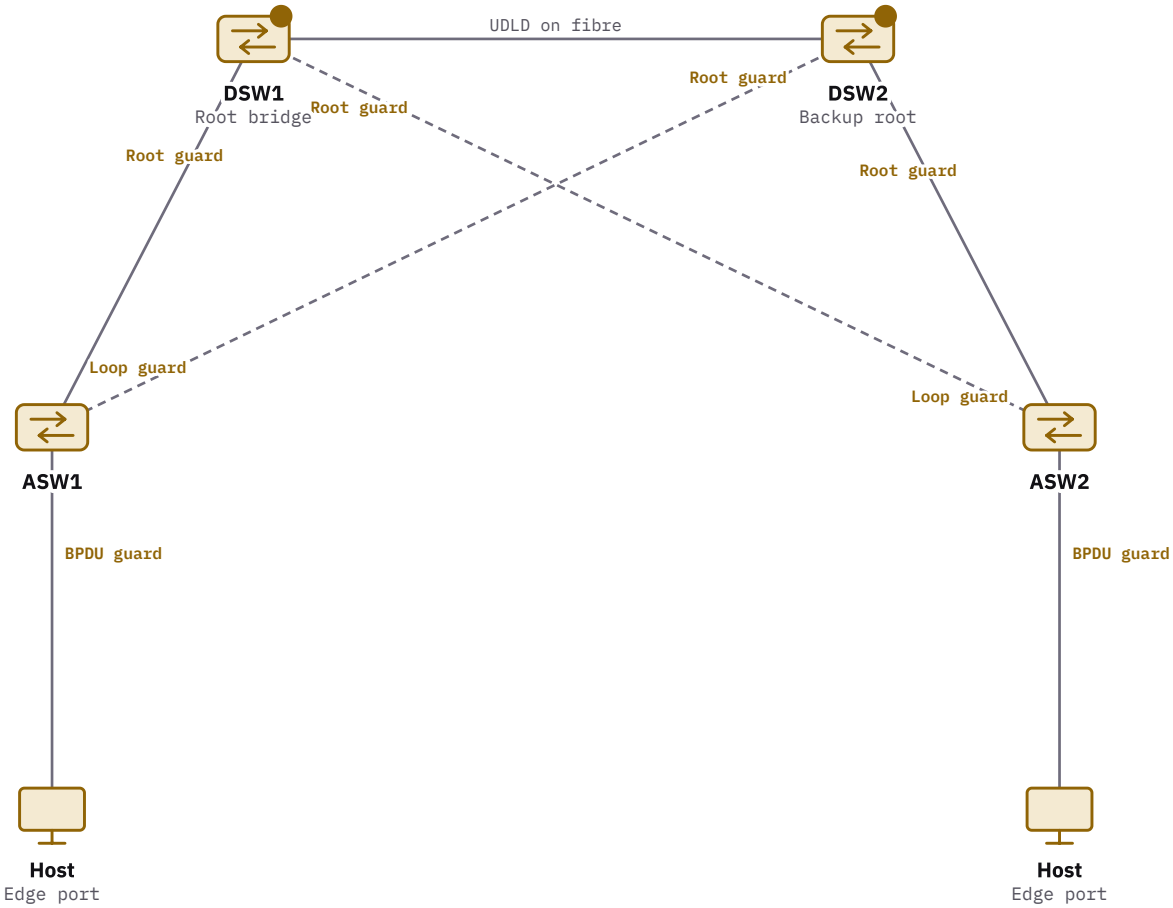
In rapid spanning tree only a non-edge port moving to forwarding generates a topology change. The bridge flushes MAC addresses learned on its other non-edge ports and floods the change. In large layer 2 domains, a single access port without edge configuration that flaps every minute causes network-wide MAC flushing and bursts of unknown unicast flooding.

Protection features and where they belong

FEATURE	APPLY ON	TRIGGERS ON	ACTION
BPDU guard	Edge ports to hosts	Any BPDU received	Error-disable the port
BPDU filter	Rarely, edge ports only	Suppresses sending and receiving BPDUs	No protection; can create silent loops
Root guard	Designated ports toward switches that must never be root	A superior BPDU	Root-inconsistent blocking until superior BPDUs stop
Loop guard	Root and alternate ports on point-to-point links	BPDUs stop arriving	Loop-inconsistent blocking instead of becoming designated
Unidirectional link detection	Fibre and long inter-switch links	Peer stops seeing our echoes	Error-disable the link
Storm control	Access ports and selected trunks	Broadcast, multicast or unknown unicast above threshold	Drop excess, trap or shut down

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Where each guard belongs



Root guard faces down from the distribution layer, loop guard protects the non-designated uplinks of access switches, and BPDU guard sits on every host-facing edge port.

Hardening access and distribution

DSW1 · IOS-style CLI

```
spanning-tree loopguard default
errdisable recovery cause bpduguard
errdisable recovery interval 300
!
interface range GigabitEthernet1/0/1 - 44
 description ACCESS-HOSTS
 switchport mode access
 spanning-tree portfast edge
 spanning-tree bpduguard enable
!
interface range TenGigabitEthernet1/1/1 - 4
 description DOWNLINKS-TO-ACCESS
 spanning-tree guard root
```

Root guard sits on the distribution side of links toward access switches, so no access device can ever take over the root. Loop guard is enabled globally but only acts on non-designated point-to-point ports.

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Reading inconsistent ports

DSW1 · IOS-style CLI

```
DSW1# show spanning-tree inconsistentports
Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------
MST0000              TenGigabitEthernet1/1/3  Root Inconsistent
MST0002              TenGigabitEthernet1/2/1  Loop Inconsistent
Number of inconsistent ports (segments) in the system : 2
DSW1# show logging | include GUARD
%STP-2-ROOTGUARD_BLOCK: Root guard blocking port Te1/1/3 on MST0000
%STP-2-LOOPGUARD_BLOCK: Loop guard blocking port Te1/2/1 on MST0002
```

Te1/1/3 is receiving superior BPDUs from an access switch, so somebody raised its priority or connected a rogue bridge. Te1/2/1 stopped receiving BPDUs on a non-designated port, which strongly suggests a unidirectional fault. Both ports block only the affected instance and recover automatically when the condition clears.

How a unidirectional fibre creates a loop

01 Fault — The fibre from A to B fails in one direction

→

02 Silence — B's alternate port stops receiving A's BPDUs

→

03 Expiry — After three missed hellos B ages out A's information

→

04 Promotion — B makes the port designated and forwarding

→

05 Loop — A still forwards toward B, so a loop forms

Timer relationships

MEASURE	FORMULA	WHAT IT TELLS YOU
Legacy worst-case convergence	`max age + 2 × forward delay = 20 + 2 × 15 = 50 s`	Why legacy timers are unacceptable and why rapid spanning tree relies on handshakes instead.
Rapid BPDU-loss detection	`3 × hello = 3 × 2 s = 6 s`	Information is aged out after three missed hellos on a rapid spanning tree port.
Legacy timer constraints	`2 × (forward delay − 1) ≥ max age ≥ 2 × (hello + 1)`	Keep tuned timers inside these limits or the protocol may converge before BPDUs have propagated.

Tune or do not tune

Worth tuning

- Root and backup root priorities
- Port cost on links that must be preferred
- Edge configuration on every host port
- Guard features at every trust boundary

Usually leave alone

- Hello, forward delay and max age in rapid domains
- Max hops, unless the region is very large
- Port priority, except to break deliberate ties

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SCENARIO · DECIDE · A CONTRACTOR'S SWITCH BECOMES ROOT

A contractor connected a small switch with bridge priority 0 to an access port in a meeting room. It became the root bridge, and traffic between two distribution switches started hairpinning through a 1 Gb/s access uplink, saturating it. You must prevent a recurrence.

(A) Set the core switches to priority 0 so they always win the root election.

(B) Enable BPDU filter on all access ports so foreign BPDUs are ignored.

(C) Enable loop guard on all access ports so they block when anomalies occur.

(D) Enable BPDU guard on all host-facing edge ports and root guard on distribution ports facing access switches.

Circle your choice. Answer and feedback: module answer key.

Spanning tree hardening

☐ Every host port configured as edge with BPDU guard

☐ Root guard on distribution ports facing access switches

☐ Loop guard or equivalent on point-to-point non-designated ports

☐ Unidirectional link detection on fibre inter-switch links

☐ BPDU filter absent, or justified and documented

☐ Root and backup root priorities set explicitly

☐ Topology change counters baselined and monitored

Key terms

Proposal and agreement
The rapid spanning tree handshake that lets a designated port forward immediately on a point-to-point link.

Edge port
A port declared to connect only to end hosts; it forwards immediately and does not generate topology changes.

Root guard
Blocks a port that receives a superior BPDU, protecting the root placement.

Unidirectional link
A link that passes traffic in only one direction, which can defeat spanning tree's loop prevention.

Root-inconsistent
The blocking state applied by root guard while superior BPDUs are received.

Sync
Blocking all non-edge designated ports before agreeing to a proposal, so no temporary loop can form.

BPDU guard
Error-disables an edge port that receives any BPDU.

Loop guard
Prevents a non-designated port from becoming designated when BPDUs stop arriving.

Topology change
A notification that triggers MAC flushing; in rapid spanning tree it is caused by a non-edge port moving to forwarding.

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REFLECT

Where in your network would a single misbehaving port cause topology changes across the whole layer 2 domain? What would you configure there first?

Key takeaways

- Rapid spanning tree converges by proposal and agreement on point-to-point links, not by timers.
- Only non-edge ports moving to forwarding cause topology changes and MAC flushes, so make every host port an edge port.
- BPDU guard on host ports, root guard toward access switches, loop guard or UDLD on point-to-point uplinks.
- BPDU filter hides loops from spanning tree; treat every use as a documented exception.

KNOWLEDGE CHECK

6. Which event generates a topology change in rapid spanning tree?

   (A) Any port transitioning from forwarding to the blocking state
   (B) An edge port moving to forwarding after its link comes up
   (C) A non-edge port transitioning to the forwarding state
   (D) A BPDU arriving with a lower root path cost than before

7. Where should root guard be configured?

   (A) On designated ports facing switches that must never become root
   (B) On the root ports of access switches that face the distribution layer
   (C) On every edge port that connects directly to an end host
   (D) On alternate ports of switches that sit in the core layer

8. A point-to-point uplink is reported as loop-inconsistent. What does it indicate?

   (A) A superior BPDU arrived from a switch with a better priority
   (B) BPDUs stopped arriving on a port that was root or alternate
   (C) The port received a BPDU while configured as an edge port
   (D) Two ports on the same switch received the same BPDU

9. Why is BPDU filter considered dangerous on host ports?

   (A) A looped cable or bridge on that port is invisible to spanning tree
   (B) It increases the BPDU rate on the port and loads the switch CPU
   (C) It forces the port into blocking whenever a host first connects
   (D) It disables the edge behaviour so every link-up becomes a topology change

10. What is the worst-case convergence with legacy default timers?

    (A) About 30 seconds
    (B) About 20 seconds
    (C) About 50 seconds
    (D) About 6 seconds

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · LAYER2 · LESSON 3 · 40 MIN

First-Hop Security at Scale

DHCP snooping, dynamic ARP inspection, IP source guard and storm control as one system

- Explain how DHCP snooping bindings feed ARP inspection and source guard
- Deploy first-hop security without breaking static hosts, relays or reloads
- Calculate and apply storm-control thresholds from line rate and frame size

The access edge is where most layer 2 attacks start: rogue DHCP servers, ARP spoofing, source address spoofing and floods. The defences are designed as a **chain**: DHCP snooping learns who legitimately owns which address, and every later feature trusts that table. Get the first link wrong and every later feature either fails open or drops legitimate traffic.

The first-hop security chain

01 **DHCP snooping**
Blocks server replies on untrusted ports and builds the binding table

→

02 **Binding table**
MAC, IP, VLAN, port and lease for every client

→

03 **ARP inspection**
Drops ARP whose sender MAC and IP do not match a binding

→

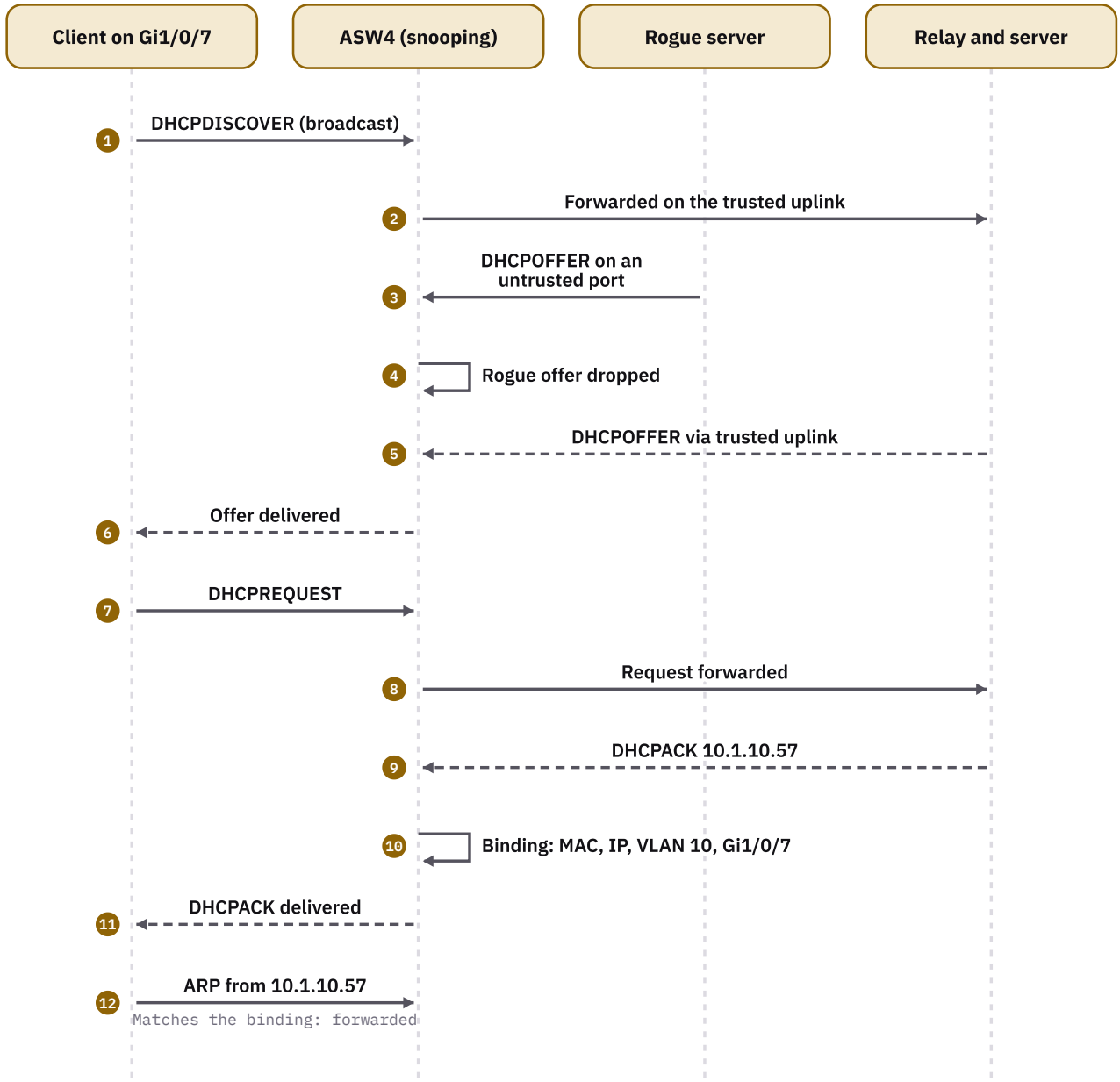
04 **IP source guard**
Filters data-plane packets per port by source IP and optionally MAC

→

05 **Storm control**
Caps broadcast, multicast and unknown unicast rates

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Snooping builds the binding that ARP inspection checks



The rogue offer dies on an untrusted port, the ACK through the trusted uplink creates the binding, and every later ARP from the client is checked against it.

What each feature validates

FEATURE	VALIDATES	DATA SOURCE	TYPICAL FAILURE WHEN MISCONFIGURED
DHCP snooping	Server messages arrive only on trusted ports	Trust configuration	Clients get no lease when the uplink is not trusted
Option 82 insertion	Port and switch identity in client requests	Snooping switch	Upstream relay discards requests with option 82 and a zero gateway address
ARP inspection	Sender MAC and IP in ARP match a binding	Binding table or ARP ACL	Static-address hosts and gateways lose connectivity

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FEATURE	VALIDATES	DATA SOURCE	TYPICAL FAILURE WHEN MISCONFIGURED
IP source guard	Source IP (and MAC) per port	Binding table	All client traffic dropped after a reload until leases renew
Storm control	Traffic class rate per second	Configured threshold	Legitimate bursts dropped, or a port shut down

Access switch first-hop security

ASW4 · IOS-style CLI

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
no ip dhcp snooping information option
ip dhcp snooping database flash:dhcp-bindings.db
ip arp inspection vlan 10,20
ip arp inspection validate src-mac dst-mac ip
!
interface TenGigabitEthernet1/1/1
 description UPLINK-TO-DSW1
 ip dhcp snooping trust
 ip arp inspection trust
!
interface range GigabitEthernet1/0/1 - 40
 ip dhcp snooping limit rate 15
 ip verify source
 storm-control broadcast level 1.00
 storm-control action trap
```

Option 82 insertion is disabled here because the distribution layer is the relay and does not trust option 82 with a zero gateway address. The binding database is stored persistently so source guard does not drop every client after a reload.

Reading bindings and drops

ASW4 · IOS-style CLI

```
ASW4# show ip dhcp snooping binding
MacAddress          IpAddress        Lease(sec)  Type           VLAN  Interface
------------------  ---------------  ----------  -------------  ----  --------------
00:50:56:AA:01:07   10.1.10.57       85912       dhcp-snooping   10   Gi1/0/7
00:50:56:AA:01:12   10.1.20.33       86104       dhcp-snooping   20   Gi1/0/12
ASW4# show ip arp inspection statistics vlan 10
 Vlan      Forwarded        Dropped     DHCP Drops      ACL Drops
 ----      ---------        -------     ----------      ---------
   10          48211           3302           3302              0
%ARP-4-DAI_DROP: Invalid ARP (Res) on Gi1/0/31, vlan 10 ([0050.56aa.0144/10.1.10.200]
```

All drops are DHCP drops: ARP from senders with no matching binding. Gi1/0/31 hosts a printer with a static address 10.1.10.200, which never appears in the binding table. That is a deployment gap, not an attack.

WATCH OUT · OPTION 82 AND THE ZERO GATEWAY ADDRESS

A snooping switch that is not itself the relay can insert option 82 into client requests while leaving the relay gateway address at zero. Some relay agents treat that combination as untrusted and silently discard it. Either disable insertion on the access switch, or configure the relay to trust option 82 on the interfaces facing your snooping switches.

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PRO TIP · STATIC HOSTS AND RELOADS

Give statically addressed hosts static bindings or ARP ACL entries before enabling inspection, and store the binding table persistently. Without persistence, a switch reload empties the table and source guard drops every client until each renews its lease, which can take hours.

WATCH OUT · IPV6 NEEDS ITS OWN FIRST-HOP SECURITY

Everything above protects IPv4. IPv6 hosts learn their gateway from router advertisements and their addresses from SLAAC or DHCPv6, so any access VLAN with IPv6 enabled, even unintentionally, needs **RA guard**, **DHCPv6 guard** and neighbour-discovery inspection or IPv6 source guard built from an IPv6 binding table. A single rogue router advertisement can redirect a whole VLAN while every IPv4 control reports a clean bill of health.

Storm-control arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Threshold bandwidth	`level % × line rate`	1.00% of 1 Gb/s is 10 Mb/s of the controlled traffic class.
Threshold in packets	`threshold bandwidth ÷ ((frame bytes + 20) × 8)`	For minimum 64-byte frames plus 20 bytes of preamble and gap: 10 Mb/s ÷ 672 bits ≈ 14,880 pps.
Measurement window	`rate averaged per 1 s interval`	Short bursts above the level inside one interval are not distinguished from sustained floods.

TEMPLATE **First-hop security rollout matrix**

Plan trust, inspection and exceptions per VLAN and port class before enabling enforcement.

VLAN	PORT CLASS	SNOOPING TRUST	ARP INSPECTION	SOURCE GUARD	STORM LEVEL	EXCEPTIONS
10 Users	*Access host*	*Untrusted*	*On*	*On*	*Broadcast 1%*	*None*
40 Printers	*Access host*	*Untrusted*	*On + ARP ACL*	*Off*	*Broadcast 1%*	*Static IPs 10.1.40.0/26*

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SCENARIO · DECIDE · PRINTERS VANISH AFTER ENFORCEMENT

After enabling ARP inspection on VLAN 40, all DHCP clients work but every statically addressed printer in the VLAN becomes unreachable. Statistics show only DHCP drops, and the logs show invalid ARP replies from the printers' own addresses.

A. Configure all access ports in VLAN 40 as trusted for ARP inspection.

B. Disable the source MAC validation check so the printers' replies are no longer rejected.

C. Add ARP ACL entries or static bindings for the printers' MAC and IP pairs.

D. Raise the ARP inspection rate limit on the printer ports to accept more ARP traffic.

Circle your choice. Answer and feedback: module answer key.

First-hop security deployment

☐ Uplinks and legitimate DHCP server ports trusted for snooping

☐ Option 82 behaviour agreed between access switch and relay

☐ Binding table stored persistently

☐ Static hosts covered by static bindings or ARP ACLs

☐ Inspection run in monitor or logging mode before enforcement where available

☐ Storm-control levels derived from measured baseline traffic

☐ Error-disable recovery and alerting configured

Key terms

DHCP snooping
A switch function that permits DHCP server messages only on trusted ports and records client leases.

Option 82
The DHCP relay agent information option that identifies the switch and port a request came from.

ARP ACL
A static list of permitted MAC and IP pairs used by ARP inspection for hosts without DHCP leases.

Storm control
Rate limiting of broadcast, multicast or unknown unicast traffic per port.

Binding table
The snooping database of client MAC, IP, VLAN, port and lease.

Dynamic ARP inspection
Validation of ARP packets on untrusted ports against bindings or ARP ACLs.

IP source guard
A per-port data-plane filter that permits only source addresses found in the binding table.

Trusted port
A port exempt from snooping and inspection checks, normally an uplink toward infrastructure.

REFLECT

List the statically addressed devices in one of your access VLANs. How would you discover the ones you do not know about before enabling ARP inspection?

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Key takeaways

- DHCP snooping is the root of the chain: trust only uplinks and legitimate server ports.
- ARP inspection and source guard trust the binding table, so cover static hosts and store bindings persistently.
- Agree option 82 behaviour between snooping switches and relays before enforcing.
- IPv6 needs its own controls: RA guard, DHCPv6 guard and neighbour-discovery inspection.

KNOWLEDGE CHECK

11. After a switch reload, all clients on ports with IP source guard lose connectivity until they renew. What prevents this?

    (A) Trusting every access port for DHCP snooping during the boot period
    (B) Raising the DHCP snooping rate limit on all of the access ports
    (C) Enabling option 82 insertion on the access switch for all VLANs
    (D) Storing the snooping binding table persistently across reloads

12. Clients get no address after snooping is enabled. The server is reached through the uplink. What is the first thing to check?

    (A) Whether ARP inspection is enabled on the same VLANs as snooping
    (B) Whether the uplink toward the server or relay is trusted for DHCP snooping
    (C) Whether storm control is dropping broadcast DHCP discover messages
    (D) Whether IP source guard is configured on the uplink interface

13. What is 1% of a 1 Gb/s port expressed in minimum-size frames per second, including preamble and gap?

    (A) About 14,900 packets per second
    (B) About 19,500 packets per second
    (C) About 10,000 packets per second
    (D) About 1,488 packets per second

14. ARP inspection drops replies from a statically addressed gateway on an access port. What is the best fix?

    (A) Disable the IP validation option of ARP inspection for the whole VLAN
    (B) Disable DHCP snooping on the VLAN that contains the gateway
    (C) Enable IP source guard on the port that connects the gateway
    (D) Add an ARP ACL or static binding for the gateway's MAC and IP pair

15. Why might an upstream relay discard DHCP requests from a snooping switch?

    (A) They arrive with a source MAC address that is already in the binding table
    (B) They exceed the per-port DHCP snooping rate limit on the uplink
    (C) They carry option 82 while the relay gateway address field is zero
    (D) They arrive on a VLAN in which ARP inspection is also enabled

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · LAYER2 · LESSON 4 · 45 MIN

Multi-Chassis Aggregation and First-Hop Redundancy

Active-active uplinks, split-brain behaviour and gateway redundancy that converges cleanly

- Explain multi-chassis link aggregation components and failure handling
- Configure VRRP with tracking, preemption delay and load sharing
- Align gateway redundancy with spanning tree and multi-chassis designs

Multi-chassis link aggregation lets a downstream device bundle links to two separate switches as if they were one. Both uplinks forward, spanning tree sees a single logical link, and a switch failure costs only bandwidth. The two peers must behave as one LACP partner and must never let a frame loop between them. Implementations differ in names and details, but the architecture is common.

Multi-chassis aggregation components

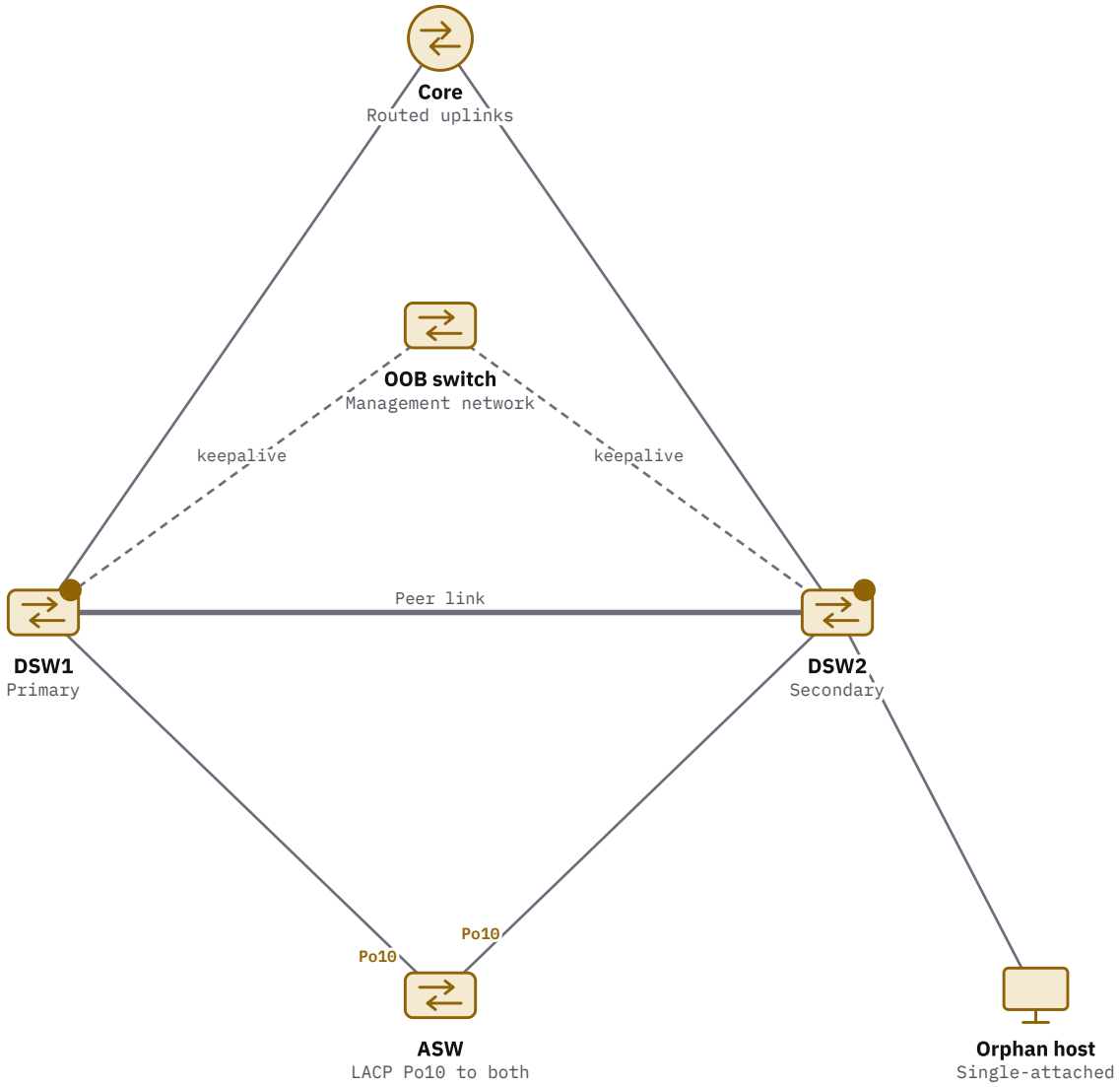
Downstream device	`standard LACP bundle` `sees one partner system ID` `hashes flows across member links`
Member ports	`one or more on each peer` `same bundle ID on both peers` `configuration must match`
Peer link	`synchronises MAC and ARP state` `carries traffic for single-attached hosts` `carries flooded traffic between peers`
Peer keepalive	`separate path, often out of band` `detects whether the peer is alive` `decides split-brain behaviour`
Control roles	`primary and secondary` `shared LACP system ID and priority` `consistency checks between peers`

INSIGHT · THE LOOP-PREVENTION RULE

A frame that arrives over the peer link is not forwarded out of a multi-chassis member port while the peer's member port in the same bundle is up. The peer has already delivered it locally. That single rule is what allows both sides to forward without spanning tree blocking anything.

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A multi-chassis pair and its failure domains



Both peers forward for Po10 and present one LACP system ID. The keepalive takes a separate out-of-band path so a peer-link failure can be told apart from a dead peer; the orphan host is the first casualty when the secondary suspends its ports.

Failure handling

FAILURE	TYPICAL BEHAVIOUR	DESIGN GUARD
Peer link down, keepalive up	Secondary suspends its member ports and often its gateway interfaces	Dual-home every device; treat single-attached ports as at risk
Keepalive down, peer link up	No traffic impact; alarm raised	Monitor keepalive; never run it over the peer link
Both down	Dual-active: both peers act alone with the same system ID	Separate physical paths and modules for peer link and keepalive
Primary reloads and returns	Member ports come up before routing converges	Restore delay keeps ports down until the control plane is ready

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FAILURE	TYPICAL BEHAVIOUR	DESIGN GUARD
Single-attached host on secondary	Isolated when the secondary suspends its ports	Avoid single attachment or move such hosts to the primary

INSIGHT · WHERE MULTI-CHASSIS AGGREGATION IS HEADING

In data-centre fabrics built today, standards-based **EVPN multihoming** (Ethernet segments signalled in BGP, covered in the Fabric module) increasingly replaces peer-link pairs: no peer link, more than two leaves per segment and mixed platforms. Peer-link pairs remain common in campus and brownfield networks, so an expert must be able to troubleshoot both.

Gateway redundancy options

VRRP version 3 (RFC 9568, which replaced RFC 5798)

- One master forwards, backups listen
- Preemption enabled by default
- Owner of the address uses priority 255
- Virtual MAC 00-00-5E-00-01-xx for IPv4
- Advertisements to 224.0.0.18, IP protocol 112

Hot-standby style FHRPs

- Active and standby with separate hello and hold timers
- Preemption often disabled by default
- Separate virtual MAC ranges per implementation
- Similar tracking and priority logic

Distributed anycast gateway

- Every switch is an active gateway
- Same virtual IP and MAC everywhere
- No election and no hairpinning
- Requires a fabric that distributes host routes

VRRP with tracking and preemption delay

DSW1 · IOS-style CLI

```
track 10 interface TenGigabitEthernet1/1/8 line-protocol
track 20 ip route 0.0.0.0 0.0.0.0 reachability
track 30 list boolean and
 object 10
 object 20
!
interface Vlan10
 ip address 10.10.10.2 255.255.255.0
 vrrp 10 ip 10.10.10.1
 vrrp 10 priority 120
 vrrp 10 preempt delay minimum 60
 vrrp 10 track 30 decrement 30
```

Object 30 is up only while both the core uplink and the default route are present. When either fails, object 30 goes down and the decrement drops the priority from 120 to 90, below the peer's default 100. The preemption delay keeps the switch from reclaiming the master role until 60 seconds after it becomes eligible, giving routing time to converge after a reload.

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Reading VRRP state

DSW1 · IOS-style CLI

```
DSW1# show vrrp brief
Interface   Grp Pri Time  Own Pre State   Master addr     Group addr
Vl10         10  90 3648       Y  Backup  10.10.10.3      10.10.10.1
Vl120        20 120 3531       Y  Master  10.10.20.2      10.10.20.1
DSW1# show track 30 brief
Track Type        Instance                  Parameter        State Last Change
30    list        boolean                   and              Down  00:02:14
```

Priority 90 on VLAN 10 shows the tracking decrement is active; the Time column is the master down interval in milliseconds, which depends on priority. VLAN 20 is still master because its group does not track object 30, so traffic for VLAN 20 is drawn to a switch with a failed uplink. That is a design gap, not a VRRP fault.

VRRP timing (version 3)

MEASURE	FORMULA	WHAT IT TELLS YOU
Skew time	`((256 − priority) × advertisement interval) ÷ 256`	Lower-priority backups wait slightly longer, so the best backup takes over first.
Master down interval	`3 × advertisement interval + skew time`	With a 1 s interval and priority 90: 3 + 0.648 = 3.648 s.
Effective priority	`configured priority − sum of active decrements`	Must fall below the peer's priority for failover to happen; check it, do not assume it.

Load sharing gateways

1. **Split by VLAN** Make each distribution switch master for half the VLANs, using priority per group.
2. **Align with spanning tree** Without multi-chassis aggregation, make the master for a VLAN also the root for its instance, so traffic does not cross the inter-switch link.
3. **Track per group** Apply tracking to every group, so a failed uplink moves all of that switch's masters.
4. **Delay preemption** Use a preemption delay longer than routing convergence after a reload.
5. **Prefer active-active where available** With multi-chassis aggregation, implementations can let both peers forward for the virtual MAC, removing the hairpin entirely.

SCENARIO · DECIDE · FORTY SECONDS OF LOSS AFTER A REBOOT

The primary distribution switch is reloaded during maintenance. Failover to the backup gateway is fast. When the primary returns, users lose connectivity for about 40 seconds even though no further fault occurs.

- (A) Disable tracking on the primary switch so its priority stays stable after the reload.
- (B) Configure a preemption delay longer than the switch's routing and uplink convergence time.
- (C) Shorten the advertisement interval so the backup detects the new master more quickly.
- (D) Raise the backup switch's priority above the primary to stop preemption entirely.

Circle your choice. Answer and feedback: module answer key.

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Gateway redundancy review

☐ Master per VLAN aligned with spanning tree root or multi-chassis design

☐ Tracking applied to every group, not only some

☐ Effective priority after decrement falls below the peer

☐ Preemption delay exceeds routing convergence after reload

☐ Peer link and keepalive on separate physical paths

☐ No critical single-attached devices on the secondary peer

☐ Restore delay configured for multi-chassis member ports

Key terms

Multi-chassis link aggregation
Bundling links from one device to two switches that present a single LACP partner identity.

Peer keepalive
A separate heartbeat path used to tell a dead peer from a failed peer link.

Orphan port
A port attached to only one multi-chassis peer.

Preemption delay
The time a higher-priority router waits before reclaiming the master role.

Master down interval
The time a VRRP backup waits without advertisements before declaring itself master.

Peer link
The inter-switch link that synchronises state and carries traffic between multi-chassis peers.

Dual-active
A split-brain condition in which both multi-chassis peers act independently.

VRRP
Virtual Router Redundancy Protocol, a standard first-hop redundancy protocol defined in RFC 5798.

Object tracking
Monitoring an interface, route or other condition and adjusting priority when it changes.

REFLECT

In your environment, which devices are attached to only one switch of a redundant pair? What would they experience during a peer-link failure?

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Key takeaways

- Multi-chassis aggregation works because frames from the peer link never leave a member port whose peer member is up.
- Keep the peer link and keepalive on separate physical paths and modules, and avoid single-attached devices on the secondary.
- VRRP preempts by default; add a preemption delay longer than routing convergence after a reload.
- Tracking must drop the master strictly below the backup, and the master for each VLAN should align with its spanning-tree root.

KNOWLEDGE CHECK

16. The peer link of a multi-chassis pair fails while the keepalive is healthy. What is the typical behaviour?

(A) Both peers keep forwarding independently with the same LACP system ID

(B) The secondary suspends its member ports to avoid dual-active

(C) The primary peer suspends its member ports and hands over its role

(D) Spanning tree blocks one of the member links in every bundle

17. A VRRP backup with priority 90 and a 1 s advertisement interval waits how long before taking over?

(A) About 3.00 seconds

(B) About 3.65 seconds

(C) About 3.35 seconds

(D) About 4.00 seconds

18. Why should the keepalive never run over the peer link?

(A) The keepalive needs more bandwidth than the peer link can reserve

(B) LACP packets would be mistaken for keepalive messages on that link

(C) A peer-link failure would then also cut the keepalive

(D) The peer link is blocked by spanning tree during normal operation

19. Tracking lowers a VRRP master's priority from 110 to 100, equal to the backup. What happens?

(A) No failover occurs, because a backup does not preempt a master of equal priority

(B) The backup becomes master, because the higher primary IP address always wins the tie

(C) Both routers become master and share traffic by virtual MAC

(D) Failover occurs because any tracked object failure forces a change

20. Without multi-chassis aggregation, why should the VRRP master for a VLAN also be the spanning tree root for it?

(A) VRRP advertisements are dropped on ports that spanning tree blocks

(B) The virtual MAC address can only be learned on the root bridge

(C) Preemption only works when the master is also the root bridge

(D) Otherwise host traffic to the gateway crosses an extra inter-switch hop

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MODULE 2 · LAYER2 · LESSON 5 · 45 MIN

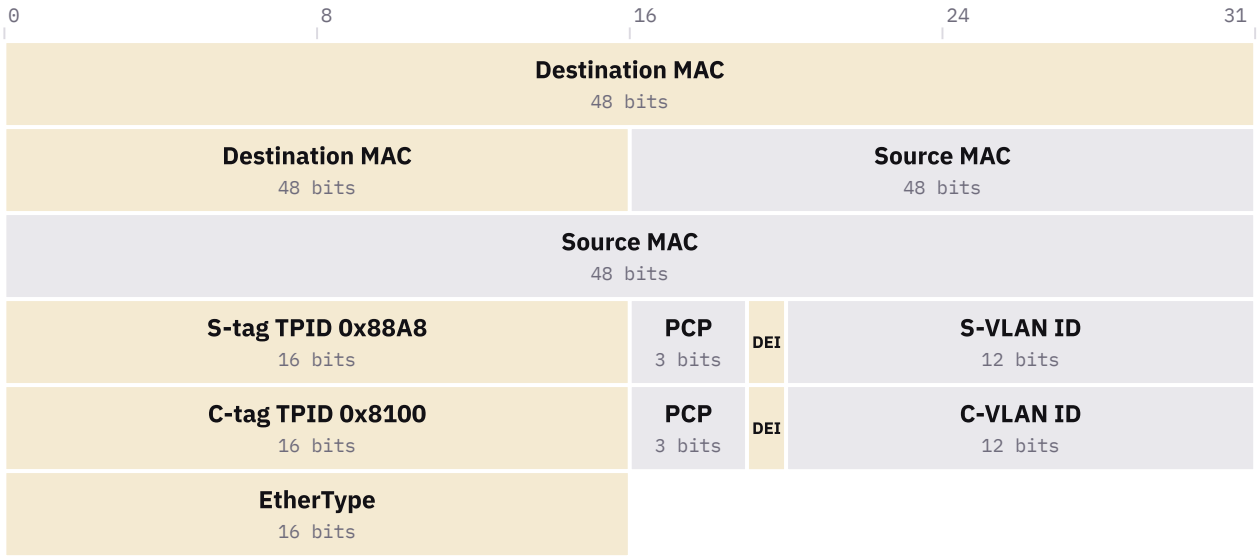
Provider Bridging (QinQ) and Layer 2 Loop Forensics

Stacked VLANs over a provider core, and how to find and stop a loop under pressure

- Explain IEEE 802.1ad tagging, MTU impact and control-protocol tunnelling
- Avoid native VLAN and MTU traps in QinQ deployments
- Run a structured loop forensic process from symptom to root cause

Provider bridging (IEEE 802.1ad, commonly called QinQ) lets a provider carry a customer's entire VLAN space across its network inside a single service VLAN. The provider edge pushes an outer **S-tag** onto every customer frame, whether or not the frame already has a customer **C-tag**. Customers keep their own VLAN numbering, and the provider needs only one VLAN per customer service.

An 802.1ad double-tagged frame header



Each tag adds 4 bytes before the customer EtherType. The standard outer TPID is 0x88A8, but some implementations still default to 0x8100, so agree it per interconnect.

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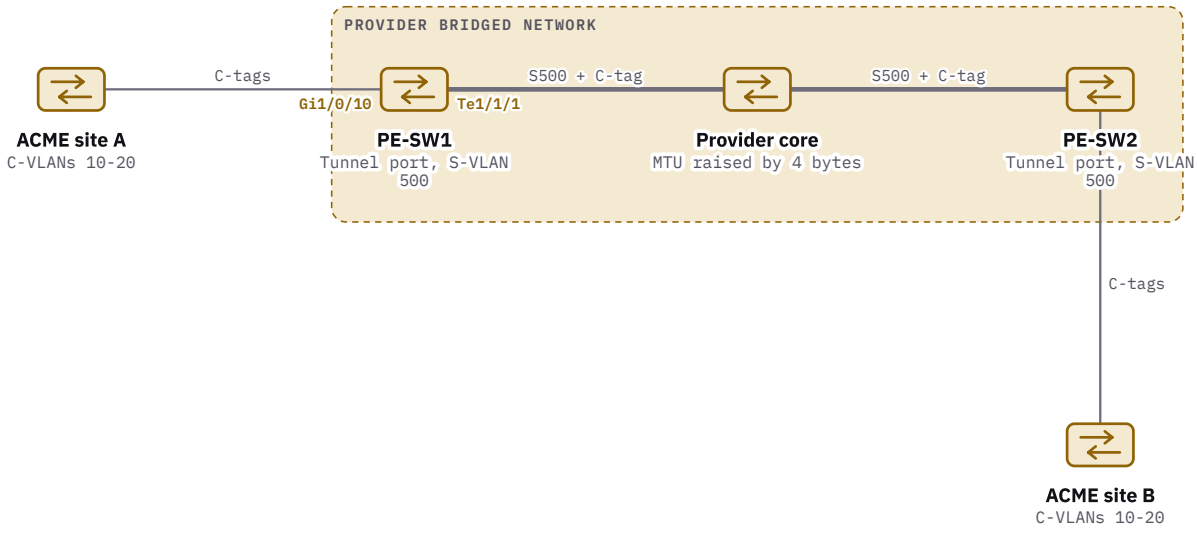
Provider edge tunnel port

PE-SW1 · IOS-style CLI

```
vlan dot1q tag native
system mtu 1504
!
vlan 500
 name CUST-ACME-SVLAN
!
interface GigabitEthernet1/0/10
 description ACME-HANDOFF
 switchport access vlan 500
 switchport mode dot1q-tunnel
 l2protocol-tunnel stp
 l2protocol-tunnel point-to-point lacp
!
interface TenGigabitEthernet1/1/1
 description PROVIDER-CORE
 switchport mode trunk
 switchport trunk allowed vlan 500-599
```

The access VLAN on the tunnel port is the S-VLAN. Native VLAN tagging stops the outer tag being stripped on core trunks, and the extra 4 bytes of MTU make room for the added tag. Tunnelling of spanning tree and LACP must match what the customer expects across the service.

One QinQ service across the provider



PE-SW1 pushes S-tag 500 on everything ACME sends and PE-SW2 pops it; every provider link must carry the extra 4 bytes, and ACME's BPDUs and LACP cross only if tunnelling is agreed.

QinQ traps

TRAP	SYMPTOM	FIX
MTU not raised for the extra tag	Large customer frames dropped silently; small ones pass	Raise MTU on every provider port by at least 4 bytes per added tag
S-VLAN equals the native VLAN of a core trunk	Outer tag stripped, inner tag misread as the service VLAN	Tag the native VLAN, or never use the native VLAN as an S-VLAN

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TRAP	SYMPTOM	FIX
TPID mismatch between providers	Frames dropped or treated as untagged at the interconnect	Agree the outer TPID per interconnect
Customer control frames consumed by the provider	Customer spanning tree or LACP behaves as if links were disconnected	Tunnel the needed protocols or agree they are not carried
Customer loop across the service	Provider core flooded by a customer's looped traffic	Storm control and MAC limits on tunnel ports

Loop forensics is what happens when every protection in the previous lessons has failed or been bypassed. A layer 2 loop has no time-to-live, so broadcast frames multiply until links and CPUs saturate. Routing adjacencies drop, gateways flap, management becomes unreachable, and the instinct to restart things destroys the evidence you need. A fixed procedure keeps you effective.

Loop forensic procedure

01 Recognise
MAC flaps, saturated links, high CPU, gateway and adjacency flaps at the same time

→

02 Contain
Break the loop: shut a known redundant link or let storm control shut ports

→

03 Locate
Follow ports with the highest broadcast rate and the MAC flap pairs

→

04 Prove
Find why spanning tree did not block: filter, unidirectional link, rogue bridge

→

05 Harden
Fix the cause and add the missing protection

Evidence from the core during a storm

CORE1 · IOS-style CLI

```
CORE1# show logging | include MACFLAP
%L2-4-MACFLAP: Host 0011.2233.4455 in vlan 30 flapping between port Te1/1/1 and port Te1/1/2
%L2-4-MACFLAP: Host 001a.2b3c.4d5e in vlan 30 flapping between port Te1/1/1 and port Te1/1/2
CORE1# show interfaces Te1/1/2 | include rate|broadcasts
  5 minute input rate 7812345000 bits/sec, 11250000 packets/sec
     Received 98234567812 broadcasts (412 IP multicasts)
CORE1# show spanning-tree vlan 30 detail | include occurred|from
  Number of topology changes 4812 last change occurred 00:00:03 ago
          from TenGigabitEthernet1/1/2
```

Two different hosts flap between the same pair of ports, Te1/1/2 is near line rate with mostly broadcasts, and topology changes keep arriving from the same port. The loop is reachable through Te1/1/2; follow it downstream and check spanning tree state and BPDU counters there.

CASE STUDY · THE TRADING-FLOOR MELTDOWN

A regional stock exchange ran its matching engines and market-data distribution on a pair of data-centre switches using multi-chassis aggregation. During a planned line-card replacement, an engineer removed the card that carried both the peer link and the peer keepalive. Both peers went dual-active with the same LACP system ID. A legacy market-data switch attached to both peers kept its bundle up and started receiving duplicate floods. That legacy switch had BPDU filter enabled on its uplinks years earlier to stop log noise. Broadcast and multicast traffic looped, CPUs on both core switches reached 100%, routing adjacencies dropped and trading was halted for 47 minutes.

1. Which design decisions turned one maintenance action into a loop?

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2. What should the first containment action have been?

3. Which evidence would prove the sequence afterwards?

4. Which changes would you make so that the same maintenance is safe?

Model answer: module answer key.

TEMPLATE **Loop forensics worksheet**

Capture evidence while containing a loop so root cause can be proven afterwards.

TIME	DEVICE AND PORT	INDICATOR	STP ROLE AND STATE	ACTION TAKEN	RESULT
14:02	*CORE1 Te1/1/2*	*11.2 Mpps input, mostly broadcast*	*Desg FWD*	*None, following downstream*	*Loop path toward DSW3*
14:05	*DSW3 Gi1/0/48*	*BPDU in 0, out 0; filter configured*	*Desg FWD*	*Shut Gi1/0/48*	*Broadcast rate falls to baseline*

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SCENARIO · DECIDE · THE STORM IS STILL RUNNING

Five minutes into a broadcast storm, management access to the core is slow but working. You have identified two candidate downstream switches by broadcast rate. A colleague proposes rebooting both core switches to stop the storm.

A. Reboot both core switches so that all MAC tables and spanning tree state are rebuilt.

B. Enable debug output for spanning tree events on the core to see which port causes the loop.

C. Shut the uplink toward the highest-rate switch and watch whether core broadcast rates fall.

D. Lower the core spanning tree priority further so the root election is forced to repeat.

Circle your choice. Answer and feedback: module answer key.

Loop readiness

☐ Containment steps documented per site with the link to shut first

☐ Out-of-band management reachable when in-band management is saturated

☐ Storm control on all access ports and ports to legacy or third-party switches

☐ MAC flap logging enabled and sent off-box

☐ Topology change counters baselined

☐ BPDU filter use inventoried and justified

Key terms

QinQ
Common name for IEEE 802.1ad provider bridging, which adds an outer service tag to customer frames.

C-tag
The inner customer VLAN tag carried transparently through the provider network.

Layer 2 protocol tunnelling
Carrying customer control frames such as spanning tree or LACP across a provider service instead of consuming them.

Containment
The first response step that stops impact quickly, such as shutting one redundant link.

S-tag
The outer service VLAN tag added by the provider edge.

TPID
Tag protocol identifier, the EtherType value that marks a VLAN tag (0x8100 or 0x88A8).

MAC flap
A MAC address repeatedly learned on alternating ports, a strong indicator of a loop.

Broadcast storm
Uncontrolled multiplication of broadcast frames caused by a layer 2 loop.

REFLECT

If a loop started in your core right now, which single link would you shut first, and how would you reach the switch to do it if in-band management failed?

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Key takeaways

- QinQ pushes an S-tag on every customer frame; raise provider MTU by 4 bytes per added tag.
- Never let an S-VLAN travel untagged as a native VLAN, and agree the outer TPID per interconnect.
- In a loop, contain first with a targeted, reversible shutdown; never reboot away the evidence.
- Follow MAC flaps, broadcast rates and topology-change sources to the port where spanning tree failed to block.

KNOWLEDGE CHECK

21. A provider core carries QinQ services with a 1500-byte MTU. Customers report that large frames are lost. Why?

(A) The customer tag is removed in the core and the frame becomes malformed

(B) Provider switches fragment tagged frames and customers drop the fragments

(C) The outer tag changes the frame check sequence to an invalid value

(D) The extra 4-byte service tag pushes full-size tagged frames beyond the core MTU

22. Why is the native VLAN dangerous on provider trunks carrying QinQ services?

(A) The native VLAN cannot carry any traffic marked with the 0x88A8 TPID

(B) Native traffic loses its outer tag and the inner tag is misread

(C) Spanning tree blocks the native VLAN on every provider trunk port

(D) The native VLAN always uses a lower MTU than tagged VLANs

23. Two hosts flap between the same pair of core ports while one port runs near line rate with broadcasts. What does this show?

(A) The two hosts have duplicate MAC addresses and should be renumbered

(B) The link aggregation hash is unbalanced between the two core ports

(C) A loop is reachable through those ports; follow it downstream

(D) The spanning tree root has moved to a switch behind one of the ports

24. What is the best first action during an active loop?

(A) Shut a known redundant link toward the suspected source of the loop

(B) Reboot the core switches to clear MAC tables and spanning tree state

(C) Enable spanning tree debugs on all switches to locate the loop

(D) Raise the storm-control level so that legitimate traffic is not dropped

25. A customer's spanning tree behaves as if the provider service were disconnected. What is the likely cause?

(A) The provider has set the S-tag TPID to 0x8100 instead of 0x88A8

(B) The provider edge is not tunnelling the customer's BPDUs

(C) The customer's C-tag uses the same VLAN ID as the provider S-tag

(D) The provider core runs MST while the customer runs per-VLAN STP

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · LAYER2

Answer key

Knowledge checks

1. A The digest of the mapping is part of region identity; any mismatch places the switches in different regions.

2. C Outside its region an MSTI has no meaning, so at the boundary all instances inherit the CIST role.

3. B MST ignores allowed VLAN lists when computing topology, so pruning inconsistent with the instance topology silently blackholes traffic.

4. D The CIST regional root is chosen by external root path cost first, which is why it sits at the region's best exit toward the CIST root.

5. D Inside a region BPDUs carry remaining hops (default 20); message age is used for the CIST outside.

6. C Only non-edge ports becoming forwarding signal a topology change; edge ports and transitions to blocking do not.

7. A Root guard protects the root placement by blocking superior BPDUs arriving on designated ports toward downstream switches.

8. B Loop guard triggers on BPDU loss on non-designated ports, a typical sign of a unidirectional fault.

9. A Filtering suppresses BPDUs entirely, so spanning tree cannot detect a loop through that port.

10. C Max age 20 s plus two forward delay periods of 15 s gives 50 s.

11. D Source guard relies on bindings; persistence restores them after a reload.

12. B Server replies arriving on an untrusted port are dropped, so no client receives an offer.

13. A 10 Mb/s divided by 84 bytes × 8 bits = 672 bits per frame gives roughly 14,880 pps.

14. D A static device needs an explicit entry; disabling validation or snooping weakens the whole VLAN.

15. C Some relays treat option 82 with a zero gateway address as untrusted and discard it.

16. B With the peer alive but unsynchronised, the secondary steps back to avoid inconsistent forwarding.

17. B 3 × 1 s + (256 − 90) ÷ 256 s = 3.648 s.

18. C The keepalive exists to distinguish a dead peer from a dead peer link, which is impossible if they share a path.

19. A A backup only preempts when its priority is strictly higher than the master's advertised priority. The address tie-break applies when two routers both claim mastership, not here, so decrements must drop the master clearly below the peer.

20. D Misalignment forces host traffic across the inter-switch link to reach the master, wasting bandwidth and adding latency.

21. D Each added tag adds 4 bytes; ethernet does not fragment, so oversize frames are dropped.

22. B If the S-VLAN is native, the outer tag is stripped and the customer tag is then interpreted as a provider VLAN.

23. C Multiple unrelated hosts flapping on the same port pair plus broadcast saturation is the signature of a loop.

24. A Targeted shutdown contains impact reversibly and confirms the location.

25. B Without layer 2 protocol tunnelling, customer BPDUs are processed or dropped by the provider edge and never reach the far site.

Scenarios & case studies

> **Scenario · Building B loses VLAN 210**
> **A** is best. Correct. The boundary makes every instance follow the CIST role, so the second uplink blocks VLAN 210. Joining the region restores per-instance roles; the trunk audit removes the fragile dependency on pruning.

> **Scenario · A contractor's switch becomes root**
> **D** is best. Correct. BPDU guard stops any bridge on host ports, and root guard ensures no access-layer device can win the root even if BPDU guard is missing somewhere.

> **Scenario · Printers vanish after enforcement**
> **C** is best. Correct. The printers have no snooping bindings because they never used DHCP. Explicit entries let inspection validate them without weakening protection for other hosts.

> **Scenario · Forty seconds of loss after a reboot**
> **B** is best. Correct. VRRP preempts by default, so the returning switch reclaimed the master role as soon as its interface came up, before its routing protocols had converged, and blackholed traffic until they did.

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Case study · The trading-floor meltdown

Three decisions combined: the peer link and keepalive shared one line card, so a single removal caused dual-active; the legacy switch was dual-homed while dual-active could occur; and BPDU filter on its uplinks hid the loop from spanning tree. The fastest containment is to shut one of the legacy switch's uplinks (or the member ports on one peer), which breaks the loop immediately; restarting the cores would have destroyed evidence and lengthened the outage. Evidence: multi-chassis role and keepalive logs with timestamps, MAC flap logs between the two peers' member ports, interface broadcast counters, and the legacy switch configuration showing the filter. Remediation: separate the peer link and keepalive across modules and paths, remove BPDU filter, enable storm control on all ports facing legacy devices, add a maintenance procedure that verifies the keepalive path before touching line cards, and rehearse the containment step.

Scenario · The storm is still running

C is best. Correct. A targeted shutdown is reversible, contains the loop, and confirms the location by the immediate effect on counters, while preserving evidence.

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03

MODULE 3 · IGP

IGP Mastery

OSPF and IS-IS databases, area and level design, fast convergence and safe redistribution

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Read OSPF and IS-IS databases to diagnose missing, unstable or unexpected routes
- Design areas, levels, virtual links, summaries and filters that behave predictably
- Tune detection, throttling and fast reroute for sub-second convergence
- Build mutual redistribution that is loop-free and path-optimal at multiple points

LESSONS

1. OSPF Database and Flooding Deep Dive · 45 min
2. OSPF Area Design Edge Cases and Virtual Links · 45 min
3. IS-IS Deep Dive: TLVs, Levels and Route Leaking · 45 min
4. Fast Convergence: Throttling, BFD and Fast Reroute · 50 min
5. Mutual Redistribution, Loop Prevention and Database Troubleshooting · 50 min

MODULE 3 · IGP · LESSON 1 · 45 MIN

OSPF Database and Flooding Deep Dive

Read the link-state database like a forensic record, not a list of routes

- Identify every common LSA type by originator, scope and content
- Decide which of two LSA instances is newer and explain flooding on broadcast segments
- Diagnose missing or unstable routes directly from database output

Every OSPF router in an area holds an identical copy of the area's link-state database and runs the same shortest-path calculation over it. Routes are only the output. When something is wrong, the database tells you **who said what, when, and how often**, which is far more than any routing table can. Expert troubleshooting in OSPF starts with the database.

LSA types at expert depth

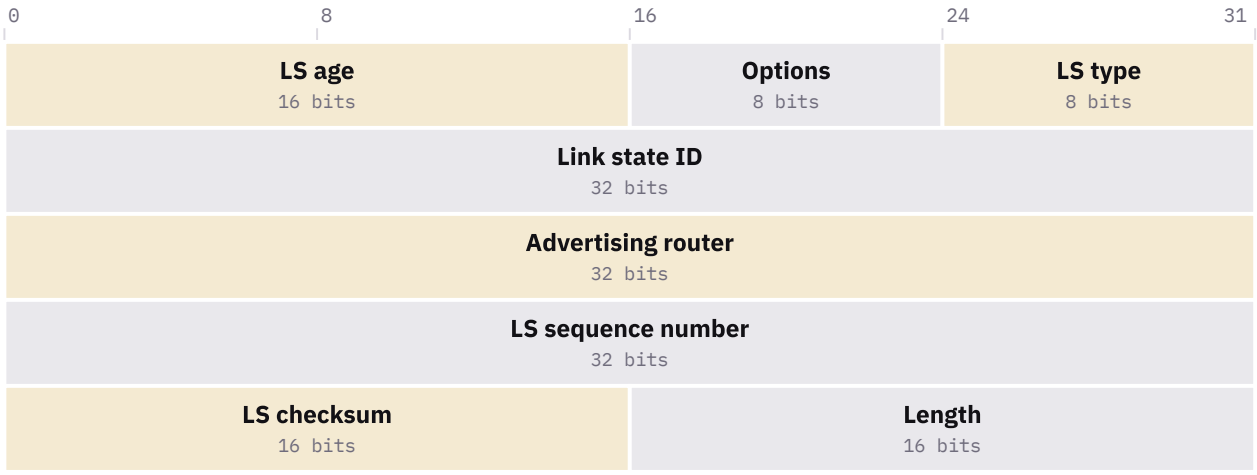
TYPE	NAME	ORIGINATED BY	FLOODING SCOPE	KEY CONTENT
1	Router	Every router	Area	Links, neighbours, stub networks, ABR and ASBR flags
2	Network	DR on a transit segment	Area	Segment mask and all attached router IDs
3	Summary (network)	ABR	Area	Inter-area prefix and cost from the ABR
4	Summary (ASBR)	ABR	Area	Cost from the ABR to an ASBR in another area
5	AS external	ASBR	AS, except stub and NSSA areas	Prefix, metric type, forwarding address, route tag
7	NSSA external	ASBR inside an NSSA	NSSA only	As type 5 plus the propagate bit; translated to type 5 by an ABR
9, 10, 11	Opaque	Any router	Link, area or AS	Extensions such as traffic engineering, segment routing and graceful restart

Fields in every LSA header

FIELD	WHY IT MATTERS
LS age	Seconds since origination; reaching MaxAge (3600 s) withdraws the LSA
Options	Capabilities such as external routing, NSSA propagate and demand circuits
LS type, link state ID, advertising router	Together they identify the LSA uniquely
LS sequence number	Increments with every new instance; starts at 0x80000001
Checksum	Integrity check; also used as a tie-breaker between instances
Length	Total size; sudden growth can indicate added links or prefixes

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The 20-byte LSA header



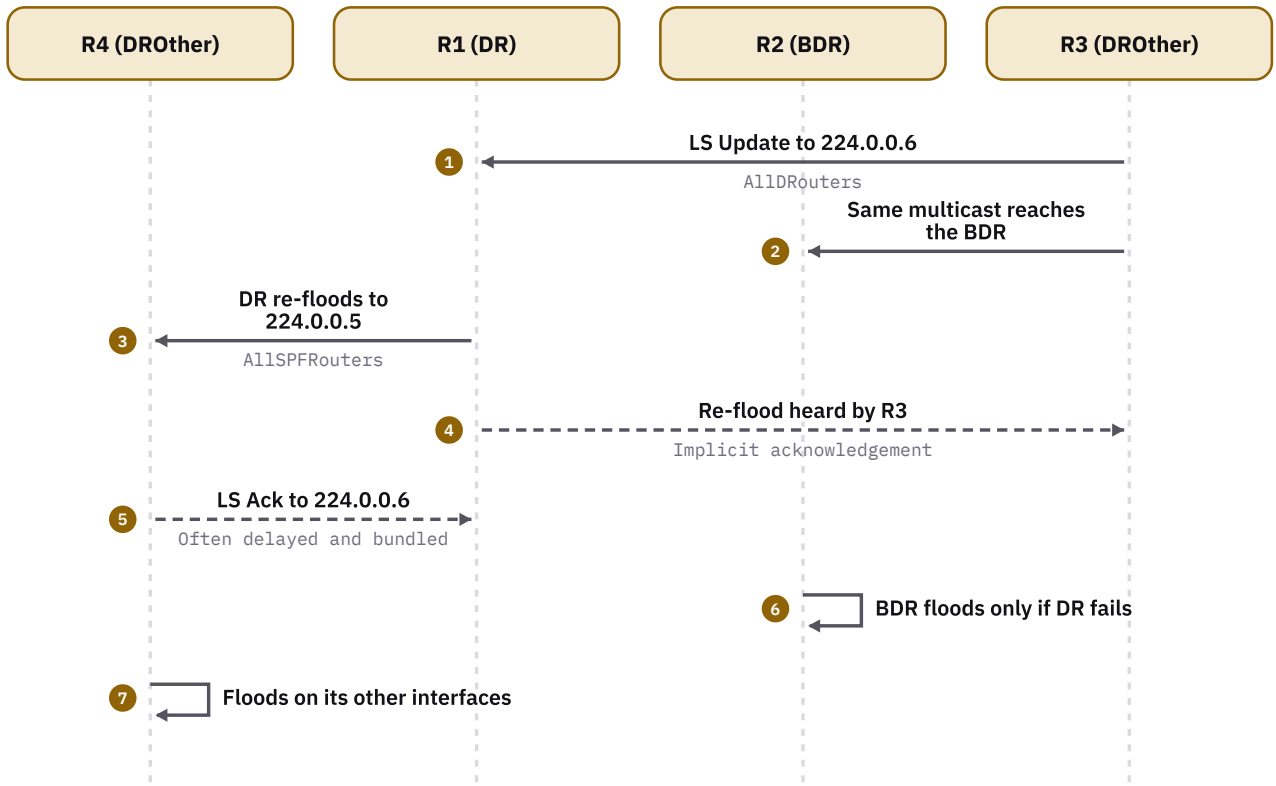
Type, link state ID and advertising router identify the LSA; sequence number, checksum and age decide which instance is newer.

Which instance is newer?

1. **Sequence number** The instance with the higher sequence number is newer.
2. **Checksum** If sequence numbers match, the instance with the larger checksum is newer.
3. **MaxAge** If both still match, an instance at MaxAge is considered newer, so withdrawals always win.
4. **Age difference** If the ages differ by more than 15 minutes (MaxAgeDiff), the younger instance is newer.
5. **Identical** Otherwise the instances are treated as the same and no flooding is needed.

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Flooding a new LSA on a broadcast segment



DROthers talk only to the DR and BDR; the DR re-floods to everyone, and that re-flood doubles as R3's acknowledgement. The BDR stays quiet unless the DR fails.

OSPF architectural constants

MEASURE	FORMULA	WHAT IT TELLS YOU
Refresh	`LSRefreshTime = 1800 s`	An originator re-floods each of its LSAs every 30 minutes even without change.
Withdrawal	`MaxAge = 3600 s`	An LSA at MaxAge is flushed from every database.
Origination pacing	`MinLSInterval = 5 s`	Minimum gap between new instances of the same LSA (many implementations replace this with tunable throttling).
Acceptance pacing	`MinLSArrival = 1 s`	New instances of the same LSA arriving faster than this are discarded.
Expected sequence growth	`uptime ÷ 1800 s`	A stable LSA should have a sequence number close to this; far higher means the LSA keeps changing.

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Reading a router LSA

R1 · IOS-style CLI

```
R1# show ip ospf database router 10.0.0.2
            OSPF Router with ID (10.0.0.1) (Process ID 1)
                Router Link States (Area 0)
  LS age: 412
  Options: (No TOS-capability, DC)
  Link State ID: 10.0.0.2
  Advertising Router: 10.0.0.2
  LS Seq Number: 80000A3C
  Checksum: 0x5B1E
  Area Border Router
  AS Boundary Router
   Number of Links: 3
    Link connected to: another Router (point-to-point)
     (Link ID) Neighboring Router ID: 10.0.0.3
     (Link Data) Router Interface address: 10.1.23.2
       TOS 0 Metrics: 10
```

R2 has been up for about six weeks, which would explain roughly 2,000 refreshes, but the sequence number 0x80000A3C means over 2,600 instances. Combined with an LS age of only 412 seconds, it suggests R2's links have been changing. Compare the sequence number again in a few minutes: if it climbs, find the flapping link.

An external route that never installs

R5 · IOS-style CLI

```
R5# show ip ospf database external 172.16.50.0
  LS Type: AS External Link
  Link State ID: 172.16.50.0 (External Network Number )
  Advertising Router: 10.0.0.7
  Network Mask: /24
        Metric Type: 2 (Larger than any link state path)
        Metric: 20
        Forward Address: 10.7.7.9
        External Route Tag: 65010
R5# show ip route 10.7.7.9
% Subnet not in table
R5# show ip route 172.16.50.0
% Subnet not in table
```

The LSA is present but the route is not. With a non-zero forwarding address, the external route is only usable if that address is reachable through an intra-area or inter-area OSPF route. 10.7.7.9 is not, so every router ignores the LSA. Advertise the forwarding address subnet into OSPF, or remove the condition that sets it.

SCENARIO · DECIDE · A ROUTE THAT IS IN THE DATABASE BUT NOT THE TABLE

On R5 the type 5 LSA for 172.16.50.0/24 is present and current, the ASBR's router LSA shows the E bit, and the ASBR is reachable. Yet the route is absent from R5's routing table and from every other router's table.

A Clear the OSPF process on the ASBR so it originates the LSA again with a new sequence number.

B Check whether the external metric type is 2, because type 2 routes are not installed while type 1 routes exist.

C Check whether the area is configured as a stub, because stub areas reject type 5 LSAs.

D Check the forwarding address in the LSA and whether it is reachable through an internal OSPF route.

Circle your choice. Answer and feedback: module answer key.

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Database-first OSPF investigation

☐ Compare database summaries across routers in the area for consistency

☐ Check sequence numbers against uptime for unstable LSAs

☐ For missing externals, verify the forwarding address and the ASBR reachability

☐ For missing inter-area routes, verify the type 3 and the ABR's own reachability

☐ Look for MaxAge LSAs that linger, which point to a vanished originator

☐ Confirm that the expected router and network LSAs exist before blaming SPF

Key terms

Link-state database
The synchronised collection of LSAs in an area from which every router computes its shortest-path tree.

Network LSA
Type 2 LSA originated by the DR describing a transit segment and its attached routers.

Forwarding address
A field in external LSAs that, when non-zero, names the address traffic should be sent to instead of the ASBR.

LS sequence number
A counter in each LSA that increases with every new instance.

Opaque LSA
Types 9, 10 and 11, used to carry application-specific information such as traffic engineering.

Router LSA
Type 1 LSA describing a router's links in an area, with ABR, ASBR and virtual-link flags.

Summary LSA
Type 3 or 4 LSA originated by an ABR to describe inter-area prefixes or ASBR reachability.

MaxAge
The LSA age (3600 s) at which an LSA is flushed from all databases.

AllDRouters
The multicast group 224.0.0.6 used to send updates to the DR and BDR.

REFLECT

Pick an OSPF router you operate. What would you expect its router LSA's sequence number to be, given its uptime? Check it, and explain any difference.

Key takeaways

- The link-state database records who said what and when; routes are only its output.
- Sequence number, then checksum, then MaxAge, then age difference decide which instance is newer.
- A sequence number far above uptime ÷ 1800 s points to an unstable originator.
- An external LSA with a non-zero forwarding address is used only if that address resolves through an internal OSPF route.

KNOWLEDGE CHECK

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1. Two instances of the same LSA have equal sequence numbers and checksums. One is at MaxAge. Which is newer?

   (A) The instance with the smaller age, because it was refreshed more recently

   (B) Neither, because equal sequence numbers mean the instances are the same

   (C) The instance at MaxAge, so that withdrawals always propagate

   (D) The instance received first, because flooding order decides ties

2. A type 5 LSA is present on every router but no router installs the route. What is the most likely cause?

   (A) Its forwarding address has no intra-area or inter-area route

   (B) The LSA uses external metric type 2 instead of type 1 for the prefix

   (C) The advertising router is not a designated router on any segment

   (D) The LSA's age is above the 1800-second refresh interval

3. On a broadcast segment, where does a DROther send a new LSA first?

   (A) To 224.0.0.5, so every OSPF router receives it

   (B) To the DR's unicast interface address only

   (C) To the ABR of the area by unicast

   (D) To 224.0.0.6, so the DR and BDR receive it

4. A router has been up for 10 days. Its router LSA sequence number is 0x80000600. What does that suggest?

   (A) The router has been refreshing its LSAs normally every thirty minutes

   (B) The router's sequence number wrapped and restarted recently

   (C) The LSA has changed far more often than the 30-minute refresh interval explains

   (D) The LSA was received from a neighbour with a newer software release

5. Which LSA type tells routers in another area how to reach an ASBR?

   (A) Type 3, the network summary originated by the ABR

   (B) Type 4, the ASBR summary originated by the ABR

   (C) Type 7, the NSSA external originated by the ASBR

   (D) Type 2, the network LSA originated by the DR

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MODULE 3 · IGP · LESSON 2 · 45 MIN

OSPF Area Design Edge Cases and Virtual Links

Stub variants, NSSA translation, virtual links, summarisation and filtering that behave as intended

- Predict which LSA types exist in each area type and where defaults come from
- Explain NSSA translation and forwarding address behaviour
- Configure and verify virtual links and inter-area filtering without blackholes

Area design decides what each router knows. Every stub variant trades knowledge for stability: fewer LSAs, smaller databases, but less precise exit selection. At expert level the interesting behaviour is at the edges: which ABR translates NSSA routes, where default routes come from, why a filter removes a route from one router but not its neighbours, and when a virtual link is the least bad option.

What each area type contains

AREA TYPE	TYPE 3	TYPE 4 AND 5	TYPE 7	DEFAULT ROUTE
Normal	Yes	Yes	No	Only if originated deliberately
Stub	Yes	No	No	Type 3 default injected by the ABR
Totally stubby	Default only	No	No	Type 3 default injected by the ABR
NSSA	Yes	No	Yes	Not automatic; configure a type 7 default on the ABR
Totally NSSA	Default only	No	Yes	Type 3 default injected by the ABR

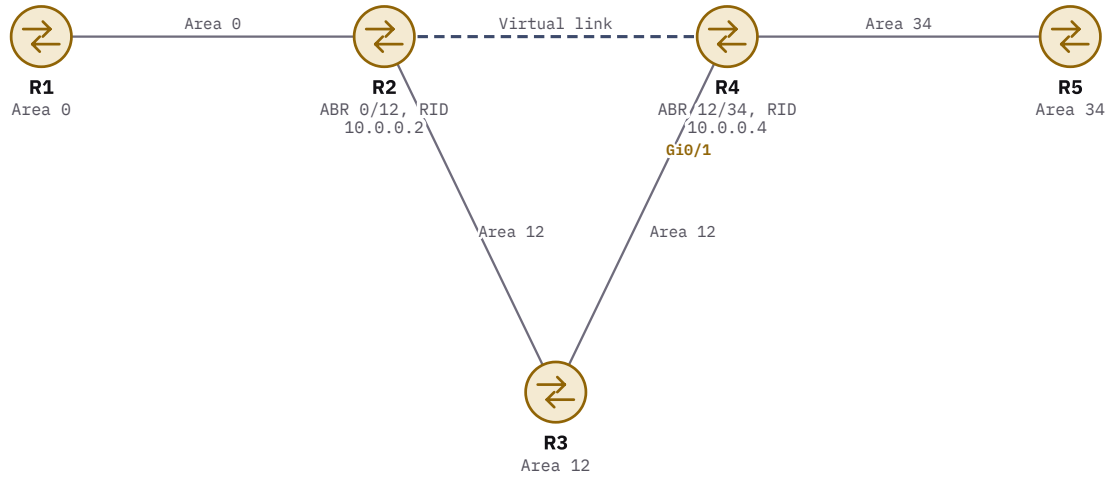
INSIGHT · NSSA TRANSLATION

Type 7 LSAs never leave the NSSA. An ABR translates them into type 5 LSAs for the rest of the domain, but only if the propagate bit is set. When several NSSA ABRs exist, the one with the highest router ID is elected translator unless one is configured to always translate. The forwarding address from the type 7 is normally kept in the type 5, so every router in the domain needs an internal route to it.

A **virtual link** connects an ABR to the backbone through a non-backbone transit area. It belongs to area 0 and runs as a point-to-point, demand-style adjacency across the transit area, using the transit area's intra-area path. It is used to attach an area that has no physical connection to area 0, or to heal a partitioned backbone. The transit area cannot be any type of stub, and area 0 authentication applies to the virtual link.

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A virtual link across transit area 12



Area 34 has no physical link to area 0, so ABRs R2 and R4 build a virtual link that belongs to area 0 and follows the intra-area path through R3. Area 12 must not be any kind of stub.

Building and verifying a virtual link

R4 · IOS-style CLI

```
R2(config)# router ospf 1
R2(config-router)# area 12 virtual-link 10.0.0.4
R4(config)# router ospf 1
R4(config-router)# area 12 virtual-link 10.0.0.2
R4# show ip ospf virtual-links
Virtual Link OSPF_VL0 to router 10.0.0.2 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 12, via interface GigabitEthernet0/1
  Topology-MTID    Cost    Disabled     Shutdown      Topology Name
        0           20         no          no            Base
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Adjacency State FULL (Hello suppressed)
```

Each end names the other end's router ID, not an interface address, and both are ABRs in area 12. The cost is the intra-area cost across area 12. Because the link runs as a demand circuit, LSAs learned over it carry the DoNotAge bit and periodic hellos are suppressed once the adjacency is full.

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Summarisation and inter-area filtering

ABR1 · IOS-style CLI

```
router ospf 1
 area 10 range 10.10.0.0 255.255.0.0 cost 50
 area 20 filter-list prefix PL-NO-LAB in
 summary-address 172.16.0.0 255.255.240.0
!
ip prefix-list PL-NO-LAB seq 5 deny 10.99.0.0/16 le 32
ip prefix-list PL-NO-LAB seq 10 permit 0.0.0.0/0 le 32
!
ABR1# show ip route 10.10.0.0 255.255.0.0 | include via|Null
  * directly connected, via Null0
```

The area range replaces area 10's prefixes with one type 3 summary at cost 50 and installs a discard route on the ABR to prevent loops toward missing components. The filter-list stops type 3 LSAs for the lab range from being generated into area 20. The summary-address command applies only to external routes this router redistributes as an ASBR.

Two very different filters

ABR type 3 filtering

- Prevents the type 3 LSA from being originated into an area
- Every router in that area loses the route consistently
- Changes the database, so it is a design decision

Inbound distribute-list

- Removes the route from the local routing table only
- The database and flooding are unchanged
- Neighbours may still route through this router and be dropped

WATCH OUT · THE DISTRIBUTE-LIST BLACKHOLE

Filtering a route from one router's table with an inbound distribute-list does not stop other routers computing paths through it, because their databases still show the prefix reachable via that router. Packets arrive and are dropped or follow a default route. Filter at the ABR, or design the filter so no other router's shortest path crosses the filtering router.

Inter-area cost

MEASURE	FORMULA	WHAT IT TELLS YOU
Inter-area route cost	`cost to the ABR + cost in the ABR's type 3 LSA`	Routers choose between ABRs using this sum.
Virtual link cost	`intra-area cost between the two ABRs across the transit area`	Not configurable directly; change transit area link costs instead.
Fixed summary cost	`area range … cost N`	Freezes the type 3 cost so summary metrics do not change when components change.

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SCENARIO · DECIDE · TRAFFIC DIES AT R3

Area 30 contains R3 and R4 in a line toward the ABR. An engineer added an inbound distribute-list on R3 to keep 10.50.0.0/16 out of R3's table. Since then, hosts behind R4 cannot reach 10.50.0.0/16; their traces stop at R3.

(A) Remove the distribute-list; if area 30 must not use the prefix, filter it at the ABR.

(B) Add the same distribute-list on R4 so both routers agree that the prefix is unreachable.

(C) Increase the cost of the R3 to R4 link so R4 prefers another path toward the prefix.

(D) Configure a default route on R3 so packets for the filtered prefix follow it instead.

Circle your choice. Answer and feedback: module answer key.

Area design review

☐ Each area type chosen deliberately, with its default route source known

☐ NSSA translator identified and forwarding addresses reachable domain-wide

☐ Virtual links documented, authenticated and justified

☐ Summaries have discard routes and fixed costs where stability matters

☐ Inter-area filtering done at ABRs, not with local distribute-lists

☐ No area relies on a single ABR without an explicit decision

Key terms

Stub area
An area that receives no type 4 or type 5 LSAs and uses a default route from the ABR.

NSSA
Not-so-stubby area: blocks type 5 LSAs but allows local external routes as type 7 LSAs.

Propagate bit
The flag in a type 7 LSA that permits translation into a type 5 LSA.

Transit area
The area whose intra-area path a virtual link uses.

Discard route
A route to a null interface installed with a summary to drop traffic for missing components.

Totally stubby area
A stub area that also receives no type 3 LSAs except a default route.

NSSA translator
The ABR that converts type 7 LSAs into type 5 LSAs for the rest of the domain.

Virtual link
A logical area 0 adjacency between two ABRs across a non-stub transit area.

Area range
ABR summarisation of an area's prefixes into one type 3 LSA.

REFLECT

Find one place in a network you know where a filter is applied with a local distribute-list. Which routers could still compute paths through that router?

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Key takeaways

- Know which LSA types exist in each area type and where every default route comes from.
- Only type 7 LSAs with the propagate bit are translated, by the highest-router-ID NSSA ABR unless one is forced.
- Virtual links belong to area 0, need a non-stub transit area and use area 0 authentication.
- Filter at the ABR so the database stays consistent; a local distribute-list creates blackholes.

KNOWLEDGE CHECK

6. Which area type needs a manually configured default route on the ABR?

   (A) A plain NSSA that still receives type 3 LSAs
   (B) A stub area that receives type 3 LSAs from the ABR
   (C) A totally stubby area with no external routes
   (D) A totally NSSA area with local external routes

7. Two ABRs connect an NSSA to area 0. Neither is configured to always translate. Which one translates?

   (A) The ABR with the lowest router ID
   (B) Both ABRs translate every type 7 LSA
   (C) The ABR nearest the originating ASBR
   (D) The ABR with the highest router ID

8. Which statement about virtual links is correct?

   (A) They belong to area 0 and cannot use any kind of stub area for transit
   (B) They belong to the transit area and use that area's authentication
   (C) They are configured with the neighbour's interface address on each end
   (D) They can be built across any area type, including totally stubby areas

9. An inbound distribute-list on R3 removes a prefix from R3's table. What happens to routers behind R3?

   (A) They remove the prefix too because R3 stops flooding the LSA
   (B) They may still send that traffic to R3, which then drops it
   (C) They receive a new type 3 LSA withdrawing the prefix from R3
   (D) They mark R3 as unreachable and recompute every route

10. What protects against loops when an ABR advertises an area range and some components disappear?

    (A) The type 4 LSA that the ABR generates for the summary range
    (B) The forwarding address placed in the summary
    (C) The discard route the ABR installs for the summary
    (D) The propagate bit set on the summary LSA

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MODULE 3 · IGP · LESSON 3 · 45 MIN

IS-IS Deep Dive: TLVs, Levels and Route Leaking

The extensible link-state protocol that underpins most large provider cores

- Describe IS-IS levels, the attached bit and how level 1 routers exit an area
- Read LSPs and identify key TLVs and flags
- Configure level 2 to level 1 route leaking safely using the up/down bit

IS-IS runs directly over layer 2, not over IP, and carries everything in **type-length-value** (TLV) records. New capabilities such as IPv6, multi-topology, traffic engineering and segment routing were added as new TLVs and sub-TLVs without changing the protocol's core. That extensibility, plus a simple two-level hierarchy, is why large service-provider cores favour it.

IS-IS and OSPF structure

IS-IS

- Runs over layer 2; immune to IP addressing errors on adjacencies
- A router belongs to one area; area borders lie on links
- Two levels: level 1 inside areas, level 2 backbone
- Extended with new TLVs

OSPF

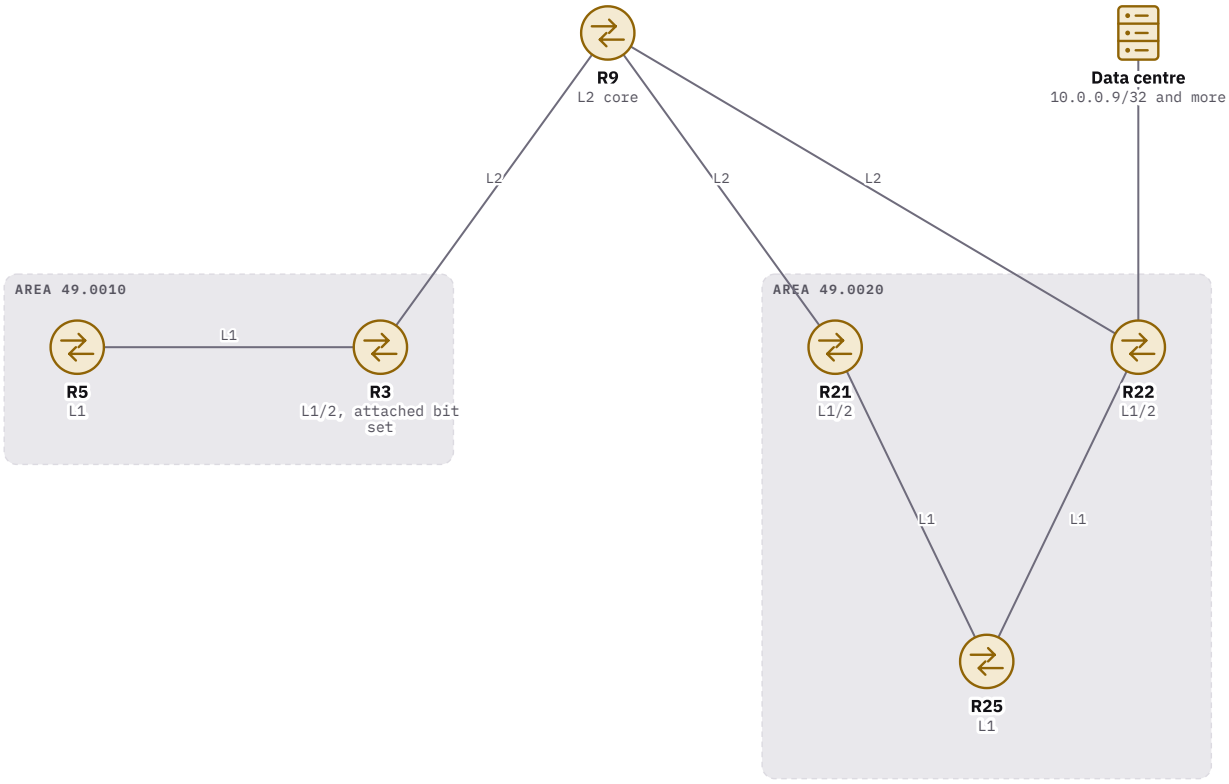
- Runs over IP
- Area borders lie on routers (ABRs)
- Area 0 backbone plus non-backbone areas
- Extended with new LSA types and opaque LSAs

The IS-IS hierarchy

Level 2 backbone	`contiguous chain of level 2 capable routers` `can cross several areas` `carries all inter-area prefixes`
Level 1/2 routers	`one adjacency set per level` `set the attached bit in level 1 LSPs` `can leak level 2 prefixes into level 1`
Level 1 routers	`know only their own area` `install a default toward the nearest attached router` `exit choice is by distance to the attached router`
Areas	`identified by area address in the NET` `a router can carry several area addresses during migrations`

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Two areas joined by the level 2 backbone



Level 1 routers know only their own area and follow a default to the nearest attached L1/2 router, so R25 may exit through R21 even though the data centre sits behind R22, unless the DC prefixes are leaked into level 1.

TLVs you should recognise

TLV	NAME	PURPOSE
1	Area addresses	Areas the router belongs to
10	Authentication	Authentication for hellos and LSPs
22	Extended IS reachability	Neighbours with wide metrics and traffic engineering sub-TLVs
128 / 130	IP internal / external reachability	Legacy narrow-metric prefixes
135	Extended IP reachability	Prefixes with wide metrics, up/down bit and sub-TLVs such as tags
137	Dynamic hostname	Maps system IDs to names in show output
229	Multi-topology	Topologies the router participates in
232 / 236	IPv6 interface address / IPv6 reachability	IPv6 support in a single topology
27	SRv6 locator	Advertises SRv6 locators and their End SIDs for segment routing over IPv6
242	Router capability	Advertises capabilities such as segment routing support

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DEFINITION · THE UP/DOWN BIT

When a level 1/2 router leaks a level 2 prefix into level 1, it sets the **up/down bit** in the extended IP reachability TLV. A prefix with the bit set must never be advertised back into level 2, which prevents a leaked route from looping between levels. Route tags carried in sub-TLVs require wide metrics.

Level 1/2 router with controlled leaking

R3 · IOS-style CLI

```
router isis CORE
 net 49.0010.0100.0000.0003.00
 is-type level-1-2
 metric-style wide
 set-overload-bit on-startup 300
 log-adjacency-changes
 passive-interface Loopback0
 redistribute isis ip level-2 into level-1 route-map RM-L2-LEAK
!
route-map RM-L2-LEAK permit 10
 match ip address prefix-list PL-DC-LOOPBACKS
!
interface GigabitEthernet0/0
 ip router isis CORE
 isis network point-to-point
 isis circuit-type level-2-only
 isis metric 100 level-2
```

The system ID 0100.0000.0003 encodes the loopback 10.0.0.3. The overload bit is set for 300 seconds after boot so transit traffic avoids this router until BGP and the FIB are ready. Only data-centre loopbacks are leaked into level 1, so level 1 routers can choose the best exit for them while everything else follows the default.

Reading a level 1 LSP

R5 · IOS-style CLI

```
R5# show isis database detail R3.00-00 level-1
IS-IS Level-1 LSP R3.00-00
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd  ATT/P/OL
R3.00-00              0x0000041A   0x8C2D        1032/1200          1/0/0
  Area Address: 49.0010
  NLPID:        0xCC
  Hostname: R3
  IP Address:   10.0.0.3
  Metric: 10         IS-Extended R5.00
  Metric: 10         IP 10.0.0.3/32
  Metric: 20         IP 10.20.0.0/16
  Metric: 130        IP-Interarea 10.0.0.9/32
```

ATT is 1, so level 1 routers in area 49.0010 install a default route toward R3. The overload bit is clear. The last prefix is marked as inter-area, which is how this output shows a leaked level 2 route with the up/down bit set: it will never be re-advertised into level 2.

Timers and election details

ITEM	TYPICAL DEFAULT	EXPERT NOTE
LSP maximum lifetime	1200 s	Raise toward 65535 s in large networks to reduce refresh flooding

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ITEM	TYPICAL DEFAULT	EXPERT NOTE
LSP refresh interval	900 s	Must stay below the lifetime
Hello interval	10 s (DIS: about 3.3 s)	The DIS sends hellos three times as often to detect its failure quickly
CSNP interval	10 s on broadcast	The DIS periodically lists all LSPs to repair any missed flooding
DIS election	Highest priority, then highest MAC address	Preemptive and with no backup; use point-to-point mode on two-router links

SCENARIO · DECIDE · LEVEL 1 ROUTERS TAKE THE LONG WAY OUT

Area 49.0020 has two level 1/2 routers, R21 and R22. Level 1 router R25 is two hops from R21 and three hops from R22. Traffic from R25 to a data centre that sits much closer to R22 in the level 2 backbone always exits through R21, adding 20 ms of latency.

- (A) Clear the attached bit on R21 so that R25 only uses R22 for every destination.
- (B) Leak the data-centre prefixes from level 2 into level 1 on both R21 and R22.
- (C) Convert R25 into a level 1/2 router so that it participates in the level 2 backbone.
- (D) Increase the metric of R25's link toward R21 so that R22 becomes the nearest attached router.

Circle your choice. Answer and feedback: module answer key.

IS-IS design review

- ☐ Wide metrics everywhere, so tags and large metrics work
- ☐ Level boundaries and the level 2 backbone are contiguous
- ☐ Point-to-point mode on two-router broadcast links
- ☐ Overload bit on startup where BGP or the FIB must settle
- ☐ Route leaking controlled by an explicit policy
- ☐ LSP lifetime and refresh tuned for network size
- ☐ Authentication configured for hellos and LSPs

Key terms

TLV
Type-length-value: the record format that carries all IS-IS information and makes the protocol extensible.

Attached bit
Set by a level 1/2 router in its level 1 LSP when it has level 2 connectivity to other areas.

Overload bit
Tells other routers not to use this router for transit traffic.

Pseudonode
The virtual node created by the DIS to represent a broadcast segment in the topology.

NET
Network entity title: the area address, system ID and selector that identify an IS-IS router.

Up/down bit
Marks a prefix leaked from level 2 into level 1 so it is never advertised back into level 2.

DIS
Designated intermediate system: the router that represents a broadcast segment with a pseudonode LSP.

CSNP
Complete sequence number PDU: a periodic list of all LSPs used to synchronise databases.

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Wide metrics
32-bit link and 32-bit prefix metrics carried in extended TLVs 22 and 135.

REFLECT

Where in your network would leaking a small set of level 2 prefixes into level 1 improve exit selection, and what would you tag them with?

Key takeaways

- IS-IS runs over layer 2 and grows by adding TLVs; use wide metrics everywhere.
- Level 1 routers exit through the nearest attached router; leak selected level 2 prefixes to improve exit choice.
- The up/down bit stops leaked prefixes returning to level 2.
- Set the overload bit on startup so transit traffic waits until BGP and the FIB are ready.

KNOWLEDGE CHECK

11. A level 1 router has no leaked routes. How does it reach a prefix in another area?

(A) It sends traffic to the DIS, which forwards it to the level 2 backbone
(B) It queries a level 1/2 router for the exact prefix before forwarding
(C) It drops the traffic because level 1 routers cannot leave their area
(D) It uses a default route toward the nearest attached level 1/2 router

12. Why is the up/down bit set on prefixes leaked from level 2 into level 1?

(A) To raise the prefix metric so that level 1 prefixes are preferred
(B) To tell level 1 routers to ignore the prefix during SPF
(C) To mark the prefix as originated by an external protocol
(D) To stop the prefix being advertised back into level 2

13. Which TLV carries IPv4 prefixes with wide metrics and route tags?

(A) TLV 22, extended IS reachability
(B) TLV 128, IP internal reachability
(C) TLV 135, extended IP reachability
(D) TLV 137, dynamic hostname

14. Why is the overload bit commonly set for a few minutes after boot?

(A) So transit traffic avoids it until BGP and FIB are ready
(B) So the router can elect itself DIS before its neighbours do
(C) So the router's LSPs are flooded with a longer lifetime
(D) So the attached bit is set before level 1 adjacencies form

15. How is the DIS on a broadcast segment chosen?

(A) Lowest system ID, with a backup DIS elected at the same time
(B) Highest priority, then highest MAC, with preemption
(C) First router to come up, and it keeps the role until it fails
(D) Highest priority, then lowest system ID, with no preemption

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MODULE 3 · IGP · LESSON 4 · 50 MIN

Fast Convergence: Throttling, BFD and Fast Reroute

From failure to forwarding in tens of milliseconds without destabilising the network

- Break convergence into detection, propagation, computation and installation
- Tune SPF and LSA throttling and BFD with an understanding of the trade-offs
- Explain LFA, remote LFA and TI-LFA protection and their coverage limits

Convergence is the sum of four phases: **detecting** the failure, **propagating** the news, **computing** new paths and **installing** them in the forwarding plane. Expert tuning attacks each phase separately, and fast reroute removes most of them from the critical path by pre-installing a backup that the forwarding plane can use immediately.

Where the time goes

01 Detect — Loss of light, BFD or hello timeout →

02 Originate — New LSA or LSP after the throttle delay →

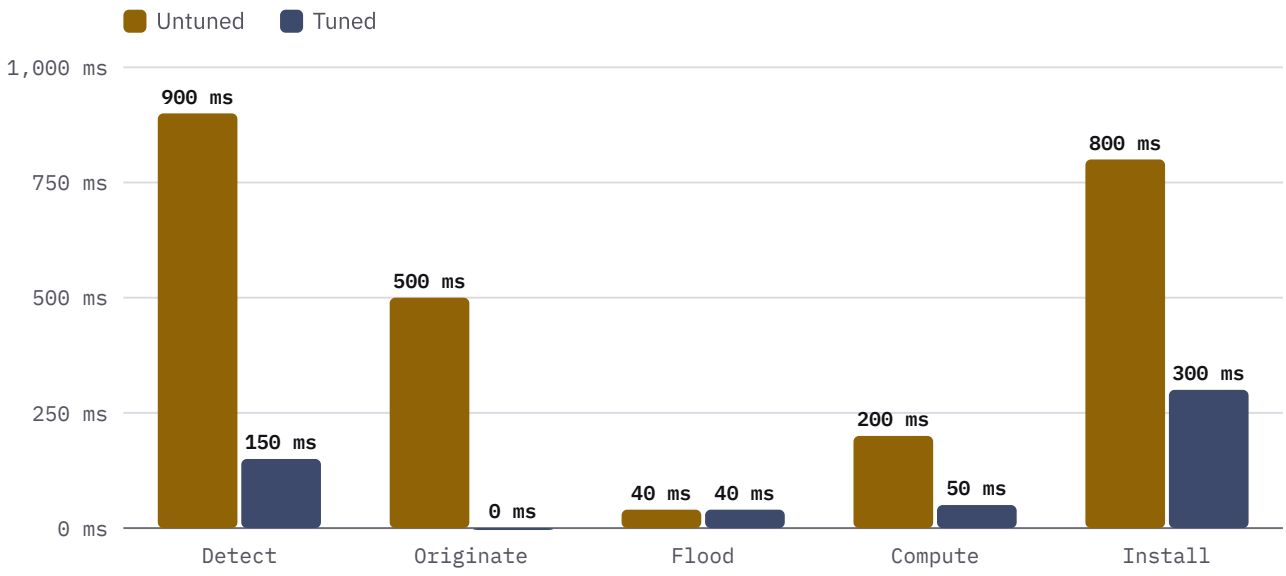
03 Flood — Propagation across the area or level →

04 Compute — SPF after its own throttle delay →

05 Install — Routing and forwarding table updates, prefix by prefix →

06 Protect — Fast reroute switches to a pre-computed backup during all of the above

Where the time goes for one link failure (illustrative)



Illustrative numbers: BFD and short throttles shrink detection and computation, but installing thousands of prefixes still dominates, which is why fast reroute and prefix prioritisation matter.

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Convergence formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
BFD detection time	`remote detect multiplier × max(local required min RX, remote desired min TX)`	Detection is governed by the slower side's agreed interval; 3 × 50 ms gives 150 ms.
Exponential throttle	$wait_1$ = initial; $wait_2$ = hold; $wait_n$ = $\min(2 \times wait_{n-1}, maximum)$	Fast reaction to the first event and growing back-off during instability.
Basic LFA condition	`D(N,D) < D(N,S) + D(S,D)`	Neighbour N will not send traffic for D back through source S.
Node-protecting LFA	`D(N,D) < D(N,E) + D(E,D)`	N's path to D does not pass through the protected next hop E.
Downstream path	`D(N,D) < D(S,D)`	N is strictly closer to D than S, which also prevents micro-loops between them.

Tuning detection and computation

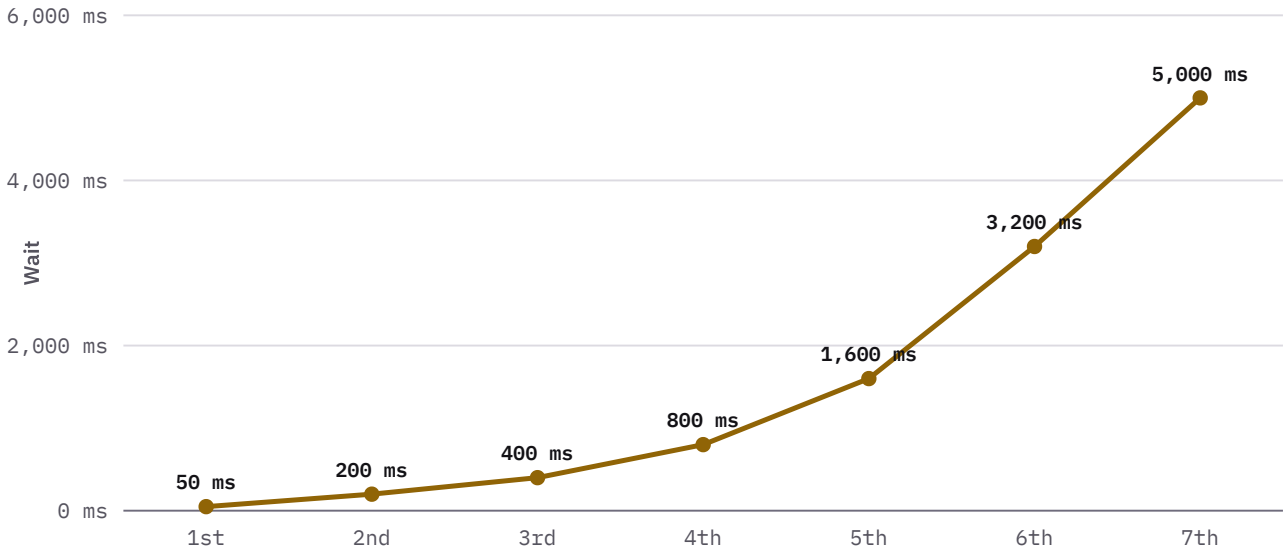
R1 · IOS-style CLI

```
interface GigabitEthernet0/0
 bfd interval 50 min_rx 50 multiplier 3
 ip ospf bfd
!
router ospf 1
 timers throttle spf 50 200 5000
 timers throttle lsa all 0 20 5000
 timers lsa arrival 15
! prioritise loopback prefixes for installation and
! enable fast reroute; both commands differ between platforms:
! enable per-prefix LFA, then remote LFA or TI-LFA where supported
```

SPF runs 50 ms after the first change, then waits 200 ms, 400 ms and so on up to 5 s while changes continue. The LSA arrival timer must be lower than the LSA hold value or neighbours discard legitimate rapid updates. Prioritising loopbacks makes BGP next hops converge before bulk prefixes.

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SPF throttle 50/200/5000 ms during a burst of changes



Computed from the configuration above: the first SPF runs 50 ms after the first change, then the wait doubles from the 200 ms hold value while changes continue, capped at 5 s, and resets after a quiet period.

A BFD session slower than configured

R1 · IOS-style CLI

```
R1# show bfd neighbors detail
Neighbor 10.1.12.2  Interface Gi0/0  State Up  Uptime 3d02h
 Local discr 7, Remote discr 12, Diag 0 (No Diagnostic)
 Local:  Desired min TX 50 ms, Required min RX 50 ms, Detect mult 3
 Remote: Desired min TX 300 ms, Required min RX 300 ms, Detect mult 3
 Negotiated TX 300 ms, Detection time 900 ms
 Clients: OSPF
```

The local side asked for 50 ms, but the neighbour will only send every 300 ms and will not accept faster than 300 ms. Detection on R1 is therefore 3 × 300 ms = 900 ms. BFD always settles on the slower requirement; fix the remote configuration or accept the longer detection time.

Fast reroute families

LFA

- Backup is a directly connected neighbour
- No tunnels needed
- Coverage depends heavily on topology; rings are poor
- Can use a path that is not the post-convergence path

Remote LFA

- Tunnels traffic to a PQ node beyond the failure
- Needs a label or tunnel to the PQ node
- Much better coverage in rings
- Extra state for tunnels to PQ nodes

TI-LFA

- Uses segment routing to encode the post-convergence path
- Full link and node protection coverage in any topology
- Backup equals the path used after convergence
- Requires segment routing in the network

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WATCH OUT · MICRO-LOOPS AND AGGRESSIVE TIMERS

Even with perfect detection, routers install new paths at slightly different times. For a few hundred milliseconds two routers can point at each other for the same prefix. Fast reroute protects the failure itself, but micro-loops can occur on unaffected paths during reconvergence; ordered updates or segment-routing-based avoidance address them. Very aggressive SPF and LSA timers in large or unstable networks can also multiply CPU load and flooding.

A safe tuning sequence

1. **Measure first** Record the current failure-to-recovery time for representative failures.
2. **Detection** Use loss of light where possible and BFD where links hide failures; choose intervals the platform can sustain in hardware.
3. **Propagation** Use a zero or near-zero initial LSA delay with exponential back-off.
4. **Computation** Set a short initial SPF delay with exponential back-off, and prioritise loopbacks.
5. **Protection** Deploy LFA, then remote LFA or TI-LFA for coverage gaps; check coverage reports.
6. **Validate** Repeat the same failures and compare against the baseline.

SCENARIO · DECIDE · COVERAGE GAPS IN THE RING

A metro network is built as a ring of eight routers. After enabling per-prefix LFA, coverage reports show only about half of destinations protected. The business requires sub-50 ms recovery for all destinations after any single link failure.

- (A) Reduce the BFD interval to 10 ms so failures are detected before traffic is lost.
- (B) Add remote LFA, or TI-LFA where available, for the unprotected destinations.
- (C) Lower the initial SPF delay to zero so the routers compute new paths immediately.
- (D) Add equal-cost paths by setting all ring link costs equal across the ring.

Circle your choice. Answer and feedback: module answer key.

Fast convergence readiness

☐ Baseline convergence measured for representative failures
☐ BFD intervals sustainable by the platform and matched on both ends
☐ LSA and SPF throttles use exponential back-off
☐ LSA arrival timer lower than the LSA hold value
☐ Loopbacks and next hops prioritised for installation
☐ Fast reroute coverage reported and gaps addressed
☐ Micro-loop exposure understood for critical flows

Key terms

BFD
Bidirectional Forwarding Detection (RFC 5880): a lightweight hello protocol that detects forwarding failures in milliseconds.

Detect multiplier
The number of missed BFD packets that declares a session down.

SPF throttling
Exponential back-off that delays SPF runs during repeated changes.

LSA throttling
Exponential back-off on originating new instances of the same LSA.

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LFA
Loop-free alternate: a pre-computed backup neighbour that will not loop traffic back.

TI-LFA
Topology-independent LFA: a segment-routing-based backup that follows the post-convergence path.

Prefix prioritisation
Installing important prefixes such as loopbacks before others during convergence.

PQ node
A node reachable from the source without the failed link and able to reach the destination without it; the target of remote LFA tunnels.

Micro-loop
A transient loop caused by routers updating their forwarding tables at different times.

REFLECT

Measure, or estimate, each convergence phase in a network you know. Which phase dominates, and which tool from this lesson attacks it?

Key takeaways

- Convergence is detect + propagate + compute + install; tune each phase separately and measure before and after.
- BFD detection uses the slower side: remote multiplier × max(local required RX, remote desired TX).
- Exponential throttles react quickly to the first event and back off under instability.
- LFA coverage depends on topology; TI-LFA covers any redundant topology on the post-convergence path but needs segment routing.

KNOWLEDGE CHECK

16. Local BFD is 50 ms × 3. The neighbour requires min RX 300 ms and sends at 300 ms with multiplier 3. What is the detection time on the local router?

(A) 150 milliseconds
(B) 450 milliseconds
(C) 900 milliseconds
(D) 300 milliseconds

17. Which inequality defines a basic loop-free alternate N for destination D from source S?

(A) D(N,D) < D(S,D)
(B) D(N,D) < D(N,S) + D(S,D)
(C) D(N,D) < D(N,E) + D(E,D)
(D) D(S,N) < D(S,D)

18. Why are rings difficult for basic LFA?

(A) Neighbours often reach destinations back through the failed link
(B) Ring links cannot carry BFD sessions at the short intervals that LFA requires
(C) SPF throttling behaves differently on ring topologies
(D) Ring routers cannot exchange the required LSA types

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19. What does an SPF throttle of 50/200/5000 ms do during a burst of changes?

(A) Runs SPF every 50 ms for as long as the network keeps changing, then waits 5 s

(B) Waits 5 s after the first change, then runs every 200 ms

(C) Runs SPF exactly three times, at 50, 200 and 5000 ms

(D) Runs 50 ms after the first change, then backs off from 200 ms up to 5 s

20. What advantage does TI-LFA have over basic LFA?

(A) It needs no forwarding plane support beyond plain IP routing

(B) It removes the need for BFD or any other failure detection

(C) Its backup is the post-convergence path, in any topology

(D) It prevents every micro-loop during network reconvergence

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MODULE 3 · IGP · LESSON 5 · 50 MIN

Mutual Redistribution, Loop Prevention and Database Troubleshooting

Two-way redistribution at multiple points without loops or suboptimal paths

- Explain how redistribution feedback loops and suboptimal routing form
- Prevent them with route tags, administrative distance and filtering
- Troubleshoot redistribution issues from database and routing table evidence

Redistribution loses information. A prefix that crosses from one protocol to another arrives with a new metric, a new origin and, by default, no memory of where it came from. With two-way redistribution at a single point this is manageable. With two or more points, a prefix can travel out of its home domain and back in, where it may beat the original route on administrative distance. The results are suboptimal paths, routes that persist after their source is gone, and loops.

How a feedback loop forms

01 **Origin**
Prefix P lives in the IS-IS domain

→

02 **Out**
R1 redistributes P into OSPF as an external route

→

03 **Preference**
R2 learns P via OSPF (distance 110) and IS-IS (115) and prefers OSPF

→

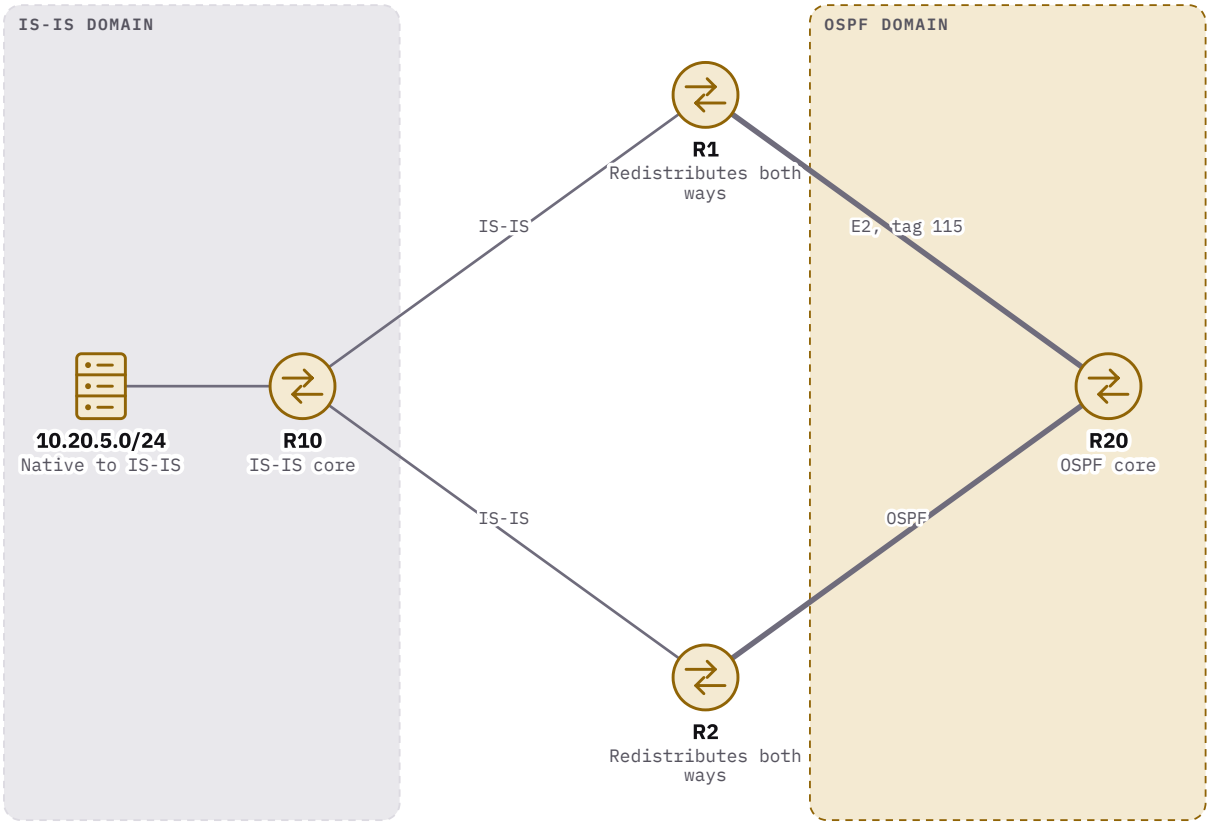
04 **Back in**
R2 redistributes its OSPF route for P back into IS-IS

→

05 **Persistence**
If P's origin fails, the copies keep each other alive and traffic loops

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Two redistribution points, one feedback path



R1 exports 10.20.5.0/24 into OSPF with tag 115. Without a deny-by-tag on R2 the copy re-enters IS-IS, and because OSPF's distance (110) beats IS-IS (115), R2 also prefers the detour through R1.

Prevention tools

Route tags

- Mark routes with their source domain on the way out
- Deny that tag on the way back in
- Works at any number of redistribution points
- IS-IS tags need wide metrics

Administrative distance

- Make each protocol's external routes less preferred than the other's internal routes
- Fixes suboptimal path selection at the redistribution routers
- Must be consistent on every redistribution router

Prefix filtering

- Explicit lists of prefixes allowed in each direction
- Very precise, high maintenance
- Good for small, static sets

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Tag-based loop prevention on R1 and R2

R1 · IOS-style CLI

```
route-map RM-ISIS-TO-OSPF deny 10
 match tag 110
route-map RM-ISIS-TO-OSPF permit 20
 set tag 115
!
route-map RM-OSPF-TO-ISIS deny 10
 match tag 115
route-map RM-OSPF-TO-ISIS permit 20
 set tag 110
!
router ospf 1
 redistribute isis CORE level-2 subnets route-map RM-ISIS-TO-OSPF
 distance ospf external 170
router isis CORE
 metric-style wide
 redistribute ospf 1 route-map RM-OSPF-TO-ISIS
```

Routes leaving IS-IS carry tag 115 and are refused on the way back into IS-IS; routes leaving OSPF carry tag 110 and are refused on the way back into OSPF. Raising OSPF external distance to 170 makes both redistribution routers prefer the native IS-IS path for IS-IS-originated prefixes, fixing suboptimal routing as well.

Evidence of a suboptimal path

R2 · IOS-style CLI

```
R2# show ip route 10.20.5.0
Routing entry for 10.20.5.0/24
  Known via "ospf 1", distance 110, metric 20, type extern 2, forward metric 30
  Tag 115, type extern 2
  Last update from 10.0.12.1 on GigabitEthernet0/1, 00:00:41 ago
  Routing Descriptor Blocks:
  * 10.0.12.1, from 10.0.0.1, 00:00:41 ago, via GigabitEthernet0/1
      Route metric is 20, traffic share count is 1
      Route tag 115
R2# show isis database detail | include 10.20.5.0
  Metric: 30         IP 10.20.5.0/24
```

The prefix is native to IS-IS (it appears in the IS-IS database) but R2 uses an OSPF external copy redistributed by R1, recognisable by tag 115, because distance 110 beats IS-IS 115. Traffic takes a detour through R1. The tags already stop the loop; the distance change is what removes the detour.

Database clues and what they mean

CLUE	LIKELY MEANING	NEXT STEP
External LSA or LSP from an unexpected advertising router	A second redistribution point exists or leaked	Check redistribution and tags on that router
External route with the other domain's tag inside its home domain	Feedback through a redistribution point	Confirm the deny-by-tag entries on every redistribution router
Rapidly increasing sequence numbers on external LSAs	Redistributed routes flapping, often due to a loop between points	Compare timestamps across both redistribution routers
MaxAge LSAs lingering for one prefix	Originator withdrew the route; another router may re-originate it	Look for re-origination from another redistribution point

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CLUE	LIKELY MEANING	NEXT STEP
Prefix present in the database, absent from the table	Lower-distance source, unreachable forwarding address or filter	Compare the routing entry's source and distance
IS-IS LSP with the overload bit set	Router intentionally avoided for transit	Check startup timers or overload configuration

CASE STUDY · THE MERGER THAT KEPT A DEAD DATA CENTRE ALIVE

Two utilities merged. Company A ran OSPF, company B ran IS-IS, and the integration team connected them at two sites with two-way redistribution and no tags. Months later, company B decommissioned a small data centre. Its prefixes were removed from IS-IS, yet they stayed reachable in the routing tables for hours, and traffic sent to them looped between the two redistribution routers until TTL expired, saturating an inter-site link. The issue cleared only when one redistribution point was disabled.

1. Explain the exact mechanism that kept the prefixes alive.
2. Why did the problem not appear while the data centre existed?
3. What would you configure at both redistribution points, and why both?
4. How would you prove the fix in a maintenance window?

Model answer: module answer key.

TEMPLATE **LSDB analysis sheet**

Record what each database entry says and whether it matches design intent, so anomalies stand out.

LSA OR LSP	TYPE OR LEVEL	ADVERTISING ROUTER	SEQ AND AGE	KEY FIELDS	EXPECTED?	FINDING
172.16.50.0/24	*Type 5*	*10.0.0.7*	*0x80000002 / 120 s*	*E2 metric 20, FA 10.7.7.9, tag 65010*	*No*	*FA unreachable; LSA unusable*
10.20.5.0/24	*Type 5*	*10.0.0.1*	*0x800000A1 / 41 s*	*E2 metric 20, tag 115*	*Partly*	*IS-IS prefix preferred via OSPF at R2*

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LSA OR LSP	TYPE OR LEVEL	ADVERTISING ROUTER	SEQ AND AGE	KEY FIELDS	EXPECTED?	FINDING

SCENARIO · DECIDE · CHOOSING THE FIX AT TWO REDISTRIBUTION POINTS

A network redistributes both ways between OSPF and IS-IS on R1 and R2. Tags are already set and denied correctly, so no loops occur. However, R2 sends traffic for several IS-IS prefixes through R1 using OSPF external routes, adding latency.

(A) Raise the distance of OSPF external routes above the IS-IS distance on both redistribution routers.

(B) Remove the tags and rely on distance alone to prevent loops at both routers.

(C) Stop redistributing IS-IS into OSPF on R1 so only R2 provides the OSPF domain's routes.

(D) Lower the IS-IS metric of the affected prefixes so their IS-IS path looks shorter.

Circle your choice. Answer and feedback: module answer key.

Mutual redistribution checklist

☐ Every redistribution point tags routes by source domain

☐ Every redistribution point denies the other domain's tag on the way back

☐ Distances make native routes win over redistributed copies

☐ Redistribution limited to intended prefix ranges

☐ Wide metrics enabled where IS-IS tags are used

☐ Failure of each redistribution point tested in turn

☐ Withdrawal of a test prefix verified to clear both domains

Key terms

Mutual redistribution
Redistributing routes in both directions between two routing domains.

Route tag
A numeric label attached to a route, used to recognise its origin during later redistribution.

Suboptimal routing
Forwarding along a longer path because a redistributed copy wins on distance.

Seed metric
The metric assigned to routes when they are redistributed into a protocol.

Feedback loop
A route redistributed out of its home domain and back in, where it can persist or loop.

Administrative distance
A per-source preference value used to choose between routes from different protocols.

Redistribution point
A router that redistributes between two routing domains.

External route
A route injected into a link-state protocol from outside it, such as an OSPF type 5 LSA.

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REFLECT

Draw a network with three redistribution points between two domains. Which routers need tags, which need distance changes, and which failure combinations would you test?

Key takeaways

- Redistribution loses origin information; with two or more points, routes can return home and loop.
- Tag routes by source domain and deny the other domain's tag at every redistribution point.
- Tags stop feedback; administrative distance decides preference, so make native routes win.
- Prove the design by withdrawing a test prefix and watching it clear from both domains.

KNOWLEDGE CHECK

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21. Tags are set and denied correctly at both redistribution points, but one router still prefers redistributed copies of the other domain's routes. What fixes the preference?

(A) Raising the external distance so native routes win at that router

(B) Adding a second tag so the routes carry both domain values

(C) Lowering the seed metric for all routes redistributed at that router

(D) Removing the deny entries so routes can be compared by metric

22. A prefix withdrawn from IS-IS persists in both domains and traffic loops. What is the most likely cause?

(A) An IS-IS LSP stuck at MaxAge in the database of one level 1 router

(B) An OSPF summary range that still covers the withdrawn prefix

(C) A BFD session that did not detect the failure of the prefix

(D) Two untagged redistribution points feeding each other's copies

23. Why must IS-IS use wide metrics in a tag-based redistribution design?

(A) Narrow metrics cannot represent the distance values used by OSPF

(B) Tags are sub-TLVs of the extended IP reachability TLV

(C) Wide metrics are required for any redistribution into IS-IS at all

(D) Narrow metrics disable the up/down bit used to stop loops

24. R2 shows a prefix as OSPF external with tag 115, while the IS-IS database also contains it. What does that indicate?

(A) R2 prefers an OSPF external copy redistributed from IS-IS by another router

(B) R2 itself redistributed the prefix from OSPF into IS-IS earlier

(C) The prefix originated in OSPF and was leaked into IS-IS by mistake

(D) The IS-IS copy has a lower metric and should be preferred by R2

25. Why should tag filtering be configured on every redistribution point, not just one?

(A) Tags are only attached when every router agrees on the same value

(B) Any unfiltered point can carry routes back to their home domain

(C) Routers ignore tags unless they are configured on both protocols

(D) OSPF discards external routes that carry an unknown tag value

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · IGP

Answer key

Knowledge checks

1. C After sequence number and checksum, an instance at MaxAge is considered newer, ensuring flushes win.

2. A A non-zero forwarding address that does not resolve through an intra-area or inter-area route disqualifies the LSA.

3. D DROthers send updates to AllDRouters; the DR then floods to AllSPFRouters.

4. C 10 days ÷ 30 minutes ≈ 480 refreshes; 0x600 is 1536 instances, so the router's links have been changing.

5. B Type 4 carries the cost from the ABR to the ASBR so external routes can be resolved across areas.

6. A Stub, totally stubby and totally NSSA areas get an automatic type 3 default; a plain NSSA does not.

7. D Translator election picks the highest router ID among eligible NSSA ABRs unless one is forced.

8. A Virtual links are area 0 links configured with router IDs and require a non-stub transit area.

9. B The database is unchanged, so downstream routers still compute paths through R3.

10. C Traffic to missing components inside the range is dropped at the ABR instead of following a default route.

11. D Level 1 routers install a default toward the closest attached level 1/2 router.

12. D The up/down bit prevents leaked routes from looping between levels.

13. C TLV 135 carries wide-metric prefixes with flags and sub-TLVs such as tags.

14. A The overload bit keeps the router out of transit paths while it converges, preventing blackholes.

15. B IS-IS DIS election is preemptive with no backup and uses priority, then MAC address.

16. C Detection = remote multiplier 3 × max(local required RX 50, remote desired TX 300) = 900 ms.

17. B The basic LFA condition ensures N's shortest path to D does not go back through S.

18. A In rings, the neighbour on the other side usually reaches the destination back through the source.

19. D Exponential back-off reacts fast to the first event and slows down under instability.

20. C TI-LFA uses segment routing to encode the post-convergence path with full coverage; it still needs detection and does not eliminate all micro-loops.

21. A Choice between protocols is by distance; tags only prevent feedback.

22. D Without tags, each redistributed copy supports the other after the origin disappears.

23. B Tags are sub-TLVs of TLV 135, which requires wide metrics.

24. A Tag 115 marks routes redistributed from IS-IS; R2 prefers that copy because OSPF distance is lower.

25. B Feedback can occur through any unprotected redistribution point.

Scenarios & case studies

> **Scenario · A route that is in the database but not the table**
> **D** is best. Correct. A non-zero forwarding address must resolve through an intra-area or inter-area route; if it does not, the external LSA is not used, which matches every router ignoring it.

> **Scenario · Traffic dies at R3**
> **A** is best. Correct. R4 still sees the prefix in the database and computes its path through R3, which has no route. Filtering at the ABR removes the prefix consistently from the whole area.

> **Scenario · Level 1 routers take the long way out**
> **B** is best. Correct. Without leaking, R25 only knows the default route toward the nearest attached router. Leaked prefixes carry the end-to-end metric, so R25 can choose R22 when its total cost is lower.

> **Scenario · Coverage gaps in the ring**
> **B** is best. Correct. Rings are the classic topology where a neighbour's shortest path loops back through the failed link. Tunnelling to a PQ node or encoding the post-convergence path closes the gap.

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Case study · The merger that kept a dead data centre alive

Each redistribution router learned the prefixes from the other protocol as externals and redistributed them back, so each copy was supported by the other. While the data centre existed, the native IS-IS routes were present, and depending on distances the loop was hidden or caused only suboptimal paths. When the origin disappeared, the two external copies kept refreshing each other, and traffic followed them in a loop. The fix is tag-based filtering on both routers in both directions (a single router can still create the loop) plus distance settings so each domain's native routes win, and redistribution limited to the intended prefix ranges. Prove it by withdrawing a test prefix from IS-IS and confirming it disappears from both domains within normal convergence time, with no externals carrying the wrong tag in either database.

Scenario · Choosing the fix at two redistribution points

A is best. Correct. Tags stop feedback, but distance still makes the OSPF copy win at R2. Making OSPF externals less preferred than native IS-IS routes restores the direct path, and doing it on both routers keeps behaviour symmetric.

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04

MODULE 4 · BGP

BGP Mastery

Reflection and confederations, community policy, best-path depth, advanced advertisement, convergence and security

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Scale iBGP with reflectors and confederations while preserving path diversity
- Build community-driven policy frameworks that prevent leaks by design
- Predict best-path outcomes, including MED subtleties, from raw show output
- Engineer conditional advertisement, add-path and fast BGP convergence
- Secure BGP sessions and routes, and diagnose complex BGP incidents

LESSONS

1. Scaling iBGP: Route Reflectors and Confederations · 45 min
2. Communities and Policy Frameworks · 50 min
3. Best-Path Selection in Depth · 45 min
4. Advanced Advertisement and Convergence Tuning · 50 min
5. BGP Security and Complex Troubleshooting · 50 min

MODULE 4 · BGP · LESSON 1 · 45 MIN

Scaling iBGP: Route Reflectors and Confederations

Reflection rules, loop prevention, path hiding and when to choose confederations

- Apply route reflection rules and explain ORIGINATOR_ID and CLUSTER_LIST
- Design redundant reflector clusters and recognise path hiding
- Configure and verify a confederation and explain how attributes cross sub-AS borders

iBGP does not re-advertise iBGP-learned routes to other iBGP peers, so a plain design needs a full mesh: every router peers with every other. That does not scale. **Route reflection** (RFC 4456) relaxes the rule for designated reflectors, and **confederations** (RFC 5065) split the AS into sub-ASes that exchange routes using eBGP-like sessions. Both trade some path visibility for scale.

Session arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Full mesh	`n × (n − 1) ÷ 2`	100 routers need 4,950 sessions; every new router adds n − 1 sessions.
Two reflectors, all others clients	`2 × (n − 2) + 1`	100 routers need 197 sessions; a new client adds two.
Paths a client sees per prefix	`one per reflector it peers with (without add-path)`	Reflectors advertise only their own best path.

Reflection rules

ROUTE LEARNED FROM	REFLECTOR SENDS IT TO
An eBGP peer	All clients and all non-clients
A client	All other clients and all non-clients (and eBGP peers)
A non-client iBGP peer	Clients only (and eBGP peers)

DEFINITION · LOOP PREVENTION ATTRIBUTES

A reflector adds **ORIGINATOR_ID**, the router ID of the iBGP speaker that injected the route, if it is not already present; a router that receives a route carrying its own router ID as originator drops it. The reflector also prepends its **cluster ID** to **CLUSTER_LIST**; a reflector that receives a route with its own cluster ID drops it. Neither attribute is sent to eBGP peers.

Cluster ID choices for a redundant pair

Same cluster ID on both reflectors

- Each reflector drops routes reflected by its partner
- Less memory on the reflectors
- Every client must peer with both reflectors for redundancy

Unique cluster IDs

- Each reflector keeps its partner's reflected copy
- More paths stored, more memory
- Survives a client losing one of its sessions more gracefully

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Route reflector configuration

RR1 · IOS-style CLI

```
router bgp 65000
 bgp router-id 10.0.0.1
 bgp cluster-id 10.0.0.100
 neighbor RR-CLIENTS peer-group
 neighbor RR-CLIENTS remote-as 65000
 neighbor RR-CLIENTS update-source Loopback0
 neighbor 10.0.0.11 peer-group RR-CLIENTS
 neighbor 10.0.0.12 peer-group RR-CLIENTS
 neighbor 10.0.0.2 remote-as 65000
 neighbor 10.0.0.2 update-source Loopback0
 !
 address-family ipv4
  neighbor RR-CLIENTS activate
  neighbor RR-CLIENTS route-reflector-client
  neighbor RR-CLIENTS send-community both
  neighbor 10.0.0.2 activate
  neighbor 10.0.0.2 send-community both
```

10.0.0.2 is the second reflector, peered as a non-client. Only the reflectors need reflection configuration; clients are ordinary iBGP speakers. Reflectors do not change the next hop, so every client still needs an IGP route to every edge router's next hop.

A client sees one exit through two reflectors

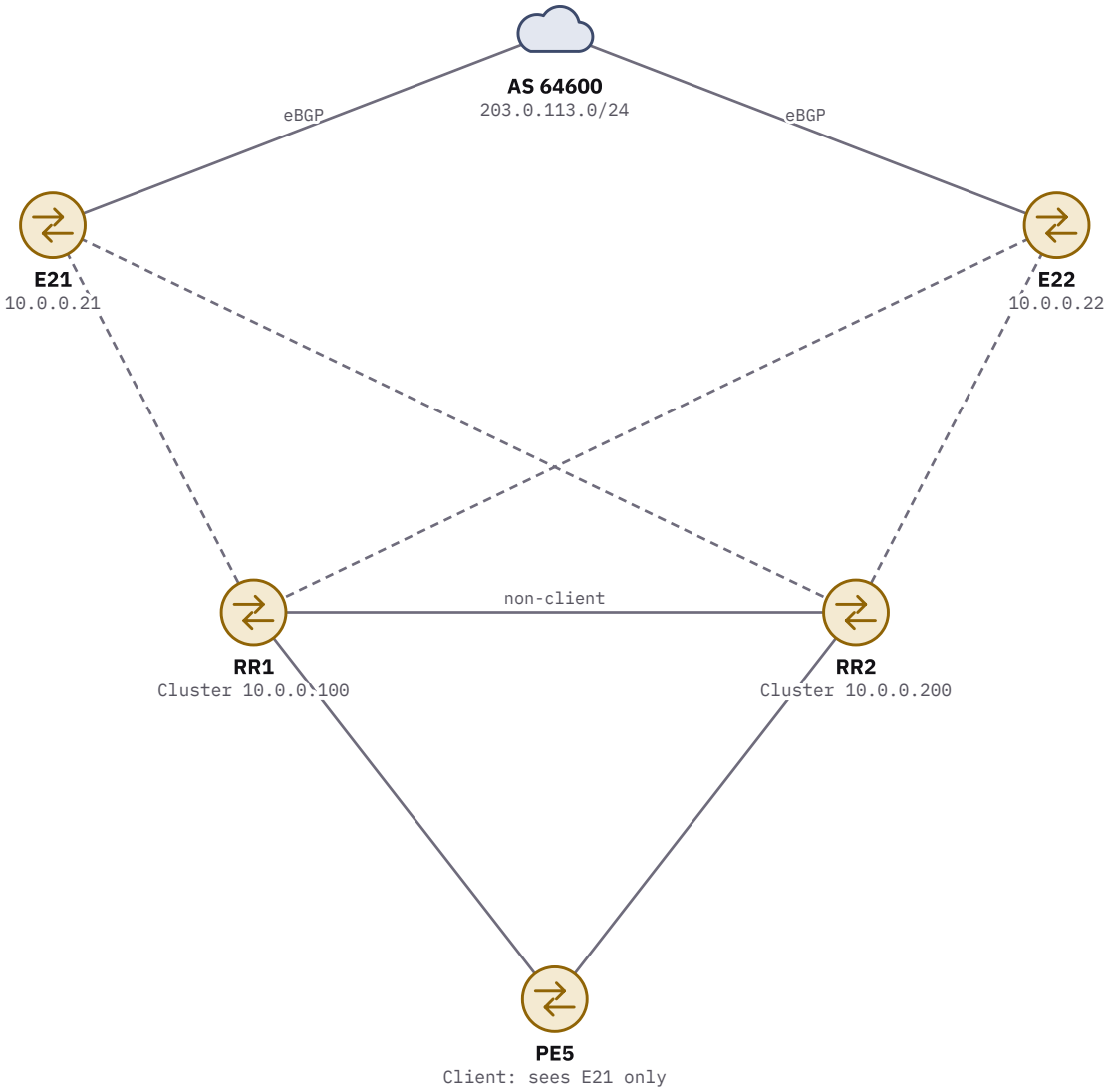
PE5 · IOS-style CLI

```
PE5# show ip bgp 203.0.113.0/24
BGP routing table entry for 203.0.113.0/24, version 8812
Paths: (2 available, best #1, table default)
  64600
    10.0.0.21 (metric 30) from 10.0.0.1 (10.0.0.1)
      Origin IGP, metric 0, localpref 200, valid, internal, best
      Originator: 10.0.0.21, Cluster list: 10.0.0.100
  64600
    10.0.0.21 (metric 30) from 10.0.0.2 (10.0.0.2)
      Origin IGP, metric 0, localpref 200, valid, internal
      Originator: 10.0.0.21, Cluster list: 10.0.0.200
```

Both reflectors (unique cluster IDs) reflect the same path through edge router 10.0.0.21. Edge router 10.0.0.22 also learns 203.0.113.0/24 from AS 64600, but neither reflector selected it, so PE5 never sees it. That is path hiding: no load sharing, no pre-computed backup, and slower convergence if 10.0.0.21 fails.

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A redundant reflector pair hides an exit



Both reflectors select the path via E21 because it is closer to them in the IGP, so PE5 never learns the exit through E22. Add-path, best-external or optimal route reflection brings it back.

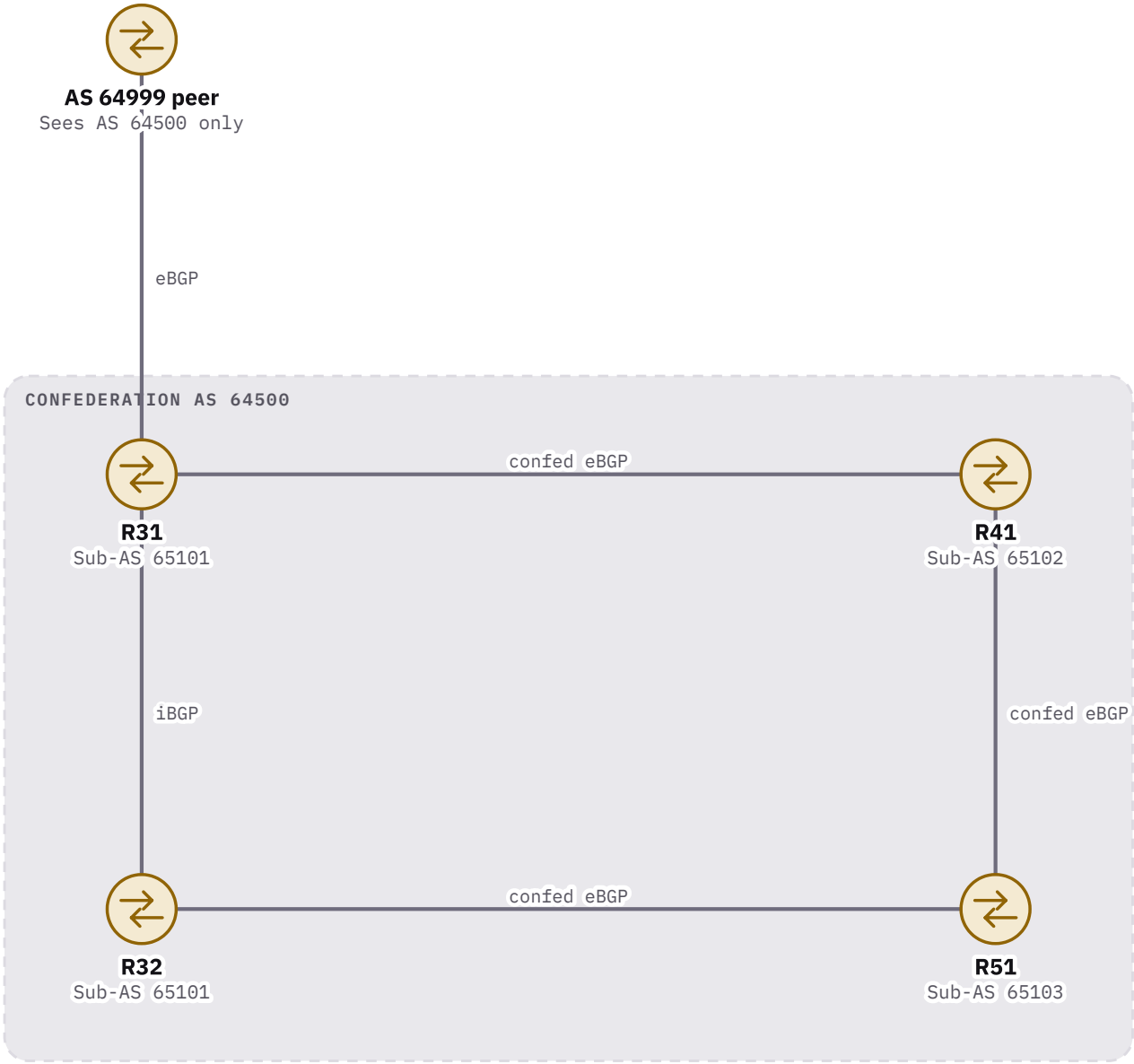
PRO TIP · REFLECTORS THAT CHOOSE FOR THE CLIENT

Optimal route reflection (RFC 9107) lets a reflector run best-path selection from the IGP position of each client or client group instead of its own, so centralised reflectors stop breaking hot-potato exits. Many large networks now run reflectors as virtual machines or containers outside the forwarding path; that is only safe when optimal route reflection or add-path compensates for their position.

A **confederation** divides an AS into sub-ASes, usually with private AS numbers. Inside a sub-AS, normal iBGP rules apply (full mesh or reflectors). Between sub-ASes, sessions behave like eBGP for loop prevention, using AS_CONFED_SEQUENCE segments, but NEXT_HOP, LOCAL_PREF and MED are preserved as if the session were iBGP. Confederation segments are not counted in AS-path length during best-path selection and are removed when routes leave the confederation, so outsiders see a single AS.

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A confederation from inside and outside



Inside AS 64500, sub-AS 65101 runs normal iBGP. Sessions between sub-ASes behave like eBGP for loop prevention but keep NEXT_HOP, LOCAL_PREF and MED, and the external peer sees only AS 64500.

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Confederation member

R31 · IOS-style CLI

```
router bgp 65101
 bgp confederation identifier 64500
 bgp confederation peers 65102 65103
 neighbor 10.0.0.41 remote-as 65102
 neighbor 10.0.0.41 update-source Loopback0
 neighbor 10.0.0.41 ebgp-multihop 2
 neighbor 10.0.0.32 remote-as 65101
 neighbor 10.0.0.32 update-source Loopback0
 neighbor 192.0.2.1 remote-as 64999
```

The router lives in sub-AS 65101; the outside world sees AS 64500. 10.0.0.41 is a confederation peer in sub-AS 65102 and, like any eBGP session between loopbacks, needs multihop. 10.0.0.32 is an iBGP peer inside the same sub-AS, and 192.0.2.1 is a true external peer.

SCENARIO · DECIDE · HOT-POTATO ROUTING BREAKS AFTER ADDING REFLECTORS

A national network moved from full mesh to two centralised route reflectors in the capital. Since then, edge routers in the south send traffic for many destinations to exits in the north, even though equally good exits exist in the south. Policy and local preference are identical at all exits.

- (A) Lower the IGP metric on all southern links so the reflectors see southern exits as closer.
- (B) Set the same cluster ID on both reflectors so they stop exchanging paths with each other.
- (C) Use add-path or regional reflectors so each client can pick its nearest exit.
- (D) Configure next-hop-self on the reflectors so that all traffic is drawn toward the capital.

Circle your choice. Answer and feedback: module answer key.

iBGP scaling review

- ☐ Reflectors redundant, with a deliberate cluster ID choice
- ☐ Every client peers with at least two reflectors
- ☐ Reflector placement considered for hot-potato exit selection
- ☐ Path diversity strategy defined (add-path, best-external or regional reflectors)
- ☐ Next hops resolvable by every client through the IGP
- ☐ send-community enabled on every iBGP session
- ☐ Confederation sub-AS sessions use multihop where peering via loopbacks

Key terms

Route reflector
An iBGP speaker allowed to re-advertise iBGP-learned routes to its clients and non-clients according to fixed rules.

Client
An iBGP peer of a reflector that receives reflected routes; it needs no special configuration.

ORIGINATOR_ID
An optional attribute holding the router ID of the route's originator inside the AS, used to drop looping routes.

CLUSTER_LIST
An optional attribute listing the cluster IDs a route has been reflected through.

Cluster ID
The identifier of a reflection cluster, by default the reflector's router ID.

Path hiding
Loss of alternative paths because a reflector advertises only its own best path.

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Confederation
An AS split into sub-ASes that exchange routes over eBGP-like sessions while appearing as one AS externally.

Hot-potato routing
Sending traffic to the nearest exit by IGP metric when other attributes are equal.

AS_CONFED_SEQUENCE
The AS-path segment listing the sub-ASes a route has crossed inside a confederation.

REFLECT

In an iBGP design you know, which exits are hidden from which routers today? What would you change first to expose them?

Key takeaways

- Reflectors pass on only their own best path, chosen from their own IGP position.
- ORIGINATOR_ID and CLUSTER_LIST stop reflection loops; a shared cluster ID makes dual peering to both reflectors essential.
- Restore path diversity with add-path, best-external, regional reflectors or optimal route reflection.
- Confederation segments are ignored in AS-path length and stripped at the edge, so outsiders see one AS.

KNOWLEDGE CHECK

1. A reflector learns a route from a non-client iBGP peer. Where does it reflect it?
   - (A) To all clients and all other non-client iBGP peers
   - (B) To non-client iBGP peers only, plus any eBGP peers
   - (C) Nowhere, because non-client routes are never reflected
   - (D) To its own clients only, plus any of its eBGP peers
2. A reflector receives a route whose CLUSTER_LIST contains its own cluster ID. What happens?
   - (A) It reflects the route only to its non-client iBGP peers
   - (B) It accepts the route and replaces the originator ID
   - (C) It discards the route as a reflection loop in its cluster
   - (D) It accepts the route but marks it as not eligible for best path
3. Two reflectors share a cluster ID. A client loses its session to RR1. What is the effect?
   - (A) RR1 still learns the client's routes, reflected to it by RR2 as usual
   - (B) The client's routes are withdrawn from every router in the AS
   - (C) RR1 loses the client's routes, as it drops RR2's reflected copies
   - (D) RR2 stops reflecting the client's routes to avoid a duplicate
4. How is AS_CONFED_SEQUENCE treated during best-path selection?
   - (A) Each sub-AS counts as one hop like any normal AS
   - (B) It is excluded entirely from the AS-path length comparison
   - (C) The whole confederation segment counts as a single hop
   - (D) It is compared only after the router ID tie-breaker

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5. Why can centralised reflectors break hot-potato routing?

(A) Clients inherit best paths chosen from the reflector's IGP position

(B) They change the next hop of every route to their own loopback address

(C) They remove the MED attribute from routes reflected to clients

(D) They set a lower local preference on routes from distant clients

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BGP · LESSON 2 · 50 MIN

Communities and Policy Frameworks

Standard, extended and large communities as the language of scalable routing policy

- Distinguish standard, extended and large communities and the well-known values
- Design a classify-act-export policy framework driven by communities
- Verify community propagation and policy outcomes per neighbour

Per-neighbour route-maps written by hand do not scale beyond a handful of peers: every new policy must be replicated, and nobody can say what the policy is without reading every line. Mature networks **classify** routes once, at ingress, by attaching communities, and then make every later decision by matching those communities. The policy becomes a small, readable vocabulary.

Community types

TYPE	DEFINED IN	SIZE AND FORMAT	TYPICAL USE
Standard	RFC 1997	32 bits, written ASN:value with 16-bit halves	Classification and action signals in 16-bit ASN networks
Extended	RFC 4360	64 bits with a type field	Route targets, route origin, transitive and non-transitive types
Large	RFC 8092	96 bits: global administrator:local data 1:local data 2	Policy signalling for networks with 32-bit ASNs
Well-known	RFC 1997 and later	Reserved standard values	NO_EXPORT, NO_ADVERTISE, NO_EXPORT_SUBCONFED, BLACKHOLE (65535:666), GRACEFUL_SHUTDOWN (65535:0)

The ZyeemZia policy framework

P1

Sanitise

At ingress, drop bogons, overly specific prefixes, RPKI-invalid routes and anything outside the neighbour's expected prefix set; strip communities in your own namespace that the neighbour may not set.

P2

Classify

Tag every route with its relationship (customer, peer, transit, internal), region and point of entry.

P3

Prefer

Set local preference from the class: customer above peer above transit.

P4

Act

Honour documented action communities from customers, such as prepend toward a region or do not announce to a named peer.

P5

Export

Advertise by class: customer routes to everyone, peer and transit routes only to customers.

P6

Scrub

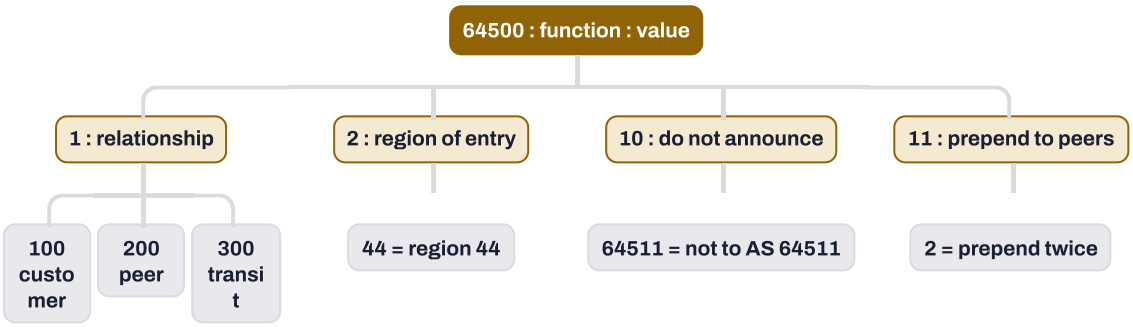
Remove internal-only communities at the edge and send only what neighbours are documented to receive.

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Example community plan (AS 64500, large communities)

COMMUNITY	MEANING	SET BY
64500:1:100	Learned from a customer	Ingress policy
64500:1:200	Learned from a settlement-free peer	Ingress policy
64500:1:300	Learned from a transit provider	Ingress policy
64500:2:44	Entered in region 44	Ingress policy
64500:10:64511	Do not announce to AS 64511	Customer action
64500:11:2	Prepend twice toward peers	Customer action

The AS 64500 large-community plan



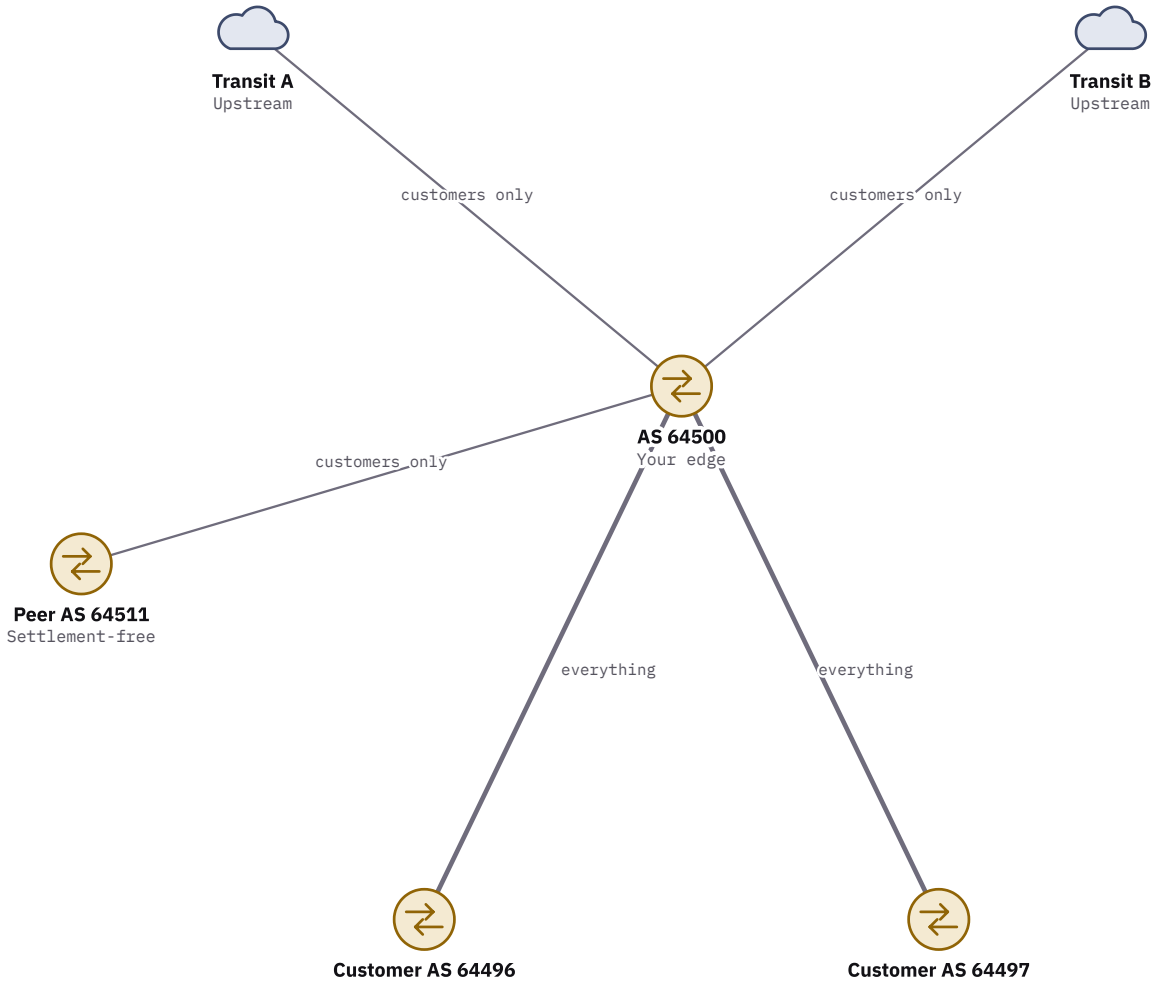
The middle field names a function, so policies match function and value instead of raw prefixes. Classification functions are set only by your ingress policy; action functions may be set by customers.

INSIGHT · EXPORT BY CLASS IS ROUTE-LEAK PREVENTION

The rule that peer and transit routes are exported only to customers is the valley-free model in practice. If every route is classified at ingress and every export policy matches on class, a misconfigured neighbour cannot turn your network into a transit path between two providers.

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Valley-free export by class



Routes learned from transits and peers go only to customers; customer routes go to everyone. Classified at ingress and exported by class, no route can turn your network into a free path between two providers.

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Classification and export with standard communities

EDGE1 · IOS-style CLI

```
ip prefix-list PL-CUST-ACME seq 5 permit 198.51.100.0/24
ip community-list standard CL-CUSTOMER permit 64500:100
ip community-list standard CL-NO-PEERS permit 64500:990
ip community-list standard CL-INTERNAL permit 64500:100
ip community-list standard CL-INTERNAL permit 64500:990
!
route-map RM-IN-CUST-ACME permit 10
 match ip address prefix-list PL-CUST-ACME
 set local-preference 200
 set community 64500:100 additive
!
route-map RM-OUT-PEER deny 5
 match community CL-NO-PEERS
route-map RM-OUT-PEER permit 10
 match community CL-CUSTOMER
 set comm-list CL-INTERNAL delete
route-map RM-OUT-PEER deny 100
```

Inbound, the customer's routes are filtered by prefix, preferred and tagged as customer. Outbound to peers, routes the customer marked with 64500:990 are suppressed, only customer-class routes are permitted, and internal tags are removed before advertisement. Everything else, including peer and transit routes, is denied.

Proving that a community survived

EDGE2 · IOS-style CLI

```
EDGE2# show ip bgp 198.51.100.0/24
BGP routing table entry for 198.51.100.0/24, version 5521
Paths: (1 available, best #1, table default)
  64511
    10.0.0.11 (metric 20) from 10.0.0.1 (10.0.0.1)
      Origin IGP, metric 0, localpref 200, valid, internal, best
      Originator: 10.0.0.11, Cluster list: 10.0.0.100
EDGE2# show ip bgp neighbors 192.0.2.77 advertised-routes | include 198.51.100
 *>i 198.51.100.0/24  10.0.0.11       0    200      0 64511 i
```

The route reached EDGE2 with local preference 200 but no Community line at all, so the customer's do-not-announce-to-peers signal and the customer class were lost on the way. EDGE2 advertises it to peer 192.0.2.77 because a looser legacy policy still applies there. Check send-community on the sessions from EDGE1 and the reflectors.

WATCH OUT · COMMUNITIES ARE NOT SENT BY DEFAULT EVERYWHERE

Many implementations do not send communities to a neighbour unless told to. A single iBGP session without community sending, often on a reflector, silently strips classification for every router behind it. Make community sending part of the session template, and verify it per session.

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SCENARIO · DECIDE · A CUSTOMER'S SIGNAL IS IGNORED

A customer tags its prefix with 64500:990, the documented do-not-announce-to-peers community. The route still appears at an internet exchange peer. On the egress router the prefix has no communities at all, while on the ingress edge router it shows 64500:100 and 64500:990.

(A) Add an explicit prefix-list deny for the customer's prefix on the peer-facing export policy.

(B) Ask the customer to use the large-community version of the signal instead of the standard one.

(C) Raise the local preference on the ingress router so the customer route is preferred everywhere.

(D) Find the iBGP session, possibly on a reflector, that is not sending communities, and fix it.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · REDESIGNING A MULTINATIONAL'S BGP POLICY

A multinational retailer runs its own AS with six transit providers and two internet exchanges across three regions. Over ten years, engineers added per-neighbour route-maps; the largest is over 400 lines. Local preference values differ by region, so traffic for the same destination exits in different continents depending on where it enters. Two incidents in a year leaked routes from one transit provider to another, and nobody can predict the effect of a change without lab testing.

1. Which framework stages are missing today?

2. How would you define the community plan?

3. How would you migrate without an outage?

4. What would you measure to prove the redesign worked?

Model answer: module answer key.

TEMPLATE **BGP policy matrix**

Define, per neighbour class, everything that ingress and egress policy must do, and how to prove it.

NEIGHBOUR CLASS	INGRESS FILTERS	TAGS SET	LOCAL PREFERENCE	EXPORTED CLASSES	ACTIONS HONOURED	PROOF COMMAND
Customer	*Prefix list from registry, max-prefix, RPKI invalid drop*	*64500:1:100, region, entry*	*200*	*All classes*	*Prepend, no-export to named peer*	*advertised-routes per peer, community per prefix*
Transit	*Bogons, too-specific, RPKI invalid drop*	*64500:1:300, region, entry*	*100*	*Customer only*	*None*	*advertised-routes shows customer routes only*

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NEIGHBOUR CLASS	INGRESS FILTERS	TAGS SET	LOCAL PREFERENCE	EXPORTED CLASSES	ACTIONS HONOURED	PROOF COMMAND

Community policy hygiene

☐ Community plan documented in one place and generated from a source of truth

☐ Every ingress session classifies every accepted route

☐ Customers cannot set internal classification communities

☐ Export policies match on class, never on raw prefixes alone

☐ Community sending enabled and verified on every session

☐ Internal communities scrubbed before external advertisement

☐ Well-known communities honoured as documented

Key terms

Standard community
A 32-bit BGP attribute value defined in RFC 1997, written as two 16-bit numbers.

Large community
A 96-bit attribute value defined in RFC 8092 that fits 32-bit ASNs in the global administrator field.

NO_ADVERTISE
A well-known community that prevents a route from being advertised to any peer.

Classification community
A community attached at ingress to record a route's relationship, region or entry point.

Scrubbing
Removing communities that should not leave the network or that a neighbour should not have set.

Extended community
A 64-bit attribute value with a type field, defined in RFC 4360, used for route targets and similar.

NO_EXPORT
A well-known community that keeps a route inside the local AS or confederation.

Action community
A community a neighbour sets to request specific treatment, such as prepending or suppression.

Valley-free export
The rule that routes from peers and transit providers are exported only to customers.

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REFLECT

If you had to describe your organisation's routing policy in ten communities, what would they be? Which current per-neighbour rules would they replace?

Key takeaways

- Classify once at ingress with communities, then make every later decision by matching them.
- Use large communities when your ASN is 32-bit or when you need function:value structure.
- Sanitise your own namespace at ingress, export by class and scrub internal tags at egress.
- Make community sending part of every session template and verify it per session.

KNOWLEDGE CHECK

6. Which community type suits a network with a 32-bit ASN that wants its own ASN in the community?

   (A) Standard communities, using only the low 16 bits of the 32-bit ASN
   (B) Extended communities of the route-target type
   (C) Well-known communities from the reserved range
   (D) Large communities, with the ASN as global administrator

7. A route carries NO_EXPORT. A router in the same AS receives it from an iBGP peer. What may it do?

   (A) Advertise it to other iBGP peers but not to external neighbours
   (B) Advertise it to external neighbours with the community removed
   (C) Keep it locally and advertise it to no neighbour at all
   (D) Advertise it only to neighbours in the same confederation sub-AS

8. Communities are present at the ingress router but missing at the egress router. What is the most likely cause?

   (A) The egress router's inbound policy does not match on communities
   (B) Communities expire after crossing more than two iBGP hops
   (C) An iBGP session in between is not configured to send any communities
   (D) The reflector replaces communities with its own cluster ID

9. Why does exporting peer and transit routes only to customers prevent route leaks?

   (A) It ensures every route you export is valid according to RPKI
   (B) It limits the number of prefixes you advertise to each neighbour
   (C) It forces customers to set action communities on their routes
   (D) You can never become a transit path between two non-customers

10. Where should classification communities be attached?

    (A) At egress, just before the route is advertised to neighbours
    (B) At ingress, on the edge router where the route first enters the network
    (C) On the reflectors, as routes are reflected to clients
    (D) On every router, so each one can confirm the class

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BGP · LESSON 3 · 45 MIN

Best-Path Selection in Depth

Every tie-breaker, the traps around MED, and reading the BGP table like an expert

- Apply the complete best-path sequence to multi-path show output
- Explain MED comparison rules, deterministic MED and missing-MED handling
- Predict how inbound and outbound policies interact with provider policies

Best-path selection is deterministic once you know the order and the exceptions. Most surprises come from three places: attributes you did not expect to be compared (or not compared), the order in which paths are compared when MED is involved, and policies applied by neighbours that override anything you send.

The decision sequence

STEP	PREFER	NOTES
0	Valid paths only	Next hop resolvable, no AS-path loop, accepted by policy; many implementations can also exclude RPKI-invalid paths
1	Highest weight	Local to the router, not a protocol attribute; implementation-specific
2	Highest LOCAL_PREF	Default 100; carried across iBGP only
3	Locally originated	Network, aggregate or redistribution on this router
4	Shortest AS_PATH	AS_SET counts as one; confederation segments are not counted
5	Lowest ORIGIN	IGP, then EGP, then incomplete
6	Lowest MED	Compared only between paths from the same neighbouring AS unless configured otherwise
7	eBGP over iBGP	Confederation-external paths count as internal
8	Lowest IGP metric to next hop	Hot-potato step
9	Multipath	If enabled, equal paths up to here may be installed together
10	Oldest eBGP path	Stability tie-breaker; often skippable by configuration
11	Lowest router ID	ORIGINATOR_ID replaces router ID for reflected paths
12	Shortest CLUSTER_LIST	Fewer reflection hops
13	Lowest neighbour address	Final tie-breaker

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Counting AS-path length

MEASURE	FORMULA	WHAT IT TELLS YOU
Length	`count of ASNs in AS_SEQUENCE segments + 1 for each AS_SET segment`	Prepending adds to the sequence count; an aggregate's AS_SET counts once however many ASNs it holds.
Confederation segments	`AS_CONFED_SEQUENCE and AS_CONFED_SET count as 0`	Routing inside a confederation does not lengthen the path.

WATCH OUT · MED IS COMPARED IN A SURPRISING WAY

MED is only compared between paths from the same neighbouring AS by default, so the result can depend on the order in which paths are compared. Path A may beat B on IGP metric, B may beat C, and C may beat A on MED: there is no single correct winner without a fixed comparison order. **Deterministic MED** groups paths by neighbouring AS, picks the best in each group, and only then compares the group winners, which makes the outcome independent of arrival order.

Why the lowest MED is not best

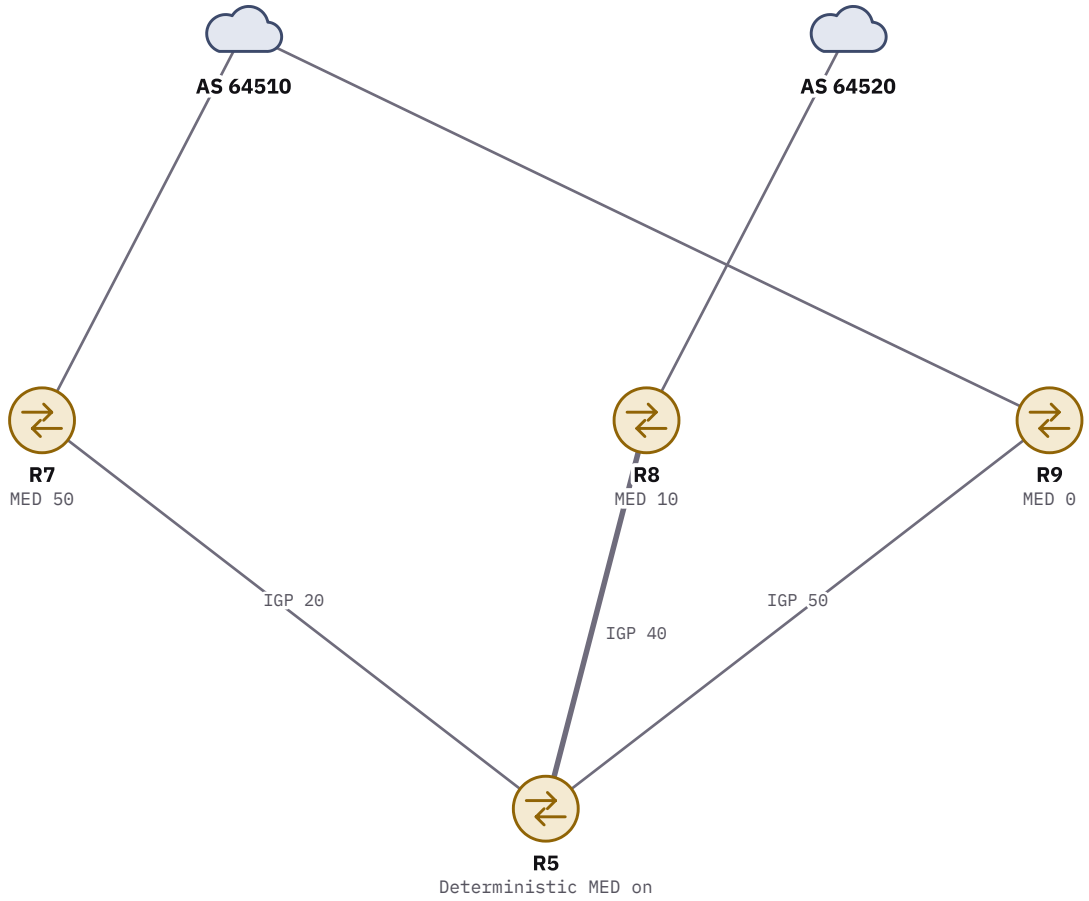
R5 · IOS-style CLI

```
R5# show ip bgp 192.0.2.0/24
BGP routing table entry for 192.0.2.0/24, version 3311
Paths: (3 available, best #2, table default)
  64510 64999
    10.0.0.7 (metric 20) from 10.0.0.7 (10.0.0.7)
      Origin IGP, metric 50, localpref 100, valid, internal
  64520 64999
    10.0.0.8 (metric 40) from 10.0.0.8 (10.0.0.8)
      Origin IGP, metric 10, localpref 100, valid, internal, best
  64510 64999
    10.0.0.9 (metric 50) from 10.0.0.9 (10.0.0.9)
      Origin IGP, metric 0, localpref 100, valid, internal
R5# show running-config | include deterministic
 bgp deterministic-med
```

In parentheses is the IGP metric to the next hop; after Origin IGP is the MED. With deterministic MED, the two AS 64510 paths are grouped and the MED 0 path via 10.0.0.9 wins the group. It is then compared with the AS 64520 path: different neighbouring AS, so MED is skipped and the IGP metric decides (40 beats 50). Without deterministic MED, comparing pairwise in the listed order would pick 10.0.0.9 instead.

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Three paths, two neighbouring ASes



Deterministic MED groups R7 and R9 (both AS 64510), where MED 0 wins. Against the AS 64520 path MED is skipped, so the IGP metric to the next hop decides and R8 wins with 40.

Handling a missing MED

Treat missing as best (protocol default)

- A path without MED is treated as MED 0
- Routes from neighbours that never set MED win MED comparisons
- Matches the base specification

Treat missing as worst (option)

- A path without MED is treated as the maximum value
- Neighbours must set MED to be preferred
- Useful when MED is a deliberate signal from a provider

Reading a multi-path entry quickly

1. **Discard the invalid** Look for inaccessible next hops, RIB-failure and policy flags first.
2. **Find the first difference** Walk the attributes in order: local preference, origination, AS-path length, origin, MED and so on, stopping at the first that differs.
3. **Check MED eligibility** Before comparing MED, check whether the paths came from the same neighbouring AS and whether deterministic or always-compare MED is on.
4. **Check for hidden paths** If you expected a path that is not listed, the reason is upstream: reflector selection or neighbour policy.
5. **Confirm in forwarding** Verify the next hop's IGP path and the FIB, not just the BGP marker.

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SCENARIO · DECIDE · INBOUND TRAFFIC IGNORES YOUR MED

Your network has two links to the same provider, in different cities. You send MED 10 on link A and MED 100 on link B so the provider sends traffic over link A. Most inbound traffic still arrives on link B. The provider confirms it receives both MED values.

(A) Prepend your AS several times on link B so the AS path through it is longer.

(B) Lower MED on link A to zero so the difference between the links is larger.

(C) Advertise more-specific prefixes on link B so the provider must prefer it for those prefixes.

(D) Use the provider's documented community to lower its local preference for your routes on link B.

Circle your choice. Answer and feedback: module answer key.

Best-path troubleshooting

☐ Every candidate path is visible, or the reason it is hidden is known

☐ Invalid and RIB-failure paths ruled out first

☐ MED comparisons checked against neighbouring AS and configuration

☐ Deterministic MED consistent across the AS

☐ Neighbours' policies that override your attributes understood

☐ Forwarding path confirmed after the BGP decision

Key terms

LOCAL_PREF
A well-known attribute carried within an AS that expresses exit preference; higher wins.

AS_PATH
The list of ASes a route has crossed, used for loop prevention and path length comparison.

MED
Multi-exit discriminator: a hint to a neighbouring AS about which entry point to prefer; lower wins.

Always-compare MED
Comparing MED between paths from different neighbouring ASes.

Multipath
Installing several equal BGP paths for load sharing after the multipath comparison step.

Weight
A router-local preference value used by some implementations before any protocol attribute.

ORIGIN
An attribute indicating how a route entered BGP: IGP, EGP or incomplete.

Deterministic MED
Grouping paths by neighbouring AS before comparing, so results do not depend on arrival order.

AS_SET
An unordered AS-path segment created by aggregation, counted as one AS in length.

REFLECT

Pick a prefix with several paths in a BGP table you can access. Walk the decision sequence and write down the step that decided it. Did it match what you expected?

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Key takeaways

- Walk the decision sequence in order and stop at the first attribute that differs.
- MED is compared only between paths from the same neighbouring AS unless always-compare is set; deterministic MED removes order dependence.
- A missing MED counts as best by default; decide explicitly whether to treat it as worst.
- Your MED and prepends act after the neighbour's LOCAL_PREF; use its documented communities to influence the earlier step.

KNOWLEDGE CHECK

11. Two paths differ only in MED (10 and 50) and come from different neighbouring ASes, without always-compare MED. What decides?

    (A) The path with MED 10, because a lower MED always wins
    (B) The next steps, starting with eBGP over iBGP and then the IGP metric
    (C) The path with MED 50, because MED is compared in reverse
    (D) The router ID of the advertising peer, ignoring every other step

12. How long is an AS path of 64510 64510 64520 {64530 64540}?

    (A) 4
    (B) 5
    (C) 3
    (D) 6

13. Why can MED comparisons give different winners on identical routers without deterministic MED?

    (A) MED values are rewritten by route reflectors in an unpredictable order
    (B) Routers round MED values differently depending on load
    (C) Pairwise comparison order decides which paths meet on MED
    (D) Older paths lose their MED attribute after a refresh

14. Which step chooses the exit closest to the router when everything before it is equal?

    (A) Oldest eBGP path among the candidates
    (B) Lowest router ID of the advertising peer
    (C) Shortest CLUSTER_LIST among reflected paths
    (D) Lowest IGP metric toward the BGP next hop

15. Inbound traffic ignores your MED on two links to the same provider. What is the most likely reason?

    (A) MED is never carried across eBGP sessions to a neighbouring AS
    (B) The provider compares MED only for paths learned from different ASes
    (C) MED only affects outbound traffic from your own network
    (D) The provider's local preference decides before MED is compared

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MODULE 4 · BGP · LESSON 4 · 50 MIN

Advanced Advertisement and Convergence Tuning

Conditional advertisement, add-path, next-hop tracking, BFD and prefix-independent convergence

- Configure and verify conditional advertisement for backup announcements
- Use add-path and best-external to restore path diversity
- Tune BGP convergence with BFD, next-hop tracking and backup path installation

Two problems dominate at expert level once policy is correct: advertising the right thing only when the right condition holds, and converging quickly when something fails. BGP was designed for stability rather than speed, so fast convergence depends on detecting failures outside BGP, reacting to next-hop changes immediately, and keeping backup paths ready in the forwarding plane.

DEFINITION · CONDITIONAL ADVERTISEMENT

An **advertise-map** names the routes to announce; an **exist-map** or **non-exist-map** names a condition route in the local BGP table. The advertise-map routes are announced only while the condition route exists (exist-map) or does not exist (non-exist-map). Implementations evaluate the condition periodically, often every 60 seconds by default, so failover is not instantaneous.

Backup announcement to a second provider

EDGE2 · IOS-style CLI

```
ip prefix-list PL-OUR-BLOCK seq 5 permit 198.51.100.0/24
ip prefix-list PL-ISP1-DEFAULT seq 5 permit 0.0.0.0/0
!
route-map RM-ADVERTISE permit 10
 match ip address prefix-list PL-OUR-BLOCK
route-map RM-ISP1-ALIVE permit 10
 match ip address prefix-list PL-ISP1-DEFAULT
!
router bgp 64500
 neighbor 203.0.113.9 remote-as 64602
 address-family ipv4
  neighbor 203.0.113.9 advertise-map RM-ADVERTISE non-exist-map RM-ISP1-ALIVE
!
EDGE2# show ip bgp neighbors 203.0.113.9 | include Condition
 Condition-map RM-ISP1-ALIVE, Advertise-map RM-ADVERTISE, status: Withdraw
```

Only the primary provider sends a default route, so its presence in the BGP table is a reliable health signal. While it exists, the block is withdrawn from the backup provider. If the condition matched something both providers send, the backup would never be announced.

Add-path (RFC 7911) lets a speaker advertise several paths for the same prefix, each with a path identifier, after both sides negotiate the capability. It restores the diversity that reflection hides, enabling multipath and pre-computed backups on clients. **Best-external** is a lighter alternative: an edge router advertises its best external path to iBGP even when its overall best is an internal path, so the alternative exit stays visible.

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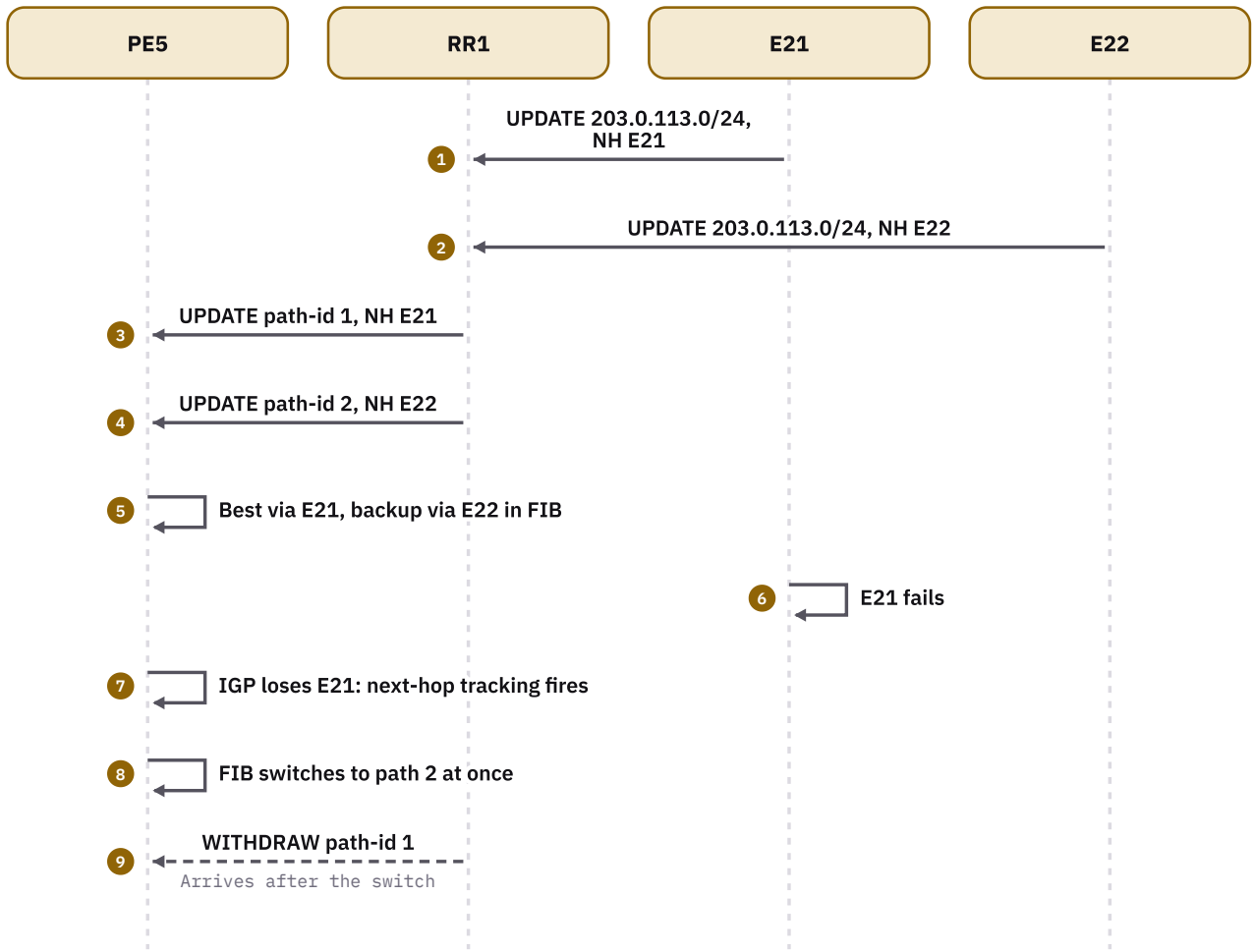
Add-path from a reflector

RR1 · IOS-style CLI

```
router bgp 65000
 address-family ipv4
  bgp additional-paths select best 2
  bgp additional-paths send receive
  neighbor RR-CLIENTS advertise additional-paths best 2
!
PE5(config-router-af)# bgp additional-paths receive
PE5(config-router-af)# bgp additional-paths install
```

The reflector selects the two best paths and sends both to clients that negotiate the capability. On the client, installing the additional path places a pre-computed backup in the forwarding table so a failed exit can be bypassed without waiting for best-path recalculation.

Add-path turns a hidden exit into a ready backup



PE5 switches in the forwarding plane as soon as the IGP removes E21's loopback; the BGP withdrawal arrives later and changes nothing urgent.

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Path diversity options

Best-external

- Configured on edge routers
- Advertises the best external path in addition to the overall best
- No capability negotiation needed

Add-path

- Several paths per prefix with path identifiers
- Must be negotiated on each session
- Memory and update cost grow with the number of paths

Regional reflectors

- Reflectors placed close to clients
- Each region's best path reflects local exits
- No protocol extension needed

BGP convergence after an exit failure

01 **Detect** — Interface down or BFD declares the eBGP neighbour dead →

02 **Invalidate** — Next-hop tracking marks the exit's next hop unreachable →

03 **Switch** — The forwarding plane moves to the pre-installed backup path →

04 **Recompute** — Best path is recalculated for affected prefixes →

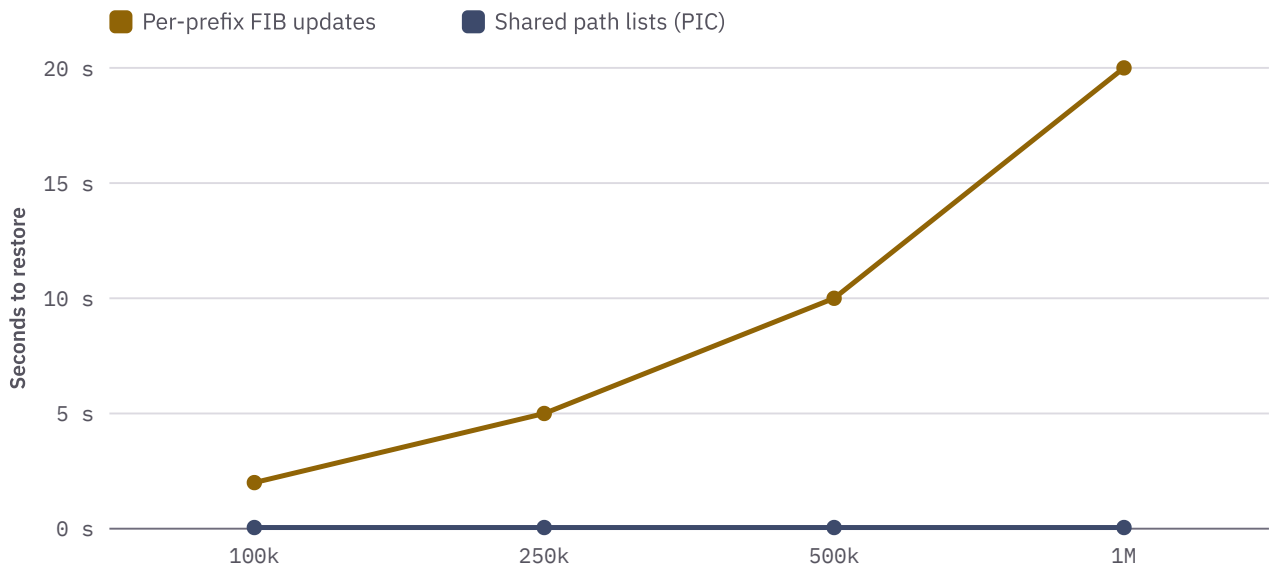
05 **Propagate** — Updates and withdrawals sent, paced by advertisement intervals

Convergence arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Negotiated hold time	`min(local hold time, remote hold time)`	Without BFD, a silent neighbour failure takes up to this long to detect (default often 180 s).
BFD detection	`multiplier × agreed interval`	300 ms × 3 = 900 ms replaces a hold time of 180 s.
FIB work without shared path lists	`proportional to affected prefixes`	One million prefixes behind a failed exit means one million updates.
FIB work with shared path lists	`one pointer change per shared path list`	Prefix-independent convergence: the switch time does not grow with table size.

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Restoration time versus table size (illustrative)



Illustrative values: with per-prefix updates the outage grows with the number of prefixes behind the failed exit; with shared path lists and a pre-installed backup one pointer change restores them all.

Convergence-oriented session settings

EDGE1 · IOS-style CLI

```
router bgp 64500
 bgp nexthop trigger delay 0
 neighbor 203.0.113.1 remote-as 64601
 neighbor 203.0.113.1 fall-over bfd
 neighbor 203.0.113.1 timers 10 30
!
interface GigabitEthernet0/0/1
 bfd interval 300 min_rx 300 multiplier 3
!
EDGE1# show ip bgp neighbors 203.0.113.1 | include hold|BFD
  Last read 00:00:02, last write 00:00:04, hold time is 30, keepalive interval is 10 seconds
  Neighbor is registered with BFD (interval 300 ms, multiplier 3)
```

BFD detects a dead neighbour in under a second and tears the session down; next-hop tracking reacts to IGP changes for internal next hops without delay. The shorter hold time is a safety net for cases BFD cannot see, such as a hung BGP process.

WATCH OUT · BFD AND GRACEFUL RESTART

Graceful restart keeps forwarding while a neighbour's control plane restarts. If BFD runs in the control plane, the restart also stops BFD, which then declares the neighbour down and defeats graceful restart. Check whether BFD is independent of the control plane on both sides before combining them, and agree the behaviour with the neighbour.

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SCENARIO · DECIDE · SESSIONS FLAP DURING PROVIDER MAINTENANCE

After enabling BFD at 50 ms × 3 on eBGP sessions carried over a provider's layer 2 service, sessions drop several times a month, each for a few seconds, always during the provider's maintenance windows. The provider reports protection switching events of about 200 ms in its network.

A. Disable BFD and rely on BGP keepalives and the hold timer for failure detection.

B. Set BFD detection above the provider's switching time, for example 100 ms × 3.

C. Enable graceful restart so the sessions survive the short BFD-triggered resets.

D. Lower the BGP hold time to 3 seconds so that sessions recover faster after each flap.

Circle your choice. Answer and feedback: module answer key.

Convergence and advertisement review

☐ Conditional advertisement conditions match only the intended signal

☐ Condition evaluation interval known and acceptable

☐ Path diversity strategy deployed where reflection hides exits

☐ BFD detection slower than underlying transport protection

☐ BFD and graceful restart interaction agreed with neighbours

☐ Next-hop tracking enabled and loopbacks prioritised in the IGP

☐ Backup paths installed in the forwarding plane where supported

Key terms

Advertise-map
A route-map naming routes whose advertisement depends on a condition.

Add-path
A BGP extension (RFC 7911) that advertises several paths per prefix with path identifiers.

Next-hop tracking
Event-driven reaction to IGP changes that affect BGP next hops.

Hold time
The negotiated time after which a silent BGP neighbour is declared down.

Non-exist-map
A condition that allows advertisement only while the named route is absent.

Best-external
Advertising an edge router's best external path to iBGP even when its overall best path is internal.

Prefix-independent convergence
A forwarding design in which prefixes share path lists, so one change repairs all of them.

Graceful restart
A mechanism that preserves forwarding while a neighbour's control plane restarts.

REFLECT

For an exit failure in your network, estimate each phase of BGP convergence today. Which single change would shorten it the most?

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Key takeaways

- Conditional advertisement needs a condition route that only the primary path provides, and it reacts on a scan interval.
- Add-path is negotiated per session and exposes several paths; best-external is the lighter alternative.
- Fast BGP convergence comes from outside BGP: BFD, next-hop tracking and pre-installed backup paths.
- Set BFD detection slower than the transport's own protection, and check how BFD interacts with graceful restart.

KNOWLEDGE CHECK

16. A non-exist-map matches a default route that both providers send. What happens to the backup announcement?

(A) It is always sent, because the condition is evaluated per provider
(B) It is never sent, because the condition route always exists
(C) It is sent only when both providers withdraw the default
(D) It alternates between sent and withdrawn every scan interval

17. What must happen before add-path can be used on a session?

(A) The reflector must use the same cluster ID as all of its clients
(B) Every path must carry a unique route distinguisher value
(C) Both peers must negotiate the add-path capability for the address family
(D) The session must use BFD with the same interval on both sides

18. Without BFD, how long can it take to detect a silent eBGP neighbour failure with timers 60/180?

(A) Up to 180 seconds
(B) Up to 60 seconds
(C) Up to 30 seconds
(D) Up to 3 seconds

19. What makes convergence independent of the number of prefixes behind a failed exit?

(A) A shorter minimum advertisement interval on every iBGP session
(B) Lower hold timers on every eBGP session at the network edge
(C) Larger update packets that carry more prefixes per message
(D) Shared forwarding path lists with pre-installed backup paths

20. BFD at 50 ms × 3 flaps sessions when the transport switches paths in 200 ms. What is the correct fix?

(A) Make BFD detection slower than the transport's switching
(B) Enable graceful restart on both sides of every affected session
(C) Shorten the BGP hold time to match the BFD detection time
(D) Disable next-hop tracking so that failures are processed later

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MODULE 4 · BGP · LESSON 5 · 50 MIN

BGP Security and Complex Troubleshooting

Session protection, origin validation, route-leak defences and hard cases from real networks

- Apply session protections: prefix limits, TTL security and authentication
- Deploy RPKI route origin validation safely and interpret validation states
- Diagnose complex BGP failures such as MTU-related flaps and route leaks
- Explain DDoS signalling with RTBH and Flowspec, and where ASPA fits in route-leak defence

BGP trusts its neighbours by default. Any session can announce any prefix with any path, and the protocol itself cannot tell a legitimate announcement from a mistake or a hijack. Defence is layered: protect the session, constrain what each neighbour may announce, validate origins cryptographically where possible, and mark relationships so leaks are caught.

Layered BGP protections

CONTROL	PROTECTS AGAINST	MECHANISM	CAVEAT
Maximum prefix	Full-table leaks, misconfiguration	Limit, warning threshold, optional automatic restart	Too low causes outages during legitimate growth
TTL security (RFC 5082)	Spoofed packets from distant attackers	Sender uses TTL 255; receiver requires a minimum TTL	Both sides must enable it
Session authenticati on	Session hijack and reset attacks	TCP-AO (RFC 5925), or the legacy TCP MD5 option	Key rotation must be planned
Prefix filtering	Announcements outside a neighbour's address space	Prefix lists built from registry data	Lists must be regenerated as customers change
Route origin validation (RFC 6811)	Wrong origin AS, many hijacks	Signed ROAs, validated cache, router checks origin and length	Does not validate the rest of the path
BGP roles and OTC (RFC 9234)	Route leaks between relationships	Roles negotiated in OPEN; Only-to-Customer attribute marks routes	Both sides must support it
DDoS signalling: RTBH and Flowspec	Volumetric attacks toward a customer	BLACKHOLE community (RFC 7999) drops all traffic to a target; Flowspec (RFC 8955, RFC 8956 for IPv6) distributes match-and-action filters in BGP	Accept them only from authorised sources; Flowspec rules consume hardware filter space
ASPA (in standardisati on)	Route leaks and forged paths that origin validation cannot see	RPKI objects in which an AS lists its authorised providers; routers check that the AS path is valley-free	Check the current IETF status and validator and router support before relying on it

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Routing security milestones

2010
TCP-AO (RFC 5925)
Successor to the TCP MD5 option for session authentication

2012
RPKI architecture (RFC 6480)

2013
Origin validation (RFC 6811)
Valid, Invalid and NotFound states

2017
RPKI-to-router v1 and BGPsec
RFC 8210; BGPsec (RFC 8205) signs paths but sees little deployment

2020
Flowspec revised (RFC 8955)
Filtering rules carried in BGP

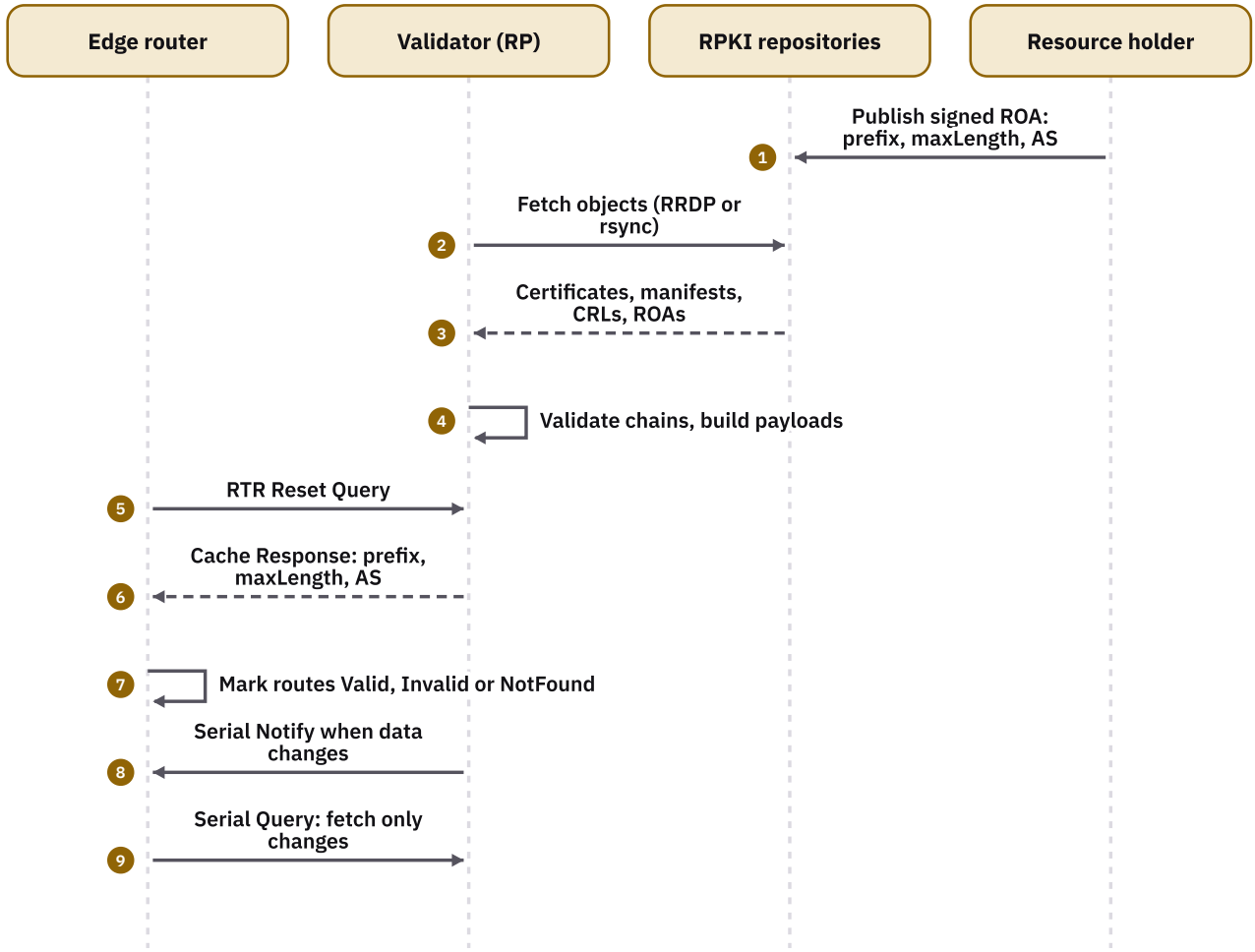
2022
Roles and OTC (RFC 9234); maxLength guidance (RFC 9319)

2024–26
ASPA progressing in the IETF
Leak and path-plausibility checks; verify the current status

Session security and origin validation are mature; path and leak protection are still arriving, so check current status before you depend on the newest items.

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Route origin validation from ROA to router



The router never talks to the repositories: a validator does the cryptography and feeds simple prefix, maxLength and AS records over the RPKI-to-router protocol, then sends only changes.

Origin validation logic

MEASURE	FORMULA	WHAT IT TELLS YOU
Covered	`some validated payload's prefix contains the route's prefix`	If nothing covers the route, its state is NotFound.
Valid	`covered and ∃ payload with origin AS = route origin and route length ≤ maxLength`	At least one matching authorisation exists.
Invalid	`covered but no payload matches both origin AS and length`	Wrong origin, or too specific for the authorised maximum length.

WATCH OUT · KEEP MAXLENGTH TIGHT

Current best practice (RFC 9319) is to avoid maxLength values longer than the prefixes you actually announce. A ROA for a /22 with maxLength 24 also authorises /24s you never announce, so an attacker who forges your origin AS can announce one, win by longest match and still be marked Valid. Publish one ROA entry per announced prefix instead.

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Session protection and origin validation

EDGE1 · IOS-style CLI

```
router bgp 64500
 bgp rpki server tcp 192.0.2.50 port 3323 refresh 600
 neighbor 203.0.113.1 remote-as 64601
 neighbor 203.0.113.1 ttl-security hops 1
 ! session authentication configured from a key chain
 address-family ipv4
  neighbor 203.0.113.1 activate
  neighbor 203.0.113.1 maximum-prefix 1200 85 restart 15
  neighbor 203.0.113.1 route-map RM-IN-PEER-64601 in
!
route-map RM-IN-PEER-64601 deny 5
 match rpki invalid
route-map RM-IN-PEER-64601 permit 10
 match ip address prefix-list PL-PEER-64601
```

Invalid routes are dropped explicitly instead of relying on a platform default, which differs between implementations. The prefix limit warns at 85% of 1,200 and tears the session down above the limit, retrying after 15 minutes. TTL security means only packets from a directly connected sender are accepted.

Monitor-mode validation results

EDGE1 · IOS-style CLI

```
EDGE1# show ip bgp 198.51.100.0/24
BGP routing table entry for 198.51.100.0/24, version 991
Paths: (2 available, best #2, table default)
  64601 64777
    203.0.113.1 from 203.0.113.1 (192.0.2.1)
      Origin IGP, localpref 100, valid, external
      RPKI State invalid
  64602 64496
    203.0.113.5 from 203.0.113.5 (192.0.2.5)
      Origin IGP, localpref 100, valid, external, best
      RPKI State valid
```

Captured during a monitor-only phase, before invalid routes were dropped. The ROA authorises AS 64496 to originate this prefix, so the path ending in 64777 is invalid: a probable hijack or misconfiguration. Monitoring first shows how much traffic would move and which legitimate routes are invalid because of stale ROAs.

Complex symptoms and their causes

SYMPTOM	LIKELY ROOT CAUSE	PROOF
Session establishes, then drops on hold timer a few minutes later, repeatedly	Path MTU blackhole: keepalives pass, full-size update segments are lost	High TCP retransmissions on the session; DF-bit ping fails at full size
Routes received but not used by iBGP peers	Next hop of external routes not reachable inside the AS	Peers show the next hop as inaccessible; fix with next-hop-self or IGP
Network statement does not originate the prefix	No exact prefix and mask match in the routing table	Exact routing-table lookup for the prefix and mask
Session down after a neighbour's growth	Maximum prefix exceeded	Neighbour log and state showing the prefix limit reason

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SYMPTOM	LIKELY ROOT CAUSE	PROOF
Traffic for a major destination enters through a small customer	Customer leaking provider routes, preferred as customer routes	AS path with a provider behind the customer; customer class on the route

Session that dies after the initial transfer

R7 · IOS-style CLI

```
R7# show ip bgp neighbors 10.0.0.9 | include state|Last reset|segment|Rcvd|Sent
  BGP state = Established, up for 00:02:51
  Last reset 00:03:02, due to BGP Notification sent, hold time expired
Datagrams (max data segment is 1460 bytes):
Rcvd: 812 (out of order: 0), with data: 3, total data bytes: 287
Sent: 24411 (retransmit: 3912, fastretransmit: 0), with data: 24398, total data bytes: 35112934
R7# ping 10.0.0.9 source Loopback0 size 1500 df-bit
.....
Success rate is 0 percent (0/5)
```

The peer has acknowledged only a small fraction of the data R7 sent, with thousands of retransmissions of 1460-byte segments, and the session resets on the hold timer every few minutes. Full-size packets do not pass. Fix the path MTU or reduce the maximum segment size, then confirm the session stays up after a full table transfer.

CASE STUDY · A NATIONAL ISP AND A LEAKED FULL TABLE

A small regional ISP buys transit from a national ISP and from an international carrier. During a router replacement, an engineer applied a template that exported every route to every eBGP neighbour. The regional ISP announced the international carrier's full table to the national ISP. The national ISP classified everything from that session as customer routes with local preference 200, above its peer and transit routes, and propagated them across its network. For about an hour, a large share of the national ISP's traffic to major content networks was drawn through the regional ISP's 10 Gb/s links, which saturated. Some leaked routes were more specific than the originals, because the international carrier carried de-aggregated prefixes from a traffic optimiser.

1. Which controls at the national ISP should have stopped the leak?
2. Which control at the regional ISP failed, and how should export policy be built instead?
3. Why did the more-specific routes make the incident worse?
4. What would you do in the first ten minutes if you were on call at the national ISP?

Model answer: module answer key.

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SCENARIO · DECIDE · A PEER STARTS ANNOUNCING A FULL TABLE

Monitoring shows the prefix count from a small settlement-free peer rising from 300 toward 900,000 within minutes. Your maximum-prefix limit for that peer is 5,000 with a warning at 80%, and the session has just been torn down by the limit. The peer's operations team is not answering.

- (A) Keep the session down until the peer confirms the fix, then verify the prefix count.
- (B) Raise the maximum-prefix limit to one million so the session can come back up immediately.
- (C) Configure automatic restart every five minutes so the session recovers on its own when the leak ends.
- (D) Remove the inbound prefix list so the peer's legitimate routes are accepted faster when it returns.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **BGP incident triage log**

Record session and route evidence during a BGP incident so decisions can be reviewed and repeated.

TIME	SESSION OR PREFIX	OBSERVATION	EVIDENCE COMMAND	DECISION	OWNER
09:14	*Peer 192.0.2.77*	*Prefix count 300 to 4,000 in 3 min*	*show ip bgp summary*	*Limit reached; keep down*	*On-call core*
09:20	*198.51.100.0/24*	*Origin AS 64777, RPKI invalid*	*show ip bgp 198.51.100.0/24*	*Already dropped by policy; notify origin owner*	*Security*

BGP security baseline

- ☐ Maximum-prefix limits on every eBGP session, sized from real counts
- ☐ TTL security on directly connected eBGP sessions
- ☐ Session authentication with a key rotation plan
- ☐ Customer prefix lists generated from registry data
- ☐ Route origin validation deployed, Invalid dropped explicitly
- ☐ Class-based export preventing transit-to-transit leaks
- ☐ Roles or equivalent leak detection on supported sessions
- ☐ Bogon and too-specific filtering on all external sessions

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Key terms

Maximum prefix
A per-neighbour limit on accepted prefixes that warns or tears down the session when exceeded.

TCP-AO
TCP Authentication Option (RFC 5925), the successor to the TCP MD5 signature option for BGP sessions.

Route origin validation
Classifying routes as Valid, Invalid or NotFound against validated ROA data (RFC 6811).

Route leak
Propagation of routes beyond their intended scope, such as from one provider to another.

Path MTU blackhole
Silent loss of full-size packets on a path whose MTU is smaller than expected, with no usable feedback.

Flowspec
A BGP address family that distributes traffic-filtering rules (match on addresses, ports, protocol; actions such as drop, rate-limit or redirect), commonly used for DDoS mitigation.

TTL security
Generalised TTL Security Mechanism (RFC 5082): accepting only packets with a TTL that proves a nearby sender.

ROA
Route origin authorisation: a signed statement that an AS may originate a prefix up to a maximum length.

maxLength
The longest prefix length a ROA authorises; more specific announcements are Invalid.

Only-to-Customer
An attribute defined with BGP roles (RFC 9234) that marks routes which may only be sent to customers.

ASPA
Autonomous System Provider Authorisation: signed RPKI objects listing an AS's upstream providers, so routers can detect AS paths that violate valley-free routing. Still maturing; check current standard and product status.

REFLECT

Which of the layered BGP protections are missing on the external sessions you know? Rank the gaps by the damage a single misconfigured neighbour could cause today.

Key takeaways

- Layer the defences: session protection, prefix limits, registry-based filters, origin validation and leak prevention.
- Drop RPKI Invalid routes by explicit policy and keep ROA maxLength equal to what you announce.
- Origin validation does not check the path; roles, Only-to-Customer and ASPA address route leaks.
- Sessions that die after the initial transfer with many retransmissions point to a path MTU problem.

KNOWLEDGE CHECK

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21. A ROA authorises AS 64496 for 198.51.100.0/22 with maxLength 23. A route for 198.51.100.0/24 originated by AS 64496 arrives. What is its state?

(A) Valid, because the origin AS matches the AS in the authorisation

(B) NotFound, because no ROA exists for the exact /24 prefix

(C) Invalid, because it exceeds the authorised maximum length

(D) Valid, because the /24 is fully covered by the authorised /22

22. A session comes up, then drops on hold timer expiry every few minutes, with thousands of retransmissions. What is the most likely cause?

(A) The neighbour's maximum-prefix limit is too low for your table

(B) Full-size TCP segments are lost on a path with a smaller MTU

(C) The neighbours have configured mismatched keepalive intervals

(D) The neighbour rejects your routes because they are RPKI invalid

23. What does TTL security require of a directly connected eBGP neighbour?

(A) It must send BGP packets with TTL 1 so that they cannot leave the link

(B) It must set the IP options field to record the route of each packet

(C) It must send packets with TTL 255, so the arrival TTL proves it is adjacent

(D) It must authenticate every packet with a key derived from the TTL

24. Which control would have stopped a small customer leaking a full transit table to you within seconds?

(A) Route origin validation that drops every route marked as NotFound

(B) A maximum-prefix limit sized to the customer's real count

(C) TTL security configured on the customer's eBGP session

(D) Session authentication using TCP-AO on the customer session

25. Why should Invalid routes be dropped by explicit policy rather than relying on default behaviour?

(A) Default handling of Invalid differs between implementations

(B) Routers always install Invalid routes unless a ROA is published

(C) Explicit policy makes RPKI validation run faster on the router

(D) Invalid routes are dropped automatically only for iBGP sessions

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BGP

Answer key

Knowledge checks

1. D Routes from non-clients are reflected to clients only; non-clients are expected to be meshed with each other.

2. C The cluster ID in CLUSTER_LIST indicates the route already passed through this cluster.

3. C With a shared cluster ID, RR1 drops RR2's reflected copies, but other clients still learn the routes through RR2.

4. B Confederation segments are excluded from AS-path length comparison.

5. A Reflectors advertise their own best path, chosen from their position in the IGP, not the client's.

6. D Large communities were defined so 32-bit ASNs can be expressed directly.

7. A NO_EXPORT keeps the route inside the AS (or confederation); iBGP propagation is allowed.

8. C Communities are only lost in transit when a session does not send them or a policy deletes them.

9. D Valley-free export stops you carrying traffic between providers and peers on their behalf.

10. B Classifying once at ingress lets every later policy rely on the tag.

11. B MED is skipped between different neighbouring ASes by default, so the following steps decide.

12. A Three ASNs in the sequence plus one for the AS_SET equals 4.

13. C Without grouping, the result depends on the order in which paths are compared.

14. D The IGP metric step implements hot-potato routing.

15. D Local preference in the provider's AS is evaluated before MED.

16. B With both providers sending the default, the route stays present unless both fail, which defeats the design. The condition must match a signal only the primary sends.

17. C Add-path is a negotiated capability; without it the extra paths are not sent.

18. A Detection relies on the hold time when no faster mechanism exists.

19. D Prefix-independent convergence changes one shared pointer instead of each prefix.

20. A Detection above the layer below's protection time avoids false failures.

21. C The route is covered but exceeds maxLength, so no payload matches both origin and length.

22. B Small keepalives pass while large update segments are dropped, so the hold timer expires.

23. C Remote attackers cannot deliver packets with a TTL that high, so they are rejected.

24. B Leaked routes usually have valid origins and arrive over a legitimate session, so only volume and content controls catch them.

25. A Making the action explicit removes ambiguity across platforms and versions.

Scenarios & case studies

> **Scenario · Hot-potato routing breaks after adding reflectors**
> **C** is best. Correct. Each reflector selects one best path using its own IGP distance to the exits, which favours exits near the capital. Clients inherit that choice. More paths to clients, or reflectors placed per region, restore true hot-potato behaviour.

> **Scenario · A customer's signal is ignored**
> **D** is best. Correct. The communities exist at ingress and are missing at egress, so they were stripped in transit. Without them the egress policy cannot recognise the signal or the class.

> **Case study · Redesigning a multinational's BGP policy**
> Classification and export by class are missing: decisions are made per neighbour on raw prefixes, so local preference is inconsistent and nothing stops a transit route from being exported to another transit. Define a large-community plan in the organisation's own ASN namespace: relationship class, region, point of entry, and a small set of action communities, documented in one place and generated from a source of truth. Migrate by first adding classification at every ingress without changing decisions, verifying that every route carries a class; then switch export policies one neighbour at a time to class-based matching, comparing advertised routes before and after; finally replace local preference rules with class-based values and remove the legacy route-maps. Measure the number of policy lines, consistency of exit selection for test prefixes from each region, zero transit-to-transit advertisements in daily audits, and change lead time.

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Scenario · Inbound traffic ignores your MED
D is best. Correct. Local preference is evaluated before MED. If the provider's own policy sets a higher local preference on link B, for example because of regional exit policy, your MED never gets compared. Provider communities act on that earlier step.

Scenario · Sessions flap during provider maintenance
B is best. Correct. A 150 ms detection time is shorter than the 200 ms the provider needs to switch, so BFD declares failure for an event the transport would recover from. Detection must be slower than the layer below's own protection.

Case study · A national ISP and a leaked full table
At the national ISP: a prefix list built from registry data for the customer cone would have rejected almost all leaked routes; a maximum-prefix limit sized to the customer's real announcement count would have shut the session within seconds; an AS-path filter rejecting known large transit networks behind a small customer, or BGP roles with the Only-to-Customer attribute where supported, would also have caught it. At the regional ISP, export policy was not class-based: routes learned from transit must carry a transit class at ingress and never match an export policy toward another transit. More-specific routes win by longest match regardless of local preference, so they attracted traffic even in networks that would have preferred the original paths. First ten minutes: confirm the leak from AS paths and the customer class, shut or filter the leaking session (a prefix limit or inbound deny to the customer's registered prefixes), verify traffic returns to normal paths, and contact the customer before re-enabling with corrected filters.

Scenario · A peer starts announcing a full table
A is best. Correct. The limit did its job. Bringing the session back without confirmation risks repeating the leak, and verification afterwards proves the peer's export policy is fixed.

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05

MODULE 5 · LABEL

Label — MPLS and Segment Routing

Build, scale, engineer and troubleshoot label-switched transport and the VPN services that ride on it.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Build and harden an LDP or SR-MPLS transport and explain every label operation along a path.
- Deliver MPLS L3VPN services end to end, including PE-CE design, route-target topologies and route reflection at scale.
- Compare inter-AS options and SRv6 with SR-MPLS, and choose the right one for a given scale and trust model.
- Engineer traffic with SR policies, colour steering and Flex-Algo, contrasting them with RSVP-TE.
- Troubleshoot broken label paths with a disciplined layer-by-layer method.

LESSONS

1. Label Switching and LDP: The Transport Foundation · 40 min
2. MPLS L3VPN End to End · 45 min
3. Scaling and Interconnecting VPNs · 45 min
4. Segment Routing: SR-MPLS, TI-LFA and SRv6 · 50 min
5. Traffic Engineering and Troubleshooting Label Paths · 45 min

MODULE 5 · LABEL · LESSON 1 · 40 MIN

Label Switching and LDP: The Transport Foundation

How labels decouple services from the core, and how LDP builds the label-switched paths underneath them

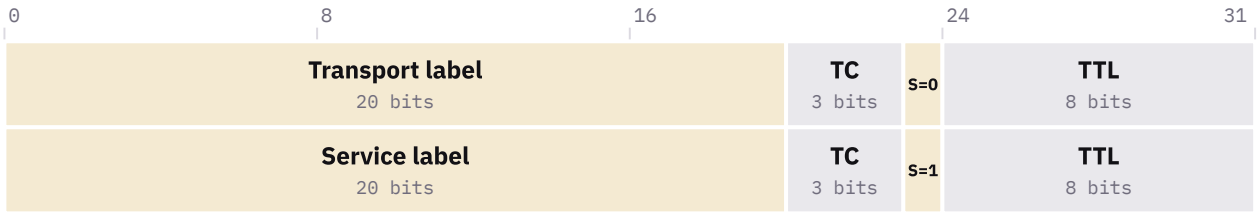
- Trace a packet through push, swap and pop operations and explain penultimate hop popping.
- Explain LDP discovery, session establishment and label distribution well enough to debug each stage.
- Harden an LDP core with IGP synchronisation, session protection, label filtering and correct MTU.

The **Label** segment of the ZCIE Mastery Ring is where enterprise and service-provider thinking meet. Labels matter in 2026 for the same reason they mattered twenty years ago: they **decouple the service from the core**. A provider router in the middle of the network does not need customer routes, Internet routes or VPN state; it only needs to know how to reach the egress edge. Every expert lab that contains a VPN, traffic engineering or fast reroute rests on a working label-switched path (LSP), so this lesson makes that foundation solid before we stack services on top.

What travels on the wire in an MPLS L3VPN

Payload	`Customer IPv4 or IPv6 packet` `Untouched by the core`
Service label (bottom, S=1)	`Identifies VRF or prefix at the egress PE` `Signalled by MP-BGP`
Transport label (top)	`Identifies the LSP to the egress PE loopback` `Signalled by LDP, RSVP-TE or the IGP (SR)`
Layer 2 header	`EtherType 0x8847 for MPLS unicast` `Rewritten hop by hop`

Two label stack entries on the wire



Each entry is exactly 32 bits: a 20-bit label, 3-bit traffic class, the bottom-of-stack bit and an 8-bit TTL. Only the innermost (service) entry has S set.

The 32-bit label stack entry and the reserved labels you must recognise

FIELD OR VALUE	SIZE OR MEANING	WHY AN EXPERT CARES
Label	20 bits (0 to 1,048,575)	Values 0 to 15 are reserved; dynamic ranges are platform-specific
Traffic Class (TC)	3 bits	Carries QoS across the core; set from DSCP on imposition or by policy

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FIELD OR VALUE	SIZE OR MEANING	WHY AN EXPERT CARES
Bottom of Stack (S)	1 bit	Set only on the innermost label; tells the egress node that IP follows
TTL	8 bits	Copied from IP by default; disabling propagation hides the core from traceroute
Label 0 / 2	IPv4 / IPv6 explicit null	Keeps TC marking visible to the egress PE instead of popping early
Label 3	Implicit null	Signalled only, never seen on the wire; asks the upstream router to pop (PHP)
Label 7 / 13	Entropy label indicator / generic associated channel	Load-balancing entropy and in-band OAM respectively

Life of a labelled packet across one LSP

01
Ingress PE: push
FIB lookup on the destination; impose the service label, then the transport label for the BGP next hop

→

02
P router: swap
Pure LFIB lookup on the top label; swap to the label the downstream neighbour advertised

→

03
Penultimate hop: pop
Egress PE advertised implicit null, so the top label is removed one hop early

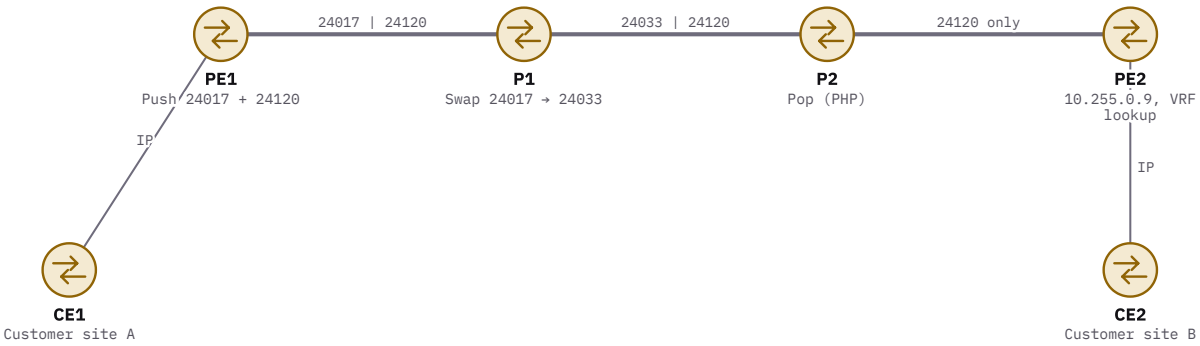
→

04
Egress PE: service lookup
Only the service label remains; it maps straight to a VRF or a CE next hop

→

05
CE: plain IP
The customer never sees a label

Label operations along one LSP



PE1 pushes P1's label for PE2's loopback over the service label; P1 swaps 24017 to 24033; P2 pops because PE2 advertised implicit null; PE2 sees only 24120 and maps it to the VRF.

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Three tables, three jobs

LIB (label information base)

- Every binding learned from every LDP neighbour
- With liberal retention, also holds unused bindings
- Enables fast switch-over when the IGP next hop changes

LFIB (label forwarding table)

- Only the binding from the current IGP next hop
- Local label to outgoing label plus interface
- Used for packets that arrive labelled

FIB (CEF-style table)

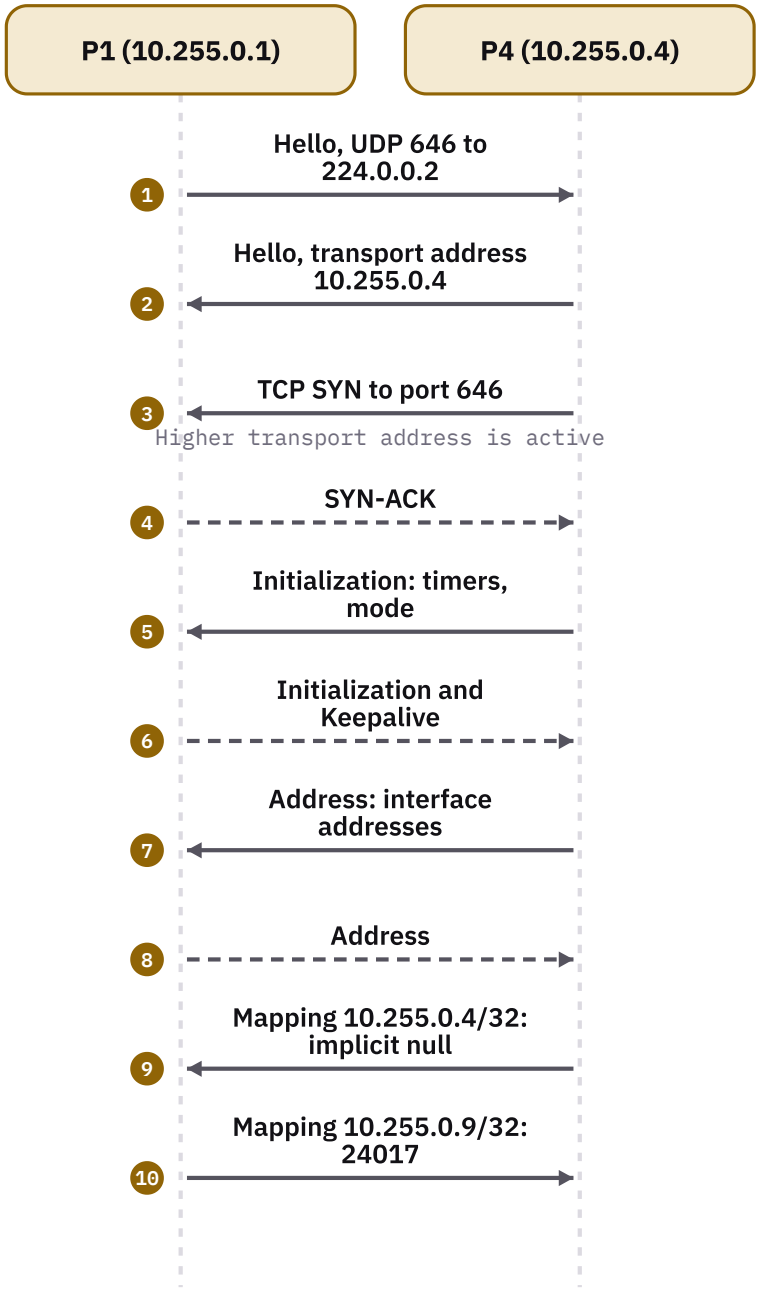
- IP prefixes with imposition instructions
- Used for packets that arrive unlabelled
- Must show a label for the BGP next hop of every VPN route

LDP from silence to label exchange

1. **Discovery** Link hellos are sent to UDP 646 at 224.0.0.2 on every LDP-enabled interface; targeted hellos are unicast for non-adjacent peers.
2. **Transport address** Each hello advertises a transport address, by default the LDP router ID. It must be reachable through the IGP, which is why a /32 loopback is used.
3. **TCP session** The peer with the higher transport address takes the active role and opens TCP to port 646.
4. **Initialisation** Session parameters such as keepalive time and label advertisement mode are negotiated; a mismatch keeps the session in a non-operational state.
5. **Address and label exchange** Peers exchange their interface addresses, then label mappings for each FEC, typically every IGP prefix or only host routes if filtered.

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LDP from hello to label mapping



Discovery is UDP, the session is TCP 646 opened by the higher transport address, and only after initialisation and address exchange do label mappings flow; the egress advertises implicit null for its own loopback.

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LDP on a provider router

P1 · IOS-style CLI

```
mpls label protocol ldp
mpls ldp router-id Loopback0 force
mpls ldp igp sync holddown 2000
mpls ldp session protection
! advertise labels for loopbacks only
ip access-list standard LDP-FEC
 permit 10.255.0.0 0.0.255.255
no mpls ldp advertise-labels
mpls ldp advertise-labels for LDP-FEC
!
interface GigabitEthernet0/1
 mtu 9100
 mpls ip
!
router isis CORE
 mpls ldp sync
 mpls ldp autoconfig level-2
```

With autoconfig the IGP enables LDP on its interfaces, so a link cannot run IS-IS without also running LDP. The per-interface `mpls ip` *is shown for clarity.*

Verifying sessions and bindings

P1 · IOS-style CLI

```
P1# show mpls ldp neighbor 10.255.0.4
    Peer LDP Ident: 10.255.0.4:0; Local LDP Ident 10.255.0.1:0
        TCP connection: 10.255.0.4.31544 - 10.255.0.1.646
        State: Oper; Msgs sent/rcvd: 1204/1199; Downstream
        LDP discovery sources:
          GigabitEthernet0/1, Src IP addr: 10.1.14.4
          Targeted Hello 10.255.0.1 -> 10.255.0.4, active, passive
P1# show mpls forwarding-table 10.255.0.9 32
Local  Outgoing   Prefix           Bytes Label  Outgoing  Next Hop
Label  Label      or Tunnel Id     Switched     interface
24017  24033      10.255.0.9/32    88412907     Gi0/1     10.1.14.4
P1# show mpls forwarding-table 10.255.0.4 32
24005  Pop Label  10.255.0.4/32    102939       Gi0/1     10.1.14.4
```

Peer 10.255.0.4 has the higher transport address, so it opened the TCP session to port 646. The targeted hello is session protection: it keeps the session and its bindings alive if the direct link fails.

WATCH OUT · THE CLASSIC BLACK HOLE: SUMMARISED LOOPBACKS

LDP binds labels to the **exact** prefixes in the routing table. If an ABR or level boundary summarises PE loopbacks into a /16, downstream routers only hold a label for the summary. The ingress PE then finds no label for the /32 BGP next hop and either drops VPN traffic or forwards it unlabelled into a core that has no customer routes. Keep PE loopbacks as /32 host routes end to end, or move to a design (BGP labelled unicast or segment routing) that explicitly handles hierarchy.

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LDP hardening kit

SYNC

LDP-IGP synchronisation

The IGP advertises maximum metric on a link until LDP is operational on it, so traffic is not drawn onto a link that cannot yet label-switch.

PROT

Session protection

A targeted hello keeps the session and retains bindings when a direct link flaps, avoiding a full relearn on restoration.

FILT

Label filtering

Allocate or advertise labels only for loopbacks; it shrinks the LIB and speeds convergence in large cores.

AUTH

Session authentication

Protect the TCP session with TCP-AO or MD5 so a rogue device cannot inject bindings.

MTU

Core MTU headroom

Size core links for the largest customer frame plus the deepest label stack you will impose.

MTU arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Required MPLS MTU	`customer IP MTU + (4 bytes × labels in stack)`	1500-byte packets with transport + VPN + TI-LFA repair labels (4 labels) need at least 1516 bytes of MPLS MTU.
Ethernet frame on the core	`MPLS MTU + 14 (+4 per 802.1Q tag)`	Configure jumbo frames on core links rather than calculating the minimum and living on the edge.

SCENARIO · DECIDE · TWENTY SECONDS OF LOSS AFTER A LINK COMES BACK

A core link between P2 and P3 is restored after maintenance. IS-IS adjacency forms in two seconds and immediately attracts traffic, but LDP takes another eighteen seconds to come up because the TCP session waits on a hello hold timer. VPN customers report about twenty seconds of loss every time a core link recovers. What is the best permanent fix?

- (A) Lower the IS-IS hello interval so the adjacency forms faster than it does today.
- (B) Disable penultimate hop popping so the egress PE always receives a label it understands.
- (C) Enable LDP-IGP synchronisation so IS-IS advertises maximum metric on the link until LDP is operational.
- (D) Summarise PE loopbacks at the area boundary so fewer LDP bindings need to be exchanged.

Circle your choice. Answer and feedback: module answer key.

Key terms

Label-switched path (LSP)
The unidirectional sequence of label operations that carries traffic from an ingress to an egress router.

Penultimate hop popping (PHP)
Removal of the top label by the router before the egress, triggered when the egress advertises implicit null.

Explicit null
Reserved label 0 (IPv4) or 2 (IPv6) that the egress receives, preserving TC bits for QoS until the last hop.

Forwarding equivalence class (FEC)
The set of packets forwarded the same way and therefore given the same label, typically all traffic to one prefix.

Implicit null
Reserved label 3, signalled by an egress router to request PHP; it never appears in a packet.

LFIB
The label forwarding table used for packets that arrive with a label; contains only bindings from the best next hop.

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Liberal label retention
Keeping bindings from every LDP neighbour, not just the current next hop, so reconvergence needs no new signalling.

LDP-IGP synchronisation
Mechanism that keeps the IGP from using a link for forwarding until LDP has converged on it.

LDP transport address
The address used for the LDP TCP session; it must be reachable and is normally the router ID loopback.

Session protection
A targeted LDP hello that preserves a session and its bindings across a direct link failure.

Key takeaways

- Labels decouple services from the core: P routers need only reachability to PE loopbacks.
- Each label entry is 32 bits; every label adds 4 bytes, so size core MTU for the deepest stack you will push.
- LDP binds labels to exact routing-table prefixes; keep PE loopbacks as /32 routes end to end.
- Harden LDP with IGP synchronisation, session protection, label filtering and authentication.

KNOWLEDGE CHECK

1. An ingress PE has a BGP VPN route whose next hop is 10.255.0.9, but its FIB shows the next hop as reachable via 10.255.0.0/16 with no outgoing label. What is the most likely cause?

   (A) The VPN route target is missing on the remote PE, so the route was never exported correctly.

   (B) PE loopbacks are summarised somewhere in the core, so no node holds a label for the /32 next hop.

   (C) The LDP session uses the wrong transport address and therefore never leaves the initialised state.

   (D) Penultimate hop popping is disabled, so the egress PE expects explicit null instead of a label.

2. Two LDP neighbours have transport addresses 10.255.0.3 and 10.255.0.7. Which statement about the session is true?

   (A) 10.255.0.3 opens the session because the lower address always takes the active role in LDP.

   (B) Both routers open TCP connections at once and the session with the lower port number is retained.

   (C) The session runs over UDP 646 because LDP only needs TCP for targeted sessions between non-neighbours.

   (D) 10.255.0.7 has the higher address, so it actively opens the TCP connection to port 646 on 10.255.0.3.

3. Why does an egress PE usually advertise implicit null for its own loopback?

   (A) It tells upstream routers to stop forwarding labelled traffic and fall back to plain IP routing.

   (B) It reserves label 3 for VPN traffic so the egress can distinguish customer packets from core traffic.

   (C) It asks the penultimate router to pop the transport label so the egress performs only one lookup.

   (D) It forces the TC bits to be preserved so QoS markings survive until the packet reaches the egress.

4. A core uses liberal label retention. What is the operational benefit after an IGP next-hop change?

   (A) The router already holds the new next hop's binding, so it can update the LFIB without new signalling.

   (B) The router can reuse its own local label for every prefix, which avoids allocating new labels at all.

   (C) The router can ignore IGP changes for a hold period, which prevents transient loops while LDP updates.

   (D) The router no longer needs LDP-IGP synchronisation because bindings exist before the adjacency forms.

5. Customer packets of 1500 bytes cross a core using transport, VPN and a TI-LFA repair label during failures. What is the minimum MPLS MTU that avoids fragmentation or drops in that state?

   (A) 1504 bytes, because only the top label counts toward the MTU on each individual core link.

   (B) 1508 bytes, because the transport label is popped before the last hop and never counts twice.

   (C) 1512 bytes, because three 4-byte labels are added to the 1500-byte IP packet.

   (D) 1500 bytes, because MPLS labels are carried inside the Ethernet header and never affect MTU.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · LABEL · LESSON 2 · 45 MIN

MPLS L3VPN End to End

VRFs, route distinguishers, route targets and MP-BGP working together across a shared core

- Explain precisely what a route distinguisher does and what a route target does, and why they are not interchangeable.
- Build and verify a working L3VPN across VRF, MP-BGP VPNv4/VPNv6 and PE-CE routing.
- Choose PE-CE protocols and loop-prevention tools for multi-site and multi-homed customers.

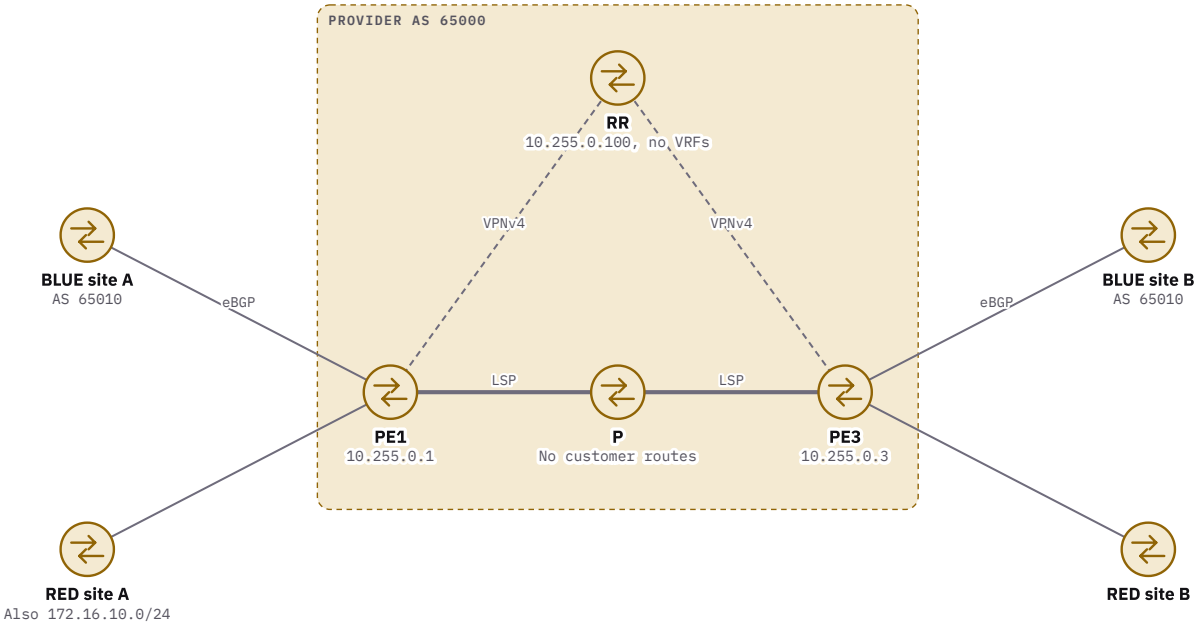
An MPLS Layer 3 VPN (RFC 4364) lets one provider core carry thousands of customers with overlapping address space. Every moving part has a single job: the **VRF** isolates a customer's routes on a PE, the **route distinguisher** makes overlapping prefixes unique inside BGP, the **route target** decides which VRFs import a route, **MP-BGP** carries the routes and their service labels between PEs, and the **transport LSP** from the previous lesson delivers packets to the right PE. Most failed lab tasks come from confusing two of these jobs, so we keep them strictly separate.

L3VPN building blocks, from customer to core

Customer edge (CE)	`Runs plain IP routing`	`Static, eBGP, OSPF or IS-IS toward the PE`
PE: VRF	`Per-customer RIB and FIB`	`Interfaces bound to the VRF`
PE: MP-BGP	`VPNv4 / VPNv6 address families`	`RD + prefix, RT communities, service label`
Route reflectors	`Distribute VPN routes`	`Need no VRFs themselves`
Transport	`LDP, SR or RSVP-TE LSPs between PE loopbacks`	`P routers hold no customer state`

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One L3VPN across a shared core



Two customers use the same 172.16.10.0/24 without conflict: RDs keep the VPN routes distinct in BGP, route targets decide which VRF imports them, and P routers carry only labels toward PE loopbacks.

Route distinguisher versus route target

Route distinguisher (RD)

- 8 bytes prepended to the prefix to form a VPN-IPv4 or VPN-IPv6 address
- Makes overlapping customer prefixes distinct in BGP
- Has no effect on import or export
- Unique RD per VRF per PE allows multipath and fast convergence across multi-homed sites

Route target (RT)

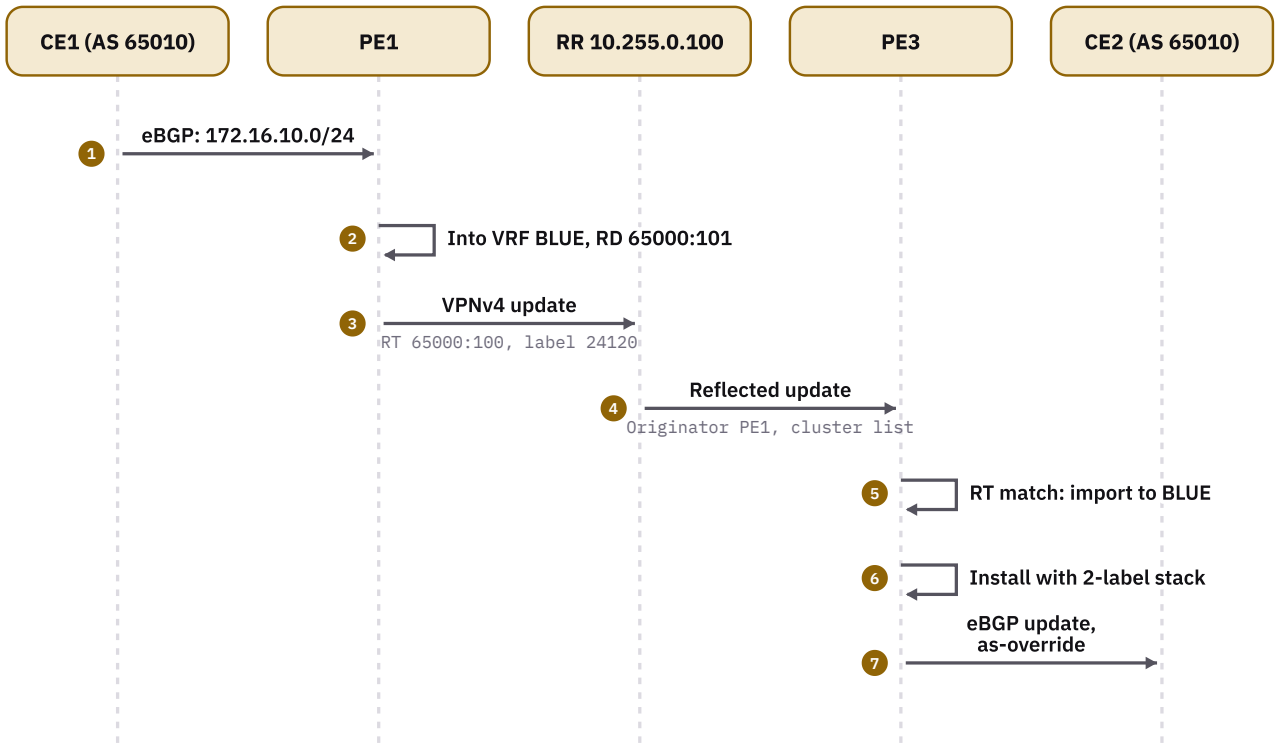
- BGP extended community attached on export
- Matched against import RTs on every PE
- Defines topology: any-to-any, hub-and-spoke, extranet
- A route may carry many RTs; a VRF may import many

Address families involved

ADDRESS FAMILY	AFI / SAFI	CARRIES
VPNv4 unicast	1 / 128	RD:IPv4 prefix, RT communities, service label
VPNv6 unicast	2 / 128	RD:IPv6 prefix, carried over an IPv4 or IPv6 transport
IPv4 labelled unicast	1 / 4	Loopbacks with labels for inter-AS option C and seamless MPLS
Route target constraint	1 / 132	Which RTs a PE wants, so reflectors send only relevant VPN routes

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Control plane: from CE route to remote VRF



The RD and service label are added at export, the reflector passes the route on without any VRF, and only the route target decides import at PE3.

PE1: VRF, PE-CE eBGP and VPNv4

PE1 · IOS-style CLI

```
vrf definition BLUE
 rd 65000:101
 address-family ipv4
  route-target export 65000:100
  route-target import 65000:100
!
interface GigabitEthernet0/2
 vrf forwarding BLUE
 ip address 192.0.2.1 255.255.255.252
!
router bgp 65000
 neighbor 10.255.0.100 remote-as 65000
 neighbor 10.255.0.100 update-source Loopback0
 address-family vpnv4
  neighbor 10.255.0.100 activate
  neighbor 10.255.0.100 send-community extended
 address-family ipv4 vrf BLUE
  neighbor 192.0.2.2 remote-as 65010
  neighbor 192.0.2.2 activate
  neighbor 192.0.2.2 as-override
```

The RD is unique to this PE and VRF; the RT is shared by every site of the customer. as-override lets other sites that also use AS 65010 accept these routes.

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Following one route from BGP to the label stack

PE3 · IOS-style CLI

```
PE3# show bgp vpnv4 unicast vrf BLUE 172.16.10.0/24
  Paths: (1 available, best #1, table BLUE)
  65010
    10.255.0.1 (metric 30) from 10.255.0.100 (10.255.0.100)
      Origin IGP, metric 0, localpref 100, valid, internal, best
      Extended Community: RT:65000:100
      Originator: 10.255.0.1, Cluster list: 10.255.0.100
      mpls labels in/out nolabel/24120
PE3# show ip cef vrf BLUE 172.16.10.0/24 detail
172.16.10.0/24, epoch 0
  recursive via 10.255.0.1 label 24120
    nexthop 10.1.35.5 GigabitEthernet0/1 label 24017-(local:24044)
```

Outer label 24017 is the transport label toward PE1's loopback; inner label 24120 is PE1's service label for the prefix. If the transport label were missing, the problem is in LDP or SR, not BGP.

PE-CE routing choices

STAT

Static routes

Simplest and most predictable; redistribute into the VRF's BGP. No dynamic failure detection unless tracked or paired with BFD.

EBGP

eBGP

The default choice. Use as-override or allowas-in when sites share an AS, and site-of-origin to stop routes re-entering a multi-homed site.

OSPF

OSPF

Routes cross the core as BGP but return as inter-area (or external) routes. Domain ID and the down bit prevent loops; a sham-link restores intra-area preference over a backdoor link.

ISIS

IS-IS or others

Possible but rarely chosen; follow the same redistribution and loop-prevention discipline.

INSIGHT · LABEL ALLOCATION MODES CHANGE BOTH SCALE AND FAILURE BEHAVIOUR

Per-prefix allocation gives each VPN prefix its own service label: simple to troubleshoot but costly at scale. Per-CE (per next hop) allocation shares one label among all prefixes from the same CE, so the egress forwards with no IP lookup. Per-VRF allocation uses one label for the whole VRF and requires an IP lookup at the egress, which also enables features such as egress filtering, but costs extra processing. Know which mode your platform uses before you reason about what a label means in a trace.

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SCENARIO · DECIDE · TWO SITES, ONE CUSTOMER AS, NO ROUTES

Customer GREEN uses AS 65010 at every site and runs eBGP to the provider. Site A's prefixes reach PE3 in VRF GREEN, and PE3 advertises them to the site B CE, but the site B CE rejects them. What is the best provider-side fix?

(A) Give site B a different route distinguisher so its routes are treated as distinct from site A's.

(B) Remove the route target from site A's export so that site B imports the routes another way.

(C) Redistribute the VRF routes into the provider IGP so site B learns them without BGP at all.

(D) Configure as-override toward the site B CE on PE3 so the customer AS in the path is replaced with the provider AS.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE

VRF and route-target plan

Plan every VRF before configuring it, so RD and RT values are consistent, unique and documented across all PEs.

VRF NAME	CUSTOMER / PURPOSE	RD SCHEME	EXPORT RT(S)	IMPORT RT(S)	TOPOLOGY	PE-CE PROTOCOL
BLUE	*Retail chain, all sites*	*65000:1<PE-id> (unique per PE)*	*65000:100*	*65000:100*	*Any-to-any*	*eBGP + as-override*
BLUE-SPOKE	*Retail branches via firewall hub*	*65000:2<PE-id>*	*65000:201 (spoke)*	*65000:202 (hub)*	*Hub-and-spoke*	*eBGP, SoO per site*

L3VPN bring-up checklist

☐ VRF defined with a unique RD on every PE and correct import/export RTs

☐ Customer interfaces bound to the VRF before addressing (binding removes the address)

☐ VPNv4/VPNv6 sessions established to the reflectors and extended communities sent

☐ PE-CE routing up and customer routes present in the local VRF table

☐ Routes present in the remote PE's VRF with the expected RT and next hop

☐ Remote next hop resolves to a /32 with a transport label in the FIB

☐ End-to-end LSP ping or trace succeeds between the PE loopbacks

☐ Ping from the VRF with large DF-set packets proves MTU across the core

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Key takeaways

- VRF isolates, RD makes prefixes unique, RT decides import, MP-BGP carries routes and labels, the LSP delivers packets.
- Only route targets control import; route distinguishers never do.
- A unique RD per PE keeps both paths of a dual-homed site visible for fast failover.
- If BGP delivered the route but the FIB lacks the outer label, the fault is in the transport, not in BGP.

KNOWLEDGE CHECK

6. Two PEs both serve customer RED and use different RDs for the same VRF. A third PE imports RED routes. What decides whether it imports them?

   (A) Only the RD, which must match the importing VRF's RD before the route can be copied into it.

   (B) Both RD and RT, because the RD selects the VRF and the RT then filters which prefixes are kept.

   (C) The BGP next hop, which must belong to a PE listed in the VRF's configured peer-PE table.

   (D) Only the route targets attached to the routes, compared with the import RTs of each local VRF.

7. Why do designers often give each PE a unique RD for the same customer VRF?

   (A) So a reflector sees distinct paths for a dual-homed prefix and can pass both on, enabling fast failover.

   (B) So the route targets can be omitted, because the unique RD already identifies the customer.

   (C) So the service label can be derived from the RD and does not need to be signalled at all.

   (D) So the P routers can separate customer traffic, since they inspect the RD when switching labels.

8. A PE shows a customer route in its VRF table, but the CEF entry has only one label. Which label is most likely missing and what does that point to?

   (A) The service label, pointing at a missing route target on the PE where the route was originated.

   (B) The transport label for the remote PE loopback, pointing at an LDP or SR problem in the core.

   (C) The service label, pointing at a missing RD on the importing PE for this particular VRF.

   (D) The transport label, pointing at a missing send-community setting toward the CE router.

9. A multihomed customer site is learning its own prefixes back from the VPN through its second PE. Which tool is designed to stop this?

   (A) allowas-in on the CE, which accepts its own AS in the path and filters the duplicate routes.

   (B) A unique RD per site, which keeps the routes in separate tables so they cannot loop back.

   (C) Site of origin, which tags routes from the site and blocks them from being sent back to it.

   (D) Per-VRF label allocation, which forces an IP lookup at the egress and drops looping packets.

10. Customer BLUE runs OSPF to its PEs and also has a backdoor link between two sites in area 0. Traffic prefers the backdoor even when the VPN path is better. Which feature restores the intended preference?

    (A) A sham-link between the PEs, so the VPN path appears as an intra-area link in the customer's area.

    (B) The down bit, so routes learned over the backdoor are ignored by the PEs during redistribution.

    (C) A matching OSPF domain ID on both PEs, so VPN routes are advertised as external type 1 routes.

    (D) A route target per site, so the backdoor routes are filtered from the VRF before they are installed.

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MODULE 5 · LABEL · LESSON 3 · 45 MIN

Scaling and Interconnecting VPNs

Route reflection, RT constraint, hub-and-spoke topologies and the inter-AS options

- Design VPN route reflection that scales without leaking unnecessary state to PEs.
- Build hub-and-spoke and extranet topologies purely with route-target policy.
- Compare inter-AS options A, B and C and select one for a given trust and scale requirement.

A five-PE lab can full-mesh VPNv4 sessions; a national backbone cannot. This lesson covers the tools that make L3VPN scale inside one AS, the route-target patterns that turn the same VPN machinery into different topologies, and the overview of how VPNs cross autonomous-system boundaries. The inter-AS material is deliberately at design level: an expert must recognise each option, know what state lives where, and pick one under pressure.

How VPN routes are filtered

On a PE

- Automatic RT filtering drops VPN routes whose RTs match no local import list
- Saves memory, but a newly added VRF needs a route refresh to learn routes
- Route refresh is automatic on most modern implementations

On a route reflector

- Holds no VRFs, so RT filtering is disabled and it keeps all VPN routes
- Uses its own resources for every VPN in the network
- Can be partitioned into planes or clusters by RT

With RT constraint (RFC 4684)

- PEs advertise which RTs they import
- Reflectors send each PE only matching routes
- Cuts update churn and memory on every PE

RT constraint in operation

1. **Enable the address family** PEs and reflectors negotiate the route-target constraint address family alongside VPNv4/VPNv6.
2. **Advertise interest** Each PE originates one membership route per import RT of its VRFs.
3. **Build filters** The reflector forms an outbound filter per client from those membership routes.
4. **Send only what matters** VPN updates go only to PEs that asked for at least one attached RT.
5. **React to change** A new VRF triggers a new membership route and the reflector sends the matching routes immediately.

Topologies built from route targets alone

TOPOLOGY	EXPORT POLICY	IMPORT POLICY	NOTES
Any-to-any	All sites export RT A	All sites import RT A	The default full-mesh VPN
Hub-and-spoke	Spokes export RT SPOKE; hub exports RT HUB	Spokes import HUB; hub imports SPOKE	Forcing spoke-to-spoke traffic through the hub needs two hub VRFs or interfaces (hub-in / hub-out)

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TOPOLOGY	EXPORT POLICY	IMPORT POLICY	NOTES
Extranet	Shared-service VRF exports RT SHARED	Consumer VRFs import SHARED; shared VRF imports each consumer's RT	Export only the needed prefixes with export maps to avoid leaking everything
Management VRF	Each customer VRF exports CE loopbacks with RT MGMT via export map	NOC VRF imports MGMT	Overlapping customer addresses need NAT or unique loopback ranges

Inter-AS options at a glance

ASPECT	OPTION A (BACK-TO-BACK VRFS)	OPTION B (EBGP VPNV4 BETWEEN ASBRS)	OPTION C (MULTIHOP VPNV4 BETWEEN RRS)
Inter-AS link carries	Plain IP, one subinterface or VRF per customer	Labelled VPN packets	Labelled packets with transport + service labels
ASBR state	Every VRF and every customer route	Every VPN route, no VRFs	Only PE loopbacks (BGP labelled unicast)
Scale	Low: per-customer configuration	Medium: ASBR holds all VPN routes	High: ASBRs hold no VPN routes
Trust and security	Strongest isolation; QoS and filtering per VRF	Shared labels; moderate trust needed	Exposes internal loopbacks and labels; high trust needed
Typical use	Few shared customers, different operators	Partner networks, mergers	Same organisation, multiple ASes

Option B: what each ASBR does to a VPN route

01 PE in AS 65000
Exports with RT and service label L1; next hop PE1

→

02 ASBR1
Keeps the route without a VRF, sets next hop self toward ASBR2, allocates new label L2

→

03 ASBR2
Receives with L2, sets next hop self toward its RR, allocates L3

→

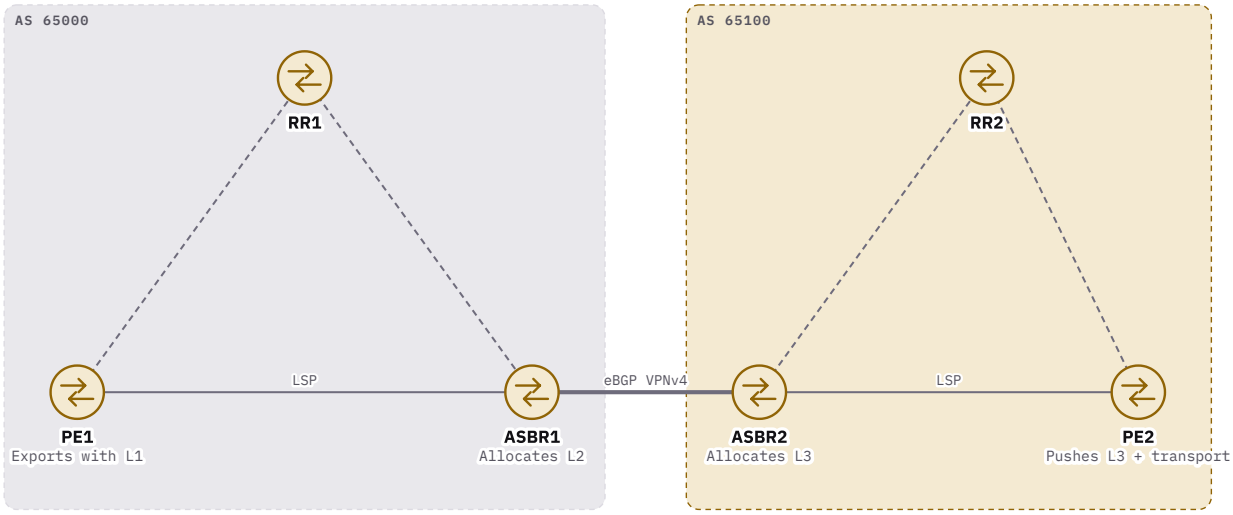
04 PE in AS 65100
Imports by RT; pushes L3 + transport label to ASBR2

→

05 Forwarding
ASBR2 swaps L3 to L2; ASBR1 swaps L2 to L1 and pushes transport to PE1

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Inter-AS option B: labels swapped at each ASBR



Each ASBR keeps every VPN route without a VRF, sets next hop self and allocates a new label (L1 → L2 → L3), so the inter-AS link carries labelled packets without LDP.

Option B ASBR in AS 65000

ASBR1 · IOS-style CLI

```
router bgp 65000
 no bgp default route-target filter
 neighbor 203.0.113.2 remote-as 65100
 neighbor 10.255.0.100 remote-as 65000
 neighbor 10.255.0.100 update-source Loopback0
 address-family vpnv4
  neighbor 203.0.113.2 activate
  neighbor 203.0.113.2 send-community extended
  neighbor 10.255.0.100 activate
  neighbor 10.255.0.100 send-community extended
  neighbor 10.255.0.100 next-hop-self
!
interface GigabitEthernet0/3
 ip address 203.0.113.1 255.255.255.252
 mpls bgp forwarding
```

Without disabling RT filtering, an ASBR with no VRFs would discard every VPN route it receives. `mpls bgp forwarding` *lets the eBGP link carry labelled packets without LDP.*

WATCH OUT · OPTION C NEEDS A TRUST CONVERSATION BEFORE A CONFIG SESSION

In option C each AS must be able to reach the other's PE loopbacks with labels, usually by exchanging them as BGP labelled unicast. That means internal addresses and label space are visible to the other side, and a misconfiguration in one AS can attract traffic in the other. Choose it when both ASes belong to the same organisation or a tightly governed partnership; otherwise option B or A is safer.

Two related patterns appear in large designs. **Carrier supporting carrier** lets a provider sell label-switched connectivity to another provider: the customer carrier's loopbacks live in a VRF of the backbone carrier, labels are exchanged on the PE-CE link (labelled BGP or LDP), and the customer carrier runs its own VPNs over the top. **Seamless MPLS** extends end-to-end LSPs across a hierarchical access, aggregation and core design by

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carrying loopbacks in BGP labelled unicast, so the IGP of each domain stays small while services still terminate end to end.

CASE STUDY · A MOBILE OPERATOR'S L3VPN BACKBONE

A mobile operator runs 14,000 cell sites over an access-aggregation-core network. Radio sites need connectivity to the mobile core for user-plane and control-plane traffic, to a separate OAM network and to timing services. The current design runs one flat IS-IS domain with LDP and full VPNv4 tables on every cell-site router, and convergence after an aggregation failure exceeds five seconds. Cell-site routers have small FIBs and leadership wants a design that supports 30,000 sites and a later move to segment routing.

1. Which VRFs would you build, and what RT topology fits radio-to-core traffic?

2. How do you stop every cell-site router from holding every VPN route?

3. How do you shrink the IGP and still keep end-to-end LSPs?

4. What would you verify first after migrating one aggregation region?

Model answer: module answer key.

SCENARIO · DECIDE · SPOKES MUST TALK ONLY THROUGH THE FIREWALL HUB

A retailer's 400 branches must reach each other only through a central firewall attached to the hub PE. Today all sites import and export the same RT and branch-to-branch traffic bypasses the firewall. What design achieves the goal with the least new configuration at branches?

(A) Branches export SPOKE and import HUB; the hub PE uses one VRF importing SPOKE toward the firewall inside interface and another exporting HUB from its outside interface.

(B) Give every branch a unique RD so branch routes no longer overlap and cannot be imported by other branches.

(C) Keep the shared RT and put an access list on every branch PE interface that drops branch-to-branch flows.

(D) Use a single hub VRF that imports SPOKE and exports HUB, and point a default route at the firewall.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Think of a network you know that interconnects two administrative domains. Which inter-AS option does it resemble today, what state sits at the boundary, and what would change if the organisations merged or separated?

Key terms

Route target constraint
An MP-BGP address family in which PEs advertise the RTs they import, so reflectors send them only relevant VPN routes.

Hub-and-spoke VPN
A topology in which spokes import only hub routes and the hub imports spoke routes, forcing traffic through the hub.

Inter-AS option A
Back-to-back VRFs on the ASBRs, exchanging plain IP per customer across the boundary.

Inter-AS option C
Reflectors exchange VPN routes over multihop eBGP while ASBRs carry only labelled PE loopbacks.

Seamless MPLS
A hierarchical design that builds end-to-end LSPs across separate IGP domains using BGP labelled unicast.

Automatic RT filtering
Default PE behaviour of discarding VPN routes whose RTs match no local import policy.

Extranet
Selective route sharing between VRFs of different customers or a shared-service VRF via RT policy.

Inter-AS option B
ASBRs exchange VPN routes over eBGP and swap service labels, without holding VRFs.

BGP labelled unicast
BGP advertisement of IPv4 or IPv6 prefixes with an MPLS label, used for hierarchical and inter-AS LSPs.

Carrier supporting carrier
A service in which one provider carries another provider's labelled traffic inside a VRF.

Key takeaways

- RT constraint lets reflectors send each PE only the VPN routes it imports.
- Hub-and-spoke, extranet and management VPNs are built from route-target policy alone.
- Option A trades scale for isolation, option B puts every VPN route on the ASBRs, option C needs high trust.
- Seamless MPLS carries loopbacks in BGP labelled unicast so each IGP domain stays small.

KNOWLEDGE CHECK

11. A new VRF is added on a PE that has automatic RT filtering, but its routes from remote sites do not appear until much later. What is the most likely reason?

(A) The new VRF needs its RD to match the remote PEs' RDs before any routes can be imported into it.

(B) The reflector holds no VRFs, so it cannot send routes for a VRF until one is configured on it too.

(C) The service labels must be reallocated network-wide before a new VRF can use the existing routes.

(D) The PE had discarded those VPN routes earlier and must request them again with a route refresh.

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12. What is the main benefit of RT constraint for a large VPN deployment?

(A) PEs can omit route targets entirely, because the reflector now assigns them to each VPN route.

(B) Reflectors send each PE only the VPN routes whose RTs it imports, cutting memory and update churn.

(C) P routers can filter labelled packets by RT, which adds an extra isolation layer in the core.

(D) Route distinguishers can be shared by all PEs, because duplicates are now resolved by the reflector.

13. Which inter-AS option keeps VPN routes off the ASBRs entirely?

(A) Option A, because each ASBR keeps customer routes inside VRFs rather than in a VPN table.

(B) Option B, because the ASBRs swap service labels without needing to install the VPN routes.

(C) Option C, because ASBRs carry only labelled PE loopbacks while reflectors exchange VPN routes.

(D) Option A combined with B, because the VRFs filter whatever routes option B would have carried.

14. An option B ASBR shows the eBGP VPNv4 session established but receives zero prefixes from its partner, while the partner sends thousands. What should be checked first?

(A) Whether the ASBR's RD values match the partner's, because mismatched RDs block VPN import.

(B) Whether RT filtering is still active on the ASBR, which has no VRFs and so discards everything.

(C) Whether LDP runs on the inter-AS link, because eBGP VPNv4 cannot start without LDP labels.

(D) Whether the IGP of each AS is redistributed into the other, so next hops resolve on both sides.

15. Why does a strict hub-and-spoke VPN that must force spoke-to-spoke traffic through a hub firewall need two VRFs on the hub PE?

(A) One VRF is needed per address family, because spokes and the hub use different route distinguishers.

(B) One VRF is needed per reflector cluster, because spokes and hub are reflected by different devices.

(C) One VRF is needed per label allocation mode, because spokes use per-prefix and the hub per-VRF labels.

(D) So spoke routes and hub routes sit in different tables and the PE cannot shortcut spoke-to-spoke traffic.

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MODULE 5 · LABEL · LESSON 4 · 50 MIN

Segment Routing: SR-MPLS, TI-LFA and SRv6

Source routing with IGP-distributed segments, full-coverage fast reroute and the IPv6 data plane

- Calculate SR-MPLS labels from SRGB and index, and explain prefix, node, anycast and adjacency SIDs.
- Plan a migration from LDP to SR-MPLS, including interworking with LDP-only nodes.
- Describe SRv6 locators, behaviours and compressed SIDs, and compare SRv6 with SR-MPLS.

Segment routing (architecture in RFC 8402) replaces label-distribution protocols with segments advertised by the IGP. The headend expresses a path as an ordered list of segments; transit nodes keep **no per-path state**. With the MPLS data plane, a segment is a label; with the IPv6 data plane (SRv6), a segment is an IPv6 address. Experts must be fluent in both, because 2026 networks routinely run SR-MPLS in the core while evaluating or deploying SRv6 at the edge or in new builds.

LDP versus SR-MPLS

LDP

- Separate protocol with its own sessions and timers
- Labels locally significant and different on every hop
- Needs IGP synchronisation to avoid black holes
- Remote LFA coverage depends on topology
- No native traffic engineering

SR-MPLS

- Labels carried in IS-IS or OSPF extensions
- Prefix SIDs are global: the same label everywhere when SRGBs match
- No separate session, so no sync problem
- TI-LFA gives full coverage in any topology
- SR policies and Flex-Algo provide TE without midpoint state

Segment types

SEGMENT	SCOPE	ALLOCATED FROM	TYPICAL USE
Prefix SID	Global	SRGB as an index	Shortest-path forwarding to a prefix
Node SID	Global	SRGB (prefix SID with the node flag on a loopback)	Identifies a router; the basic building block of paths
Anycast SID	Global	SRGB, same index on several routers	Reach the nearest of a group, such as a pair of ABRs
Adjacency SID	Local	SRLB or dynamic range	Force traffic over a specific link; used by TI-LFA repair paths
Binding SID	Local	SRLB or dynamic range	Represents an entire SR policy as one label for stitching or steering
BGP peering SIDs	Local	Dynamic range	Egress peer engineering toward a chosen external peer or link

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SR-MPLS label arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Local label for a prefix SID	`own SRGB base + index`	With SRGB 16000-23999 and index 4, the router installs local label 16004.
Outgoing label	`next hop's SRGB base + index`	If the next hop uses SRGB 900000-907999, the router sends 900004. Matching SRGBs everywhere keep labels identical and troubleshooting simple.
SRGB sizing	`highest index + 1 ≤ SRGB size`	An 8,000-label SRGB fits node indexes 0-7999; plan index ranges per region and per Flex-Algo up front.

Enabling SR-MPLS and TI-LFA in IS-IS

P2 · hierarchical IOS-style CLI

```
segment-routing
 global-block 16000 23999
!
router isis CORE
 is-type level-2-only
 net 49.0001.0102.5500.0002.00
 address-family ipv4 unicast
  metric-style wide
  segment-routing mpls
 interface Loopback0
  passive
  address-family ipv4 unicast
   prefix-sid index 2
 interface GigabitEthernet0/0/0/1
  point-to-point
  address-family ipv4 unicast
   fast-reroute per-prefix
   fast-reroute per-prefix ti-lfa
```

Index 2 on the loopback becomes label 16002 on every router that uses the same SRGB. Adjacency SIDs are allocated automatically per IS-IS adjacency.

Reading the result

PE1 · hierarchical IOS-style CLI

```
PE1# show mpls forwarding prefix 10.255.0.4/32
Local  Outgoing    Prefix           Outgoing     Next Hop     Bytes
Label  Label       or ID            Interface                 Switched
16004  16004       SR Pfx (idx 4)   Gi0/0/0/0    10.1.12.2    5539012
       16004       SR Pfx (idx 4)   Gi0/0/0/1    10.1.13.3    0        (!)
PE1# show isis fast-reroute 10.255.0.4/32 detail
  Primary: via 10.1.12.2, Gi0/0/0/0, P2
  Backup (TI-LFA): via 10.1.13.3, Gi0/0/0/1, P3
    Repair label stack (top to bottom): 16006, 24001, 16004
    Repair path: node SID of P6, then P6-to-P4 adjacency SID
```

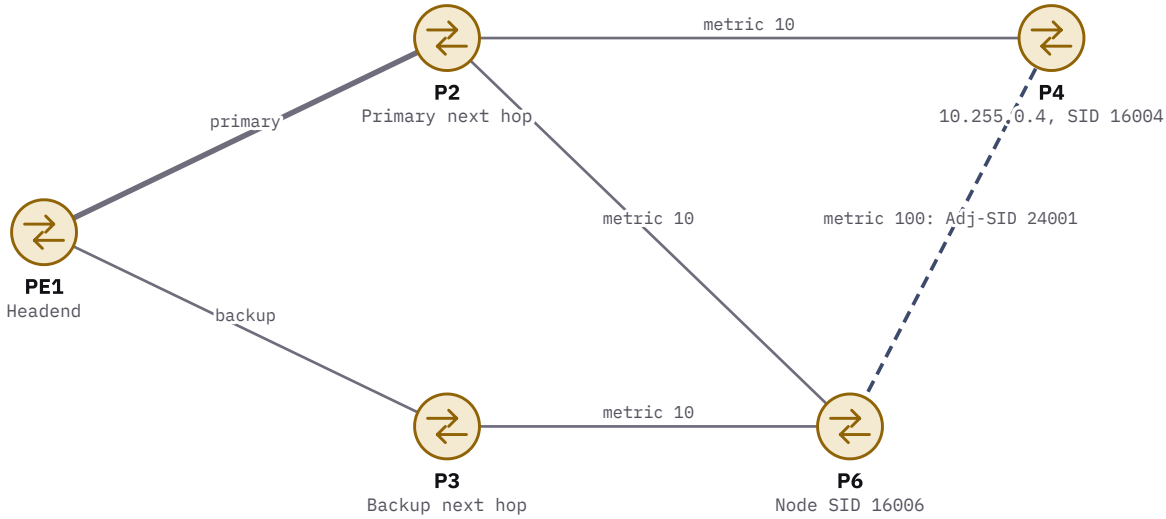
The (!) marks the pre-installed backup path. The repair pushes P6's node SID, then P6's adjacency SID toward P4, then the destination SID, which matches the post-convergence route.

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INSIGHT · WHY TI-LFA CHANGED THE FAST-REROUTE CONVERSATION

Classic LFA works only when a neighbour happens to have a loop-free path, and remote LFA adds tunnels but still leaves gaps. TI-LFA computes the **post-convergence path** for each destination and encodes it with node and adjacency SIDs, so every destination is protected in any topology with redundancy, and traffic follows the path it will use after the IGP converges. Pair it with **microloop avoidance**, which briefly steers traffic along an explicit segment list while the rest of the network catches up, and you remove most transient loss after failures and restorations.

TI-LFA repair for 10.255.0.4



If P2 fails (node protection), the post-convergence path is P3 → P6 → P4, but P6's current shortest path to P4 still runs through P2 (10 + 10 beats the metric-100 link). The repair stack is P6's node SID 16006, P6's adjacency SID 24001 toward P4, then 16004.

Migrating an LDP core to SR-MPLS

1. **Prepare** Agree SRGB and index plan, check platform support and maximum SID depth, and confirm that MTU allows the extra labels TI-LFA may push.
2. **Enable SR alongside LDP** Turn on SR in the IGP and assign prefix SIDs. LDP labels remain preferred, so forwarding does not change yet.
3. **Interwork with LDP-only nodes** Configure an SR mapping server to advertise SIDs on behalf of LDP-only prefixes; boundary routers stitch SR and LDP LSPs.
4. **Prefer SR** Switch imposition preference to SR on the headends, region by region, verifying LSP ping and trace after each step.
5. **Remove LDP** Once every PE loopback is reachable over SR, disable LDP and remove the synchronisation and session-protection configuration it needed.

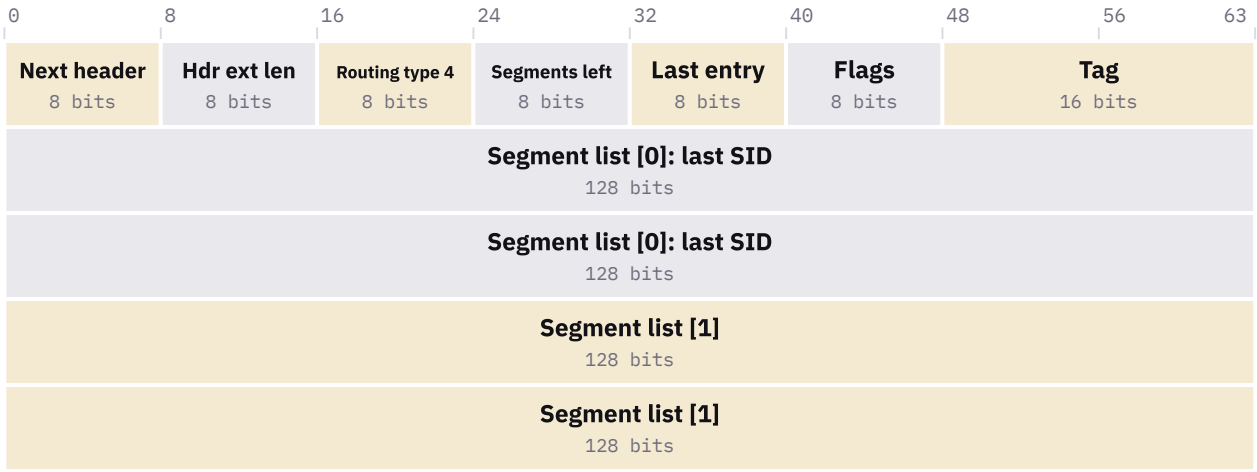
SRv6 behaviours you must recognise

BEHAVIOUR	WHAT THE NODE DOES	TYPICAL ROLE
End	Process the SRH and move to the next segment	Transit waypoint, equivalent to a node SID

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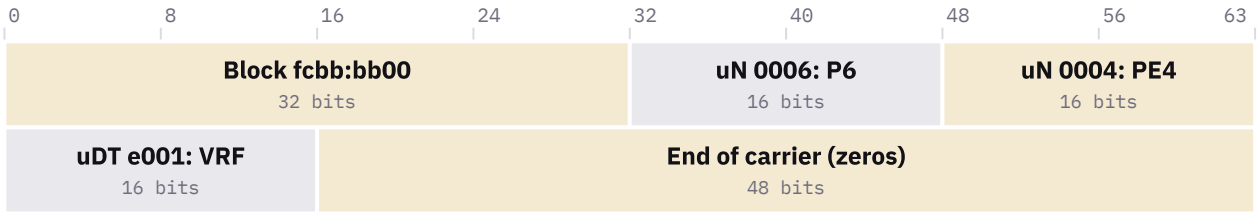
BEHAVIOUR	WHAT THE NODE DOES	TYPICAL ROLE
End.X	As End, then forward over a specific adjacency	Adjacency segment for TE and TI-LFA
uN / uA (micro-SID flavours)	Shift the next 16-bit micro-SID into place and forward, optionally over a given adjacency	Compressed node and adjacency segments in a micro-SID carrier
End.DT4 / End.DT6 / End.DT46	Decapsulate and look up the inner packet in a given VRF table	L3VPN service SID, typically per VRF
End.DX4 / End.DX6	Decapsulate and forward to a specific next hop	Per-CE L3VPN service
End.DT2U / End.DX2	Decapsulate and bridge in a Layer 2 table or to an interface	EVPN and pseudowire services
End.B6.Encaps	Bind to an SR policy and encapsulate in a new SRH	Binding SID for stitching domains

The segment routing header (routing type 4)



The segment list is stored in reverse order and Segments Left points to the active SID, which is copied into the IPv6 destination address at each segment endpoint. Every full SID costs 16 bytes, which is why micro-SIDs matter.

An SRv6 micro-SID carrier (32-bit block, 16-bit micro-SIDs)



Destination fcbb:bb00:6:4:e001:: steers through P6 to PE4 and into a VRF with no SRH at all: each node consumes its own micro-SID and shifts the rest left. A classic SID has the same logic in one address: locator (block + node), function and optional argument.

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SRv6 locator and L3VPN service

PE1 · hierarchical IOS-style CLI

```
segment-routing
 srv6
  encapsulation
   source-address 2001:db8:255::1
  locators
   locator MAIN
    micro-segment behavior unode psp-usd
    prefix fcbb:bb00:1::/48
!
router isis CORE
 address-family ipv6 unicast
  segment-routing srv6
   locator MAIN
!
router bgp 65000
 vrf BLUE
  address-family ipv4 unicast
   segment-routing srv6
    locator MAIN
    alloc mode per-vrf
```

The locator is advertised in IS-IS as an ordinary IPv6 prefix, so non-SRv6 routers can still forward toward it. BGP now advertises an SRv6 service SID (End.DT4-style) instead of an MPLS service label.

SR-MPLS versus SRv6 in 2026 designs

SR-MPLS

- Small 4-byte segments
- Mature hardware support and tooling
- Reuses existing MPLS services and OAM
- Needs MPLS on every transit node

SRv6

- Plain IPv6 transit: non-SRv6 routers forward on the locator prefix
- Service and transport in one address
- Full 128-bit SIDs are heavy; compressed micro-SIDs (the IETF NEXT-C-SID flavour) pack several segments into one address
- Requires careful filtering of the SID block at domain edges and solid IPv6 operations

SCENARIO · DECIDE · RETIRING LDP ON A MIXED CORE

Your core has twelve SR-capable routers and two legacy PEs that only support LDP. You want to disable LDP on the P routers, but the legacy PEs still terminate VPN services. What must be in place first?

(A) Nothing further: SR headends will automatically fall back to plain IP forwarding toward any LDP-only PE.

(B) An SR mapping server advertising prefix SIDs for the legacy PE loopbacks, with LDP kept only on the boundary routers next to them for stitching.

(C) A larger SRGB on the legacy PEs so that they can learn SR labels from their SR-capable neighbours.

(D) RSVP-TE tunnels from every SR headend to each legacy PE, configured in addition to the SR policies.

Circle your choice. Answer and feedback: module answer key.

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Key terms

Segment
An instruction in a source-routed path: a label in SR-MPLS or an IPv6 address in SRv6.

SRLB
Segment Routing Local Block: a label range for locally significant segments such as adjacency SIDs.

Adjacency SID
A local segment that forces forwarding over a specific IGP adjacency.

TI-LFA
Topology-independent loop-free alternate: fast reroute that pre-computes the post-convergence path for every destination.

SRv6 locator
The routable IPv6 prefix of a node from which its SRv6 SIDs are allocated.

Micro-SID
A compressed SRv6 format that packs several short segments into one 128-bit address.

SRGB
Segment Routing Global Block: the label range a node reserves for global segments such as prefix SIDs.

Prefix SID index
The offset added to a node's SRGB base to form the label for a prefix segment.

Anycast SID
The same prefix SID advertised by several routers, so traffic goes to the nearest one.

Mapping server
A function that advertises SIDs on behalf of LDP-only prefixes to allow SR and LDP interworking.

Segment routing header (SRH)
An IPv6 routing extension header that carries the SRv6 segment list.

End.DT4
SRv6 behaviour that decapsulates and performs an IPv4 lookup in a specific VRF table.

Key takeaways

- Segment routing puts the path in the packet; transit nodes keep no per-path state.
- A prefix SID label is the next hop's SRGB base plus the index; identical SRGBs keep labels the same everywhere.
- TI-LFA encodes the post-convergence path with node and adjacency SIDs for full coverage; check maximum SID depth and MTU.
- SRv6 SIDs are routable IPv6 addresses: compress them with micro-SIDs and filter the SID block at domain edges.

KNOWLEDGE CHECK

16. Router R5 has SRGB 16000-23999 and advertises its loopback with index 5. Its neighbour R4 uses SRGB 20000-27999. Which label does R4 send toward R5 for that prefix, and which label does R4 install as its local label?

(A) R4 sends 16005, which is R5's SRGB base plus index, and installs 20005 locally from its own SRGB.

(B) R4 sends 20005 and installs 16005, because the sender always uses its own SRGB for outgoing labels.

(C) R4 sends and installs 16005, because the prefix owner's SRGB is the only one that is used anywhere.

(D) R4 sends and installs 20005, because SR labels are always taken from the local SRGB on every hop.

17. What does TI-LFA protect against that classic LFA often cannot?

(A) Node failures of the egress PE itself, because TI-LFA can redirect traffic to any other PE in the VPN.

(B) Control-plane failures of BGP, because TI-LFA re-signals VPN routes over a pre-computed backup session.

(C) Failures in topologies where no neighbour has a loop-free path, by encoding the repair path with segments.

(D) Congestion on the primary path, because TI-LFA moves traffic away whenever link utilisation crosses a threshold.

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18. During an LDP to SR migration, prefix SIDs are configured everywhere but traces still show LDP labels. What explains this?

    (A) LDP remains the preferred imposition method until SR preference is explicitly enabled on the headends.

    (B) The SRGB is too small, so routers silently fall back to LDP for any prefix outside the label range.

    (C) SR labels are only used for traffic-engineered paths and never for ordinary shortest-path forwarding.

    (D) Adjacency SIDs have not been configured manually, so the IGP cannot advertise any prefix SIDs yet.

19. Which SRv6 behaviour would a PE bind to a per-VRF L3VPN service SID for IPv4 and IPv6 customers in the same VRF?

    (A) End.X, which forwards the packet over a specific adjacency toward the customer edge router.

    (B) End.B6.Encaps, which binds the VRF to an SR policy and encapsulates traffic in a new header.

    (C) End, which processes the routing header and continues to the next segment in the segment list.

    (D) End.DT46, which decapsulates and looks up the inner packet in the VRF's IPv4 or IPv6 table.

20. Why do SRv6 designs filter the SID block at the domain boundary?

    (A) SIDs use private label space, so leaking them would cause label collisions with partner networks.

    (B) SIDs are routable IPv6 addresses, so outside packets aimed at them could invoke network functions.

    (C) SIDs require MPLS on every hop, so external networks would drop them and generate error traffic.

    (D) SIDs change every few minutes, so external routers would cache stale entries and black-hole traffic.

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MODULE 5 · LABEL · LESSON 5 · 45 MIN

Traffic Engineering and Troubleshooting Label Paths

RSVP-TE, SR policies and Flex-Algo, then a disciplined method for broken LSPs

- Compare RSVP-TE with SR-TE and choose between explicit SR policies, on-demand policies and Flex-Algo.
- Configure colour-based automated steering of VPN traffic into an SR policy.
- Troubleshoot a failing label path layer by layer with LSP ping, trace and table inspection.

Traffic engineering moves traffic off the IGP shortest path to meet latency, diversity or capacity goals. The industry has shifted from **RSVP-TE**, which signals and holds state at every midpoint, toward **SR-TE**, where only the headend holds the path. Experts still meet RSVP-TE in brownfield cores and must compare the two calmly. The second half of the lesson is the payoff of the whole module: a repeatable method for finding why labelled traffic does not arrive.

RSVP-TE versus SR-TE

RSVP-TE

- PATH and RESV messages signal every LSP hop by hop
- Every midpoint holds per-LSP state and refresh timers
- Native bandwidth reservation and admission control
- Facility FRR protects links and nodes with bypass tunnels
- Scale limited by midpoint state (n^2 tunnels in full meshes)

SR-TE

- Path is a segment list imposed at the headend
- No midpoint state; the core only knows SIDs
- Bandwidth managed by a PCE or controller, not by signalling
- TI-LFA protects every segment
- Scales to large meshes and on-demand policies

Anatomy of an SR policy

KEY

Headend, colour, endpoint

A policy is identified by where it starts, a colour that represents intent (for example 100 = low latency) and the endpoint address.

CP

Candidate paths

One or more paths with a preference; the highest valid preference is active, the others are standby.

SL

Segment lists

Explicit lists of SIDs, or dynamic paths computed locally or by a PCE against metric and constraints.

BSID

Binding SID

A label (or SRv6 SID) representing the whole policy, so remote nodes or other domains can steer into it.

AS

Automated steering

BGP routes carrying a colour extended community are steered into the policy whose colour and endpoint match their next hop.

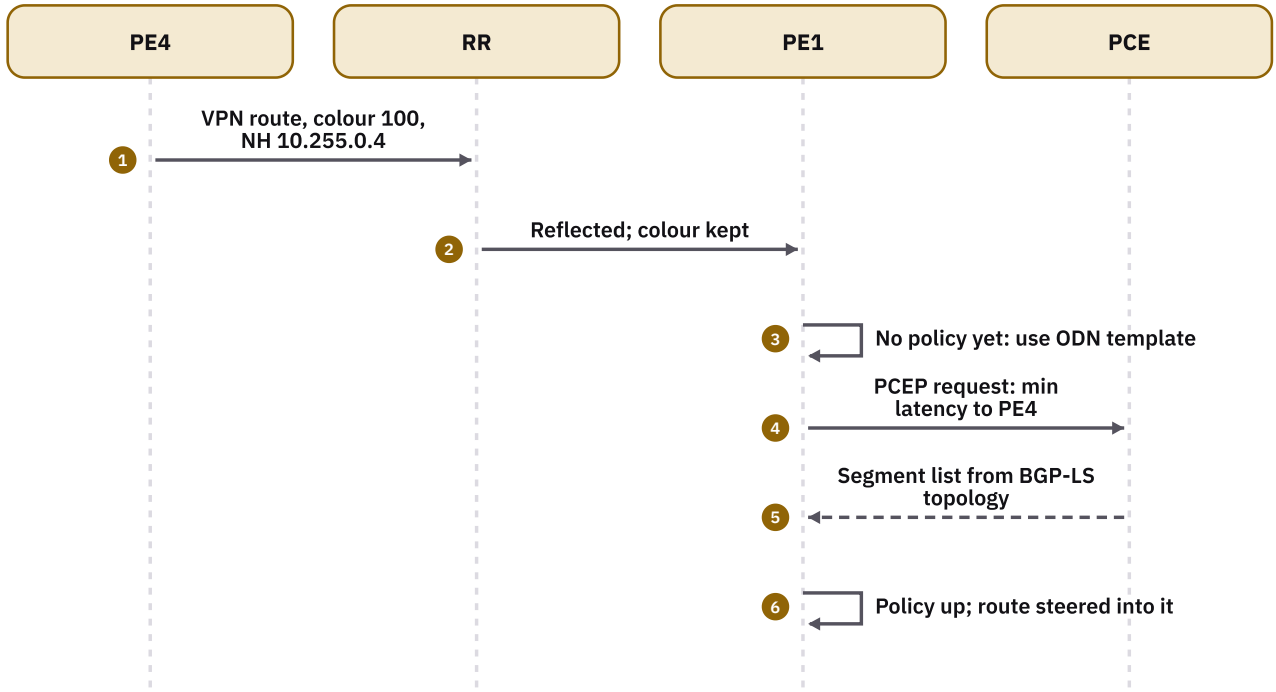
ODN

On-demand next hop

The headend instantiates a policy automatically when it receives a coloured route for an endpoint with no existing policy.

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On-demand next hop: from coloured route to steered traffic



No policy was configured for PE4: the colour on the route triggers the template, the PCE returns a latency-optimised segment list, and only PE1 ever holds the policy state.

Low-latency VPN traffic with colour steering

PE1 · hierarchical IOS-style CLI

```
segment-routing
 traffic-eng
  on-demand color 100
   dynamic
    metric
     type latency
  policy GOLD-TO-PE4
   color 100 end-point ipv4 10.255.0.4
   candidate-paths
    preference 200
     explicit segment-list VIA-P6
    preference 100
     dynamic
      metric
       type latency
!
! On PE4: colour set used by the GOLD VRF export policy
extcommunity-set opaque COLOR-100
  100
end-set
```

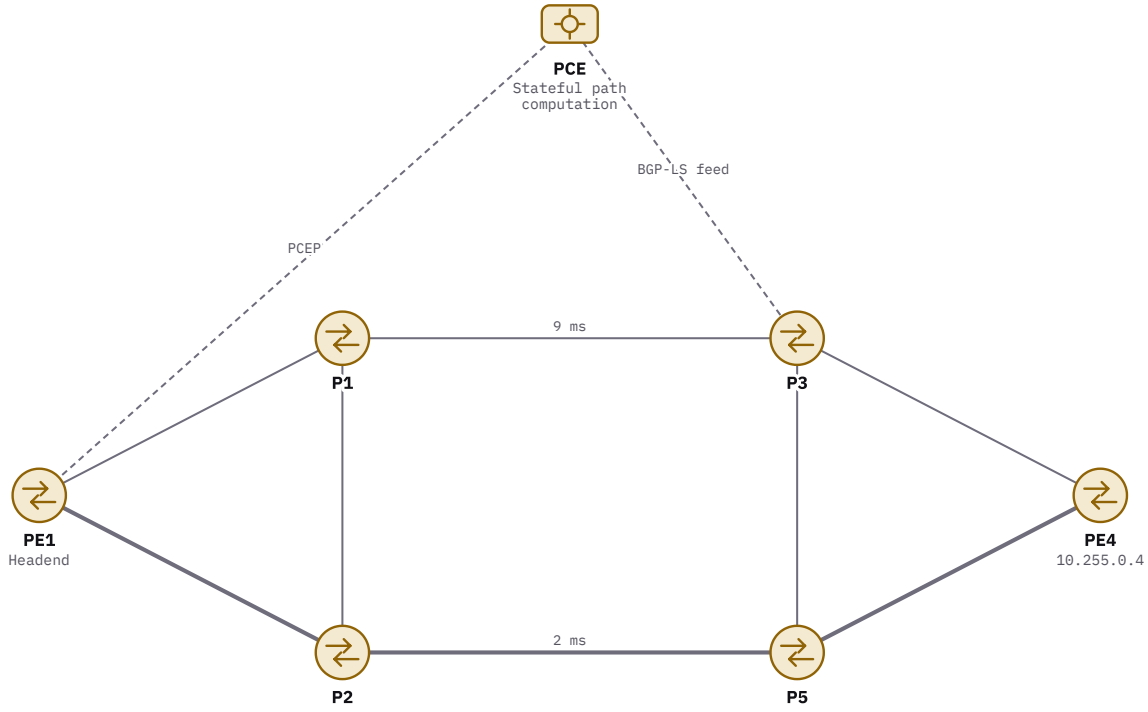
PE4 attaches colour 100 to the GOLD VRF routes it exports, using its export route policy. PE1 then steers those routes into the colour-100 policy toward 10.255.0.4: the explicit list when valid, otherwise the dynamic low-latency path.

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INSIGHT · FLEX-ALGO: TRAFFIC ENGINEERING WITHOUT EXPLICIT POLICIES

Flexible Algorithm lets the IGP compute additional topologies, numbered 128 to 255, each with its own metric type (IGP, delay or TE) and constraints such as excluding links with a given affinity. Every router that participates in an algorithm advertises an extra prefix SID for it. A headend that wants the low-latency topology simply pushes the algorithm-specific SID of the destination. Combined with automated steering, Flex-Algo delivers latency-optimised or plane-disjoint paths with one SID and no controller.

A controller-assisted SR-TE domain



The PCE learns topology, delay and SIDs through BGP-LS, computes the low-latency segment list and hands it to PE1 over PCEP; P routers keep no per-policy state.

Troubleshooting a broken label path, layer by layer

01
1 Underlay
Is the egress /32 loopback in the IGP with the expected next hop?

→

02
2 Transport label
Does every hop hold a label (LDP binding or SR SID) for that /32?

→

03
3 LFIB
Do local and outgoing labels line up hop by hop, with no untagged entries?

→

04
4 Service
Is the VPN route imported with the expected RT, next hop and service label?

→

05
5 Data plane
Do LSP ping and LSP trace succeed to the egress loopback?

→

06
6 MTU and QoS
Do large packets and marked traffic survive the full label stack?

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An LSP trace that reveals the fault

PE1 · IOS-style CLI

```
PE1# traceroute mpls ipv4 10.255.0.4/32
Codes: '!' - success, 'L' - labeled output interface,
       'D' - DS map mismatch, 'U' - unlabeled output, '.' - timeout
  0 10.1.12.1 MRU 1500 [Labels: 16004 Exp: 0]
L 1 10.1.12.2 MRU 1500 [Labels: 16004 Exp: 0] 2 ms
U 2 10.1.26.6 MRU 1500 [No Label] 3 ms
! 3 10.1.46.4 4 ms
```

The LSP breaks at P6, which forwards unlabelled toward PE4. A ping to the loopback still succeeds, but VPN traffic is dropped at P6 because a BGP-free core has no VRF routes. Check the IGP-to-label configuration on P6 first: SR not enabled in its IGP instance, LDP missing on the P6-PE4 link, or a label filter. Also note MRU 1500: the core links are not jumbo.

Symptom to cause

SYMPTOM	MOST LIKELY CAUSE	PROVE IT WITH	FIX
VPN route present, FIB shows no transport label	Next hop summarised or not bound	FIB entry and LIB/SID table for the /32	Keep /32 loopbacks; fix binding or filters
Small pings work, large transfers stall	MTU too small for the label stack	Ping in the VRF with DF set at 1500 and 1472	Raise core MTU; check MRU in LSP trace
Route missing in remote VRF	RT mismatch or RT filtering	VPN table on the RR and remote PE	Align RTs; route refresh
Loss for seconds after link restoration	IGP uses link before LDP is ready	Timestamps of IGP adjacency vs LDP session	LDP-IGP sync or move to SR
Traceroute in VRF shows core hops	TTL propagation enabled	Customer traceroute output	Disable TTL propagation for forwarded traffic if policy requires

SCENARIO · DECIDE · LARGE TRANSFERS STALL IN ONE VPN

Customer ORANGE reports that web pages load slowly and file copies stall between two sites. Pings with default size succeed. The path recently moved onto an SR policy with a four-label stack across a newly added link whose MPLS MTU is 1500. What is the most precise next step?

(A) Clear the BGP sessions on both PEs so that the VPN routes are re-advertised with fresh service labels.

(B) Change the customer's route distinguisher so that its routes are recalculated through the original path.

(C) Disable TI-LFA on the core because repair labels make every packet too large for the existing links.

(D) Ping inside the VRF with the DF bit set at 1500 bytes, then check the MRU reported by an LSP trace along the policy path.

Circle your choice. Answer and feedback: module answer key.

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Label path troubleshooting checklist

☐ Identify ingress PE, egress PE, VRF, prefix and BGP next hop before touching anything
☐ Confirm the next hop is a /32 in the IGP on every hop toward the egress
☐ Confirm a label binding or prefix SID exists for the /32 on every hop
☐ Walk the LFIB hop by hop, or run LSP trace, looking for untagged or no-label entries
☐ Confirm the service route: RT, RD, service label and best path on the ingress PE
☐ Test MTU with DF-set packets in the VRF and compare with the MRU in the LSP trace
☐ Check TE: which policy or tunnel the traffic is steered into and whether it is up
☐ Record findings and the fix, then re-run the same tests to prove recovery

REFLECT

Recall the last time a VPN or tunnel failed in a network you supported. Map the investigation onto the six-layer flow in this lesson. Which layer did you check first, which layer held the root cause, and how much time would the disciplined order have saved?

Key terms

RSVP-TE
Signalling protocol that establishes traffic-engineered LSPs with per-hop state and optional bandwidth reservation.

Colour
A numeric intent value attached to an SR policy and, as an extended community, to BGP routes.

On-demand next hop (ODN)
Automatic creation of an SR policy when a coloured route arrives for an endpoint without one.

PCE
Path computation element: a controller that computes paths for headends, often over PCEP.

MRU
Maximum receive unit for labelled packets on a hop, reported by LSP trace.

SR policy
A source-routed path identified by headend, colour and endpoint, with one or more candidate paths.

Automated steering
Placing BGP traffic into the SR policy whose colour and endpoint match the route's colour and next hop.

Flexible Algorithm
IGP extension that computes extra constraint-based topologies, each with its own prefix SIDs.

LSP ping and trace
MPLS echo tests (RFC 8029) that validate a label path end to end and hop by hop.

BGP-LS
BGP link-state: an address family that exports IGP topology, TE attributes and SIDs from the network to controllers and PCEs (current specification RFC 9552).

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Key takeaways

- SR-TE keeps path state only at the headend; RSVP-TE keeps it at every midpoint.
- An SR policy is headend, colour and endpoint, with candidate paths, segment lists and a binding SID.
- Colour communities with automated steering and ODN put VPN routes on intent-based paths without per-route configuration.
- Troubleshoot label paths layer by layer: IGP /32, labels, LFIB, service route, LSP trace, then MTU and QoS.

KNOWLEDGE CHECK

21. Why does SR-TE scale better than RSVP-TE in a large full-mesh design?

(A) SR-TE reserves bandwidth on every link through the IGP, so no separate signalling is needed.

(B) Only the headend holds path state, so midpoints keep no per-LSP state or refresh timers.

(C) SR-TE paths are computed by each midpoint, which spreads the computing load around the core.

(D) SR-TE tunnels are bidirectional, so each pair of routers needs only half as many tunnels.

22. A PE receives VPN routes carrying colour 100 from PE4 but has no policy configured for colour 100 to PE4. With an on-demand template for colour 100, what happens?

(A) The PE drops the coloured routes, because steering requires a policy to exist before routes arrive.

(B) The PE installs the routes on the IGP shortest path and ignores the colour until a policy is added.

(C) The PE asks PE4 to create a matching policy in the reverse direction and waits for its binding SID.

(D) The PE instantiates a colour-100 policy to PE4 from the template and steers the routes into it.

23. An LSP trace shows 'U' at the third hop with [No Label]. What does that tell you?

(A) The third-hop router forwards the traffic unlabelled, so it has no outgoing label for the destination.

(B) The third-hop router dropped the packet because the imposed label stack exceeded its maximum supported depth.

(C) The third-hop router is the egress PE and has correctly popped the last label for that FEC.

(D) The third-hop router received the packet with a DS map mismatch and returned it to the source.

24. Which Flex-Algo property lets a headend steer into a low-latency topology with a single label?

(A) Each router advertises a binding SID per algorithm, which represents the full explicit path.

(B) Each router floods its delay values in BGP, so the headend can build an explicit segment list.

(C) Each participating router advertises a prefix SID per algorithm, so pushing that SID selects the topology.

(D) Each router reserves bandwidth for the algorithm, so the label carries an admission guarantee.

25. Customer traceroutes across the VPN show provider core addresses, and the security team objects. What is the usual change?

(A) Enable explicit null on all egress PEs, so the core addresses are replaced with the PE address.

(B) Disable TTL propagation into the label stack for forwarded customer traffic at the ingress PEs.

(C) Enable PHP on every P router, so the core hops no longer decrement the TTL of labelled packets.

(D) Filter ICMP on the CE routers, so time-exceeded messages from the core never reach customers.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · LABEL

Answer key

Knowledge checks

1. B LDP binds labels to exact routing-table prefixes; a summary hides the /32 and leaves the next hop unlabelled. An RT problem would stop the route existing at all, a broken session would remove all labels, and PHP mode does not remove a binding.

2. D The peer with the higher transport address is active. LDP discovery uses UDP 646 but the session itself is always TCP 646, for link and targeted sessions alike.

3. C Implicit null triggers PHP, saving the egress a label lookup. Preserving TC bits is the job of explicit null, and label 3 never appears on the wire.

4. A Liberal retention stores bindings from all neighbours, so reconvergence is a local LFIB rewrite. It does not replace synchronisation, which protects a newly restored link that has no session yet.

5. C Every label in the stack adds 4 bytes: 1500 + 3 × 4 = 1512. PHP removes a label only at the penultimate hop, so the worst-case stack must fit everywhere else.

6. D RTs control import and export. The RD only makes the prefix unique in BGP and plays no role in VRF selection.

7. A With identical RDs a reflector picks one best VPN path and hides the other. Unique RDs keep both paths visible for multipath and quick convergence. P routers never see RDs.

8. B If BGP delivered the route, it delivered the service label with it. A missing outer label means the next hop has no LSP. Communities toward a CE have nothing to do with labels.

9. C SoO marks the routes with the site identity; PEs refuse to advertise them to a CE with the same SoO. allowas-in does the opposite, and RDs and label modes do not filter routes.

10. A VPN routes normally arrive as inter-area, which loses to intra-area backdoor routes. A sham-link makes the VPN path intra-area so cost decides. The down bit and domain ID solve different loop and route-type problems.

11. D RT filtering drops routes nobody imports; after adding a VRF the PE must ask its peers to resend them. Reflectors never need VRFs, and RDs do not control import.

12. B RT constraint moves filtering upstream to the reflector. It does not remove RTs, involve P routers or change RD design.

13. C Option C moves VPN route exchange to multihop sessions between reflectors. Option A holds every customer route in VRFs; option B holds every VPN route without VRFs.

14. B Without VRFs, default RT filtering drops all VPN routes. Option B uses BGP-assigned labels on the link rather than LDP, and redistributing IGPs between ASes is neither required nor wise.

15. D With one hub VRF the PE would know specific spoke routes and switch spoke-to-spoke traffic itself. Splitting the hub VRF forces traffic through the firewall. The other reasons are invented.

16. A The outgoing label uses the next hop's SRGB; the local label uses the router's own SRGB. That is why identical SRGBs are recommended: the label is then the same everywhere.

17. C TI-LFA guarantees coverage by expressing the post-convergence path with node and adjacency SIDs. Egress PE failure needs BGP-level protection such as PIC, and TI-LFA is not congestion-driven.

18. A By default most implementations keep LDP labels when both exist. SR is used for shortest-path forwarding too, and adjacency SIDs are allocated automatically. An undersized SRGB would cause SID errors, not a silent preference for LDP.

19. D End.DT46 is the dual-stack table-lookup decapsulation behaviour. End and End.X are transport behaviours; End.B6.Encaps represents a policy.

20. B An SRv6 SID is just an IPv6 address bound to a behaviour, so spoofed packets from outside could trigger it. SRv6 does not use MPLS labels, and SIDs are stable.

21. B Removing midpoint state is the scaling win. SR-TE does not reserve bandwidth in the IGP, midpoints do not compute the path, and SR policies are unidirectional.

22. D On-demand next hop creates policies when coloured routes arrive. Without ODN the routes would normally follow the default path, and policies are never created in the reverse direction automatically.

23. A U means unlabelled output: the label path breaks there. A correct egress returns '!', a mismatch is 'D', and depth problems produce other error codes.

24. C Flex-Algo prefix SIDs are per algorithm, and every node forwards them along that algorithm's topology. It is computed by the IGP, not BGP, and does not reserve bandwidth.

25. B With TTL propagation disabled, the label TTL starts at 255 and core hops never expire customer packets. Explicit null and PHP affect label popping, and CE filtering would break legitimate diagnostics.

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Scenarios & case studies

Scenario · Twenty seconds of loss after a link comes back
C is best. Correct. Synchronisation keeps the IGP from preferring the link until labels exist on it, removing the black hole without slowing failure detection.

Scenario · Two sites, one customer AS, no routes
D is best. Correct. The CE drops the route because its own AS 65010 appears in the AS path. as-override on the PE solves it without changing the customer's routers.

Case study · A mobile operator's L3VPN backbone
Build separate VRFs for user plane, control plane and OAM, each **hub-and-spoke**: cell sites export a spoke RT and import only the core hub RT, while the core data-centre PEs import all spokes; there is no business need for site-to-site traffic, so blocking it reduces state and attack surface. Enable **RT constraint** on the reflectors and place regional reflectors in each aggregation domain so cell-site routers receive only hub routes; a cell-site router then needs a handful of prefixes per VRF, not thousands. Split IS-IS into separate access, aggregation and core domains and carry PE loopbacks across boundaries with **BGP labelled unicast** (seamless MPLS), with aggregation routers acting as inline reflectors and next-hop-self; access routers may even use a default route plus labels on demand. This also prepares the move to segment routing, where TI-LFA gives sub-50 ms protection in each domain. After migrating a region, verify loopback reachability with labels end to end (LSP ping from cell-site to core PE), route counts per VRF on a sample of cell-site routers, failover timing during a controlled aggregation link failure, and MTU with the deepest label stack.

Scenario · Spokes must talk only through the firewall hub
A is best. Correct. Two hub VRFs force every spoke-to-spoke flow up into the firewall and back down, while branches only change their RT policy.

Scenario · Retiring LDP on a mixed core
B is best. Correct. The mapping server lets SR headends build paths to LDP-only prefixes, and the boundary routers stitch SR to LDP so services keep working.

Scenario · Large transfers stall in one VPN
D is best. Correct. This confirms or rules out an MTU problem caused by the deeper label stack on the new link, which matches every symptom.

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06

MODULE 6 · FABRIC

Fabric — Overlays, VXLAN EVPN and SD-WAN

Design, build, scale and migrate overlay networks in the data centre and across the WAN.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain the VXLAN data plane and every EVPN route type, and use them to reason about forwarding.
- Build tenant routing with symmetric IRB, anycast gateways and type-5 routes, including border connectivity.
- Design multihoming, a robust leaf-spine underlay and AI/ML back-end fabrics with correct EVPN control-plane roles.
- Architect SD-WAN overlay routing, policy, application steering and security integration.
- Plan and execute migrations from legacy WAN and data-centre networks with safe coexistence.

LESSONS

1. VXLAN Data Plane and the EVPN Control Plane · 45 min
2. Routing in the Fabric: IRB, Anycast Gateway and Type-5 Routes · 45 min
3. Multihoming and Underlay Design · 45 min
4. SD-WAN Architecture Deep Dive · 45 min
5. Migrating Legacy WANs and Data-Centre Networks · 45 min

MODULE 6 · FABRIC · LESSON 1 · 45 MIN

VXLAN Data Plane and the EVPN Control Plane

What the encapsulation does, what each EVPN route type carries and how hosts are learned

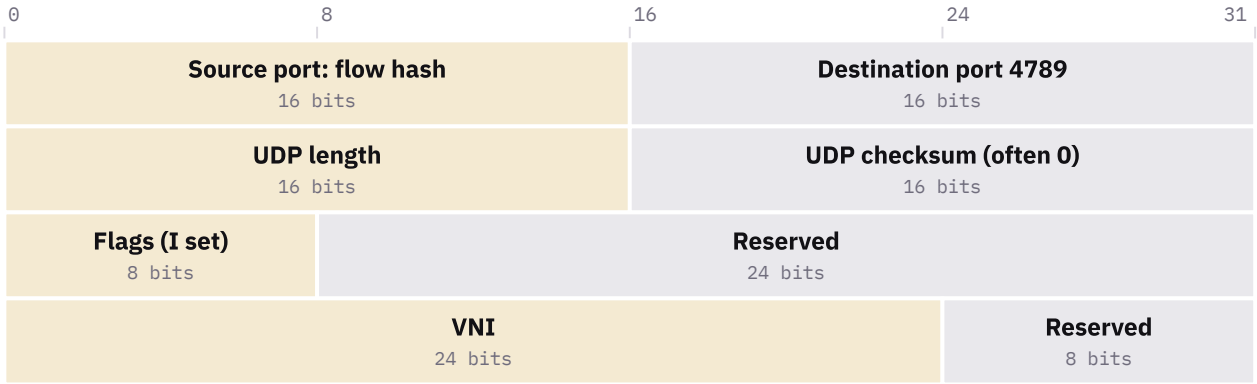
- Describe the VXLAN encapsulation, its overhead and how it provides entropy for ECMP.
- Explain EVPN route types 1 to 5 and identify which one solves which problem.
- Configure a leaf for Layer 2 VNIs with BGP-signalled ingress replication and verify host learning.

The **Fabric** segment of the Mastery Ring covers overlays: networks built on top of a simple routed underlay. In the data centre the dominant pattern is **VXLAN with an EVPN control plane** (EVPN is defined in RFC 7432 for MPLS; RFC 8365 specifies it over VXLAN and other overlays, and RFC 9135 and RFC 9136 cover integrated routing and IP-prefix routes). VXLAN supplies the encapsulation; EVPN replaces flood-and-learn with BGP advertisement of MAC addresses, IP addresses and flooding lists. In the WAN the same idea appears as SD-WAN. This lesson nails down the data plane and the route types, because every later topic, from symmetric IRB to multihoming, is just a particular use of them.

A VXLAN packet, outermost header first

Outer Ethernet	`Leaf to spine next hop` `Rewritten at every underlay hop`
Outer IP	`Source and destination are VTEP addresses` `DSCP copied from the inner packet by default on most platforms`
Outer UDP	`Destination port 4789` `Source port derived from a hash of the inner headers, giving ECMP entropy`
VXLAN header (8 bytes)	`I flag set` `24-bit VNI: about 16 million segments`
Inner frame	`Original Ethernet frame without FCS` `Tenant VLAN tag normally stripped and mapped to the VNI`

Outer UDP and VXLAN headers



Sixteen bytes sit between the outer IP header and the tenant frame. The source port is a hash of the inner headers for ECMP entropy, port 4789 identifies VXLAN, and the I flag marks the 24-bit VNI as valid.

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Overlay arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
VXLAN overhead (IPv4 underlay)	`14 outer Ethernet + 20 IP + 8 UDP + 8 VXLAN = 50 bytes`	Add 4 for an outer 802.1Q tag; an IPv6 underlay adds 20 more because its header is 40 bytes.
Minimum underlay IP MTU	`tenant MTU + 14 inner Ethernet + 8 VXLAN + 8 UDP + 20 outer IP`	A 1500-byte tenant MTU needs at least 1550 on underlay interfaces; real fabrics run 9216-byte frames and give tenants 9000 or so.
Segment space	`2^24 = 16,777,216 VNIs`	Far beyond the 4,094 usable VLAN IDs, which is why VLANs become locally significant per leaf.

Flood-and-learn versus EVPN

Flood-and-learn VXLAN

- Unknown unicast and ARP flooded to every VTEP in the VNI
- Remote MACs learned from the data plane
- Needs underlay multicast or static peer lists
- No integrated routing, no mobility detection, poor multihoming

EVPN control plane

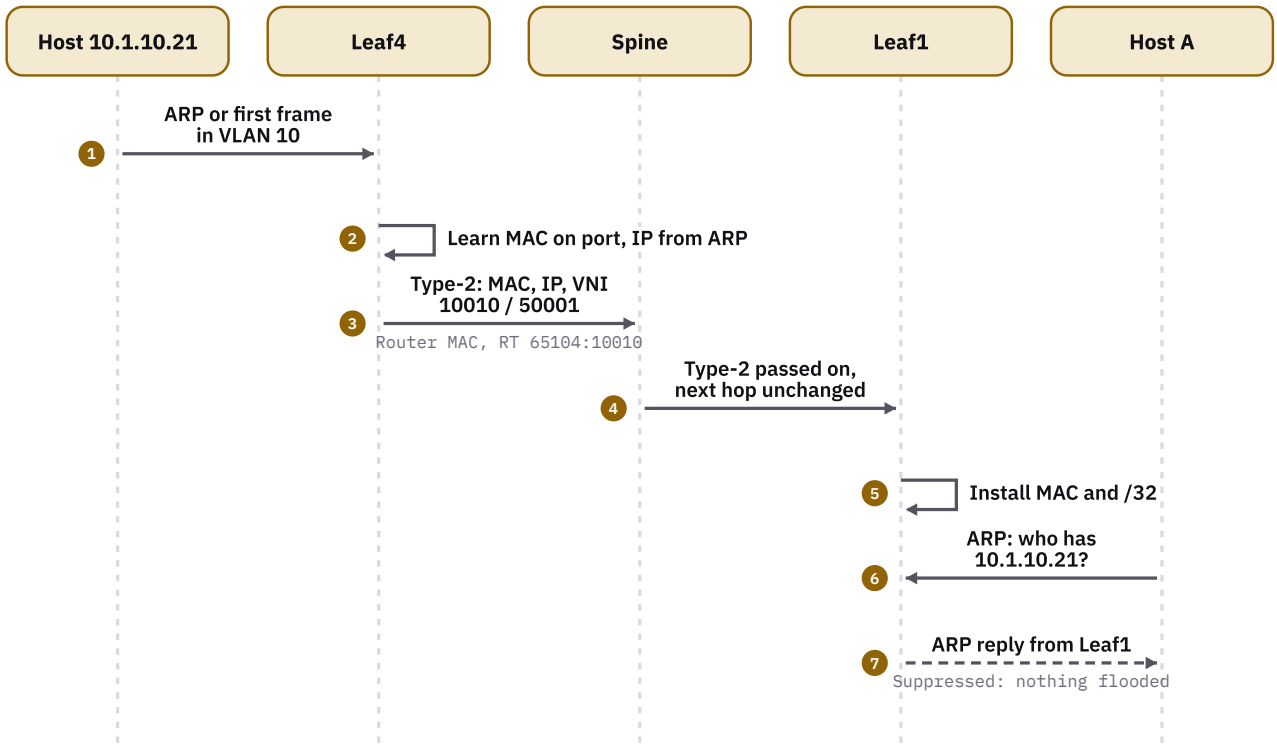
- MAC and IP learned locally, advertised in BGP
- Remote VTEPs discovered from route type 3
- ARP suppression answers locally from known bindings
- Built-in mobility, multihoming and routing (IRB, type 5)

EVPN route types

TYPE	NAME	WHAT IT DOES	KEY FIELDS
1	Ethernet auto-discovery	Per ES: mass withdrawal on link failure. Per EVI: aliasing, so remote leaves load-balance to all multihomed peers	ESI, Ethernet tag, label
2	MAC/IP advertisement	Advertises a host MAC and optionally its IP; drives remote MAC tables, ARP suppression and host routes	MAC, IP, L2 VNI (label 1), L3 VNI (label 2), router MAC, MAC mobility
3	Inclusive multicast Ethernet tag	Announces that a VTEP participates in a VNI; builds the ingress-replication flood list	Originating VTEP, PMSI tunnel attribute
4	Ethernet segment	Discovers peers on the same ES and drives designated-forwarder election	ESI, originator, ES-import RT
5	IP prefix	Advertises IP prefixes independent of any MAC: subnets, external routes, summaries	Prefix, L3 VNI, router MAC, gateway IP

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How a host becomes reachable across the fabric



Leaf4 learns locally and advertises once; every other leaf installs the MAC and host route, and Leaf1 answers Host A's ARP itself, so nothing is flooded.

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Leaf with a Layer 2 VNI and BGP ingress replication

Leaf1 · IOS-style CLI (fabric)

```
vlan 10
 vn-segment 10010
!
interface nve1
 no shutdown
 source-interface loopback1
 host-reachability protocol bgp
 member vni 10010
  suppress-arp
  ingress-replication protocol bgp
!
evpn
 vni 10010 l2
  rd auto
  route-target import auto
  route-target export auto
!
router bgp 65101
 neighbor 10.255.0.1 remote-as 65000
  update-source loopback0
  ebgp-multihop 3
  address-family l2vpn evpn
   send-community extended
```

Auto RTs are derived from the local AS and VNI. With a different AS per leaf (eBGP underlay design), auto-derived RTs differ between leaves, so either configure explicit RTs or use an implementation option that ignores the AS when importing.

Reading a type-2 route

Leaf1 · IOS-style CLI (fabric)

```
Leaf1# show bgp l2vpn evpn route-type 2 10.1.10.21
Route Distinguisher: 10.255.1.4:32777
*>e[2]:[0]:[0]:[48]:[0050.56a1.0021]:[32]:[10.1.10.21]/272
      10.255.2.4            0        65000 65104 i
      Extcommunity: RT:65104:10010 RT:65104:50001 ENCAP:8
                    Router MAC:5254.0012.0004
      Received label 10010 50001
Leaf1# show l2route evpn mac-ip all | include 10.1.10.21
10    0050.56a1.0021 BGP  --   0   10.1.10.21   10.255.2.4
```

Label 10010 is the L2 VNI for bridging, label 50001 the L3 VNI for routing, ENCAP:8 means VXLAN, and 10.255.2.4 is Leaf4's VTEP address on loopback1.

> **INSIGHT · MAC MOBILITY AND DUPLICATE DETECTION**
> When a VM moves from Leaf4 to Leaf7, Leaf7 advertises the same MAC with a **MAC mobility** extended community carrying a higher sequence number, and every leaf, including Leaf4, prefers it. If the same MAC keeps moving back and forth, typically more than five moves in 180 seconds, leaves declare it a **duplicate** and freeze advertisement, which protects the control plane from a looped or misconfigured host. In a lab, a frozen duplicate MAC is a classic hidden fault after a temporary loop.

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SCENARIO · DECIDE · BROADCAST FAILS TO REACH ONE LEAF

Hosts on Leaf1 and Leaf2 in VNI 10010 communicate normally. A new host on Leaf3 in the same VNI can ping hosts whose MAC addresses are already known, but DHCP (broadcast) from Leaf3 never reaches the server behind Leaf1, and silent hosts on Leaf1 are unreachable from Leaf3. Leaf1 does not list Leaf3's VTEP in its flood list for VNI 10010. What is the most likely cause?

(A) Leaf3 is using a different route distinguisher from Leaf1, so the type-2 routes are ignored by Leaf1.

(B) The underlay MTU is too small, so the broadcast frames are fragmented and dropped by the spines.

(C) Leaf3 is not originating a type-3 route for VNI 10010, for example because ingress replication is not set to BGP for that VNI.

(D) ARP suppression is enabled, so all broadcast traffic in the VNI is blocked at every leaf by design.

Circle your choice. Answer and feedback: module answer key.

Key terms

VTEP
VXLAN tunnel endpoint: the device that encapsulates and decapsulates VXLAN, identified by a loopback address.

EVI
EVPN instance: the set of Ethernet segments and bridge domains that make up one EVPN service.

Type-3 route
EVPN inclusive multicast route announcing VTEP membership in a VNI for BUM replication.

ARP suppression
Answering ARP requests locally on a leaf from EVPN-learned IP-to-MAC bindings.

BUM traffic
Broadcast, unknown-unicast and multicast frames that must be flooded within a segment.

VNI
VXLAN network identifier: a 24-bit segment ID that maps to a bridge domain (L2 VNI) or a VRF (L3 VNI).

Type-2 route
EVPN MAC/IP advertisement carrying a host MAC and optional IP with L2 and L3 VNI labels.

Ingress replication
Head-end copying of BUM traffic as unicast to every remote VTEP in the flood list.

MAC mobility
EVPN extended community with a sequence number that lets the newest location of a moved MAC win.

Key takeaways

- VXLAN adds 50 bytes on an IPv4 underlay; run jumbo frames on the underlay.
- The hashed UDP source port gives underlay ECMP its entropy.
- Type 2 carries hosts, type 3 builds flood lists, types 1 and 4 handle multihoming, type 5 carries prefixes.
- MAC mobility sequence numbers track moves; rapid moves freeze a MAC as a duplicate.

KNOWLEDGE CHECK

1. Why does the VXLAN source UDP port vary between flows?

(A) It identifies the tenant VRF, so the remote VTEP can select the right routing table without using the VNI.

(B) It is incremented for every packet, so the receiving VTEP can detect loss and reordering.

(C) It encodes the inner VLAN tag, so the remote VTEP can rebuild the original 802.1Q header.

(D) It is derived from a hash of the inner headers, so underlay ECMP can spread different tenant flows.

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2. A tenant uses 1500-byte packets and the underlay is IPv4 without outer VLAN tags. What is the minimum underlay IP MTU?

   (A) 1524 bytes, adding only the VXLAN and UDP headers because inner Ethernet is stripped.

   (B) 1500 bytes, because VXLAN fragments packets at the VTEP before they enter the underlay.

   (C) 1550 bytes, adding the inner Ethernet header, VXLAN, UDP and the outer IP header.

   (D) 1554 bytes, because the outer Ethernet header is always counted as part of the IP MTU.

3. Which EVPN route type lets a remote leaf load-balance toward both leaves of an all-active multihomed server even if only one of them learned the MAC?

   (A) Type 1 per EVI, which provides aliasing for every leaf attached to the Ethernet segment.

   (B) Type 3, which lists every VTEP in the VNI and therefore every possible path to the server.

   (C) Type 4, which elects a designated forwarder and shares the MAC table with the non-DF leaf.

   (D) Type 5, which advertises the server subnet so remote leaves can use any leaf as next hop.

4. In a type-2 route, what does the second label (for example 50001) represent in a symmetric IRB fabric?

   (A) The L3 VNI of the tenant VRF, used when the remote leaf routes traffic to that host.

   (B) The ESI label, used for split horizon when the host sits on a multihomed segment.

   (C) The MAC mobility sequence number, used to decide which advertisement is most recent.

   (D) The L2 VNI of the gateway subnet, used when the host is reached through the default gateway.

5. After a temporary loop, one host's MAC is no longer advertised by any leaf and the host is unreachable from remote leaves. What should be checked first?

   (A) Whether the host's route distinguisher was reset by the loop and must be configured again.

   (B) Whether the spine route reflectors dropped the type-3 routes because of the loop.

   (C) Whether the leaves have flagged the MAC as a duplicate after rapid moves and frozen its advertisement.

   (D) Whether the L3 VNI was removed automatically because the loop overloaded the control plane.

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MODULE 6 · FABRIC · LESSON 2 · 45 MIN

Routing in the Fabric: IRB, Anycast Gateway and Type-5 Routes

Where routing happens, how gateways stay local everywhere and how prefixes enter the overlay

- Contrast symmetric and asymmetric IRB and justify symmetric IRB for most fabrics.
- Configure a tenant VRF with an L3 VNI and distributed anycast gateway.
- Use type-5 routes for silent hosts, external prefixes and border connectivity.

Bridging across a fabric is only half the story; tenants route between subnets and to the outside world. EVPN supports **integrated routing and bridging (IRB)** on every leaf, so a packet is routed at the first leaf it meets instead of hairpinning to a central gateway. How that routing is split between the ingress and egress leaf, symmetric or asymmetric, decides how much state each leaf must hold.

Symmetric versus asymmetric IRB

Asymmetric IRB

- Ingress leaf routes and bridges into the destination VNI
- Egress leaf only bridges
- Every leaf must have every VLAN, VNI and gateway it might route into
- No L3 VNI needed; simpler packet walk
- State grows with tenant subnets times leaves

Symmetric IRB

- Ingress leaf routes into the tenant L3 VNI; egress leaf routes again to the host
- Both leaves route, hence symmetric
- Leaves need only local subnets plus one L3 VNI per VRF
- Supports type-5 prefixes and external routes naturally
- The default choice for production fabrics in 2026

Symmetric IRB packet walk: 10.1.10.11 on Leaf1 to 10.1.20.25 on Leaf4

01 Ingress route
Leaf1 receives the frame on its anycast gateway MAC and looks up 10.1.20.25 in VRF TENANT-A

→

02 Encapsulate
Host route via Leaf4: VXLAN with VNI 50001, inner destination MAC = Leaf4 router MAC

→

03 Underlay
Spines forward on the outer IP like any other packet

→

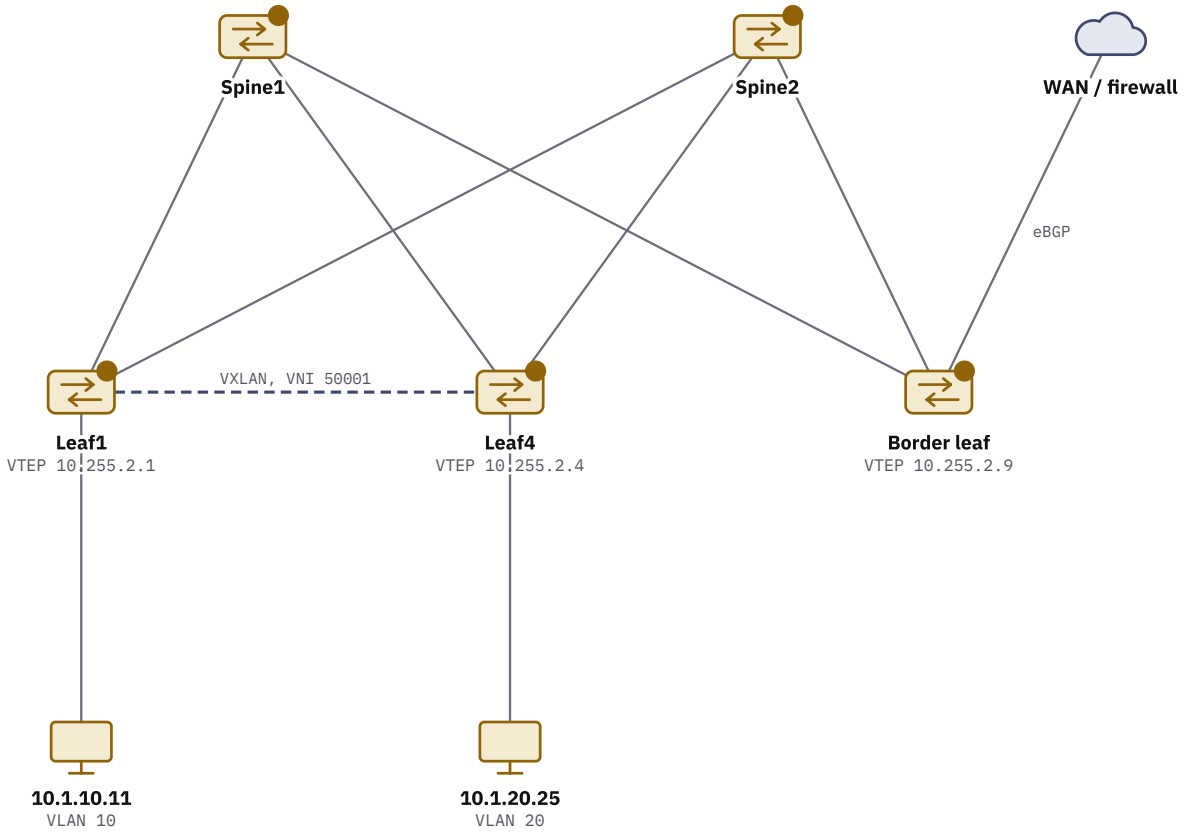
04 Egress route
Leaf4 maps VNI 50001 to TENANT-A, looks up the host and resolves its MAC

→

05 Deliver
Frame bridged into VLAN 20 toward the host

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Symmetric IRB across the fabric



Leaf1 routes 10.1.10.11 → 10.1.20.25 into L3 VNI 50001 toward Leaf4's router MAC, and Leaf4 routes again into VLAN 20. Spines forward only the outer IP; the border leaf injects the default route as type 5.

Tenant VRF, L3 VNI and anycast gateway

Leaf1 · IOS-style CLI (fabric)

```
fabric forwarding anycast-gateway-mac 0000.2222.3333
vlan 3001
 vn-segment 50001
vrf context TENANT-A
 vni 50001
 rd auto
 address-family ipv4 unicast
  route-target both auto evpn
interface Vlan3001
 vrf member TENANT-A
 ip forward
 no shutdown
interface Vlan20
 vrf member TENANT-A
 ip address 10.1.20.1/24
 fabric forwarding mode anycast-gateway
interface nve1
 member vni 50001 associate-vrf
```

VLAN 3001 and its interface exist only to give the L3 VNI a local routing context; they carry no hosts. Every leaf with VLAN 20 uses the identical gateway IP and the identical anycast gateway MAC.

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DEFINITION · DISTRIBUTED ANYCAST GATEWAY

Every leaf that serves a subnet owns the same gateway IP and the same virtual MAC. Hosts always find their gateway one hop away, a moved VM keeps working without re-ARPing, and no first-hop redundancy protocol runs anywhere. The trade-off is that gateway-sourced traffic, such as ARP probes the leaf sends to hosts, needs care: implementations use the anycast address with special handling, or a unique per-leaf address, so replies return to the right leaf.

Type 2 or type 5?

NEED	ROUTE TYPE	WHY
Reach a known host anywhere in the fabric	Type 2 with IP (host route)	Exact /32 or /128 follows the host and supports mobility
Reach a subnet that exists only on some leaves, including silent hosts	Type 5 for the subnet	Remote leaves route to a leaf that owns the subnet, which then resolves the host with ARP
Inject external or summary routes from a border leaf	Type 5	Prefixes have no MAC; carries L3 VNI and router MAC only
Advertise routes learned from a firewall or load balancer inside a VRF	Type 5	Treated like any IP prefix with a VTEP next hop

Type-5 prefixes in the tenant routing table

Leaf1 · IOS-style CLI (fabric)

```
Leaf1# show ip route vrf TENANT-A
0.0.0.0/0, ubest/mbest: 2/0
    *via 10.255.2.9%default, [20/0], bgp-65101, external, tag 65109
       (evpn) segid: 50001 tunnelid: 0xaff0209 encap: VXLAN
    *via 10.255.2.10%default, [20/0], bgp-65101, external, tag 65110
10.1.20.0/24, ubest/mbest: 1/0, attached
    *via 10.1.20.1, Vlan20, [0/0], direct
10.1.30.0/24, ubest/mbest: 2/0
    *via 10.255.2.4%default, [20/0], bgp-65101, external, tag 65104
       (evpn) segid: 50001 encap: VXLAN
    *via 10.255.2.5%default, [20/0], bgp-65101, external, tag 65105
```

The default route arrives as type 5 from two border leaves; 10.1.30.0/24 arrives as type 5 from the two leaves that host it. Both are ECMP across VTEPs in the default (underlay) VRF.

Connecting the fabric to the outside world

BL

Border leaf

A leaf pair that peers with WAN routers or firewalls; it is also an ordinary VTEP for tenant traffic.

VRF

VRF-lite handoff

One subinterface and eBGP session per tenant VRF toward the external router; simple and widely supported.

FW

Firewall insertion

Place a firewall between VRFs (inter-tenant) or between a VRF and the outside; advertise its routes back as type 5.

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LEAK

Route leaking

Import a shared-services VRF's RT into tenant VRFs for DNS or backup, with export filters to avoid leaking everything.

DCI

Data-centre interconnect

Multi-site EVPN with border gateways re-originating routes, so each site's fabric stays a separate failure domain.

SCENARIO · DECIDE · SILENT HOSTS IN A SUBNET THAT IS NOT EVERYWHERE

In a symmetric IRB fabric, subnet 10.1.20.0/24 exists only on Leaf4 and Leaf5. A backup appliance at 10.1.20.25 behind Leaf4 never initiates traffic. Hosts on Leaf1, which has no VLAN 20, cannot reach it; hosts that do send traffic in the same subnet are reachable. What is the best fix?

(A) Configure VLAN 20 and its anycast gateway on Leaf1 as well, so that Leaf1 can broadcast ARP for the host itself.

(B) Advertise 10.1.20.0/24 as a type-5 route from Leaf4 and Leaf5 so Leaf1 forwards toward them and they ARP for the silent host.

(C) Disable ARP suppression in VNI 10020 so that all ARP requests flood throughout the fabric again.

(D) Add a static MAC entry for the appliance on Leaf1 pointing at Leaf4's VTEP address.

Circle your choice. Answer and feedback: module answer key.

Key terms

IRB
Integrated routing and bridging: routing between subnets on the same device that bridges them, here on every leaf.

Asymmetric IRB
Model in which the ingress leaf routes and bridges into the destination VNI, and the egress leaf only bridges.

Router MAC
EVPN extended community carrying the egress VTEP's MAC, used as inner destination MAC for routed traffic.

Type-5 route
EVPN IP prefix route used for subnets, external routes and summaries without a MAC.

Symmetric IRB
Model in which both the ingress and egress leaf route, using a per-VRF L3 VNI between them.

L3 VNI
A VNI that identifies a tenant VRF for routed traffic between VTEPs.

Anycast gateway
Identical gateway IP and MAC on every leaf serving a subnet.

Border leaf
A leaf that connects tenant VRFs to external networks, firewalls or other sites.

Key takeaways

- Symmetric IRB routes at both the ingress and egress leaf through one L3 VNI per VRF.
- Every leaf shares the same anycast gateway IP and MAC, so hosts always find their gateway one hop away.
- Use type-5 routes for subnets with silent hosts and for external or summary prefixes.
- Routed VXLAN frames are addressed to the egress leaf's router MAC.

KNOWLEDGE CHECK

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6. A fabric has 200 tenant subnets but each leaf hosts only about 20 of them. Why is symmetric IRB preferable here?

   (A) Each leaf can drop the underlay routing protocol entirely, because routing is handled by the L3 VNI alone.

   (B) Each leaf needs only its local subnets plus one L3 VNI per VRF, rather than every subnet in the fabric.

   (C) Each leaf can avoid BGP entirely, because symmetric IRB learns remote hosts in the data plane.

   (D) Each leaf can use a unique gateway MAC per subnet, which removes the need for ARP suppression.

7. In symmetric IRB, which inner destination MAC does the ingress leaf use for a routed packet?

   (A) The destination host's MAC, because the ingress leaf already holds the full ARP entry for it.

   (B) The anycast gateway MAC, because every leaf owns it and accepts routed traffic addressed to it.

   (C) The broadcast MAC, because the egress leaf identifies the VRF from the VNI and ignores the MAC.

   (D) The egress leaf's router MAC, learned from the router MAC extended community in EVPN.

8. Why must every leaf use the same anycast gateway MAC?

   (A) So that spines can recognise gateway traffic and route it locally instead of forwarding it on.

   (B) So that the L3 VNI can be derived from the MAC address and not configured on each leaf by hand.

   (C) So that hosts and moved VMs keep a valid gateway ARP entry no matter which leaf they connect to.

   (D) So that type-5 routes can be advertised without a router MAC extended community attached.

9. A border leaf pair learns a default route from two WAN routers. How should it reach the other leaves?

   (A) As a type-5 route in each tenant VRF, with the border leaves' VTEPs as ECMP next hops.

   (B) As a type-2 route with the WAN routers' MAC addresses, so leaves can bridge directly to them.

   (C) As a type-3 route, so the default route is flooded to every VTEP that participates in the VNI.

   (D) As an underlay IGP route, so tenant traffic follows the default route without any encapsulation.

10. With asymmetric IRB, a leaf routes from VLAN 10 to a host in VLAN 30 on a remote leaf. What must exist on the ingress leaf?

    (A) Only the L3 VNI of the tenant VRF, because the egress leaf always performs the final routing step itself.

    (B) Only a type-5 route for VLAN 30's subnet, because the egress leaf will resolve the host.

    (C) Only the router MAC of the remote leaf, because routed traffic is addressed to that MAC.

    (D) VLAN 30 and its VNI, because the ingress leaf bridges into the destination segment after routing.

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MODULE 6 · FABRIC · LESSON 3 · 45 MIN

Multihoming and Underlay Design

Ethernet segments, designated forwarders and the routed underlay that carries everything

- Explain how ESI multihoming uses route types 1 and 4 for aliasing, mass withdrawal and DF election.
- Compare EVPN multihoming with peer-link based multi-chassis link aggregation.
- Design an underlay: protocol, addressing, MTU, ECMP, and the role of spines in the EVPN control plane.
- Describe how AI/ML back-end fabrics differ: rail-optimised, non-blocking and lossless for RoCEv2

Servers, firewalls and legacy switches attach to two leaves for resilience. EVPN multihoming achieves this with standard BGP signalling and no peer link: the two leaves share an **Ethernet segment (ES)** identified by a 10-byte **ESI**. Beneath every overlay sits an underlay whose only job is fast, equal-cost IP reachability between VTEP loopbacks; a good underlay is boring by design.

EVPN multihoming mechanics

ESI

Ethernet segment identifier

Same 10-byte ESI and same LACP system ID on every leaf attached to the bundle, so the server sees one partner.

RT4

ES route and DF election

Leaves discover each other on the ES and elect a designated forwarder per VLAN, which alone floods BUM from the fabric to the ES.

RT1E

Type 1 per ES

Advertised with the ES; withdrawing it tells remote leaves to drop every MAC on that ES at once (mass withdrawal).

RT1V

Type 1 per EVI

Lets remote leaves load-balance to all leaves on the ES even if only one advertised the MAC (aliasing).

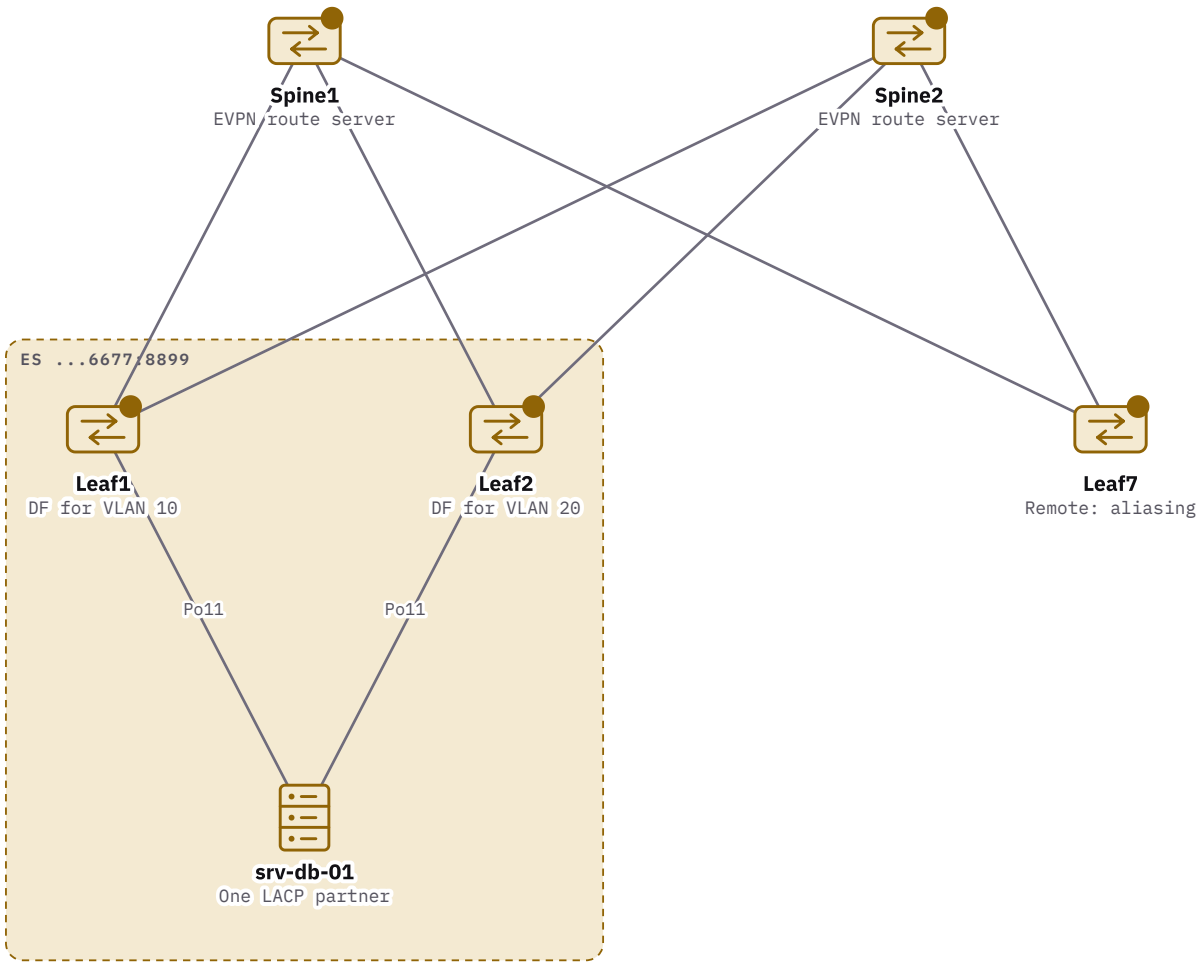
SH

Split horizon

With VXLAN, local bias: a leaf does not forward BUM back to an ES it shares with the sending VTEP.

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An all-active Ethernet segment



Both leaves share the ESI and LACP system ID, so the server sees one partner. Leaf7 load-balances to both through type-1 aliasing, and per-VLAN DF election stops duplicate BUM delivery.

All-active Ethernet segment toward a server

Leaf1 · IOS-style CLI (fabric)

```
interface port-channel11
 description srv-db-01 dual-homed
 switchport mode trunk
 switchport trunk allowed vlan 10,20
 evpn ethernet-segment
  identifier 0011:2233:4455:6677:8899
  redundancy all-active
  designated-forwarder election preference 200
 lacp system-id 02aa.bb00.0011
!
Leaf1# show evpn ethernet-segment 0011:2233:4455:6677:8899
  ESI: 0011.2233.4455.6677.8899  Interface: port-channel11  State: Up
  Peers: 10.255.2.1, 10.255.2.2   Mode: all-active
  DF VLANs: 10   Non-DF VLANs: 20
```

Configure the identical ESI and LACP system ID on Leaf2. The DF output shows per-VLAN election: Leaf1 floods BUM for VLAN 10 toward the server, Leaf2 for VLAN 20.

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Failure events and the route that handles them

EVENT	WHAT THE AFFECTED LEAF DOES	WHAT REMOTE LEAVES DO
Server link to Leaf1 fails	Withdraws type 1 per ES for the ESI	Remove Leaf1 from every MAC on that ES in one step
Leaf1 fails entirely	Nothing; it is gone	Underlay withdraws the VTEP loopback; BGP and BFD remove its routes
New VLAN added on the ES	Re-runs DF election for that VLAN	Add the ES leaves as aliasing next hops for MACs in that VLAN
Leaf1 restored	Waits a hold timer before taking DF role	Resume load-balancing once type 1 routes return

EVPN multihoming versus peer-link MLAG

EVPN ESI multihoming

- Standards-based BGP signalling
- No peer link; leaves can be different platforms
- Scales beyond two leaves per segment
- Per-VLAN DF election and fast mass withdrawal

Peer-link MLAG

- Proprietary state sync between a fixed pair
- Peer link carries orphan and failure traffic
- Shared virtual VTEP address
- Mature and familiar, but pairs are tightly coupled

A leaf-spine fabric as layers

Tenant services	`VRFs, L2 and L3 VNIs, anycast gateway, policy`
Overlay control plane	`BGP EVPN; spines as route reflectors or route servers`
Underlay routing	`eBGP per leaf or IS-IS/OSPF, BFD, ECMP across all spines`
Physical	`Leaf-spine Clos, point-to-point /31 or unnumbered links, jumbo MTU`

Underlay protocol choice

eBGP underlay

- One AS per leaf (or per leaf pair), shared AS on spines
- Policy-rich and scales to very large fabrics (RFC 7938 pattern)
- Needs allowas-in or AS planning for some designs
- EVPN often on separate multihop sessions to the spines

IS-IS or OSPF underlay

- Simple, fast and familiar link-state
- iBGP EVPN with spines as route reflectors
- Good choice up to large single-pod sizes
- Clear separation: IGP for underlay, BGP only for overlay

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Spine as EVPN route server in an eBGP design

Spine1 · IOS-style CLI (fabric)

```
route-map NH-UNCHANGED permit 10
 set ip next-hop unchanged
router bgp 65000
 address-family l2vpn evpn
  retain route-target all
 template peer LEAF-EVPN
  update-source loopback0
  ebgp-multihop 3
  address-family l2vpn evpn
   send-community extended
   route-map NH-UNCHANGED out
 neighbor 10.255.1.1 remote-as 65101
  inherit peer LEAF-EVPN
```

Spines hold no VRFs, so they must retain all route targets, and they must not rewrite the next hop: the next hop has to remain the originating leaf's VTEP address.

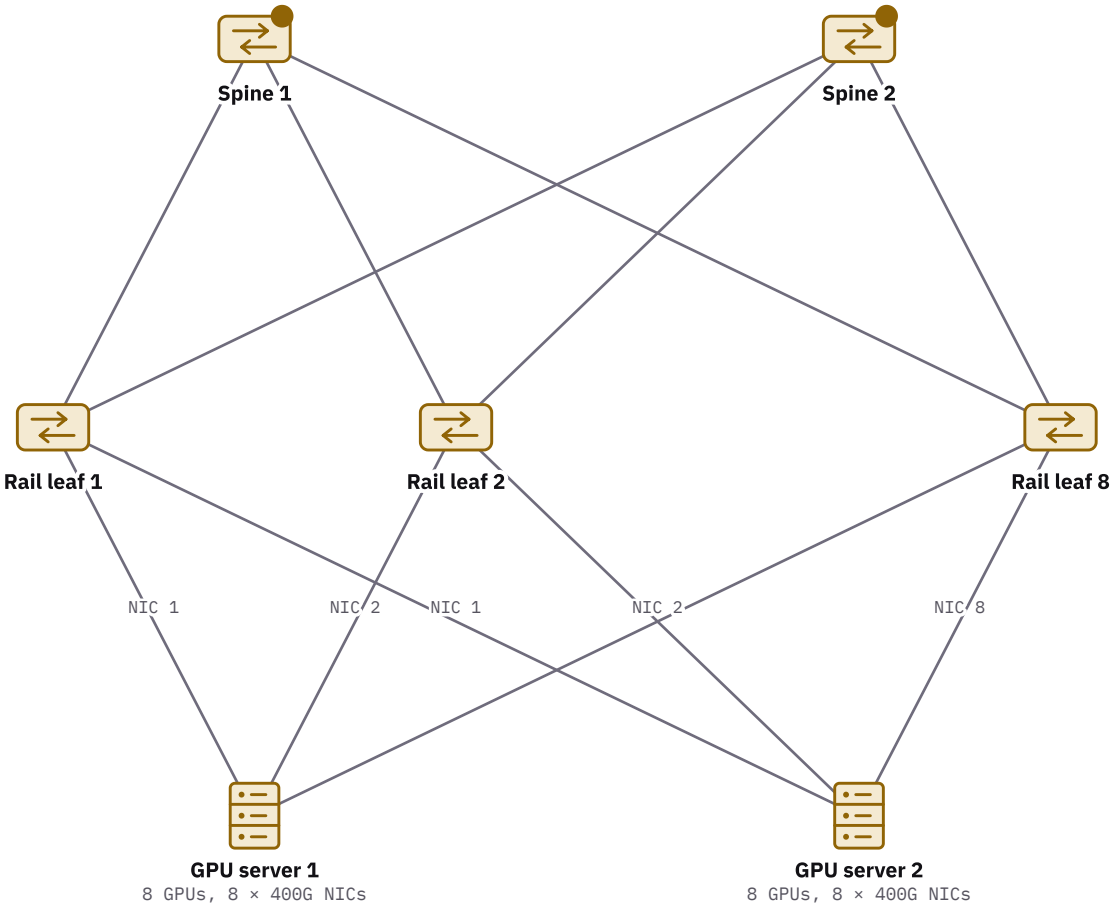
Sizing checks

MEASURE	FORMULA	WHAT IT TELLS YOU
Leaf oversubscription	`server-facing bandwidth ÷ spine-facing bandwidth`	48 × 100G down and 8 × 400G up gives 4,800 ÷ 3,200 = 1.5:1; storage and AI back-end fabrics are built non-blocking (1:1).
Spine count for ECMP	`uplinks per leaf = spines (one link each)`	Losing one of four spines removes 25% of fabric capacity; plan failure headroom.

AI and ML back-end fabrics are the fastest-growing data-centre design and break several habits. GPU servers attach with 400G or 800G NICs, often one per GPU, in a **rail-optimised** layout: NIC 1 of every server connects to rail leaf 1, NIC 2 to rail leaf 2, and so on, so most collective traffic stays one switch hop away. The fabric is non-blocking, carries a few enormous flows rather than millions of small ones, and the slowest flow decides job completion time. GPU-to-GPU memory traffic usually runs as **RoCEv2** (RDMA over UDP port 4791), which degrades sharply under loss, so these fabrics are engineered to be lossless or near-lossless with PFC and ECN (see the Stream module), and flow-hash ECMP is supplemented by adaptive routing or packet spraying because a handful of elephant flows polarise easily. The **Ultra Ethernet Consortium** published its first specification in 2025, defining a new Ethernet transport with multipath packet spraying and its own congestion control; treat product support as emerging and check its current status.

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A rail-optimised AI back-end fabric



NIC n of every GPU server lands on rail leaf n, so traffic between same-rank GPUs crosses one switch and spines carry only cross-rail traffic. Links are 400G or 800G and the fabric is non-blocking.

TEMPLATE

EVPN fabric design sheet

Capture each tenant and fabric-wide decision once, so leaves are configured consistently and audits are quick.

TENANT / VRF	L3 VNI	VLAN : L2 VNI : SUBNET	GATEWAY (ANYCAST)	RT SCHEME	MULTIHOMED ESS	BORDER / EXTERNAL HANDOFF
TENANT-A	*50001*	*10 : 10010 : 10.1.10.0/24; 20 : 10020 : 10.1.20.0/24*	*10.1.x.1, MAC 0000.2222.3333*	*Explicit 65000:VNI*	*ESI …8899 srv-db-01 (Leaf1/2)*	*Border leaves 9/10, eBGP VRF-lite to FW*

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SCENARIO · DECIDE · REMOTE LEAVES NEVER USE THE SECOND LEAF

A server is dual-homed to Leaf1 and Leaf2 in an all-active bundle. Remote leaves send all traffic for the server's MAC only to Leaf1, and during a test both leaves flooded the same broadcast to the server, which logged duplicate packets. What is the most likely cause?

(A) The ESI was mistyped on one leaf, so the leaves see two different segments: no aliasing and no shared DF election.

(B) The spines are rewriting the next hop to themselves, so remote leaves cannot tell Leaf1 and Leaf2 apart.

(C) ARP suppression is enabled on Leaf2, so it never advertises the server's MAC address to the fabric.

(D) The server's NIC teaming is in active-standby mode, so only one leaf can ever learn its MAC address.

Circle your choice. Answer and feedback: module answer key.

Underlay readiness checklist

☐ All fabric links up with the agreed MTU (for example 9216) verified by DF-set pings between loopbacks

☐ Every VTEP loopback reachable from every leaf with the expected number of ECMP paths

☐ BFD running on underlay sessions with agreed timers

☐ Underlay and overlay sessions use separate loopbacks if the design requires it

☐ Spines retain all route targets and leave EVPN next hops unchanged (eBGP) or reflect correctly (iBGP)

☐ Identical ESI and LACP system ID on both leaves of every multihomed bundle

☐ DF election per VLAN verified on each Ethernet segment

☐ Graceful insertion and removal (maintenance mode) tested on one spine and one leaf

Key takeaways

- EVPN multihoming uses one ESI and one LACP system ID across leaves, with no peer link.
- Type 1 per ES gives mass withdrawal, type 1 per EVI gives aliasing, type 4 drives DF election.
- Keep the underlay boring: routed point-to-point links, jumbo MTU, ECMP and BFD; spines must not rewrite EVPN next hops.
- AI back-end fabrics are rail-optimised, non-blocking and lossless for RoCEv2, with smarter load balancing than flow hashing.

KNOWLEDGE CHECK

11. A server link to Leaf1 fails in an all-active ES carrying 3,000 MACs. Which mechanism lets remote leaves react quickly?

    (A) Leaf1 withdraws all 3,000 type-2 routes individually, which remote leaves process in sequence.

    (B) Leaf1 withdraws its type-1 per-ES route, so remote leaves remove Leaf1 for all those MACs at once.

    (C) Leaf1 sends a type-3 withdrawal, which removes it from the flood list and all MAC entries.

    (D) Leaf1 lowers its DF preference, which signals remote leaves to stop sending unicast to it.

12. What does designated-forwarder election control on an Ethernet segment?

    (A) Which leaf advertises the device's MAC address, while the other leaves stay silent about it.

    (B) Which leaf receives all known unicast traffic for the device from remote leaves in the fabric.

    (C) Which leaf forwards BUM traffic from the fabric toward the multihomed device for each VLAN.

    (D) Which leaf runs LACP with the device, while the other leaf keeps its links in standby.

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13. Why must spines acting as eBGP EVPN route servers keep the next hop unchanged?

(A) The next hop must stay unchanged so that the spines can perform symmetric IRB on behalf of all leaves.

(B) The next hop must stay unchanged so the route distinguisher can be derived from it by each leaf.

(C) The next hop must stay unchanged because eBGP never modifies next hops for the EVPN address family.

(D) The next hop must stay the originating leaf's VTEP, because spines are not VTEPs for tenant traffic.

14. A leaf has 48 × 100G server ports and 4 × 400G uplinks. What is its oversubscription ratio?

(A) 3:1, because 4,800 Gbit/s of server capacity shares 1,600 Gbit/s of uplink capacity.

(B) 2:1, because only half of the server ports are expected to transmit at the same time.

(C) 1.2:1, because each 400G uplink can carry four 100G flows with some headroom left.

(D) 12:1, because there are twelve server ports for each individual uplink port on the leaf.

15. Which statement about EVPN multihoming compared with peer-link MLAG is accurate?

(A) EVPN multihoming requires both leaves to share a single virtual VTEP address and a peer link.

(B) EVPN multihoming signals the segment in BGP and needs no peer link between the leaves.

(C) EVPN multihoming limits each Ethernet segment to exactly two leaves of the same platform.

(D) EVPN multihoming forwards BUM traffic from every leaf, relying on the server to drop duplicates.

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MODULE 6 · FABRIC · LESSON 4 · 45 MIN

SD-WAN Architecture Deep Dive

Overlay routing, policy, application-aware steering and integrated security across many transports

- Describe the planes and components of an SD-WAN and how an edge joins the overlay.
- Design control, data and application-aware routing policies with measurable SLA classes.
- Integrate direct internet access with local or cloud-delivered security.

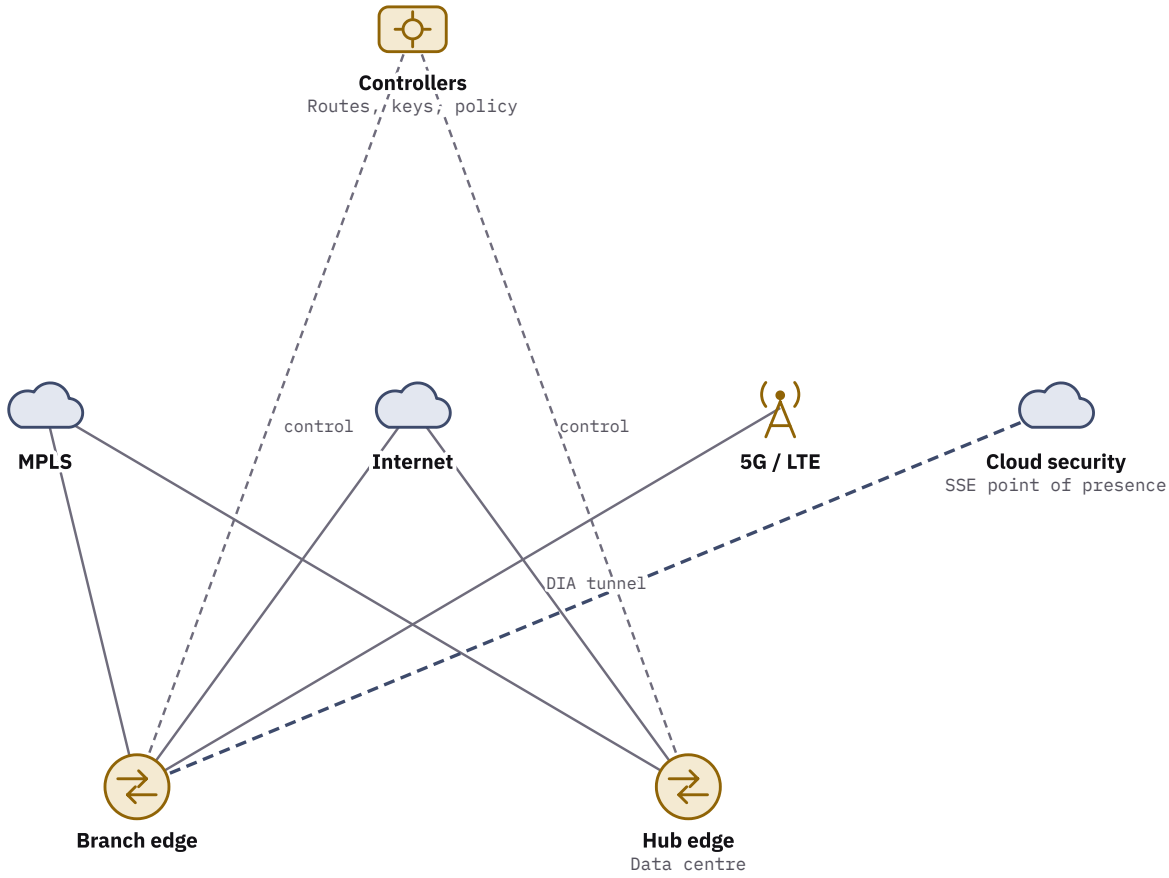
SD-WAN is the WAN's version of the overlay idea: a controller-driven overlay of encrypted tunnels across any mix of MPLS, broadband, 5G and satellite, with routing and forwarding decisions driven by central policy and live path measurements. Products differ in vocabulary, but the architecture is consistent across them. An expert describes it in neutral terms, configures intent rather than per-device detail, and keeps the transports, the overlay and the policy clearly separate when troubleshooting.

SD-WAN planes

Orchestration and management	`Central GUI and API` `Templates, software images, monitoring and analytics`
Control	`Controllers authenticate edges and distribute overlay routes, transport locators, keys and policy` `Optional route reflection hierarchy for scale`
Data	`Edges build IPsec tunnels between transport locators` `Probes measure loss, latency and jitter per tunnel`
Transport	`MPLS, internet, 5G/LTE, satellite` `Each identified by a colour or transport label`

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One SD-WAN site, three transports



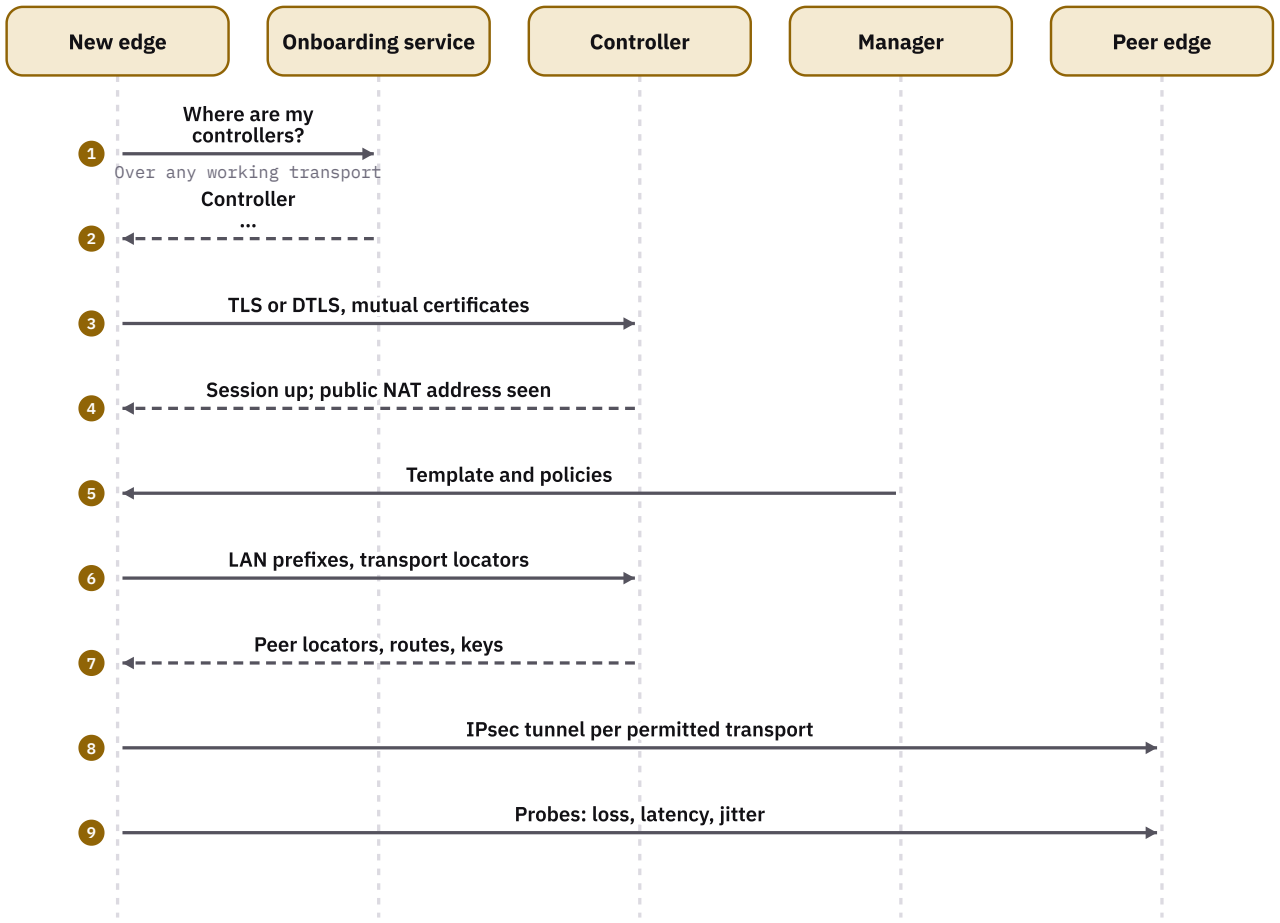
The edge builds IPsec tunnels per transport to permitted peers, while controllers distribute routes, locators and keys over separate control sessions; internet-bound traffic can break out to a nearby cloud security point of presence.

Core components and their roles

COMPONENT	ROLE	EXPERT DETAIL
Onboarding service	Zero-touch provisioning: tells a new edge where its controllers are	Device identity is a hardware certificate, not a password
Controller	Overlay routing and policy distribution	Edges learn remote prefixes with next hop = transport locator(s) of the owning edge
Manager / orchestrator	Configuration templates, inventory and monitoring	Treat its API as the source of change for automation
Edge	Forwards traffic, enforces policy, runs local routing to the LAN	Keeps forwarding with last-known state if controllers are briefly unreachable
Transport locator	Endpoint of tunnels on one transport: public IP, private IP, colour, encapsulation	Behind NAT, the public address is discovered via the controllers

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An edge joins the overlay



Identity comes first, configuration second and tunnels last; the controllers also tell the edge its post-NAT public address, which other edges need to reach it.

Policy types

CTRL

Control (topology) policy

Shapes which routes and transport locators each site receives: full mesh, hub-and-spoke, regional meshes, or service insertion.

DATA

Data policy

Matches flows and sets actions: drop, NAT to local internet, steer to a service, set DSCP.

AAR

Application-aware routing

Chooses tunnels per application based on live loss, latency and jitter versus an SLA class.

QOS

QoS

Queuing and shaping per transport, often with per-tunnel shaping toward small sites.

SEC

Security

Zone firewall, IPS, URL filtering, or redirection to cloud security for direct internet access.

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Live tunnel measurements against SLA classes

Edge E12 · SD-WAN edge CLI (generic)

```
E12# show overlay tunnel-sla remote-site HUB1
Local-TLOC   Remote-TLOC  Loss%  Latency  Jitter  Meets SLA
mpls         mpls         0.0    18 ms    2 ms    VOICE BUSINESS DEFAULT
biz-inet     biz-inet     1.6    39 ms    17 ms   BUSINESS DEFAULT
lte          biz-inet     0.4    71 ms    9 ms    BUSINESS DEFAULT
E12# show overlay app-route policy VOICE
SLA VOICE: loss <= 1%, latency <= 150 ms, jitter <= 30 ms
Preferred colour: mpls  Backup: lte  Action if none meet: best-tunnel
```

biz-inet fails VOICE on loss, so voice uses mpls and would fail over to lte, not to the lossy internet path. Measurements are averaged over a configurable window, which sets how fast steering reacts.

Steering responsiveness

MEASURE	FORMULA	WHAT IT TELLS YOU
Detection window	`probe interval × probes per bucket × buckets averaged`	1 s probes, 60 per bucket and 6 buckets give a 6-minute rolling average: stable, but slow to react to brownouts.
Tunnel liveness	`hello interval × multiplier`	1 s × 7 = 7 s to declare a tunnel down; shorter values react faster but risk flaps on lossy links.

Internet-bound traffic from branches

Backhaul to hub

- Central security stack and logging
- Adds latency and consumes hub bandwidth
- Poor experience for SaaS and video

Direct internet with on-box security

- Local firewall, IPS and URL filtering on the edge
- Low latency; security updated from the manager
- Edge sizing must include security throughput

Direct internet to cloud security

- Tunnels from edge to a cloud security service (SSE/SASE)
- Consistent policy for branch and remote users
- Choose the nearest point of presence and monitor it like a transport

WATCH OUT · APPLICATION IDENTIFICATION IS NOT INSTANT

Deep packet inspection may need several packets before it recognises an application, by which time the flow has already been placed on a path. Unless the platform uses first-packet techniques, such as learning from DNS responses or cached application-to-address mappings, the first flow of an application may stick to the default path. When testing steering policies, look at new flows after the cache is warm, not at the first connection.

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SCENARIO · DECIDE · SAAS IS SLOW AT EVERY BRANCH

A retailer's 300 branches send all internet traffic, including collaboration and SaaS, through two hub data centres for inspection. Users complain of slow SaaS and choppy video calls; hub internet links run at 85% at peak. Security requires consistent web filtering and threat inspection. What is the best design change?

(A) Double the hub internet bandwidth and keep backhauling all traffic so that security stays centralised.

(B) Send all internet traffic directly from branches without inspection, because SaaS providers already secure their applications.

(C) Prioritise SaaS traffic in the hub QoS policy, so it is protected at the congested internet links.

(D) Use direct internet access for trusted SaaS and collaboration, steered by application-aware policy, with inspection by on-box security or a nearby cloud security service.

Circle your choice. Answer and feedback: module answer key.

Key terms

Transport locator
An SD-WAN tunnel endpoint on one transport, defined by addresses, colour and encapsulation.

Application-aware routing
Per-application path selection based on measured loss, latency and jitter against SLA classes.

Zero-touch provisioning
Automatic onboarding of a new device using a hardware identity and a cloud discovery service.

SSE / SASE
Cloud-delivered security edge services, combined with SD-WAN in a secure access service edge architecture.

Colour
Label identifying a transport type, such as mpls, biz-inet or lte, used in policy.

SLA class
A set of loss, latency and jitter thresholds that a path must meet for an application.

Direct internet access
Sending internet-bound traffic out of a local branch transport rather than backhauling it.

Control policy
Policy that manipulates which routes and transport locators sites receive, shaping the overlay topology.

Key takeaways

- Keep transport, overlay and policy separate when you design and when you troubleshoot.
- Edges authenticate with device certificates; controllers distribute routes, transport locators, keys and policy.
- Control policy shapes the topology; data policy and application-aware routing act on flows against SLA classes.
- Direct internet access still needs inspection, on the box or in a nearby cloud security service.

KNOWLEDGE CHECK

16. An edge behind a NAT router must build tunnels to other edges. How does it learn its public transport address?

(A) From a static entry in its configuration template, which administrators must update for each site.

(B) From the IPsec peer, which sends its translated address after the first tunnel has been built.

(C) Through its sessions with the controllers, which observe the translated address and distribute it.

(D) From the DHCP server on the broadband link, which hands out the public address as an option.

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17. Voice must use MPLS while it meets the SLA, fail over to LTE, and never use internet broadband. Which policy type implements this?

(A) Control policy that removes internet broadband locators from every site's routing table.

(B) Application-aware routing with a VOICE SLA class, preferred colour MPLS and LTE as backup.

(C) Data policy that sets DSCP EF on voice so the underlay providers prioritise it.

(D) QoS policy that places voice in the priority queue on every transport interface.

18. Tunnel SLA averages are calculated over six minutes. Users report brief brownouts that steering never reacts to. What is the most direct adjustment?

(A) Lengthen the hello multiplier so tunnels stay up during brownouts and keep forwarding traffic.

(B) Raise the SLA thresholds so that the degraded path continues to meet the class during brownouts.

(C) Move the affected application into the default class so it is excluded from any SLA steering.

(D) Shorten the measurement window so averages react faster, accepting more frequent path changes.

19. What is the main architectural difference between a control policy and a data policy in SD-WAN?

(A) Control policy shapes the routes and locators sites learn; data policy acts on traffic flows themselves.

(B) Control policy runs on the edges only; data policy runs on the controllers and is never sent to the edges.

(C) Control policy applies to internet traffic only; data policy applies to traffic between branch sites.

(D) Control policy encrypts traffic; data policy chooses which encryption algorithm each tunnel uses.

20. A new edge cannot join the overlay and logs a certificate validation error to the controllers. Everything else, including internet reachability, works. What is most likely wrong?

(A) The edge's LAN routing protocol has not yet formed adjacencies with the local branch switches and routers.

(B) The edge's application-aware routing policy has no SLA class defined for the VOICE application.

(C) The edge's device certificate is not in the authorised device list or its trust chain is not accepted.

(D) The edge's LTE modem has no data plan, so it cannot provide a backup transport for the tunnels.

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MODULE 6 · FABRIC · LESSON 5 · 45 MIN

Migrating Legacy WANs and Data-Centre Networks

Coexistence routing, loop prevention, gateway moves and cutover discipline

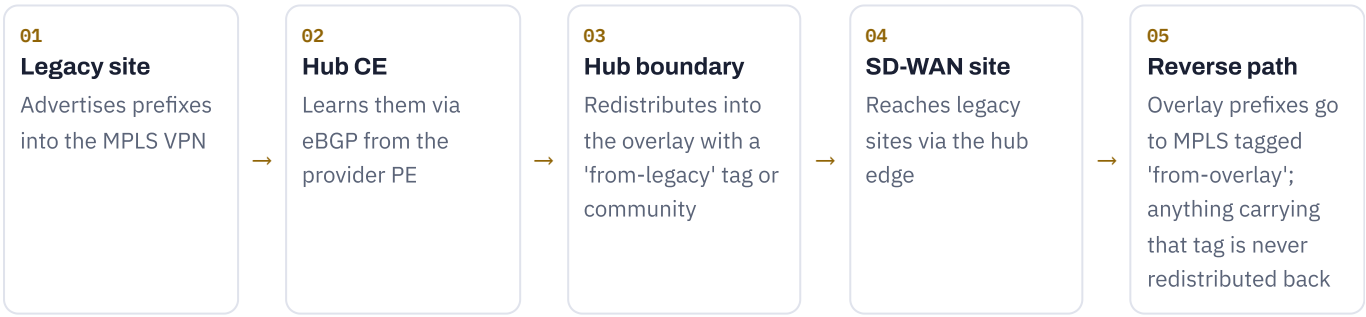
- Plan a phased migration from MPLS WAN to SD-WAN with safe coexistence routing.
- Prevent redistribution loops and suboptimal routing with tags, communities and preference.
- Migrate a Layer 2 data centre to an EVPN fabric, including the default-gateway move.

Most expert work is not greenfield. Migrations succeed when the old and new networks coexist safely for weeks or months, every site can reach every other site during the transition, and each step can be rolled back. The technical heart of both WAN and data-centre migrations is the same: a controlled **boundary** where routes and, sometimes, Layer 2 domains cross from old to new.

Phased WAN migration

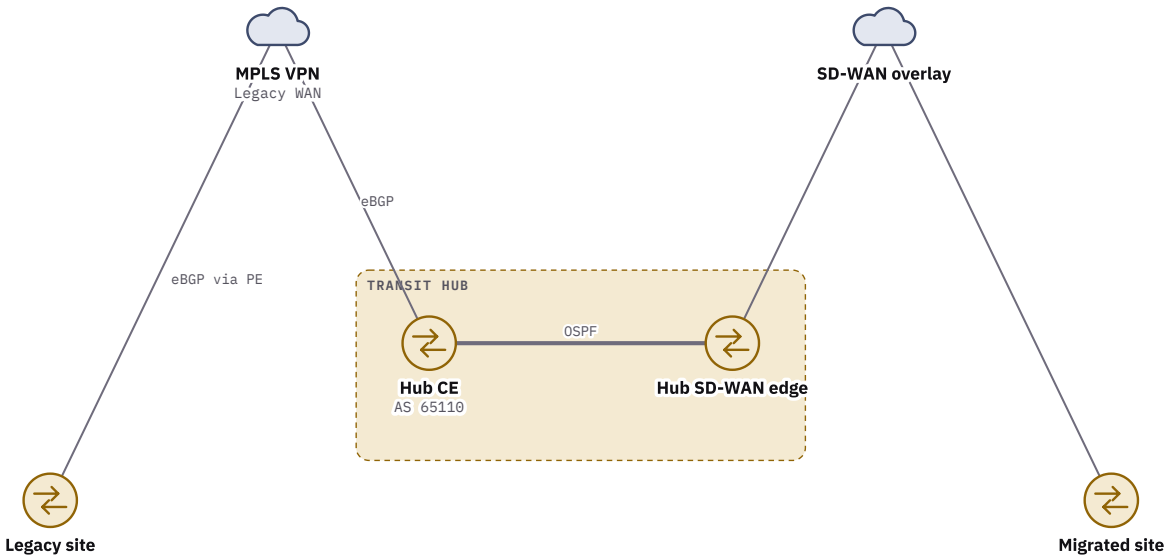
1. **Discover** Inventory sites, transports, applications and current routing; baseline performance and peak utilisation.
2. **Design coexistence** Choose transit sites (usually hubs or data centres) that run both the legacy CE and an SD-WAN edge and will exchange routes.
3. **Pilot** Migrate a few representative sites; test failover, application steering and support processes.
4. **Wave migration** Move sites in waves by region; each wave has a cutover window, rollback plan and acceptance tests.
5. **Optimise and retire** Introduce direct internet access and new policies, then decommission legacy circuits once traffic is gone.

Coexistence routing through a transit hub



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Coexistence through a transit hub



Routes from MPLS are tagged 65110 as they enter OSPF and refused on the way back; overlay routes carry community 65000:900 into MPLS and are refused if they return. A second hub must apply exactly the same marks.

Loop-safe mutual redistribution at a hub

HUB-CE1 · IOS-style CLI

```
route-map OVERLAY-TO-MPLS deny 10
 match tag 65110
route-map OVERLAY-TO-MPLS permit 20
 set community 65000:900 additive
!
ip community-list standard FROM-OVERLAY permit 65000:900
route-map MPLS-TO-OVERLAY deny 10
 match community FROM-OVERLAY
route-map MPLS-TO-OVERLAY permit 20
 set tag 65110
!
router bgp 65110
 address-family ipv4
  redistribute ospf 1 route-map OVERLAY-TO-MPLS
router ospf 1
 redistribute bgp 65110 route-map MPLS-TO-OVERLAY
```

In this design the SD-WAN edge peers with the hub CE over OSPF. Routes that came from MPLS are tagged 65110 entering OSPF and are refused on the way back; routes from the overlay carry community 65000:900 into MPLS and are refused if they ever return. This relies on the provider passing communities across the VPN; confirm it, or use another marker such as an AS-path filter.

WATCH OUT · TWO HUBS DOUBLE THE RISK

With two transit hubs, a route can leave the overlay at hub A, cross the MPLS VPN and re-enter the overlay at hub B. Tags or communities must be applied consistently at both hubs, and the preference between the two worlds must be explicit: for example, prefer the overlay path for migrated sites by setting local preference or administrative distance, so a dual-connected hub never chooses a hairpin through the legacy cloud.

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Data-centre migration strategies

Gateway stays on legacy first

- Extend VLANs from legacy core into the fabric with a Layer 2 handoff on border leaves
- Move hosts rack by rack while routing stays on the old core
- Move the gateway last, per VLAN, in one maintenance window

Gateway moves to fabric early

- Anycast gateway on the fabric as soon as the VLAN is extended
- Legacy FHRP shut down for that VLAN at the same moment
- Hosts still on legacy hairpin through the Layer 2 link to the fabric gateway

Data-centre migration risk register

RISK	EFFECT	MITIGATION
Spanning-tree event on the Layer 2 extension	Outage across both worlds	Single logical link (bundle) to the fabric, BPDU guard and storm control at the border
Two active gateways for one VLAN	Asymmetric routing, firewall drops	Move gateway per VLAN in one change; disable legacy FHRP immediately
Stale ARP entries for the gateway MAC	Hosts lose off-subnet reachability	Reuse the legacy virtual MAC as anycast gateway MAC, or send gratuitous ARP
MTU mismatch on the extension	Large frames dropped silently	Verify jumbo end to end before moving storage or backup traffic
Missing firewall or load-balancer policies	Applications fail after the gateway move	Map every VLAN to its security policy before the cutover

CASE STUDY · DATA-CENTRE EVPN MIGRATION AT A PAYMENTS PROCESSOR

A payments processor runs a ten-year-old three-tier data centre: two core switches with FHRP gateways for 140 VLANs, spanning tree everywhere and firewall clusters attached to the core. A new 24-leaf EVPN fabric with symmetric IRB, border leaves and an all-active ES toward the legacy core has been built. Regulators require no unplanned downtime for card-authorisation services and evidence that every change was tested. Hosts include bare-metal databases that cannot be re-addressed.

1. How do you connect the old and new worlds safely?
2. In what order do you move hosts and gateways, and why?
3. How do you avoid ARP problems when the gateway moves?
4. What evidence would you collect for the regulator?

Model answer: module answer key.

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SCENARIO · DECIDE · HOSTS LOSE OFF-SUBNET REACHABILITY AFTER A GATEWAY MOVE

VLAN 140 is moved from the legacy FHRP gateway to the fabric's anycast gateway with the same IP address. Hosts in the VLAN can reach each other, but most cannot reach other subnets for several minutes; a few recovered immediately. The fabric uses its default anycast MAC, different from the legacy virtual MAC. What would have prevented this?

(A) Configuring the fabric to use the legacy virtual MAC for the VLAN's gateway, or sending gratuitous ARP at cutover.

(B) Lowering the IGP timers on the border leaves so the new gateway is advertised to the rest of the fabric faster.

(C) Moving the gateway before migrating any hosts, so that the fabric holds every ARP entry from the start.

(D) Enabling ARP suppression on the VLAN so the fabric answers host ARP requests from its own tables.

Circle your choice. Answer and feedback: module answer key.

Cutover readiness checklist

☐ Change scope limited to one wave of VLANs or sites with a written rollback

☐ Pre-checks captured: routes, MACs, ARP, sessions, application tests

☐ Boundary protections in place: tags or communities, BPDU guard, storm control

☐ Gateway MAC strategy agreed (reuse legacy MAC or gratuitous ARP)

☐ Security policies mapped and tested for every moved VLAN or site

☐ Monitoring dashboards and alert suppression windows prepared

☐ Post-checks identical to pre-checks and compared automatically

☐ Stakeholders informed and an explicit go or no-go point before each irreversible step

REFLECT

Pick a migration you have been part of or have watched. Where was the boundary between old and new, what protected it against loops and duplicate gateways, and what single extra check would most have reduced risk?

Key takeaways

- Migrations succeed through a controlled boundary that can coexist for months and roll back at every step.
- Mark routes by origin at every boundary and make the preference between old and new explicit.
- In data-centre moves, migrate hosts first and gateways last, one VLAN per change.
- Reuse the legacy virtual MAC or send gratuitous ARP when the gateway moves.

KNOWLEDGE CHECK

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21. During WAN migration, a prefix from an SD-WAN site appears in the overlay twice: once natively and once re-entering through the second hub from MPLS. What is the root cause?

(A) The SD-WAN controller cannot support more than one transit hub, so the second hub re-advertises everything.

(B) Routes redistributed into MPLS at one hub are not tagged or filtered when re-entering at the other hub.

(C) The MPLS provider changed the route distinguisher, which makes the prefix look like a new route.

(D) The SD-WAN site advertised the prefix with two transport locators, which doubles every route.

22. Why do many data-centre migrations move hosts first and gateways last?

(A) Gateways cannot be configured on an EVPN fabric until every host in the VLAN has been migrated.

(B) Hosts must learn the fabric's router MAC before the gateway moves, or symmetric IRB will fail.

(C) Gateways moved first would require re-addressing every host, which the fabric cannot do automatically.

(D) Routing and security stay unchanged while hosts move, so each move is a low-risk Layer 2 change.

23. Which protection is most important on the Layer 2 link between a legacy core and a new fabric?

(A) A single logical bundle with BPDU guard and storm control, extending only the VLANs being migrated.

(B) Multiple independent links running spanning tree, so the legacy core can block the redundant ones.

(C) A routed link with no VLANs at all, because EVPN fabrics cannot extend Layer 2 to legacy switches.

(D) An extension of every VLAN at once, so that later waves do not need any further changes.

24. In mutual redistribution at a hub, what does tagging routes from MPLS before they enter the overlay achieve?

(A) It marks their origin so they can be refused if they try to return to MPLS through any boundary.

(B) It sets their administrative distance so that the overlay always prefers them to native routes.

(C) It encrypts their attributes so that the MPLS provider cannot read customer routing information.

(D) It summarises them into a single route so the overlay receives fewer prefixes from the legacy WAN.

25. What is the best evidence that a migration wave succeeded?

(A) A screenshot of the fabric dashboard showing all devices green at the end of the maintenance window.

(B) A statement from the engineer that no alarms were raised during the change and users did not complain.

(C) Automated post-checks that match the pre-check baseline, plus application tests, stored with the change.

(D) A count of BGP sessions that are up, compared with the number of sessions expected in the original design.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · FABRIC

Answer key

Knowledge checks

1. D The hash-derived source port gives the underlay entropy for load balancing. The VNI identifies the segment; VXLAN carries no sequence numbers or VLAN tag in the port.

2. C 1500 + 14 + 8 + 8 + 20 = 1550. The inner Ethernet header is carried, not stripped; VTEPs normally do not fragment; and the outer Ethernet header is not part of the IP MTU.

3. A Aliasing comes from type 1 per EVI: the remote leaf associates the MAC's ESI with every leaf advertising that ES. Type 4 handles DF election, type 3 flood lists and type 5 prefixes.

4. A Label 1 is the L2 VNI and label 2 is the L3 VNI for routed traffic. Mobility uses an extended community and split horizon in VXLAN uses local bias rather than a label.

5. C Rapid MAC moves trigger duplicate detection, which stops advertisement until the condition clears or is reset. Hosts have no RDs, and type-3 or L3 VNI loss would affect many hosts.

6. B With asymmetric IRB every leaf must hold every VNI it might bridge into. Symmetric IRB still needs the underlay and BGP EVPN, and uses one shared anycast MAC.

7. D Routed VXLAN frames are addressed to the egress VTEP's router MAC. The host MAC is used only after the egress leaf routes; the anycast MAC is host-facing.

8. C A consistent gateway MAC is what makes mobility seamless. Spines are pure underlay routers, VNIs are configured, and type-5 routes still carry the router MAC.

9. A External prefixes enter the overlay as type 5. Type 2 is for hosts, type 3 for flood lists, and leaking tenant routes into the underlay breaks isolation.

10. D In asymmetric IRB the ingress leaf routes then bridges into the destination VNI, so it needs that VNI locally. The other answers describe symmetric IRB.

11. B Mass withdrawal through type 1 per ES makes convergence independent of the MAC count. Withdrawing type 2 routes one by one is slow; type 3 and DF affect BUM only.

12. C The DF prevents duplicate BUM delivery. Both leaves may advertise MACs, both receive unicast through aliasing, and both run LACP in all-active mode.

13. D eBGP would normally set next hop self, which would make spines the VXLAN tunnel endpoint. Spines do not route tenant traffic, RDs are not derived this way, and eBGP does change next hops by default.

14. A 48 × 100 = 4,800 and 4 × 400 = 1,600, giving 3:1. Ratios compare bandwidth, not port counts, and do not assume traffic patterns.

15. B EVPN multihoming is standards-based and peer-link free, supports more than two leaves and prevents duplicates with DF election. The shared virtual VTEP and peer link describe MLAG.

16. C Controllers see the post-NAT address and advertise it in the transport locator, enabling NAT traversal. Static entries do not scale and the other mechanisms do not exist for this purpose.

17. B Application-aware routing selects paths per application against SLA thresholds. Control policy would remove broadband for all traffic; DSCP and queuing do not choose paths.

18. D Long averaging windows smooth out short events. A higher multiplier or looser thresholds make steering even less sensitive, and removing the class abandons protection.

19. A Control policy changes the topology by filtering or modifying route and locator advertisements; data policy matches packets and sets forwarding actions. Both are distributed from the control plane.

20. C Onboarding relies on mutual certificate authentication; a missing allow-list entry or chain problem stops it. LAN routing, SLA classes and backup transports come later.

21. B Consistent tagging at every boundary is what stops re-entry. Controllers support multiple hubs, RDs are invisible to the customer, and multiple locators are one route with several next hops.

22. D Keeping the gateway in place isolates host moves from routing and firewall changes. Fabrics can host gateways anytime; hosts never need the router MAC; gateway moves do not require re-addressing.

23. A One bundle avoids spanning-tree interaction, and limited VLANs limit the blast radius. Fabrics can extend VLANs to legacy switches, but extending everything at once maximises risk.

24. A Tags and communities record origin for loop prevention. They do not change distance by themselves, encrypt anything or summarise routes.

25. C Comparing post-checks with a baseline and testing applications proves function. Dashboards, absence of complaints and session counts are useful but incomplete.

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Scenarios & case studies

Scenario · Broadcast fails to reach one leaf
C is best. Correct. Without a type-3 route from Leaf3, other leaves do not add it to their flood lists, so BUM traffic never crosses between them while known unicast still works.

Scenario · Silent hosts in a subnet that is not everywhere
B is best. Correct. Type-2 host routes exist only for hosts that have been learned. A type-5 subnet route gives remote leaves a path, and the egress leaf glean-resolves the silent host.

Scenario · Remote leaves never use the second leaf
A is best. Correct. Different ESIs mean no common segment: each leaf believes it is the sole DF and remote leaves cannot alias. LACP may still bundle if the system ID matches, hiding the error.

Scenario · SaaS is slow at every branch
D is best. Correct. Breaking out locally removes the hub detour and congestion, while keeping inspection meets the security requirement.

Case study · Data-centre EVPN migration at a payments processor
Connect the legacy core to a pair of border leaves with a **single LACP bundle** forming one all-active Ethernet segment, so spanning tree sees one logical link; apply BPDU guard, storm control and an explicit VLAN allow-list, extending only VLANs in the current wave. Move **hosts first, gateways last**: migrate racks VLAN by VLAN while the legacy FHRP remains the gateway, so routing and firewall paths are unchanged during host moves. When all or most hosts of a VLAN are on the fabric, move its gateway in one change: shut the legacy FHRP and enable the anycast gateway. Configure the anycast gateway to answer for the **same virtual MAC** that the legacy FHRP used (or generate gratuitous ARP), so hosts and databases keep valid ARP entries and no re-addressing is needed. Insert firewalls on the border leaves as type-5 next hops before the first gateway move so security policy is identical. For evidence, run automated pre- and post-checks for each wave (MAC and ARP counts, gateway reachability, transaction test scripts, firewall session counts), store them with the change record, and rehearse the cutover and rollback in a lab with the same configuration templates.

Scenario · Hosts lose off-subnet reachability after a gateway move
A is best. Correct. Hosts kept the old gateway MAC in their ARP caches; with the same MAC, or an immediate gratuitous ARP update, they would have switched seamlessly.

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07

MODULE 7 · STREAM

Stream — Multicast and QoS

Deliver one-to-many traffic reliably and give every class of traffic the treatment the business needs.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Build and explain PIM-SM operation from IGMP/MLD membership to shortest-path trees.
- Design RP placement and discovery with static, BSR and anycast RP, and choose SSM where it fits.
- Troubleshoot multicast quickly with RPF-centred, hop-by-hop methods.
- Apply classification, marking, queuing, policing, shaping and congestion avoidance correctly.
- Design and verify an end-to-end QoS policy for voice, video and business-critical applications.

LESSONS

1. Multicast Foundations: Group Management and PIM Sparse Mode · 45 min
2. RP Design: Static, BSR, Anycast RP and SSM · 45 min
3. Multicast Troubleshooting: RPF and Beyond · 40 min
4. The QoS Toolkit: Classify, Mark, Queue, Police, Shape, Avoid · 45 min
5. Designing End-to-End QoS for Voice, Video and Business-Critical Traffic · 45 min

MODULE 7 · STREAM · LESSON 1 · 45 MIN

Multicast Foundations: Group Management and PIM Sparse Mode

How receivers signal interest, and how PIM-SM builds shared and shortest-path trees

- Explain IGMPv2/v3 and MLDv1/v2 operation, querier election and the role of snooping.
- Walk through PIM-SM joins, source registration and switchover to the shortest-path tree.
- Read multicast routing table entries and flags with confidence.

The **Stream** segment of the Mastery Ring covers traffic that is not simple unicast best effort: one-to-many multicast and differentiated treatment with QoS. Multicast powers IPTV, market-data feeds, video surveillance, software distribution and many industrial systems. It fails in ways unicast engineers find strange, because it is **receiver-driven** and every forwarding decision depends on a reverse-path check. This lesson builds the model; the next two cover rendezvous-point design and troubleshooting.

Multicast responsibilities by layer

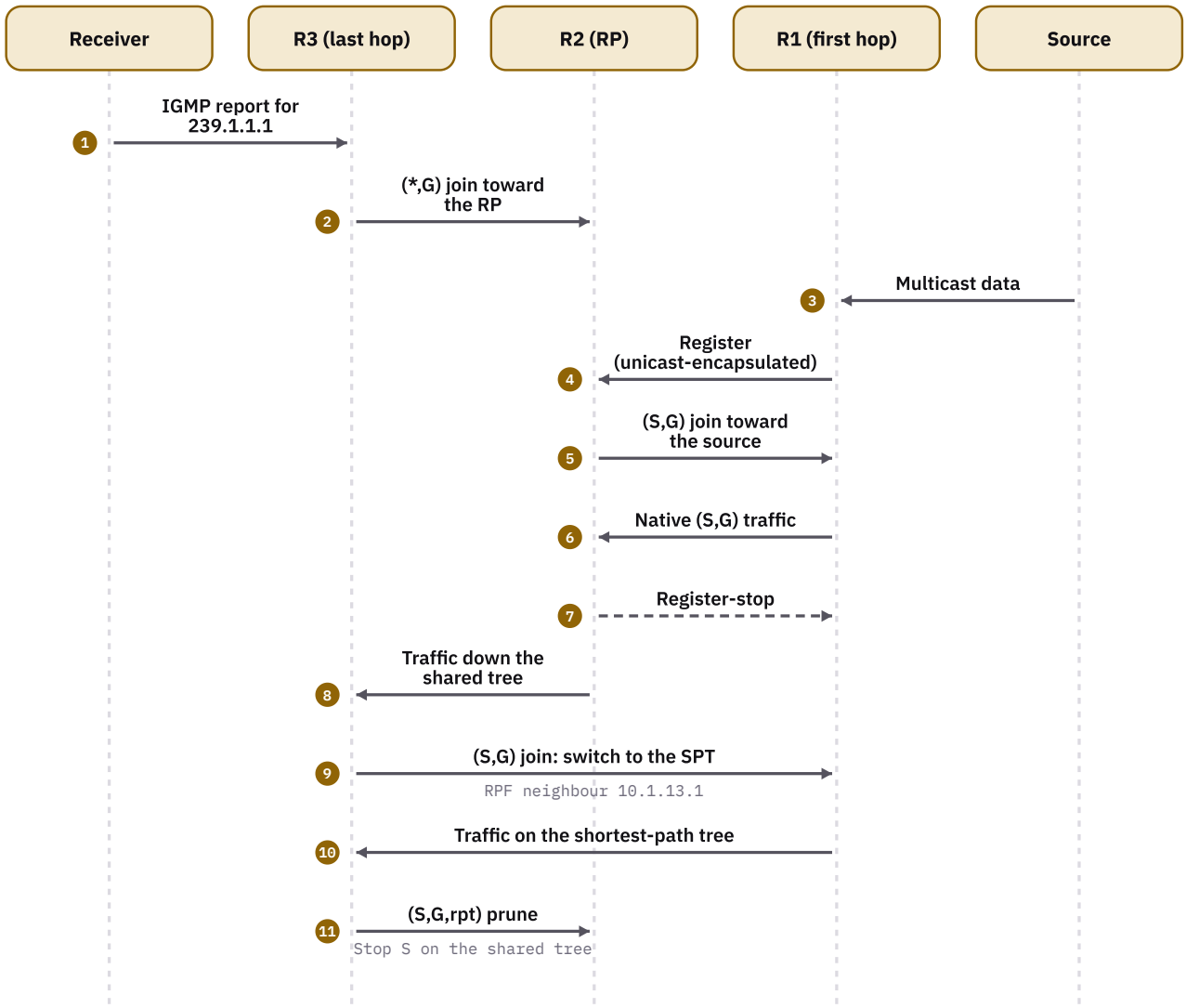
Application	`Source sends to group G; receivers open sockets for G or (S,G)`
Host signalling	`IGMP (IPv4) or MLD (IPv6) between receivers and the last-hop router`
Layer 2 optimisation	`IGMP/MLD snooping limits flooding to ports with receivers or routers`
Routing	`PIM builds distribution trees between routers using the unicast RIB for RPF`
Inter-domain	`MSDP shares active sources between RPs; multiprotocol BGP can carry RPF routes`

Group management versions

PROTOCOL	KEY BEHAVIOUR	EXPERT NOTES
IGMPv2	Joins and leaves per group; group-specific query on leave	Lowest IP address on the segment becomes querier
IGMPv3	Source filtering: INCLUDE or EXCLUDE lists; reports sent to 224.0.0.22	Required for SSM; enables explicit tracking and fast leave
MLDv1	IPv6 equivalent of IGMPv2, carried in ICMPv6	Uses link-local addresses; same querier logic
MLDv2	IPv6 equivalent of IGMPv3 with source filtering	Required for IPv6 SSM
Snooping	Switch inspects reports to build per-port group membership	Needs a querier on the VLAN, or memberships time out

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PIM-SM from first packet to shortest-path tree



Receivers and sources meet at the RP, the RP joins the source and stops the registers, and the last-hop router then joins the source directly and prunes it off the shared tree.

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Enabling PIM-SM with a static RP

R3 · IOS-style CLI

```
ip multicast-routing
ip pim rp-address 10.255.0.100
!
interface GigabitEthernet0/0
 description to core
 ip pim sparse-mode
!
interface Vlan20
 description receivers
 ip address 10.20.0.1 255.255.255.0
 ip pim sparse-mode
 ip pim dr-priority 200
 ip igmp version 3
```

PIM must be enabled on every interface along the path, including the RP's loopback where it lives. DR priority makes the intended router the designated router on the receiver VLAN.

Reading the multicast routing table

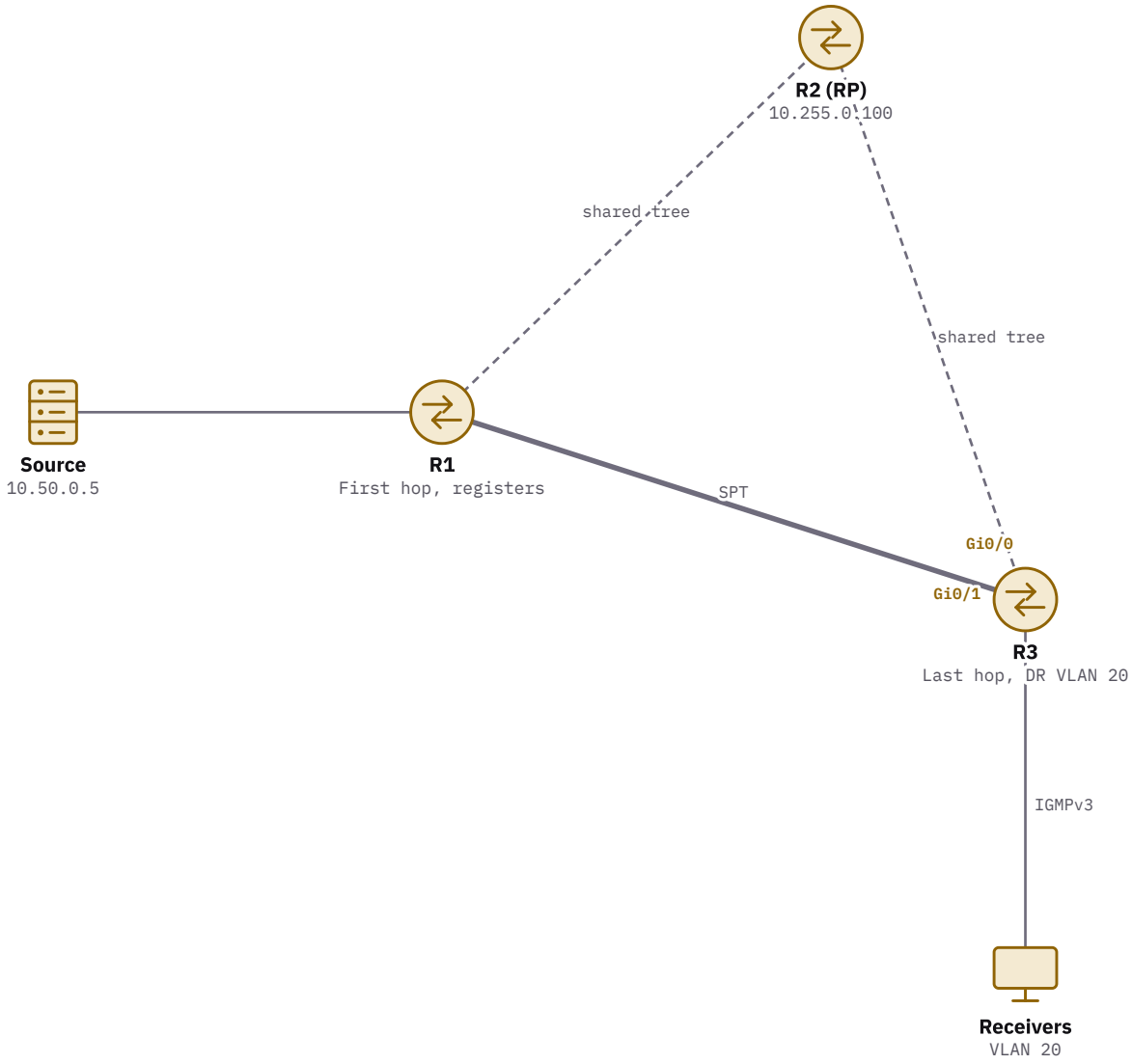
R3 · IOS-style CLI

```
R3# show ip mroute 239.1.1.1
(*, 239.1.1.1), 00:12:01/stopped, RP 10.255.0.100, flags: SJC
  Incoming interface: GigabitEthernet0/0, RPF nbr 10.1.23.2
  Outgoing interface list:
    Vlan20, Forward/Sparse, 00:12:01/00:02:41
(10.50.0.5, 239.1.1.1), 00:11:58/00:02:57, flags: JT
  Incoming interface: GigabitEthernet0/1, RPF nbr 10.1.13.1
  Outgoing interface list:
    Vlan20, Forward/Sparse, 00:11:58/00:02:41
```

S = sparse, J = join SPT, C = connected receiver, T = traffic flowing on the shortest-path tree. The (S,G) entry has a different incoming interface from the (,G) entry because the source path does not run through the RP.*

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Shared tree and shortest-path tree



R3's (*,G) entry points up Gi0/0 toward the RP via R2 (10.1.23.2); after switchover its (S,G) entry arrives on Gi0/1 from R1 (10.1.13.1), which is why the two entries show different incoming interfaces.

Common mroute flags

FLAG	MEANING
S	Sparse mode entry
C	Directly connected receiver (IGMP/MLD membership)
L	Local: the router itself joined the group
J	Join to SPT triggered for this group
T	SPT bit: traffic is arriving on the shortest-path tree
P	Pruned: no outgoing interfaces
F	Register flag: this router is registering a directly connected source

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FLAG	MEANING
R	RP bit: (S,G) entry pruned off the shared tree
A	Candidate for MSDP advertisement

INSIGHT · ASSERTS AND DESIGNATED ROUTERS ON SHARED SEGMENTS

Two PIM roles on a multi-access segment are often confused. The **DR**, elected by priority then highest IP, sends joins for receivers and registers for sources on that segment. The **assert winner** is decided only when two routers both forward the same flow onto a segment: the router with the better administrative distance and then metric to the source wins, and the loser prunes. Traffic arriving in duplicates on a receiver LAN for a second or two is often an assert, not a loop.

SCENARIO · DECIDE · RECEIVERS ON A VLAN LOSE THE STREAM EVERY FEW MINUTES

Receivers on an access VLAN join group 239.10.10.10 and watch video for a few minutes, then the stream stops until they rejoin. The VLAN has IGMP snooping enabled on the switches but the router interface for the VLAN was moved to a different switch stack, and no PIM router is now attached to the VLAN. What is the best fix?

(A) Lower the PIM hello interval on the core routers so that the multicast tree is refreshed more frequently.

(B) Enable an IGMP snooping querier on the VLAN or restore a router interface, so general queries keep memberships refreshed.

(C) Disable IGMP snooping on the switches so the stream is flooded to every port in the VLAN at all times.

(D) Configure a static RP on the access switches so they can build the shared tree for the VLAN themselves.

Circle your choice. Answer and feedback: module answer key.

Key terms

IGMP / MLD
Protocols hosts use to report multicast group membership to the local router, for IPv4 and IPv6 respectively.

IGMP snooping
A switch feature that forwards multicast only to ports where membership reports or routers were seen.

Rendezvous point (RP)
The root of the shared tree where sources and receivers first meet in PIM-SM.

Shortest-path tree
The (S,G) distribution tree rooted at the source's first-hop router.

Assert
PIM election that removes duplicate forwarders onto a shared segment.

Querier
The router or switch that sends general queries on a segment; with IGMP the lowest IP address wins.

Designated router (DR)
The PIM router on a segment that sends joins for receivers and registers for sources.

Shared tree
The (*,G) distribution tree rooted at the RP.

Register message
Unicast encapsulation of multicast packets from the source DR to the RP until native forwarding begins.

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Key takeaways

- Multicast is receiver-driven: IGMP or MLD signals interest and PIM builds the tree.
- Snooping switches need a querier on the VLAN, or memberships expire.
- PIM-SM starts on the shared tree through the RP and normally switches to the source tree.
- Duplicates for a moment on a LAN are usually an assert, not a loop.

KNOWLEDGE CHECK

1. Why does the last-hop router normally switch from the shared tree to the shortest-path tree?

   (A) To stop the RP from sending register-stop messages toward the first-hop router.

   (B) To allow IGMPv2 receivers to join source-specific channels without upgrading to IGMPv3.

   (C) To reduce the number of PIM neighbours it must maintain on its upstream interfaces.

   (D) To receive traffic on the most direct path from the source instead of through the RP.

2. An (S,G) entry shows flags 'JT' and an (*,G) entry shows 'SJC'. What is happening?

   (A) The router is registering a local source, and the RP has not yet sent a register-stop.

   (B) A connected receiver joined, and traffic for S now arrives on the shortest-path tree.

   (C) The entry is pruned, and the router is waiting for the RP to send a fresh join for it.

   (D) The router is the RP, and it is forwarding encapsulated register messages to the receivers.

3. Which IGMP version must receivers use to join a source-specific (S,G) channel without any translation on the router?

   (A) IGMPv2, because leave messages identify the source the host wants to leave.

   (B) IGMPv1, because its reports are sent to the group address and carry the source.

   (C) Any version, because the router derives the source from the unicast routing table.

   (D) IGMPv3, because it carries INCLUDE source lists in membership reports.

4. Two routers on a receiver LAN both forward the same (S,G) stream for a moment, then one stops. Which mechanism caused this?

   (A) The DR election, which is repeated whenever duplicate traffic is seen on the LAN segment.

   (B) The PIM assert process, which elects a single forwarder based on distance and metric to the source.

   (C) The IGMP querier election, which decides which router forwards each group to the segment.

   (D) The SPT switchover, which always produces duplicates until the RP sends a prune message.

5. Where is the RP address needed in a static-RP PIM-SM design?

   (A) On every PIM router that may serve receivers or sources, so each can join or register to it.

   (B) Only on the RP itself, because other routers learn its address from PIM hello messages.

   (C) Only on first-hop routers, because last-hop routers always build shortest-path trees directly.

   (D) Only on the receivers, because IGMP reports carry the RP address to the last-hop router.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · STREAM · LESSON 2 · 45 MIN

RP Design: Static, BSR, Anycast RP and SSM

Placing and discovering rendezvous points, making them redundant, or removing them entirely

- Compare static RP, BSR, anycast RP with MSDP or PIM, and embedded RP.
- Configure anycast RP with MSDP and verify that active sources are shared.
- Design SSM services and decide when SSM is the better answer.

The RP is the meeting place of PIM sparse mode, so its placement, discovery and redundancy decide how a multicast network behaves under failure. An expert can configure every option, but more importantly can say which one fits: some networks need dynamic discovery, some need sub-second RP failover, and many modern services need no RP at all because they use **source-specific multicast (SSM)**.

RP discovery and redundancy options

OPTION	HOW ROUTERS LEARN THE RP	REDUNDANCY	BEST FIT
Static RP	Configured on every router	Only with anycast addressing	Small or tightly automated networks
BSR (RFC 5059)	Elected bootstrap router floods the RP set hop by hop; routers hash groups to RPs	Multiple candidate RPs and BSRs; failover takes seconds	Multi-vendor networks needing dynamic RP sets
Anycast RP with MSDP	Same RP address on several routers (static or BSR advertised)	Unicast routing picks the nearest RP; failover at IGP speed	Enterprise and provider IPv4 networks
Anycast RP with PIM (RFC 4610)	Same as above, sources shared by PIM register forwarding	As above, without MSDP	Networks that also need IPv6 anycast RP
Embedded RP (IPv6)	RP address encoded in the group address	Depends on the RP address chosen	IPv6 inter-domain ASM without configuration

Anycast RP with MSDP in action

01 **Shared address** — RP1 and RP2 both own 10.255.0.100/32 on a loopback, advertised in the IGP

→

02 **Nearest RP** — Each first- and last-hop router reaches the closest instance of 10.255.0.100

→

03 **Source registers** — A source near RP1 registers to RP1 only

→

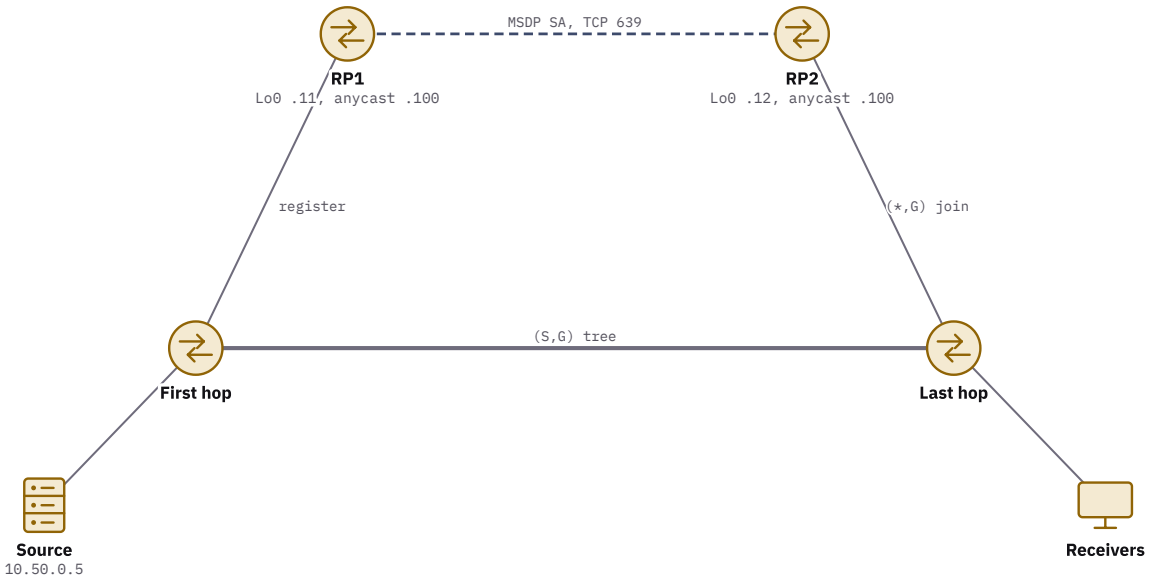
04 **SA exchange** — RP1 sends an MSDP source-active message to RP2 over a TCP 639 session using unique addresses

→

05 **Join source** — RP2, which has receivers, joins (S,G) toward the source; traffic flows

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Anycast RP with MSDP



Both RPs own 10.255.0.100, so each router registers or joins at the nearest instance. RP1 tells RP2 about the source in a source-active message sent from its unique loopback; RP2 then joins toward the source.

Anycast RP pair with MSDP

RP1 · IOS-style CLI

```
interface Loopback0
 description unique router ID
 ip address 10.255.0.11 255.255.255.255
interface Loopback100
 description anycast RP address
 ip address 10.255.0.100 255.255.255.255
 ip pim sparse-mode
!
ip pim rp-address 10.255.0.100
ip msdp peer 10.255.0.12 connect-source Loopback0
ip msdp originator-id Loopback0
!
RP1# show ip msdp sa-cache
MSDP Source-Active Cache - 2 entries
(10.50.0.5, 239.1.1.1), RP 10.255.0.12, AS ?, 00:04:11/00:05:39, Peer 10.255.0.12
```

The MSDP session and the originator ID must use the unique loopback; if SA messages named the shared anycast address as their RP, the receiving RP would see its own address as the originator, peer-RPF checks would fail and the SAs would be discarded. The SA cache entry proves RP2 is telling RP1 about a source registered at RP2.

WATCH OUT · THE ROUTER ID TRAP

On any router holding the anycast RP address, make sure the IGP, BGP and PIM router IDs do not choose the anycast loopback. Two routers with the same OSPF router ID cause adjacency failures or flapping that look unrelated to multicast. Configure router IDs explicitly on every RP.

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Any-source versus source-specific multicast

ASM (PIM-SM with RP)

- Receivers join (*,G) without knowing the source
- Needs RP placement, discovery and redundancy
- Register and SPT switchover add complexity
- Supports many-to-many applications such as some discovery protocols

SSM

- Receivers join (S,G) with IGMPv3 or MLDv2
- No RP, no registers, no shared tree
- Default range 232.0.0.0/8 and ff3x::/32
- Ideal for IPTV, market data and one-to-many video

Enabling SSM

R3 · IOS-style CLI

```
ip pim ssm default
!
interface Vlan20
 ip pim sparse-mode
 ip igmp version 3
!
R3# show ip mroute 232.1.1.1
(10.50.0.5, 232.1.1.1), 00:03:20/00:02:51, flags: sTI
  Incoming interface: GigabitEthernet0/1, RPF nbr 10.1.13.1
  Outgoing interface list:
    Vlan20, Forward/Sparse, 00:03:20/00:02:39
```

Lowercase s marks an SSM entry and I indicates an IGMPv3 source-specific report. There is no (,G) entry and no RP for 232.0.0.0/8.*

RP placement principles

TOP

Central in the topology

Place RPs where they are close, in IGP terms, to most sources and receivers to keep shared trees short.

PAIR

Always redundant

At least two anycast RPs in different failure domains.

ROLE

Dedicated control role

RPs should not be on a congested edge; register decapsulation and state must not compete with heavy data forwarding.

SCOP

Scoped groups

Use administratively scoped groups (239.0.0.0/8) per region with boundaries, so local streams stay local.

SSM

Prefer SSM where possible

Every flow moved to SSM is a flow that no longer depends on RP health.

INSIGHT · BEYOND PIM: BIER

Bit Index Explicit Replication (RFC 8279) removes per-flow tree state from the core: the ingress router writes a bit string naming every egress router that needs the packet, and transit routers replicate by bitmask. It suits providers with many channels but needs data-plane support, so most networks still run PIM SSM or ASM, while EVPN fabrics carry tenant multicast with their own optimised replication.

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SCENARIO · DECIDE · CHOOSING A DESIGN FOR A BROADCAST VIDEO SERVICE

A broadcaster will deliver 400 live channels from two encoder sites to 3,000 set-top boxes over its own network. Set-top boxes support IGMPv3, the source addresses are known and fixed, and leadership wants the simplest design with the fewest failure points. What should you recommend?

(A) ASM with a single static RP placed in the primary data centre close to both encoder sites.

(B) ASM with anycast RP and MSDP across both encoder sites, using BSR to advertise the RP address.

(C) SSM with IGMPv3, so each box joins (S,G) channels and no RP is required.

(D) Dense mode, because every set-top box should receive every channel at all times anyway.

Circle your choice. Answer and feedback: module answer key.

Key terms

BSR
Bootstrap router: an elected router that floods the set of candidate RPs to all PIM routers.

Anycast RP
Several RPs sharing one address so unicast routing selects the nearest and provides failover.

Source-active message
MSDP announcement that a source is actively sending to a group via a given RP.

Embedded RP
IPv6 technique that encodes the RP address inside the multicast group address.

Candidate RP
A router that offers itself to the BSR as an RP for a range of groups.

MSDP
Multicast Source Discovery Protocol: shares active-source information between RPs over TCP 639.

SSM
Source-specific multicast: receivers join (S,G) channels directly; no RP is used.

Administrative scoping
Use of 239.0.0.0/8 groups with boundaries so streams stay inside a region or organisation.

Key takeaways

- RP choice decides behaviour under failure: static, BSR, anycast RP with MSDP or PIM, or embedded RP.
- Anycast RPs must share sources and use unique addresses for MSDP and router IDs.
- SSM needs no RP at all; prefer it whenever receivers can name the source.
- Scope groups with administrative boundaries so local streams stay local.

KNOWLEDGE CHECK

6. Two anycast RPs share 10.255.0.100. A source registers with RP1, but receivers near RP2 get nothing. What is the most likely missing piece?

(A) A BSR election, so that both RPs agree which of them is responsible for the group range.

(B) A higher DR priority on the receiver VLAN, so that joins are sent to RP1 instead of RP2.

(C) Source sharing between the RPs, through MSDP or PIM anycast RP, so RP2 learns about the source.

(D) A static mroute on RP2 pointing to RP1, so the shared tree is rooted at RP1 for all groups.

7. Why must the MSDP originator ID on an anycast RP be a unique address rather than the anycast address?

(A) MSDP peers cannot establish TCP sessions to an address that is configured on more than one router in the same AS.

(B) SA messages name the originating RP; a shared address looks self-originated, fails peer-RPF and is discarded.

(C) The anycast address is reserved for IGMP and cannot be used by any routing protocol that runs on the RP.

(D) The originator ID is used as the BSR priority, and identical values would cause a BSR election loop.

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8. Which statement about SSM is correct?

   (A) SSM uses the RP only for the first packet and then switches to the shortest-path tree immediately.

   (B) SSM requires MSDP between domains, because each domain must learn the active sources from peers.

   (C) SSM builds only (S,G) trees and needs no RP; receivers must signal the source with IGMPv3 or MLDv2.

   (D) SSM works with any IGMP version because routers derive the source from the group address itself.

9. An RP pair runs OSPF, and after anycast RP was configured the two RPs started flapping their OSPF adjacencies with neighbours. What is the most likely cause?

   (A) Both RPs selected the anycast loopback as OSPF router ID, creating a duplicate router ID.

   (B) MSDP sessions consume the OSPF hello packets and cause dead-timer expiry on the RPs.

   (C) PIM sparse mode on the anycast loopback disables OSPF on that interface automatically.

   (D) BSR messages flood the anycast address into OSPF as a type-5 external route every minute.

10. When is BSR a better choice than static RP configuration?

    (A) When the network needs SSM, because BSR advertises the 232.0.0.0/8 range to every receiver.

    (B) When sources are in another AS, because BSR exchanges active source information between domains.

    (C) When the RP must be selected per receiver, because BSR runs inside IGMP membership reports.

    (D) When a multi-vendor network needs RP sets to change dynamically without touching every router.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · STREAM · LESSON 3 · 40 MIN

Multicast Troubleshooting: RPF and Beyond

A source-to-receiver method that finds the broken link in minutes

- Explain the RPF check and diagnose RPF failures caused by asymmetric routing.
- Apply a structured troubleshooting sequence from receiver to source.
- Fix RPF problems with the least disruptive tool: IGP change, static mroute or multicast BGP.

Most multicast outages reduce to one of a handful of causes: no membership, no PIM neighbour, no RP mapping, a failed RPF check, a TTL or filter problem at the source, or a boundary dropping the group. The method below works hop by hop from the receiver toward the source, because each router's state tells you what it believes about the next upstream hop.

DEFINITION · REVERSE PATH FORWARDING (RPF) CHECK

A router accepts a multicast packet only if it arrives on the interface the router would use to send unicast traffic **back to the source** (for (S,G)) or **to the RP** (for (*,G)). The lookup uses the unicast RIB, or a separate multicast RIB when one is populated by static mroutes or multicast BGP. Packets that fail the check are dropped silently, which prevents loops but also makes asymmetric routing invisible until you look.

Troubleshooting from receiver to source

01 **Membership** — Does the last-hop router show the group (and source) from the receiver? Is a querier present?

→

02 **Last-hop state** — Is there a (*,G) or (S,G) with the receiver interface in the outgoing list?

→

03 **RPF and neighbours** — What is the RPF interface and neighbour, and is that neighbour a PIM neighbour?

→

04 **Walk upstream** — Repeat on the RPF neighbour until you reach the RP or first-hop router

→

05 **Source side** — Is the source sending, with sufficient TTL, and is the first-hop router registering or forwarding?

→

06 **Boundaries and filters** — Check scoped boundaries, ACLs, and rate limits along the path

Finding an RPF failure

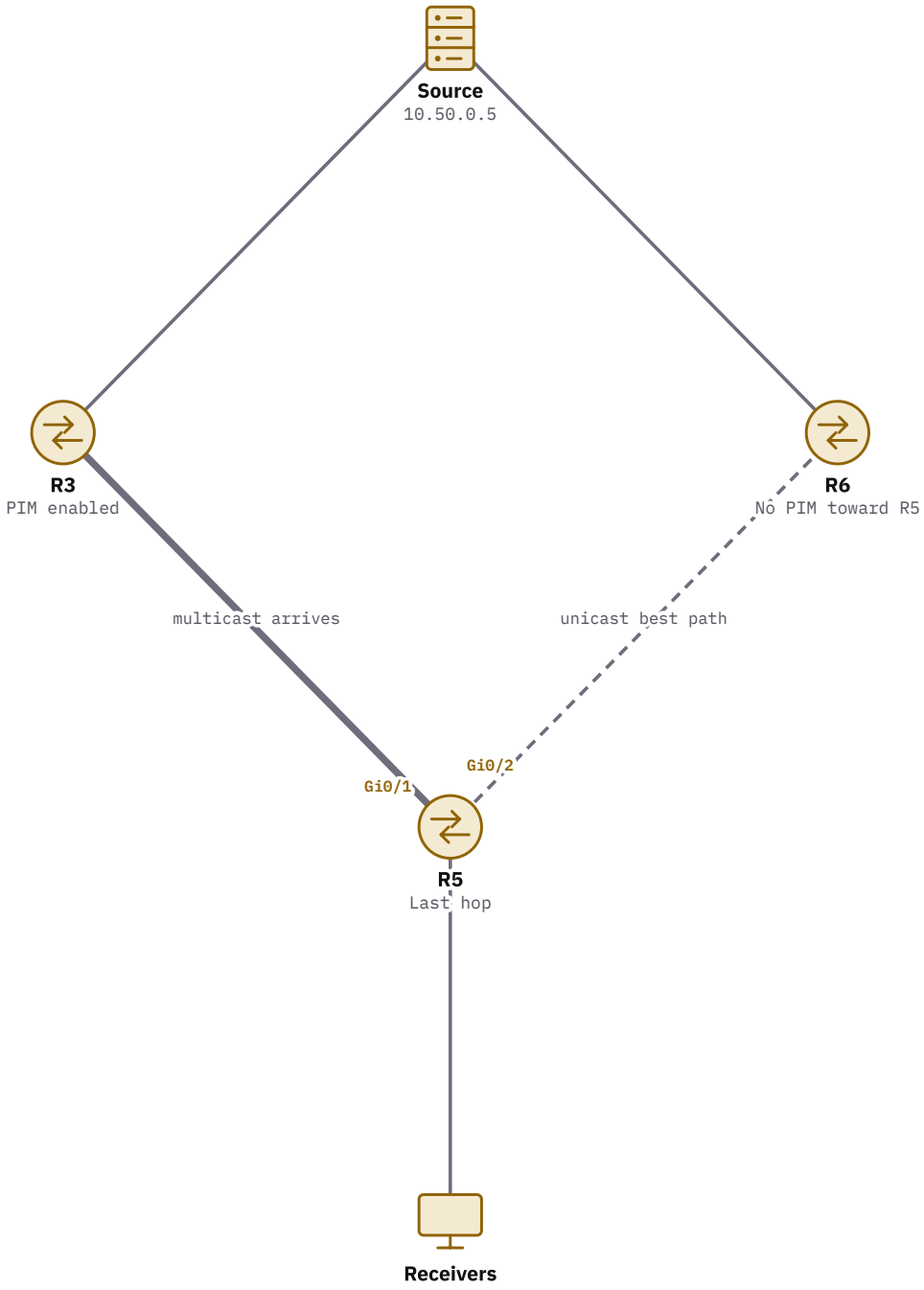
R5 · IOS-style CLI

```
R5# show ip rpf 10.50.0.5
RPF information for ? (10.50.0.5)
  RPF interface: GigabitEthernet0/2
  RPF neighbor: ? (10.1.56.6)
  RPF route/mask: 10.50.0.0/16
  RPF type: unicast (ospf 1)
R5# show ip pim neighbor GigabitEthernet0/2
PIM Neighbor Table
Neighbor Address  Interface   Uptime/Expires  Ver  DR Prio/Mode
R5# show ip mroute 10.50.0.5 239.1.1.1 count
  Source: 10.50.0.5/32, Forwarding: 0/0/0/0, Other: 4183/4183/0
```

The unicast path to the source now points out Gi0/2, where no PIM neighbour exists, so R5 sends its join into nowhere. 'Other' counters equal to total packets mean every packet arriving on the old interface fails RPF and is dropped.

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An RPF failure after a unicast change



R5's unicast route to 10.50.0.0/16 now points out Gi0/2 toward R6, where no PIM neighbour exists. Packets still arriving on Gi0/1 fail the RPF check, and joins sent toward R6 go nowhere.

Fixing RPF problems

TOOL	HOW IT WORKS	USE WHEN
Enable PIM on the new path	Make the unicast path multicast-capable	The unicast change is intended and the link should carry multicast
Static mroute	Overrides the RPF interface for a source prefix, in the multicast RIB only	A small, stable exception such as a tunnel or a one-off asymmetric link

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TOOL	HOW IT WORKS	USE WHEN
Multicast BGP (IPv4 multicast SAFI)	Carries separate RPF routes in BGP	Large or inter-domain designs that need multicast paths to differ from unicast
IGP metric change	Aligns unicast with the multicast-capable path	Unicast and multicast should share the same topology anyway

Two RPF fixes, IOS-style

R5 · IOS-style CLI

```
! Option 1: make the new unicast path multicast-capable
interface GigabitEthernet0/2
 ip pim sparse-mode
! Option 2: keep multicast on the old path for this source
ip mroute 10.50.0.0 255.255.0.0 10.1.35.3
R5# show ip rpf 10.50.0.5
  RPF interface: GigabitEthernet0/1
  RPF neighbor: ? (10.1.35.3)
  RPF type: multicast (static)
```

Prefer option 1 when the topology change is permanent. A static mroute is invisible to unicast routing, so document it, or it will confuse the next engineer.

Quick fault catalogue

SYMPTOM	LIKELY CAUSE	CHECK
Receivers on the source's own VLAN get traffic, nobody else does	Source TTL of 1	Capture at the first-hop router: TTL field of the stream
(*,G) exists, no (S,G) anywhere	Registers not reaching or not accepted by RP	RP mapping on the first-hop router; register path and filters
Stream works through the RP but stops after SPT switchover	RPF toward the source fails on the SPT path	show ip rpf for the source on the last-hop router
Some groups fail, others in the same range work	Different RP mapping per group or scoped boundary	RP mapping for each group; boundary ACLs
Stream stops every few minutes on a VLAN	No querier; snooping memberships expire	Querier status on the VLAN

CASE STUDY · AN IPTV OUTAGE AFTER A CORE UPGRADE

A regional operator delivers 220 IPTV channels with SSM from head-end encoders in 10.50.0.0/16. Overnight, a new 400G link was added between two core routers, C2 and C5, and IS-IS metrics were tuned to move unicast traffic onto it. By morning, 40% of subscribers, all behind aggregation routers fed by C5, have no TV, while internet service is fine. On C5, show ip mroute for a channel shows the (S,G) with incoming interface Null and the RPF neighbour as C2's address on the new link. The change ticket listed only IS-IS and interface configuration.

1. What is the root cause?

2. Why is internet service unaffected?

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3. What is the immediate fix, and what is the permanent fix?

4. What process change prevents a repeat?

Model answer: module answer key.

SCENARIO · DECIDE · ONLY THE SOURCE'S OWN SUBNET RECEIVES THE STREAM

A new trading application sends market data to 239.20.1.1. Receivers on the same VLAN as the source receive it; receivers elsewhere, with correct joins and PIM state up to the RP, receive nothing. The RP shows no (S,G) and no registers from the first-hop router. What should you check first?

(A) The IGMP version on the receiver VLANs, because IGMPv2 receivers cannot join groups in the 239.0.0.0/8 range.

(B) The MSDP sessions between the RPs, because the RP will not accept registers until MSDP is established.

(C) The SPT threshold on the last-hop routers, because switching to the shortest-path tree too early drops the stream.

(D) The TTL of the multicast packets leaving the source, because an application default of 1 stops routing at the first hop.

Circle your choice. Answer and feedback: module answer key.

Multicast troubleshooting checklist

☐ Identify group, source, receivers and the expected service model (ASM or SSM)

☐ Verify membership and querier on the receiver segment

☐ Verify (*,G) and (S,G) state and the outgoing interface list on the last-hop router

☐ Check RPF interface and neighbour for the source and the RP on each hop

☐ Confirm PIM neighbours on every RPF interface

☐ Confirm consistent RP mapping for the group on all routers (ASM)

☐ Confirm the source is sending with an adequate TTL and the first-hop router is registering or forwarding

☐ Check boundaries, ACLs and rate limits along the path

Key takeaways

- Every multicast forwarding decision depends on the RPF check against the unicast or multicast RIB.
- Troubleshoot from receiver to source: membership, last-hop state, RPF neighbour, then walk upstream.
- Fix RPF with the least disruptive tool: enable PIM on the new path, align metrics, or use a documented static mroute or multicast BGP.
- If nobody beyond the source subnet receives traffic, check the source TTL first.

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KNOWLEDGE CHECK

11. A router receives (S,G) traffic on Gi0/1 but its unicast route to S points out Gi0/2. What happens and why?

(A) The packets are forwarded normally, because RPF applies only to traffic on the shared tree from the RP.

(B) The packets are forwarded and an assert is sent on Gi0/2 to move the source traffic to that interface.

(C) The packets fail the RPF check and are dropped, because they arrive on an interface not used to reach S.

(D) The packets are re-encapsulated as registers and sent to the RP, which forwards them down the shared tree.

12. A static mroute fixes an RPF failure. What is its main operational drawback?

(A) It disables PIM on the original interface, so other groups using that path stop working.

(B) It must be configured on every router in the domain, because RPF decisions are made globally.

(C) It forces all groups into SSM mode, because the static entry contains a source prefix.

(D) It changes multicast RPF only, so unicast and multicast topologies silently diverge unless documented.

13. On a last-hop router the (S,G) entry shows incoming interface Null and the RPF neighbour as 10.1.56.6. What is the most likely cause?

(A) There is no PIM neighbour at 10.1.56.6 on the RPF interface, so the join cannot be sent upstream.

(B) The source has stopped transmitting, so the router has removed the incoming interface after a timeout.

(C) The RP has pruned the source, so the router no longer knows which interface faces the source.

(D) The receiver left the group, so the router cleared the incoming interface but kept the entry.

14. Which first check best separates a source-side problem from a network problem when nobody beyond the source subnet receives traffic?

(A) Whether the receivers are running IGMPv3, because only v3 receivers can join streams from remote sources.

(B) Whether the first-hop router sees the stream with a TTL above 1 and creates (S,G) state or registers.

(C) Whether the RP is reachable from the receivers, because remote receivers must ping the RP before joining.

(D) Whether MSDP is configured on the last-hop router, because it must learn the source before joining.

15. In the IPTV case, why was internet service unaffected while TV failed?

(A) Unicast forwarding does not depend on PIM, so only the RPF-dependent multicast trees broke.

(B) Internet traffic uses MPLS labels, which are exempt from the RPF checks applied to IP packets.

(C) Internet traffic was protected by QoS priority, so it was moved to the old link automatically.

(D) Internet traffic uses SSM, which is not affected by changes to IS-IS metrics in the core.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · STREAM · LESSON 4 · 45 MIN

The QoS Toolkit: Classify, Mark, Queue, Police, Shape, Avoid

Each QoS mechanism, what it can and cannot do, and how it behaves under congestion

- Design a DSCP plan with trust boundaries and consistent mapping into MPLS and overlays.
- Configure priority and weighted queuing, and explain why queuing only matters during congestion.
- Choose between policing and shaping and calculate committed bursts and intervals.
- Apply congestion avoidance with WRED, ECN and L4S, and explain when PFC makes a class lossless.

QoS does not create bandwidth; it decides **who suffers** when there is not enough. Every mechanism acts at a specific point: classification and marking at the edge, queuing and congestion avoidance at congested egress interfaces, policing at contract boundaries and shaping wherever a fast interface feeds a slower service. Expert labs test whether you know which tool solves which problem and whether your configuration survives a detailed show command.

Where each QoS tool acts

01 Classify — Identify traffic at the trust boundary: ports, ACLs, application recognition, or trusted markings →

02 Mark — Write DSCP (and CoS or MPLS TC where needed) once, at the edge →

03 Police — Enforce a rate at ingress or contract points; drop or re-mark the excess →

04 Queue — At congested egress: priority for real-time, weighted shares for the rest →

05 Avoid — Drop or ECN-mark early and selectively before queues overflow →

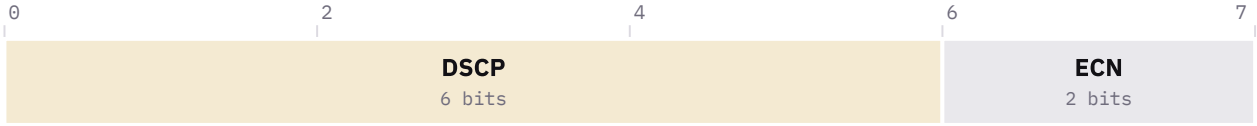
06 Shape — Buffer to a sub-rate so the provider's policer never drops

A practical enterprise DSCP plan

CLASS	DSCP	DECIMAL	TREATMENT
Network control	CS6	48	Small guaranteed share; never policed aggressively
Voice	EF	46	Strict priority, policed to its share
Interactive video	AF41 (AF42/43 for excess)	34	Guaranteed share, low drop
Call signalling	CS3	24	Small guaranteed share
Business-critical / transactional	AF21	18	Guaranteed share, WRED by drop precedence
Bulk data	AF11	10	Guaranteed small share, WRED
Best effort	DF	0	Remaining bandwidth, fair queuing
Scavenger	CS1 or LE (1)	8 / 1	Minimal share; first to be starved

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The DiffServ field in the IP header



The old type-of-service byte now holds a 6-bit DSCP and a 2-bit ECN field. ECN-capable senders set 01 or 10, and a congested queue rewrites them to 11 (Congestion Experienced) instead of dropping; L4S uses ECT(1) to select its low-latency queue.

PRO TIP · TRUST BOUNDARIES AND REMARKING

Trust markings only from devices you control: IP phones identified by the access switch, servers in managed segments, and core links. Everything else is re-marked at ingress. In MPLS cores decide the DiffServ tunnelling model explicitly: **uniform** copies core TC changes back to the customer DSCP, **pipe** and **short-pipe** preserve the customer's DSCP and apply provider treatment only to the TC bits, differing in which marking the egress PE uses for its own queuing. In VXLAN fabrics the outer DSCP is normally copied from the inner header, so the underlay can queue correctly.

Classification, LLQ and CBWFQ

WAN-R1 · IOS-style CLI

```
class-map match-any VOICE
 match dscp ef
class-map match-any VIDEO
 match dscp af41 af42 af43
class-map match-any CRITICAL
 match dscp af21 af22 af23
class-map match-any SCAVENGER
 match dscp cs1
!
policy-map WAN-QUEUING
 class VOICE
  priority percent 10
 class VIDEO
  bandwidth remaining percent 30
 class CRITICAL
  bandwidth remaining percent 35
  random-detect dscp-based
 class SCAVENGER
  bandwidth remaining percent 1
 class class-default
  bandwidth remaining percent 34
  fair-queue
```

Priority traffic is served first and policed to 10% during congestion, so voice cannot starve other classes. 'bandwidth remaining' shares divide what is left after the priority queue.

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Policing versus shaping

Policing

- Measures against a token bucket and drops or re-marks the excess
- No buffering, so no added delay
- Works inbound or outbound
- TCP sees drops and backs off sharply, causing sawtooth throughput
- Right for contracts, scavenger limits and voice queue protection

Shaping

- Buffers excess and releases it at the target rate
- Adds delay and needs queue memory
- Outbound only
- Smooths bursts so a downstream policer never drops
- Right for sub-rate handoffs such as 200 Mbit/s over a 1G port

Token bucket and delay arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Committed burst	`Bc = CIR × Tc`	Shaping to 200 Mbit/s with Tc = 4 ms gives Bc = 800,000 bits (100 KB) per interval.
Interval	`Tc = Bc ÷ CIR`	Smaller Tc means smoother output and less burst latency for voice behind the shaper; 10 ms or less is typical where real-time traffic exists.
Serialisation delay	`frame bits ÷ link rate`	A 1500-byte frame on a 2 Mbit/s link takes 6 ms; on 1 Gbit/s only 12 µs, which is why fragmentation matters only on very slow links.
G.711 call on Ethernet	`(160 payload + 40 IP/UDP/RTP + 18 Ethernet) bytes × 8 × 50 pps`	About 87.2 kbit/s per call at 20 ms packetisation; size the priority class from concurrent calls, not from the codec rate of 64 kbit/s.

Congestion avoidance

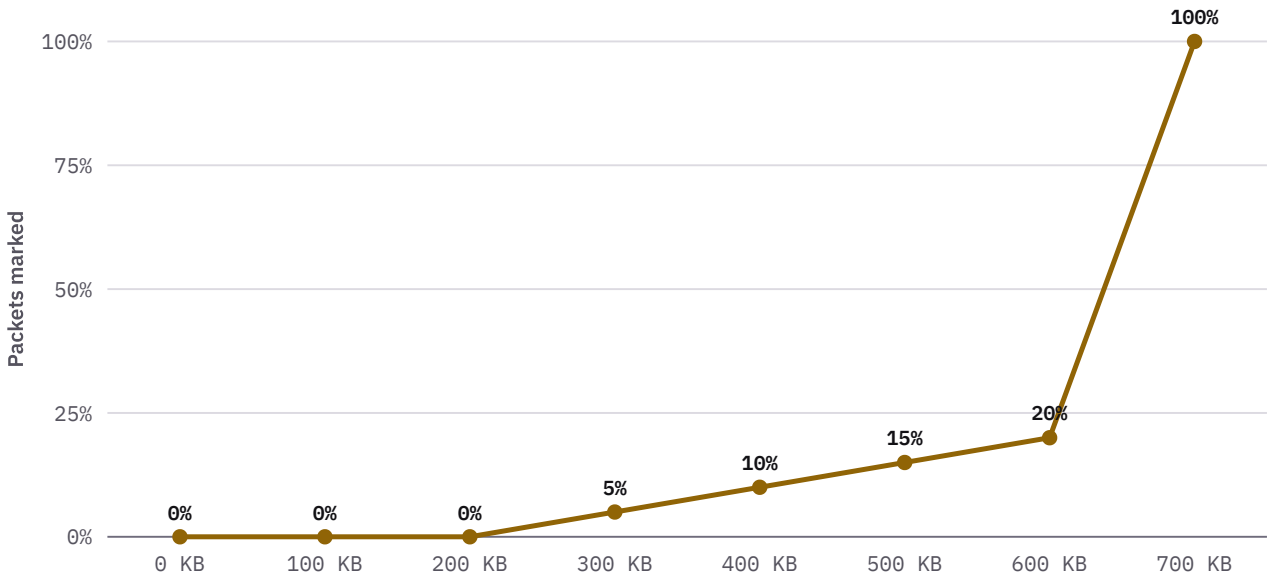
MECHANISM	BEHAVIOUR	NOTES
Tail drop	Queue full, every new packet dropped	Causes TCP global synchronisation and unfairness
WRED	Random early drops between minimum and maximum thresholds, steeper for higher drop precedence	AFx3 dropped before AFx2 before AFx1 within a class
ECN	Marks Congestion Experienced instead of dropping for ECN-capable flows	Requires both ends to negotiate ECN; WRED with ECN marks rather than drops
Fair queuing	Separate flow queues within a class	Stops one elephant flow from dominating best effort
L4S with dual-queue AQM	ECT(1)-marked traffic gets a shallow queue with immediate ECN marking; classic traffic keeps its own queue	Standardised in 2023 (RFC 9330 to 9332); needs scalable congestion control at the endpoints
PFC (IEEE 802.1Qbb)	Pauses one priority on the upstream link when a buffer fills, making that class lossless	Only for dedicated lossless classes such as RoCEv2; can spread congestion through pause storms and head-of-line blocking

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INSIGHT · LOSSLESS ETHERNET FOR AI AND STORAGE

RoCEv2 traffic in GPU clusters and some storage networks degrades sharply when packets are lost, so its class is engineered to be **lossless**: ECN marking, with an endpoint congestion-control loop such as DCQCN, slows senders early, and **PFC** pauses that one priority hop by hop only as a last resort. The ECN threshold must sit well below the PFC pause threshold, with headroom buffer sized for link speed and cable length; otherwise PFC fires first and spreads congestion backwards through the fabric. Keep every other class lossy.

ECN marking profile for a lossless class (illustrative)



Illustrative values: no marking below 200 KB of queue, a linear rise to 20% at 600 KB, then every packet marked. The PFC pause threshold should sit above this whole curve so ECN acts first.

SCENARIO · DECIDE · BULK TRANSFERS WRECK INTERACTIVE TRAFFIC

A branch has a 100 Mbit/s service delivered on a 1 Gbit/s Ethernet handoff. The provider polices at 100 Mbit/s. The branch router has an LLQ/CBWFQ policy applied directly to the 1G interface, but voice quality collapses whenever backups run. Interface counters show no drops on the router. What is the best fix?

- (A) Apply a hierarchical policy: a parent shaper at 100 Mbit/s with the existing queuing policy as its child.
- (B) Increase the priority percentage for voice from 10% to 30% on the existing interface policy.
- (C) Ask the provider to shape instead of police, so their device buffers the excess and protects voice for you.
- (D) Police the backup traffic to 50 Mbit/s at ingress from the LAN so the total stays under the contract.

Circle your choice. Answer and feedback: module answer key.

Key terms

DSCP
Differentiated Services Code Point: the 6-bit field in the IP header used to mark per-hop behaviour.

Assured Forwarding
Four AF classes with three drop precedences each, for example AF41, AF42, AF43.

EF
Expedited Forwarding (DSCP 46): low-latency, low-loss behaviour for real-time traffic such as voice.

Trust boundary
The point in the network where markings start to be accepted rather than rewritten.

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LLQ
Low-latency queuing: a strict-priority queue with an implicit policer, added to weighted queuing.

CBWFQ
Class-based weighted fair queuing: guaranteed bandwidth shares per class during congestion.

Policer
A rate enforcer that drops or re-marks traffic exceeding a token bucket, without buffering.

Shaper
A rate enforcer that buffers excess traffic and releases it at a target rate.

WRED
Weighted random early detection: probabilistic early drops weighted by drop precedence.

ECN
Explicit Congestion Notification: marking packets instead of dropping them to signal congestion.

Hierarchical QoS
Nested policies, typically a parent shaper with a child queuing policy.

Key takeaways

- QoS decides who suffers under congestion; it does not create bandwidth.
- Classify and mark once at a trust boundary, and keep DSCP, CoS, MPLS TC and overlay markings consistent.
- Queuing acts only on a congested interface; shape to the contract so your queues decide instead of the provider's policer.
- ECN and modern AQM such as L4S signal congestion without drops; PFC belongs only to dedicated lossless classes.

KNOWLEDGE CHECK

16. A router's LLQ policy is configured on an interface that never reaches its line rate. When does the priority queue protect voice?

(A) Always, because priority traffic is moved to the front of the hardware queue even without congestion.
(B) Only when the implicit policer of the priority class is exceeded by voice traffic on that interface.
(C) Never on that interface, because queuing policies take effect only when the interface is congested.
(D) Only for traffic in class-default, because the other classes have guaranteed bandwidth already.

17. Within AF class 2, which packets does WRED drop first when the queue begins to fill?

(A) AF21, because it has the lowest DSCP value in the class and the least priority.
(B) AF22, because WRED always starts with the middle drop precedence of each class.
(C) All three equally, because WRED treats every packet in one queue identically.
(D) AF23, because it has the highest drop precedence in the class.

18. A provider polices a 50 Mbit/s service. Which configuration at the customer edge best avoids random provider drops?

(A) Police outbound traffic to 50 Mbit/s so the customer edge drops excess traffic before the provider does.
(B) Shape outbound traffic to 50 Mbit/s with a child queuing policy, so excess is buffered by class.
(C) Mark all traffic EF so that the provider's policer treats it as priority and does not drop it.
(D) Enable WRED on the physical interface so that early drops occur before the provider policer.

19. Using Bc = CIR × Tc, what is Bc when shaping to 20 Mbit/s with Tc = 10 ms?

(A) 2,000,000 bits, which is 250,000 bytes per interval.
(B) 20,000 bits, which is 2,500 bytes per interval.
(C) 200,000 bytes, which is 1,600,000 bits per interval.
(D) 200,000 bits, which is 25,000 bytes per interval.

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20. In an MPLS core using the short-pipe model, what does the egress PE use to choose the outgoing queue toward the CE?

(A) The provider's TC bits from the label, because they always overwrite the customer DSCP at the egress.

(B) The outer DSCP of the VXLAN header, because MPLS cores re-encapsulate traffic before the egress PE.

(C) The customer's original DSCP, because the provider TC marking is not copied back into the IP header.

(D) The IP precedence of the provider's own control traffic, which is copied into every customer packet.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · STREAM · LESSON 5 · 45 MIN

Designing End-to-End QoS for Voice, Video and Business-Critical Traffic

From business priorities to a consistent, verifiable policy across LAN, WAN, overlay and data centre

- Translate business requirements into a class model with latency, jitter and loss budgets.
- Build a QoS policy matrix that stays consistent across LAN, WAN, SD-WAN and data centre.
- Verify QoS behaviour with counters and controlled congestion tests.

A QoS design is a contract between the business and the network: which applications are protected, how much they are allowed to consume and what happens to everything else under congestion. It must be **consistent end to end**, because a single hop that ignores markings, or a provider policer you forgot, undoes every careful policy elsewhere.

ZCIE QoS design sequence

BIZ
Business priorities
List applications that must be protected, and those that may be starved, with sponsors.

BUD
Budgets
One-way targets: voice about 150 ms latency, 30 ms jitter, 1% loss; interactive video similar on latency, slightly more tolerant of loss.

CLS
Class model
Choose 4, 8 or 12 classes; fewer classes are easier to keep consistent across domains and providers.

MAP
Mapping
DSCP to CoS, MPLS TC, provider classes, SD-WAN queues and wireless access categories.

POL
Per-hop policies
Edge marking, WAN hierarchical shaping and queuing, core and data-centre queuing.

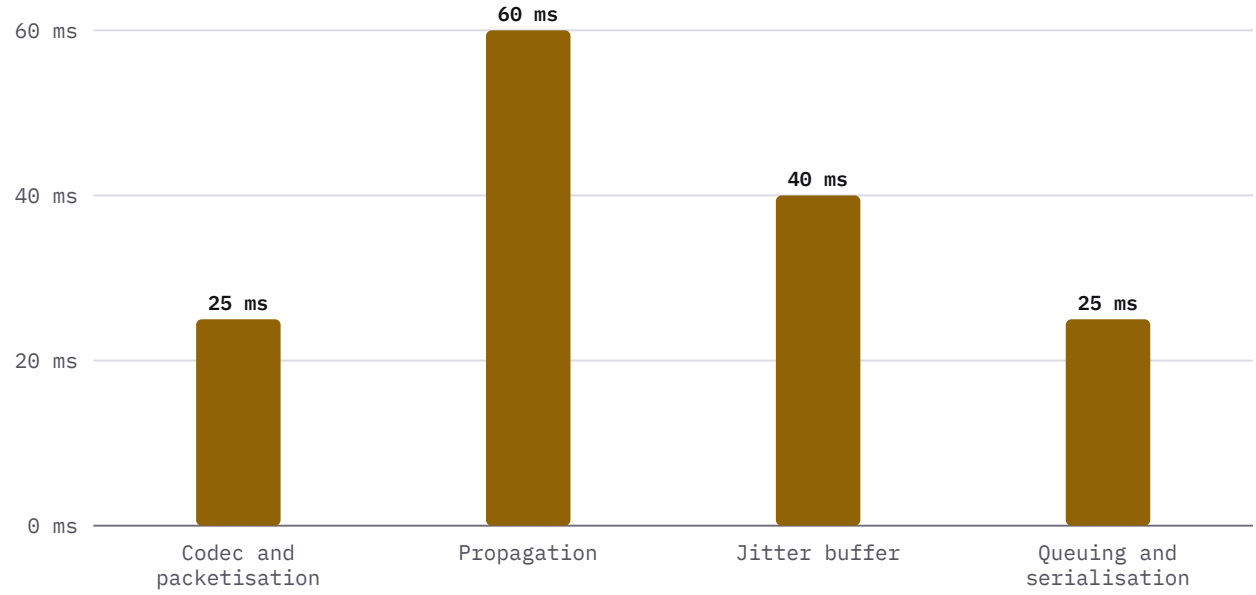
VER
Verification
Controlled congestion tests and per-class counters, repeated after every change.

Budgets and sizing

MEASURE	FORMULA	WHAT IT TELLS YOU
One-way delay budget	`codec + packetisation + serialisation + propagation + queuing + jitter buffer`	If propagation is 60 ms, the jitter buffer 40 ms and codec plus packetisation 25 ms, only about 25 ms remain for all queuing and serialisation within a 150 ms target.
Priority class size	`concurrent calls × per-call rate + video in priority (if any)`	200 concurrent G.711 calls at 87.2 kbit/s need about 17.5 Mbit/s; keep total priority traffic at roughly a third of the link or less.
Provider class mapping	`your classes → provider's classes`	Six internal classes may collapse into four provider classes; decide which share a class and re-mark to the provider's expected DSCPs at the CE.

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A 150 ms one-way voice budget (illustrative)



Illustrative split matching the formula above: once propagation, the jitter buffer and the codec are paid for, only about 25 ms remain for every queue and serialisation delay on the path.

TEMPLATE ## QoS policy matrix

One row per class: the single source for every device policy, provider mapping and test case.

CLASS	DSCP (AND COS/TC)	APPLICATIONS	QUEUE AND SHARE	DROP / POLICE BEHAVIOUR	PROVIDER / SD-WAN MAPPING	TEST
Voice	*EF (CoS 5, TC 5)*	*IP telephony media*	*Priority, 10% (policed)*	*Police excess in priority; no WRED*	*Provider real-time; SD-WAN VOICE SLA*	*200-call load at 100% link*
Business-critical	*AF21 (CoS 2, TC 2)*	*Payments, ERP, order entry*	*35% of remaining*	*WRED dscp-based*	*Provider critical data; SD-WAN BUSINESS SLA*	*Throughput floor under backup load*

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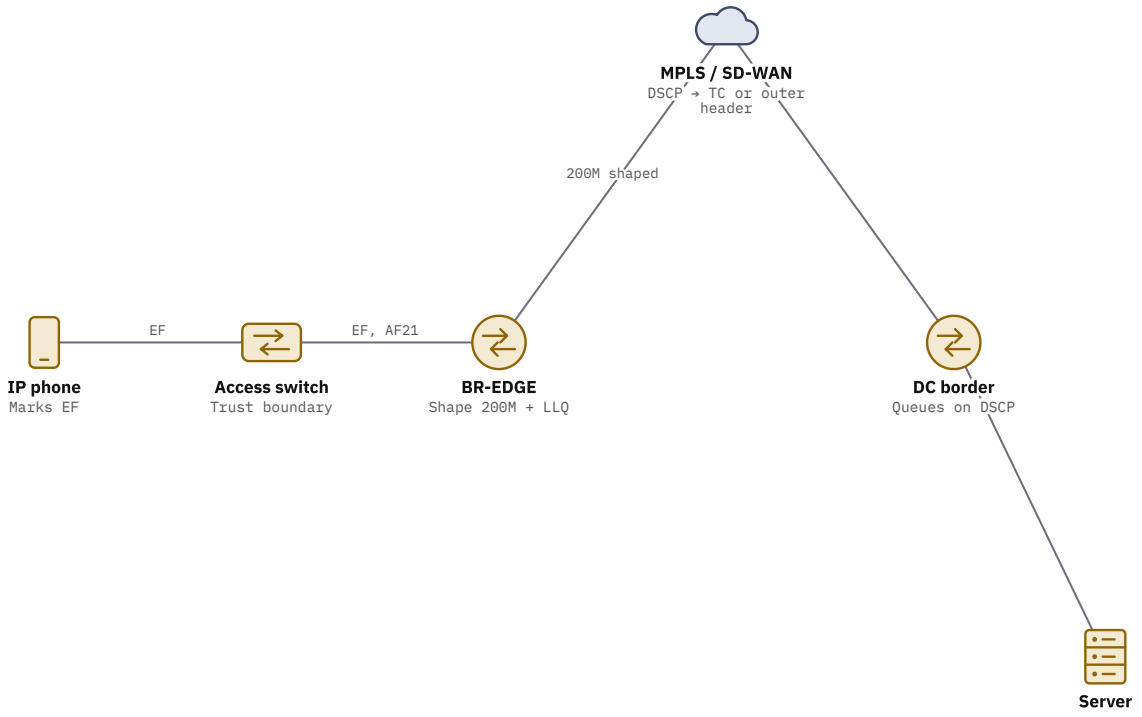
Hierarchical policy at a 200 Mbit/s branch

BR-EDGE · IOS-style CLI

```
policy-map WAN-SHAPE-200M
 class class-default
  shape average 200000000
  service-policy WAN-QUEUING
!
policy-map LAN-INGRESS-MARK
 class VOICE-PHONES
  set dscp ef
 class CRITICAL-APPS
  set dscp af21
 class class-default
  set dscp default
!
interface GigabitEthernet0/0
 description WAN handoff, 200M service on 1G port
 service-policy output WAN-SHAPE-200M
interface GigabitEthernet0/1
 service-policy input LAN-INGRESS-MARK
```

The child policy is the WAN-QUEUING policy from the previous lesson; its percentages now apply to the 200 Mbit/s shaped rate, not to 1 Gbit/s. Marking on LAN ingress makes this router the trust boundary for unmanaged devices.

One QoS policy, five domains



The same class model is enforced at every hop: marked at the trust boundary, shaped and queued at the WAN edge, mapped into MPLS TC or the tunnel's outer DSCP, and queued again in the data centre.

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Proving the policy works under congestion

BR-EDGE · IOS-style CLI

```
BR-EDGE# show policy-map interface GigabitEthernet0/0 output
  Service-policy output: WAN-SHAPE-200M
    Class-map: class-default (match-any)
      shape (average) cir 200000000, bc 800000, be 800000
      Service-policy : WAN-QUEUING
        Class-map: VOICE (match-any)
          Priority: 10% (20000 kbps), burst bytes 500000, b/w exceed drops: 0
          30 second offered rate 15233000 bps, drop rate 0000 bps
        Class-map: CRITICAL (match-any)
          (total drops/no-buffer drops) 312/0
          30 second offered rate 61022000 bps, drop rate 18000 bps
        Class-map: class-default (match-any)
          (total drops/no-buffer drops) 48213/0
          30 second offered rate 171230000 bps, drop rate 46200000 bps
```

Under a deliberate backup load, voice has zero drops, critical data loses a little through WRED, and best effort absorbs the congestion. This is the shape of a healthy policy.

Keeping policy consistent across domains

DOMAIN	WHAT TO CONFIGURE	COMMON MISTAKE
Access LAN	Trust boundary, marking, priority queue on uplinks	Trusting PCs that mark their own traffic EF
WAN edge	Hierarchical shaping and queuing, provider re-marking	Queuing on an uncongested physical interface
MPLS provider core	DSCP-to-TC mapping and DiffServ tunnelling model	Assuming the provider honours all eight of your classes
SD-WAN overlay	Copy DSCP to the tunnel header; per-tunnel shaping; SLA steering	Encrypted tunnels marked default, so the underlay cannot differentiate
Data-centre fabric	Queuing on leaf uplinks; lossless classes only where storage requires them	Applying WAN-style policies where congestion is microbursts in buffers

SCENARIO · DECIDE · VIDEO CALLS DEGRADE ONLY ACROSS SD-WAN

After moving branches to SD-WAN over broadband and MPLS, video calls between branches degrade at busy hours. Inside each branch, DSCP AF41 is set correctly. Captures on the underlay show the encrypted tunnel packets all marked DSCP 0. What is the most likely cause and fix?

(A) The provider does not support AF41, so all branch policies must be changed to mark video EF instead.

(B) The edges are not copying the inner DSCP to the outer tunnel header; enable DSCP copy so the underlay and provider can queue correctly.

(C) The SD-WAN SLA class for video is too strict, so video is kept on the broadband path at busy hours.

(D) The branch LAN switches are not trusting the endpoints, so video should be re-marked at the WAN edge.

Circle your choice. Answer and feedback: module answer key.

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QoS verification checklist

☐ Markings captured at the trust boundary match the policy matrix

☐ Markings captured on the WAN, MPLS and overlay headers match the mapping

☐ Shapers match contracted rates, not physical rates

☐ Priority class sized from concurrent sessions, and kept to roughly a third of the link or less

☐ Controlled congestion test shows zero voice loss and expected drops elsewhere

☐ Per-class counters exported to telemetry and baselined

☐ Provider class mapping confirmed in writing and tested

☐ Policy matrix version stored with the device configurations

REFLECT

Write down the four applications in your organisation whose users complain first when the network is busy. Do they share a class today? What would you change in the policy matrix to protect each one without starving the others?

Key takeaways

- Start from business priorities and delay, jitter and loss budgets, then choose a small class model.
- One policy matrix drives every device policy, provider mapping and test case.
- Shape to the contracted rate and size the priority class from busy-hour concurrency.
- Prove QoS with controlled congestion tests and per-class counters, and copy DSCP into tunnel headers.

KNOWLEDGE CHECK

21. Within a 150 ms one-way voice budget, propagation is 70 ms and the jitter buffer 40 ms. Codec and packetisation take 25 ms. How much remains for queuing and serialisation along the whole path?

    (A) 40 ms, because the jitter buffer can be counted twice, once as delay and once as protection for queuing.

    (B) 15 ms, which is tight and argues for strict priority queuing and shaping on every congestion point.

    (C) 55 ms, because codec and packetisation are performed in the handset and do not count toward it.

    (D) 80 ms, because only propagation counts against the end-to-end budget for interactive voice.

22. Your design has eight internal classes but the provider offers four. What is the right approach?

    (A) Decide which internal classes share each provider class and re-mark to its DSCPs at the CE.

    (B) Keep all eight markings and let the provider's network decide how to map them into its classes.

    (C) Mark everything as best effort toward the provider, because only four classes cannot protect eight.

    (D) Request eight separate provider VPNs, one per class, so markings no longer matter in the core.

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23. A healthy QoS policy under deliberate congestion should show which pattern?

(A) No priority drops, small drops in critical data, and most drops in best effort or scavenger.

(B) Equal drop percentages in every class, because fair treatment shows the policy is balanced.

(C) No drops in any class, because a correct policy prevents congestion rather than managing it.

(D) Drops only in the priority class, because its implicit policer should always be the first to act.

24. Why size the priority class from concurrent sessions rather than from the total number of phones?

(A) Phones that are not on a call send keepalives in EF that would otherwise overflow the queue.

(B) Only active calls consume bandwidth, and over-sizing the priority queue risks starving other classes.

(C) The implicit policer counts registered endpoints, so the total phone count is already included.

(D) The priority queue is shared with network control, which consumes whatever voice leaves unused.

25. Where is QoS most likely to fail silently in an SD-WAN deployment?

(A) At the LAN switch, because SD-WAN edges reset DSCP on all traffic arriving from the branch.

(B) At the controller, because control-plane traffic is always placed ahead of user voice traffic.

(C) At the jitter buffer, because SD-WAN tunnels disable jitter buffering in voice endpoints.

(D) At tunnel encapsulation, if inner DSCP is not copied to the outer header used by the underlay.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · STREAM

Answer key

Knowledge checks

1. D SPT switchover optimises the path and offloads the RP. Register-stop is between RP and source DR, IGMP version is unrelated, and neighbour count does not change.

2. B C indicates a connected receiver; J and T mean SPT join and traffic on the SPT. Registering would show F, pruned would show P.

3. D Only IGMPv3 (and MLDv2 for IPv6) can express a source. Earlier versions are group-only; some routers offer mapping for legacy hosts, but that is a translation.

4. B Asserts resolve duplicate forwarders. DR election uses priority and IP and is not triggered by duplicates; the querier controls queries, not forwarding.

5. A With static RP every relevant router needs the mapping. PIM hellos do not carry RP information, last-hop routers initially join toward the RP, and hosts never know the RP.

6. C Anycast RP splits sources between RP instances; without source-active sharing each RP sees only its own sources. BSR, DR priority and static mroutes do not share source information.

7. B SA messages carry the originating RP address; an SA claiming RP 10.255.0.100 looks like RP2's own advertisement and is discarded. Unique addresses are also used for the TCP session itself; the other options describe behaviours that do not exist.

8. C SSM removes the RP entirely and depends on source-aware membership. MSDP is an ASM tool; legacy IGMP needs explicit mapping configuration.

9. A Router IDs are often chosen from the highest loopback; a shared anycast /32 then becomes a duplicate RID. Configure router IDs explicitly. The other explanations do not occur.

10. D BSR distributes candidate RPs automatically. SSM needs no RP, inter-domain source exchange is MSDP's job, and BSR does not involve IGMP.

11. C The RPF check applies to (S,G) and (*,G) traffic; failure means silent drop. Asserts resolve duplicate forwarders on a LAN, and registers come only from the source's first-hop router.

12. D Static mroutes live in the multicast RIB and are invisible to unicast troubleshooting. They are local to the router, leave PIM untouched and do not change the service model.

13. A A Null incoming interface with a defined RPF neighbour points to an unusable upstream: typically PIM is not enabled there. Source silence, prunes or leaves affect timers and outgoing lists instead.

14. B Checking the first-hop router confirms the stream is routable and entering the multicast network. IGMPv2 receivers can join remote ASM sources, receivers never ping the RP, and MSDP runs only between RPs.

15. A The new link was fine for unicast; multicast failed because RPF pointed at a link without PIM. Labels, QoS and SSM do not explain the asymmetry.

16. C Without congestion there is no queue to reorder; the problem then occurs somewhere else, often at a provider policer. The other options misdescribe LLQ.

17. D Higher drop precedence (AFx3) has lower thresholds and is dropped first. DSCP-based WRED differentiates within a class by design.

18. B Shaping turns the provider's drop point into a queue the customer controls. Policing still drops without class protection, marking everything EF breaks differentiation, and WRED on an uncongested interface does nothing.

19. D 20,000,000 × 0.01 = 200,000 bits. The other values slip a decimal place or confuse bits with bytes.

20. C Short pipe preserves customer DSCP and the egress PE queues on it; pipe uses the provider marking for the egress queue; uniform rewrites customer DSCP. VXLAN is unrelated here.

21. B 150 − 70 − 40 − 25 = 15 ms. Every component counts toward one-way delay; ignoring the codec or jitter buffer understates the problem.

22. A Explicit mapping keeps behaviour predictable; unknown markings are often treated as best effort. Separate VPNs per class are wasteful and unnecessary.

23. A QoS decides who suffers; the protected classes should suffer least. Equal drops mean no differentiation; zero drops mean the test did not congest the link.

24. B Busy-hour concurrency determines real demand; an oversized priority queue can take bandwidth that other classes need. The other statements are incorrect.

25. D Encryption hides the inner header from the underlay, so missing DSCP copy flattens all classes. The other statements are not general behaviours.

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Scenarios & case studies

> **Scenario · Receivers on a VLAN lose the stream every few minutes**
> **B** is best. Correct. Without a querier, snooping memberships age out because hosts only re-report in response to queries, so the switch stops forwarding.

> **Scenario · Choosing a design for a broadcast video service**
> **C** is best. Correct. Known fixed sources and IGMPv3 support make SSM the simplest and most robust design, with no RP to place, protect or troubleshoot.

> **Case study · An IPTV outage after a core upgrade**
> The new C2-C5 link carries the unicast best path to the encoders, so C5's RPF interface for 10.50.0.0/16 moved onto it, but **PIM was not enabled on the new interfaces**. C5 sends (S,G) joins toward C2 on a link with no PIM neighbour, nothing is built, and the entry shows a Null incoming interface. Unicast does not need PIM, so internet traffic simply follows the new path. The immediate fix is to enable PIM sparse mode on both ends of the new link (fast and aligned with the intended design); if that cannot happen within the window, temporarily restore the old IS-IS metrics or add a static mroute on C5 for 10.50.0.0/16 via the old path. The permanent fix is the PIM configuration plus verification that BFD and any multicast boundaries are consistent. For prevention, add multicast to the standard link-addition template, include 'PIM neighbour up and RPF check for head-end prefixes' in automated post-checks, and monitor a sample of channels end to end so a loss of joins raises an alarm within minutes, not at breakfast.

> **Scenario · Only the source's own subnet receives the stream**
> **D** is best. Correct. A TTL of 1 means the first-hop router cannot forward or register the packets, which matches every symptom including local delivery.

> **Scenario · Bulk transfers wreck interactive traffic**
> **A** is best. Correct. The 1G interface never congests, so the router's queues never engage and the provider's policer drops indiscriminately. Shaping to the contract moves congestion into the router where the queuing policy can protect voice.

> **Scenario · Video calls degrade only across SD-WAN**
> **B** is best. Correct. Once encrypted, the underlay sees only the outer header; DSCP 0 means every video packet competes as best effort on the provider and transport links.

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08

MODULE 8 · CODE

Code — Expert Automation and Lab Mastery

Automate safely at scale, close the loop with telemetry, and bring every ring segment together under exam conditions.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Use YANG with NETCONF, RESTCONF and gNMI for safe, transactional configuration and streaming telemetry.
- Build a source of truth, templates and verification code that prove the network matches intent.
- Design CI pipelines, closed-loop remediation and AI-assisted operations with strong guardrails.
- Complete an integrated mock practical lab across all eight Mastery Ring segments and score it honestly.
- Apply an exam-day strategy and plan the ZCIE capstone deliverables.

LESSONS

1. Model-Driven Interfaces at Scale: YANG, NETCONF, RESTCONF and gNMI · 45 min
2. Source of Truth, Templating and Verification as Code · 45 min
3. CI Pipelines, Closed-Loop Operations and AI-Assisted Operations · 50 min
4. Mock Practical Lab: A Walk-Through of the Full Mastery Ring · 60 min
5. Exam-Day Strategy and the ZCIE Capstone · 35 min

MODULE 8 · CODE · LESSON 1 · 45 MIN

Model-Driven Interfaces at Scale: YANG, NETCONF, RESTCONF and gNMI

Structured configuration and state through data models, transactions and streaming

- Read YANG models and choose between native, OpenConfig and IETF models.
- Use NETCONF transactions safely with candidate datastores, validation and confirmed commits.
- Compare NETCONF, RESTCONF and gNMI and pick the right one for configuration and telemetry.

The **Code** segment closes the Mastery Ring. An expert who configures a hundred PEs by hand is not an expert at scale. Model-driven interfaces replace screen-scraping with structured data: a **YANG** model (RFC 7950) defines what can be configured and observed, and a protocol moves that data with transactions and error handling. This lesson focuses on the protocols and the habits that keep automated changes safe; the next lessons build pipelines on top of them.

The model-driven stack

Automation	`Scripts, Ansible, controllers, CI pipelines, closed-loop systems`
Protocol	`NETCONF (RFC 6241)` `RESTCONF (RFC 8040)` `gNMI`
Encoding	`XML` `JSON / JSON_IETF` `Protobuf`
Transport	`SSH (NETCONF, port 830)` `HTTPS (RESTCONF)` `gRPC over HTTP/2 with TLS (gNMI)`

Data models	`Native vendor models: complete coverage`
	`OpenConfig: operator-defined, consistent across platforms`
	`IETF: standard building blocks`

Reading a YANG model as a tree

Linux host

```
$ pyang -f tree ietf-interfaces.yang
module: ietf-interfaces
  +--rw interfaces
     +--rw interface* [name]
        +--rw name                 string
        +--rw description?         string
        +--rw type                 identityref
        +--rw enabled?             boolean
        +--ro oper-status          enumeration
        +--ro statistics
           +--ro in-octets?        yang:counter64
           +--ro in-errors?        yang:counter32
```

rw nodes are configuration, ro nodes are operational state. The asterisk marks a list keyed by name, and a question mark marks an optional leaf.

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NETCONF, RESTCONF and gNMI

NETCONF

- RPCs over SSH with XML encoding
- Multiple datastores: running, candidate, startup
- Locks, validate, commit and confirmed commit
- Best for transactional configuration of many devices

RESTCONF

- HTTP methods over HTTPS with JSON or XML
- Resource URLs map to the YANG tree
- No candidate commit in typical use; each call applies directly
- Best for simple integrations and web tooling

gNMI

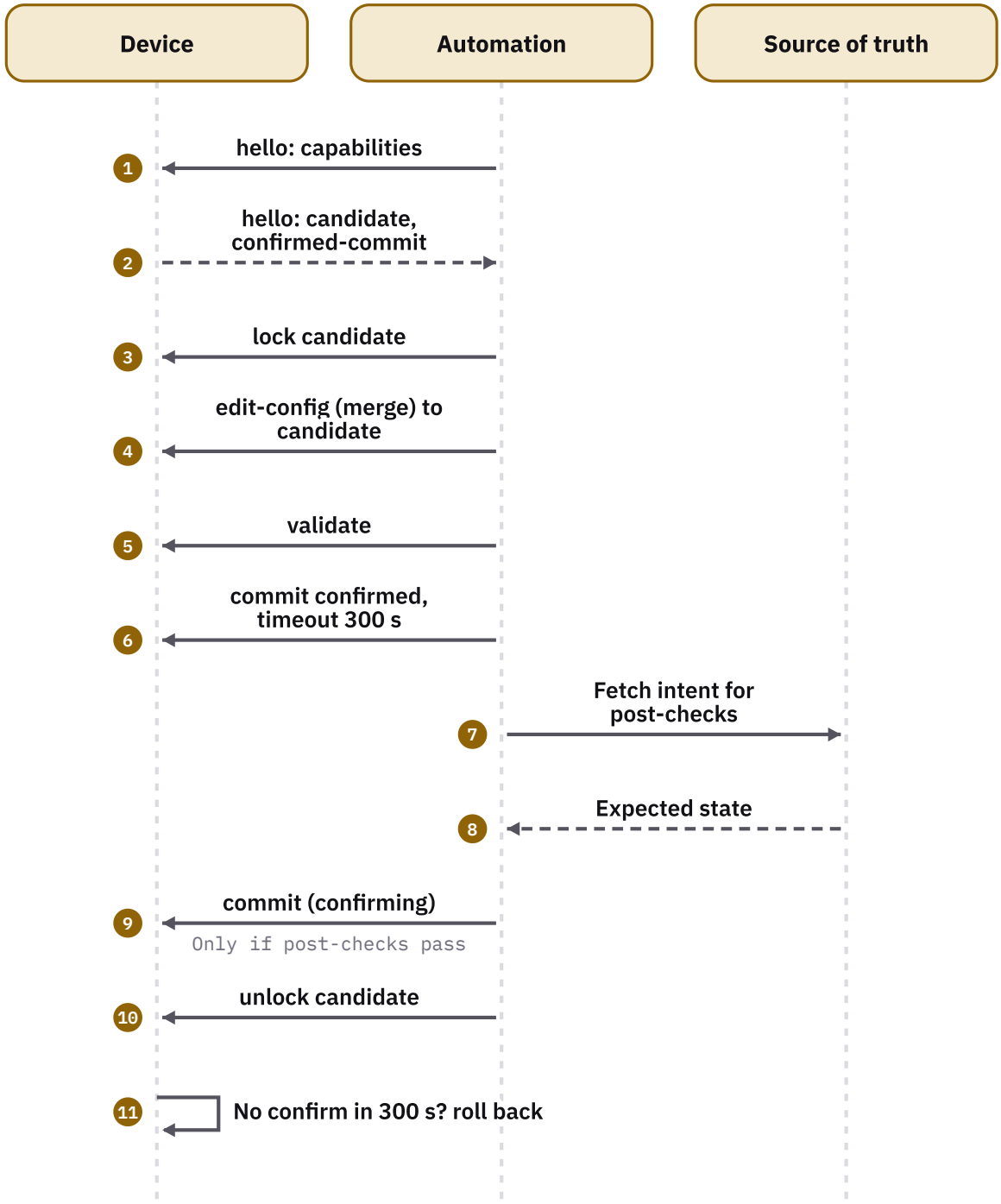
- gRPC with protobuf or JSON encoding
- Get, Set and Subscribe RPCs
- Streaming telemetry: sample, on-change, once and poll modes
- Best for high-volume telemetry; Set is atomic per call

NETCONF operations that matter for safety

OPERATION	PURPOSE	SAFE-CHANGE HABIT
lock / unlock	Prevent concurrent changes to a datastore	Lock the candidate before editing, always unlock in a finally block
edit-config (candidate)	Stage changes without affecting forwarding	Use merge for additive changes; replace only on fully owned subtrees
validate	Check the candidate for errors before commit	Fail the run before anything reaches running
commit confirmed	Apply with automatic rollback unless confirmed	Confirm only after post-checks pass; otherwise let it roll back
get-data (NMDA)	Read a specific datastore, including operational	Compare intended and operational state after the change

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A safe NETCONF change, step by step



The confirmed commit is the safety net: if post-checks fail or the session is cut off, no confirming commit arrives and the device rolls back by itself.

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A confirmed NETCONF change

NETCONF RPC payload (XML)

```
<rpc message-id="101" xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <edit-config>
    <target><candidate/></target>
    <default-operation>merge</default-operation>
    <config>
      <interfaces xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces">
        <interface>
          <name>GigabitEthernet0/0/0/3</name>
          <description>to PE7 - change CHG-4711</description>
        </interface>
      </interfaces>
    </config>
  </edit-config>
</rpc>
<rpc message-id="102" xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <commit><confirmed/><confirm-timeout>300</confirm-timeout></commit>
</rpc>
```

If no confirming commit arrives within 300 seconds, for example because the change cut off management access, the device rolls back by itself.

Streaming interface counters with gNMI

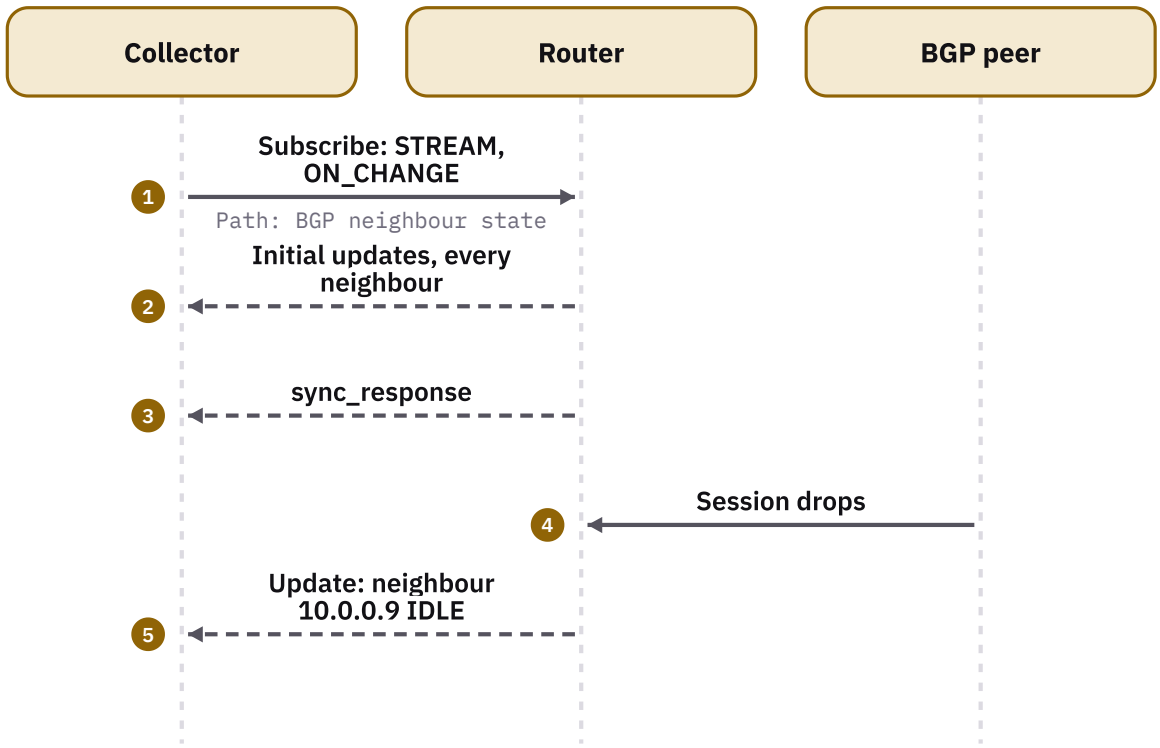
Linux host · gnmic (open source)

```
$ gnmic -a pe1.lab:9339 -u telemetry --tls-ca ca.pem subscribe \
    --path "/interfaces/interface[name=GigabitEthernet0/0/0/3]/state/counters" \
    --mode stream --stream-mode sample --sample-interval 10s
{
  "source": "pe1.lab:9339",
  "timestamp": 1792396800000000000,
  "updates": [ { "Path": "interfaces/interface[name=GigabitEthernet0/0/0/3]/state/counters",
      "values": { "in-octets": "918273645512", "in-errors": "0", "out-discards": "12" } } ]
}
```

Sample mode sends every 10 seconds whether or not values change; on-change mode suits state such as oper-status or BGP session state. The password is prompted for, never written into scripts or history.

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gNMI on-change subscription



After the initial snapshot and sync marker, the router sends nothing until a state really changes, which is why on-change suits session state across thousands of devices.

Patterns for automation at scale

IDEM
Idempotent intent
Running the same job twice yields the same result and no extra changes.

TXN
Transactions
Stage, validate and commit per device; roll back automatically on failure.

DIFF
Dry-run diffs
Show the exact configuration diff for review before any commit.

BATCH
Batches and canaries
Change a small canary group, verify, then expand in batches with a stop condition.

CONC
Bounded concurrency
Limit parallel sessions so controllers and devices are not overwhelmed.

ERR
Explicit error handling
Treat partial failure as failure: report, roll back and stop the batch.

WATCH OUT · PUT REPLACES, PATCH MERGES
In RESTCONF, PUT replaces the target resource with exactly what you send, and POST creates a child resource. PATCH merges. A PUT to the whole interfaces container with one interface in the body can delete the configuration of every other interface. Target the narrowest resource, use PATCH for additive changes, and test destructive methods only in a lab.

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SCENARIO · DECIDE · A RESTCONF CALL THAT REMOVED TOO MUCH

An engineer wanted to update the description on one interface of 400 access routers. The script sent a PUT to /restconf/data/ietf-interfaces:interfaces with a body containing only that interface. On the first ten routers, all other interface configuration disappeared. What should the script do instead?

A. Keep using PUT but add every other interface to the body, copied from a backup taken last month.

B. Send a POST to the interfaces container, because POST never deletes any existing configuration.

C. Send a PATCH to the specific list entry, interface=<name>, containing only the description leaf.

D. Switch to screen-scraping the CLI, because model-driven interfaces are too risky for production use.

Circle your choice. Answer and feedback: module answer key.

Key terms

YANG
Data modelling language (RFC 7950) describing configuration and operational data as a tree.

RESTCONF
HTTP-based protocol that exposes YANG data as resources, usually with JSON.

Candidate datastore
A scratch configuration that can be edited and validated before being committed to running.

OpenConfig
Operator-led, vendor-neutral YANG models for configuration and telemetry.

On-change subscription
A telemetry mode that sends updates only when a value changes.

NETCONF
XML-based network configuration protocol over SSH with datastores, locks and transactions.

gNMI
gRPC network management interface providing Get, Set and Subscribe, widely used for streaming telemetry.

Confirmed commit
A commit that rolls back automatically unless a confirming commit arrives before a timeout.

NMDA
Network Management Datastore Architecture: separates intended configuration from operational state.

Key takeaways

- YANG defines the data; NETCONF, RESTCONF and gNMI move it with different strengths.
- Use candidate, validate and confirmed commit so a change that cuts you off rolls itself back.
- Use gNMI on-change for state and sampling for counters.
- RESTCONF PUT replaces and PATCH merges: target the narrowest resource.

KNOWLEDGE CHECK

1. A change might cut off the automation server's management access. Which NETCONF feature limits the damage best?

   A. A lock on the running datastore, so no other session can change the configuration meanwhile.

   B. A validate operation, so the device checks the syntax before it applies the new configuration.

   C. Commit confirmed with a timeout, so the device rolls back if no confirming commit arrives.

   D. A get-config operation, so the script stores a backup of the running configuration first.

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2. Which protocol and mode suit tracking BGP session state changes across 2,000 routers with minimal load?

   (A) RESTCONF GET every 10 seconds, so the collector always has the latest state of every session.

   (B) NETCONF get-config every minute, so the collector compares configured peers against the intent.

   (C) gNMI Subscribe in sample mode every second, so no state change can ever be missed by the collector.

   (D) gNMI Subscribe in on-change mode, so updates are sent only when a session state changes.

3. In a YANG tree, what does '+--ro oper-status' tell you?

   (A) It is a configuration leaf that is optional and has a default value.

   (B) It is a list key that must be supplied when creating a new entry.

   (C) It is operational state that can be read but not configured.

   (D) It is a deprecated node that will be removed in the next revision.

4. Why might a team use OpenConfig models for telemetry but native models for some configuration?

   (A) OpenConfig gives consistent paths across platforms, while native models cover features OpenConfig lacks.

   (B) OpenConfig models are read-only by design, so all configuration changes must use native models on every platform.

   (C) Native models cannot stream telemetry, so OpenConfig is the only option for subscriptions.

   (D) OpenConfig requires RESTCONF, while native models are limited to NETCONF on most platforms.

5. A NETCONF script edits the candidate on 50 routers, but 3 fail validation. What should a well-designed job do?

   (A) Commit on the 47 that validated, then retry the 3 failures in a loop until each of them finally succeeds.

   (B) Discard the candidate on the failed routers, report them, and stop or roll back according to the batch policy.

   (C) Commit on all 50 using a forced commit, because validation errors are usually only warnings.

   (D) Leave the candidate on the 3 routers unchanged, so the next engineer can commit it manually.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · CODE · LESSON 2 · 45 MIN

Source of Truth, Templating and Verification as Code

From structured intent to rendered configuration, and from show commands to assertions

- Design a source of truth that separates intent data from device configuration.
- Render configurations with Jinja2 templates that fail loudly on missing data.
- Write verification in Python and Ansible that asserts intended state rather than printing output.

Automation fails when data is scattered: addresses in a spreadsheet, VRFs in someone's notes, BGP peers only in running configs. A **source of truth (SoT)** holds intended state as structured data; **templates** turn that data into configuration; **verification code** proves the network matches the intent. Experts write verification first, because a change pipeline without automated checks merely breaks things faster.

From intent to proof

01 Intent
Design decisions captured as data: sites, VRFs, peers, RTs, QoS classes

→

02 Source of truth
IPAM/DCIM tool (for example the open-source NetBox) plus version-controlled YAML

→

03 Render
Jinja2 templates produce full or partial configurations per device

→

04 Validate
Schema checks on data, static analysis on rendered configs

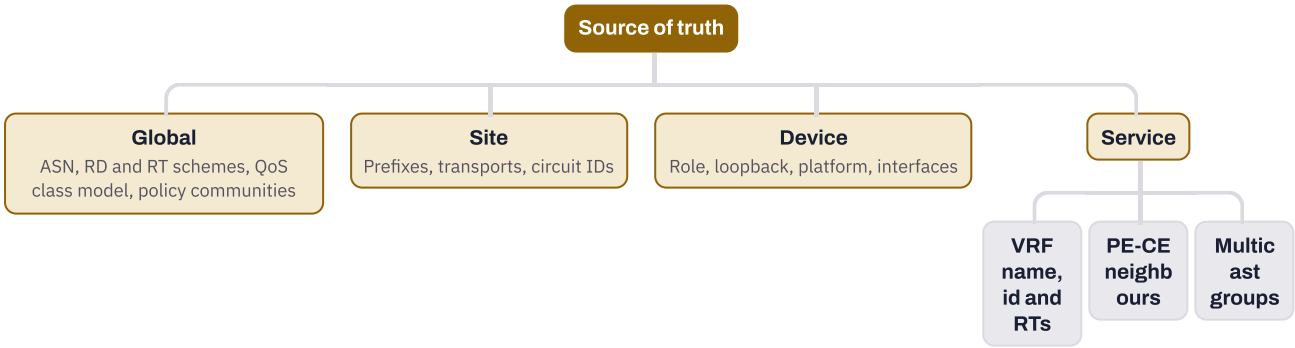
→

05 Deploy
Transactions with diffs and confirmed commits

→

06 Verify
Assertions on operational state compared with intent

Where each decision lives in the source of truth



Each decision has exactly one home; templates read from it and tests assert against it. Anything that exists only in a running configuration is drift.

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Intent data for one PE

intent/pe1.yml (YAML)

```
hostname: PE1
loopback: 10.255.0.1
asn: 65000
vrfs:
  - name: BLUE
    id: 101
    export_rts: ["65000:100"]
    import_rts: ["65000:100"]
    ce_neighbors:
      - {ip: 192.0.2.2, asn: 65010, as_override: true}
  - name: GREEN
    id: 102
    export_rts: ["65000:200"]
    import_rts: ["65000:200", "65000:999"]
    ce_neighbors: []
```

Jinja2 template for VRFs and PE-CE BGP

templates/l3vpn.j2 (Jinja2)

```
{% for vrf in vrfs %}
vrf definition {{ vrf.name }}
 rd {{ loopback }}:{{ vrf.id }}
 address-family ipv4
{% for rt in vrf.export_rts %}  route-target export {{ rt }}
{% endfor %}{% for rt in vrf.import_rts %}  route-target import {{ rt }}
{% endfor %}!
{% endfor %}
router bgp {{ asn }}
{% for vrf in vrfs if vrf.ce_neighbors %}
 address-family ipv4 vrf {{ vrf.name }}
{% for n in vrf.ce_neighbors %}  neighbor {{ n.ip }} remote-as {{ n.asn }}
  neighbor {{ n.ip }} activate
{% if n.as_override %}  neighbor {{ n.ip }} as-override
{% endif %}{% endfor %}{% endfor %}
```

Render with StrictUndefined so a missing variable stops the build instead of producing a silent blank. The RD is derived from the loopback, giving every PE a unique RD per VRF, as recommended in the Label module.

WATCH OUT · TEMPLATE PITFALLS THAT REACH PRODUCTION

Undefined variables that render as empty strings, whitespace control that glues two commands onto one line, defaults hidden in templates instead of the data, and logic so complex that nobody can review it. Keep templates simple, put decisions in data, enable strict undefined handling, and add a test that renders every device and compares the output with a known-good snapshot.

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Verification as tests (pytest)

tests/test_bgp.py (Python)

```
import pytest, yaml
from collector import get_bgp_neighbors  # thin wrapper over RESTCONF or gNMI
INTENT = yaml.safe_load(open("intent/bgp_expectations.yml"))
@pytest.mark.parametrize("dev", INTENT["devices"], ids=lambda d: d["name"])
def test_bgp_sessions(dev):
    state = get_bgp_neighbors(dev["host"])
    for peer, minimum in dev["peers"].items():
        assert peer in state, f"{dev['name']}: peer {peer} not configured"
        assert state[peer]["session_state"] == "ESTABLISHED", f"{dev['name']}: {peer} down"
        assert state[peer]["prefixes_received"] >= minimum, f"{dev['name']}: {peer} too few
prefixes"
```

Each assertion encodes an expectation from intent. A failed test names the device and peer, which turns a wall of show output into an actionable finding.

Verification with Ansible

verify_ldp.yml (Ansible playbook)

```
- name: Verify LDP peers after change
  hosts: pe_routers
  gather_facts: false
  tasks:
    - name: Collect LDP neighbours
      ansible.netcommon.cli_command:
        command: show mpls ldp neighbor
      register: ldp
    - name: Every expected peer is present
      ansible.builtin.assert:
        that: "('Peer LDP Ident: ' ~ item ~ ':0') in ldp.stdout"
        fail_msg: "LDP peer {{ item }} missing on {{ inventory_hostname }}"
      loop: "{{ expected_ldp_peers }}"
```

String matching on CLI output is fragile; prefer structured data from a model-driven interface or a parser. It is shown here because many brownfield networks start this way.

Configuration versus verification tasks

AREA	CONFIGURATION FROM SOT	VERIFICATION ASSERTION
IGP	Interfaces, metrics, authentication	Expected adjacencies up; loopbacks reachable with expected ECMP
Label	LDP or SR, VRFs, RTs	Labels for every PE loopback; VPN prefixes per VRF within expected range
Fabric	VLAN-to-VNI, ESIs, anycast gateway	Type-3 peers per VNI; ES peers and DF state; no duplicate MACs
Stream	PIM, RP, QoS policies	PIM neighbours on RPF paths; zero priority-class drops in tests
Security	AAA, ACLs, management plane	Only approved sessions and services reachable

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SCENARIO · DECIDE · THE EMERGENCY FIX THAT DISAPPEARED

During an outage an engineer hand-configured a static route on two PEs. The next night, the automated pipeline re-rendered configurations from the source of truth and removed the route, and the outage returned. What is the best process fix?

(A) Stop running the pipeline at night, so manual fixes have time to be reviewed by the engineering team.

(B) Exclude static routes from automation, so engineers can always manage them by hand on the devices.

(C) Require emergency changes to be added to the source of truth immediately afterward, and run drift detection that flags differences before each deployment.

(D) Give the pipeline a merge-only mode, so it never removes configuration that is not in the templates.

Circle your choice. Answer and feedback: module answer key.

Source-of-truth and verification checklist

☐ Every configurable decision has exactly one authoritative home

☐ Data is validated against a schema before rendering

☐ Templates use strict undefined handling and are snapshot-tested

☐ Rendered configurations are diffed against running before deployment

☐ Verification tests exist for every intent area changed

☐ Out-of-band changes are detected and reconciled before deployment

☐ Credentials come from a vault, never from repositories or scripts

☐ Test output names the device and expectation that failed

Key takeaways

- Keep intent as structured data with exactly one authoritative home per decision.
- Render with strict undefined handling and snapshot-test the templates.
- Verification asserts intended state per device and names what failed.
- Detect and reconcile drift before every deployment.

KNOWLEDGE CHECK

6. Why render with strict undefined handling in Jinja2?

(A) It speeds up rendering by skipping any blocks whose variables have not been defined in the intent data.

(B) It encrypts variables during rendering so that credentials never appear in the output files.

(C) A missing variable stops the build instead of silently producing an incomplete configuration line.

(D) It converts undefined variables into vendor defaults so the configuration is always complete.

7. What makes a verification check valuable in a pipeline?

(A) It asserts an expected state derived from intent and fails with a message naming device and expectation.

(B) It prints the complete show output from every device so engineers can read through the results later.

(C) It runs only after the change window closes so that it never delays the deployment of the change.

(D) It checks that the configuration was sent successfully, because acceptance proves the network works.

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8. Where should the decision 'PE1 imports RT 65000:999 into VRF GREEN' live?

   (A) In the Jinja2 template, as a conditional block that adds the RT when the hostname is PE1.

   (B) In the source-of-truth data for PE1, so the template renders it and tests can assert it.

   (C) In the running configuration only, because the device is the final authority on its own state.

   (D) In the change ticket, because tickets are where engineers look first for design decisions.

9. A team scrapes CLI text in Ansible assertions. What is the main risk?

   (A) Ansible cannot connect to network devices without an agent installed on every single router first.

   (B) CLI commands cannot be run in check mode, so every assertion applies configuration.

   (C) String assertions always pass because Ansible ignores the output of show commands.

   (D) Output format changes between software releases can break or falsely pass string matches.

10. Which pairing of configuration and verification is correct for an EVPN fabric?

    (A) Configure ESIs on every leaf; assert that the number of BGP sessions on the spines has increased.

    (B) Configure the anycast gateway MAC; assert that every leaf has a unique gateway MAC address.

    (C) Configure route targets; assert that each leaf advertises its own RD to the other leaves as a type-5 route.

    (D) Configure VLAN-to-VNI and ingress replication; assert that every leaf sees the expected type-3 peers per VNI.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · CODE · LESSON 3 · 50 MIN

CI Pipelines, Closed-Loop Operations and AI-Assisted Operations

Testing changes before production, acting on telemetry safely, and using AI assistance with guardrails

- Design a network CI pipeline with data validation, static analysis, virtual lab tests and staged deployment.
- Build telemetry-driven closed-loop automation with preconditions, blast-radius limits and verification.
- Define guardrails for AI-assisted operations so suggestions are useful and never unsafe.

A pipeline is the bridge between intent and production. It runs the same checks on every change, regardless of who wrote it or how urgent it is. Closed-loop operations extend the idea: telemetry detects a condition, automation acts and then verifies. AI-assisted tooling, now common in 2026 operations, belongs **inside** these guardrails, never around them.

A network CI/CD pipeline

01 Lint and schema — YAML lint, schema validation of intent data, policy rules such as unique RDs

→

02 Render — Generate configurations for every affected device

→

03 Static analysis — Model the network offline (for example the open-source Batfish) and test reachability, ACLs and BGP policy

→

04 Virtual lab — Deploy to a container-based lab (for example containerlab) and run verification tests

→

05 Approve — Human review of the diff and test report

→

06 Canary and batches — Confirmed commits on a small group, verify, then expand

→

07 Post-checks — Automated verification; automatic rollback on failure

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Pipeline definition

.ci/pipeline.yml (generic CI runner)

```
stages: [validate, render, analyse, lab, deploy, verify]
validate-data:
  stage: validate
  script: ["yamllint intent/", "python -m pytest tests/schema"]
render-configs:
  stage: render
  script: ["python render.py --out build/"]
static-analysis:
  stage: analyse
  script: ["python -m pytest tests/model --snapshot build/"]
lab-test:
  stage: lab
  script: ["containerlab deploy -t lab/core.clab.yml", "python deploy.py --target lab", "python -m
pytest tests/verify --target lab"]
deploy-canary:
  stage: deploy
  when: manual
  script: ["python deploy.py --target prod --scope canary --confirm-timeout 300"]
verify-canary:
  stage: verify
  script: ["python -m pytest tests/verify --target prod --scope canary"]
```

The manual gate sits after all automated evidence is available, so the approver reviews results, not intentions.

What each test layer catches

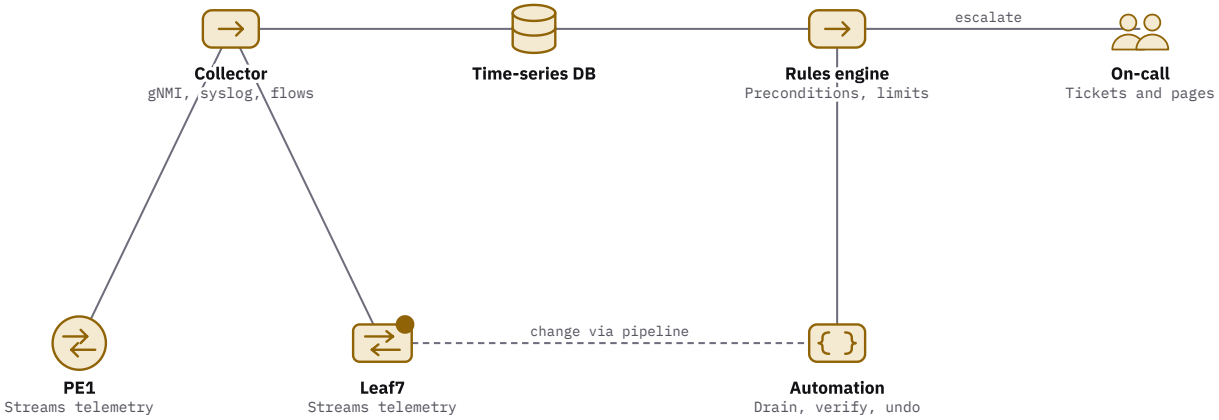
LAYER	CATCHES	MISSES
Schema and policy checks	Duplicate RDs, missing fields, forbidden values	Behaviour of the routing protocols
Snapshot tests of rendered configs	Unintended template changes	Whether the config works
Static network analysis	Reachability, ACL and route-policy errors, BGP and IGP outcomes	Software bugs and timing behaviour
Virtual lab	Real protocol behaviour, convergence, many integration errors	Hardware-specific features and scale
Canary in production	Real hardware and traffic	Problems that appear only at full scale

Telemetry-driven operations

Action	`Remediation runbooks, tickets, change requests`
Decision	`Rules and thresholds, anomaly detection, correlation`
Storage and query	`Time-series database (for example Prometheus or InfluxDB), logs, events`
Collection	`gNMI collectors (for example Telegraf or gnmic), syslog, flow records`
Network	`Devices streaming counters, state and events`

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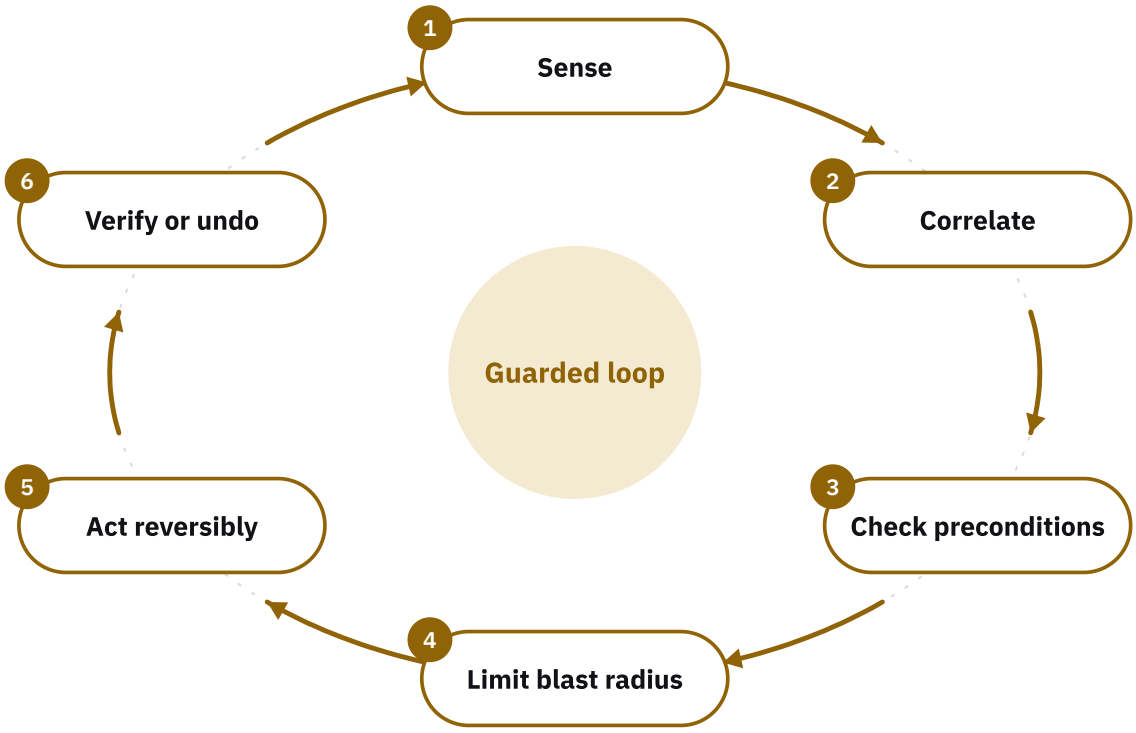
Telemetry-driven operations



Telemetry flows up from the devices; the rules engine decides, the automation layer acts through the same pipeline as any change, and anything outside its guardrails escalates to people.

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Closed loop with guardrails



1. **Sense** — Stream the metrics and states that define the condition, with timestamps
2. **Correlate** — Confirm it persists and is not explained by a change in progress
3. **Check preconditions** — Redundancy exists, no change freeze, target healthy enough
4. **Limit blast radius** — One action per device, N per network per hour, never both sides of a pair
5. **Act reversibly** — Drain with metrics or the overload bit rather than shutting
6. **Verify or undo** — Confirm the effect by a deadline; roll back and escalate if not

Every turn must pass the guardrails before acting and must verify afterwards; a loop that cannot verify its effect undoes the action and hands over to people.

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A remediation rule with preconditions

remediation/uplink-errors.yml (rule definition)

```
rule: drain-erroring-uplink
trigger:
  metric: interface_in_errors_per_second
  condition: "> 50 for 5m"
preconditions:
  - peer_uplinks_up(device) >= 1
  - change_freeze == false
  - actions_last_hour(network) < 3
  - actions_last_day(device) == 0
action:
  - set_isis_metric(interface, 16777214)
  - open_ticket(severity: 3, attach: [metrics, lldp_neighbor])
verify:
  - traffic_on(interface) < 5% within 120s
on_verify_fail: [restore_metric, page_on_call]
```

Draining with a near-maximum metric keeps the link up for diagnosis and is instantly reversible. The device and network rate limits stop a bad rule from cascading.

AI-assisted operations in 2026

Good uses

- Summarising incidents from logs, telemetry and tickets
- Drafting configurations or tests for human review
- Explaining unfamiliar output and suggesting next diagnostic steps
- Searching runbooks and past incidents

Required guardrails

- Read-only access by default; no direct writes to devices
- AI agents that call tools get read-only tools by default; any write goes through the change pipeline with human approval
- Every suggested change goes through the same pipeline and approval
- Outputs cite their sources; engineers verify syntax and facts
- Sensitive data minimised and governed; prompts and actions logged

WATCH OUT · PLAUSIBLE IS NOT CORRECT

Generated configurations can look perfect and still use a command that does not exist on your platform, an RT from another customer or a wrong mask. Treat any generated content like a change from a new team member: it must pass schema checks, static analysis, lab tests and peer review. Never paste credentials, customer data or full configurations into tools that are not approved for that data.

CASE STUDY · A BANK'S AUTOMATED CHANGE PIPELINE

A retail bank operates 1,200 branches over SD-WAN, two data centres with EVPN fabrics and an MPLS core. Change failure rate is 14%, mostly typos and inconsistent route maps, and change windows run long because every change needs manual pre- and post-checks. Regulators demand evidence of testing, segregation of duties and the ability to roll back. The operations team wants to pilot an AI assistant that drafts configurations from tickets.

1. What pipeline stages would you introduce first, and why in that order?

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2. How do you satisfy segregation of duties and evidence requirements?

3. How should the AI assistant be integrated?

4. Which metrics show the programme is working?

Model answer: module answer key.

SCENARIO · DECIDE · AUTO-REMEDIATION MADE IT WORSE

A closed-loop rule shuts any uplink whose CRC error rate exceeds a threshold. During a faulty optics batch, the rule shut both uplinks of three leaf switches within minutes, isolating dozens of servers. What is the most important change to the rule?

(A) Raise the CRC threshold so that the rule triggers less often during periods of optics problems.

(B) Replace the rule with an alert to the on-call engineer and remove all automated remediation.

(C) Make the rule re-enable interfaces automatically after ten minutes so outages are always short.

(D) Add preconditions and blast-radius limits: act only if a healthy redundant uplink remains, rate-limit actions, and drain rather than shut.

Circle your choice. Answer and feedback: module answer key.

REFLECT

If an AI assistant proposed a change to your production network tomorrow, which checks would it pass through today before reaching a device? List the gaps and decide which single guardrail you would add first.

Key terms

Source of truth
The authoritative, structured record of intended network state from which configurations are generated.

Virtual lab
A software-based replica of part of the network used to test changes before production.

Drift
A difference between the source of truth and the running network.

Static network analysis
Offline modelling of configurations to predict routing, reachability and policy outcomes.

Canary deployment
Applying a change to a small subset first and verifying before expanding.

Closed-loop automation
Automation that detects a condition from telemetry, acts and verifies the result.

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Blast radius
The scope of impact if an automated action is wrong.

Guardrail
A control that limits what automation or AI assistance can do, such as preconditions, approvals and rate limits.

Key takeaways

- A pipeline runs the same checks on every change: schema, rendering, static analysis, virtual lab, approval, canary and post-checks.
- Closed-loop automation needs preconditions, blast-radius limits, reversible actions and verification.
- Drain rather than shut, and never let automation remove the last path.
- AI assistance drafts and explains inside the pipeline; it never bypasses checks or approval.

KNOWLEDGE CHECK

11. Which pipeline stage would catch a route map that accidentally denies a customer's default route, before any device is touched?

(A) Static network analysis of the rendered configurations, which evaluates route-policy outcomes offline.

(B) YAML linting of the intent data, which checks that every route map name follows the convention.

(C) Canary deployment, which applies the route map to a small set of production routers and watches the result.

(D) Snapshot testing, which shows that the rendered route map differs from the previous version.

12. Why is draining a link with a maximum metric often preferred to shutting it in closed-loop remediation?

(A) It removes the link from the routing table immediately, which converges faster than a shutdown.

(B) It moves traffic away while keeping the link up for diagnosis, and it is quickly reversible.

(C) It disables error counters on the link, which stops the remediation rule from triggering again.

(D) It prevents LDP from running on the link, which avoids label black holes during the incident.

13. A closed-loop rule fires 40 times in an hour across the network. Which guardrail should have stopped it?

(A) A per-interface threshold that is unique for every link in the network design.

(B) A longer telemetry sampling interval so the rule sees fewer data points per hour.

(C) A network-wide action rate limit that halts automation and escalates to humans when exceeded.

(D) A dashboard that shows how many times each rule has fired during the last week.

14. How should an AI assistant that drafts configurations be integrated into a regulated change process?

(A) It applies changes directly in a maintenance window, because it is faster than a human engineer.

(B) It submits drafts as merge requests that pass the same checks and human approval as any change.

(C) It bypasses static analysis, because its output is already validated by the model that produced it.

(D) It approves its own changes, because segregation of duties applies only to human engineers.

15. What is the main limitation of a virtual lab compared with a production canary?

(A) It cannot run real routing protocols, so convergence behaviour cannot be observed at all.

(B) It cannot apply configuration through NETCONF, so pipelines must fall back to CLI screen-scraping in the lab.

(C) It cannot run tests automatically, so engineers must check results manually after deployment.

(D) It cannot reproduce hardware-specific behaviour or full scale, so some faults only show in production.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · CODE · LESSON 4 · 60 MIN

Mock Practical Lab: A Walk-Through of the Full Mastery Ring

One integrated lab touching all eight ring segments, solved with expert method

- Read an integrated lab, map task dependencies and plan time boxes before configuring.
- Work through tasks across all eight ring segments with verification after each.
- Score your own attempt honestly with a structured scoring sheet.

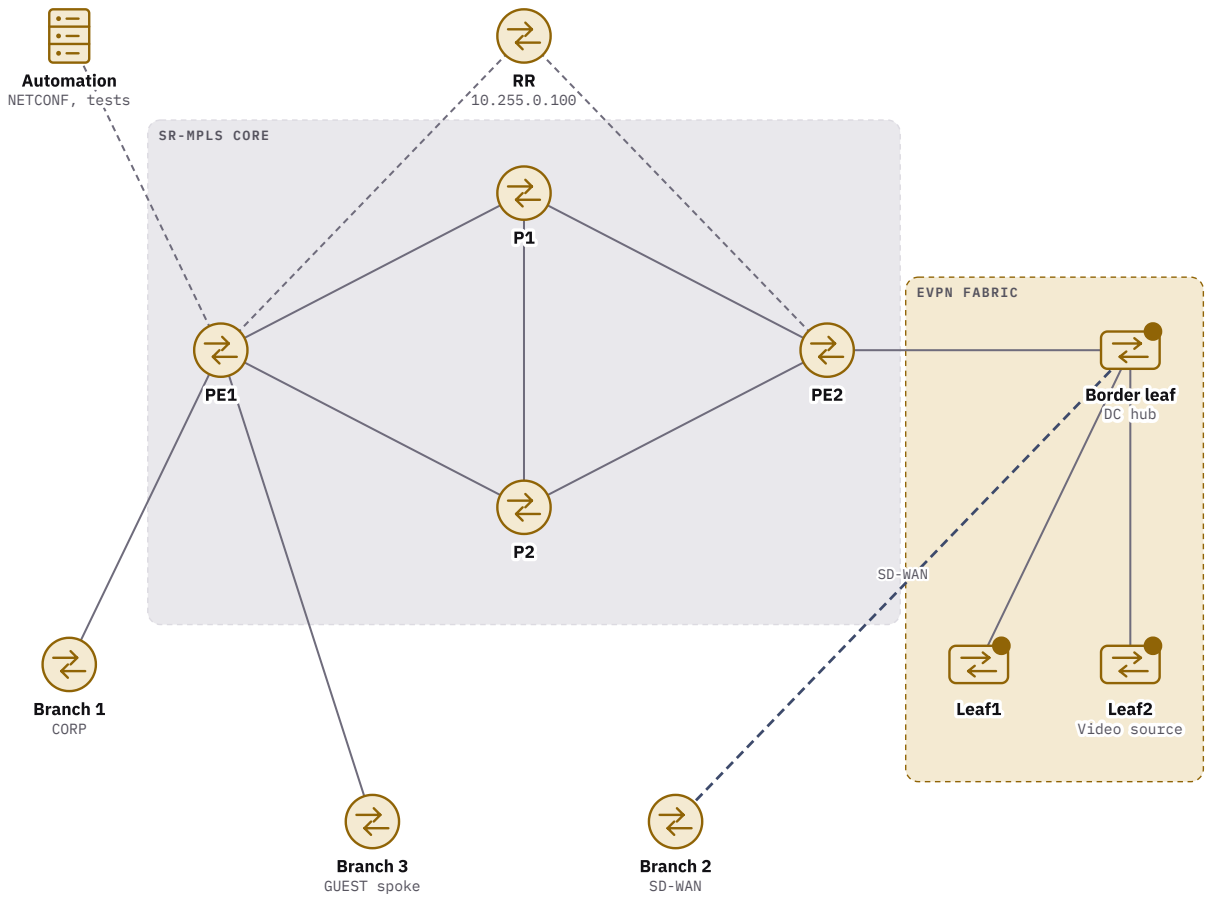
This lesson walks through a complete mock practical lab built around the ZCIE Mastery Ring. The topology is a fictional company, **Kestrel Freight**: an MPLS core with two PEs and two P routers, a data-centre EVPN fabric with two leaves and a border leaf, three branch sites (one on SD-WAN), an IPTV-style training-video service and an automation host. The mock runs for six hours and is worth 100 points. Read the whole walk-through once, then attempt the lab yourself in a virtual environment.

Mock lab topology by ring segment

Code	`Automation host with NETCONF/RESTCONF access to PEs` `Verification scripts in a repository`
Stream	`SSM video from DC to branches` `WAN QoS policy at branch edges`
Fabric	`Two leaves + border leaf, VXLAN EVPN symmetric IRB` `One SD-WAN branch reaching the DC via a hub`
Label	`PE1, PE2, P1, P2 with SR-MPLS and TI-LFA` `VRFs CORP and GUEST, hub-and-spoke for GUEST`
BGP and IGP	`IS-IS level 2 core; OSPF at branches` `VPNv4 via route reflector; eBGP PE-CE`
Layer2 and Method	`Branch access switching with LACP and spanning tree` `Dependency map and verification plan`

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Kestrel Freight mock lab



Every task maps onto this drawing: the SR-MPLS core and reflector (T2 to T4), the EVPN fabric behind PE2 (T5), the SD-WAN branch reaching the DC hub (T6), SSM video from the DC to branches (T7) and the automation host (T8).

Task list and points

TASK	RING SEGMENT	POINTS	DEPENDS ON	VERIFICATION
T1 Branch access bundles and STP root placement	Layer2	8	-	Bundle up, root as designed
T2 IS-IS core with SR and TI-LFA	IGP / Label	14	-	Prefix SIDs, backup paths present
T3 VPNv4 RR and PE-CE eBGP for CORP	BGP / Label	16	T2	Site routes in remote VRFs, two-label FIB
T4 GUEST hub-and-spoke via firewall	Label	10	T3	Spoke-to-spoke trace via hub
T5 EVPN fabric with type-5 to core	Fabric	16	T3	Type-2/3/5 present, host reachability
T6 SD-WAN branch via hub, app policy	Fabric	8	T3	Voice on preferred transport
T7 SSM video and branch QoS	Stream	14	T3, T5	(S,G) state, zero voice drops under load

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TASK	RING SEGMENT	POINTS	DEPENDS ON	VERIFICATION
T8 NETCONF change plus verification script	Code	14	T3	Confirmed commit, tests pass

The walk-through

1. **First 25 minutes: read and map** Read every task. Draw the dependency map: T2 underpins T3, which underpins T4 to T8. Note every verification statement; they define 'done'. Flag ambiguous wording and the assumptions you will make.
2. **T1 and T2: foundations (60 minutes)** Configure bundles and root bridges, then IS-IS with SR. Verify adjacencies, prefix SIDs and TI-LFA backups before moving on, because every later task rides on them.
3. **T3: the core service (45 minutes)** VRF CORP with unique RDs per PE, VPNv4 to the RR, eBGP to CEs with as-override where sites share an AS. Verify routes in remote VRFs and the two-label stack in the FIB.
4. **T4 to T6: topologies and overlays (90 minutes)** Build the GUEST hub-and-spoke with two hub VRFs. Bring up the fabric and advertise DC prefixes as type 5 via the border leaf. Join the SD-WAN branch through the hub and set the voice SLA policy.
5. **T7: stream (40 minutes)** Enable PIM and SSM along the RPF path from the DC source, IGMPv3 at the branch. Apply the hierarchical QoS policy and prove it under a load test.
6. **T8: code (40 minutes)** Push the requested change through NETCONF with a confirmed commit, then run and fix the verification script until it passes.
7. **Final 20 minutes: re-verify everything** Re-run every verification from T1 to T8, because late tasks can break early ones. Save configurations.

A quick re-verification sweep

PE1 · IOS-style CLI

```
PE1# show isis neighbors | count Up
Number of lines which match regexp = 2
PE1# show bgp vpnv4 unicast all summary | include 10.255.0.100
10.255.0.100    4 65000   8812   8790   1442    0    0 2d01h        38
PE1# show ip cef vrf CORP 172.16.30.0/24 | include label
  recursive via 10.255.0.2 label 24210
    nexthop 10.1.11.11 GigabitEthernet0/1 label 16002-(local:16002)
PE1# show ip mroute 232.5.5.5 | include flags
(10.60.0.10, 232.5.5.5), 00:21:14/00:02:44, flags: sT
```

Four commands confirm four layers: IGP adjacencies, the VPNv4 session and prefix count, the transport plus service label, and the SSM tree. Build your own sweep for every lab and run it twice: halfway and at the end.

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Troubleshooting ticket inside the mock

```
PE2 · IOS-style CLI

! Ticket: Branch 3 (GUEST) cannot reach the guest portal behind the hub.
PE2# show bgp vpnv4 unicast vrf GUEST-SPOKE 10.90.0.0/24
% Network not in table
PE2# show bgp vpnv4 unicast all | include 10.90.0.0
 *>i 10.255.0.1:202   10.90.0.0/24   10.255.0.1   0   100   0 65090 i
PE2# show run vrf GUEST-SPOKE | include route-target
  route-target export 65000:201
  route-target import 65000:201
```

The hub route exists in BGP but is not imported: the spoke VRF imports its own spoke RT instead of the hub RT 65000:202. One line fixes it, and the spoke-to-spoke traffic must still be checked to traverse the hub.

Dependency-map rules

ROOT

Find the roots

Tasks with no dependencies and many dependants (here T2) are done first and verified hardest.

PATH

Critical path

The longest chain of dependent tasks sets the minimum time; protect it.

LEAF

Leaves are optional

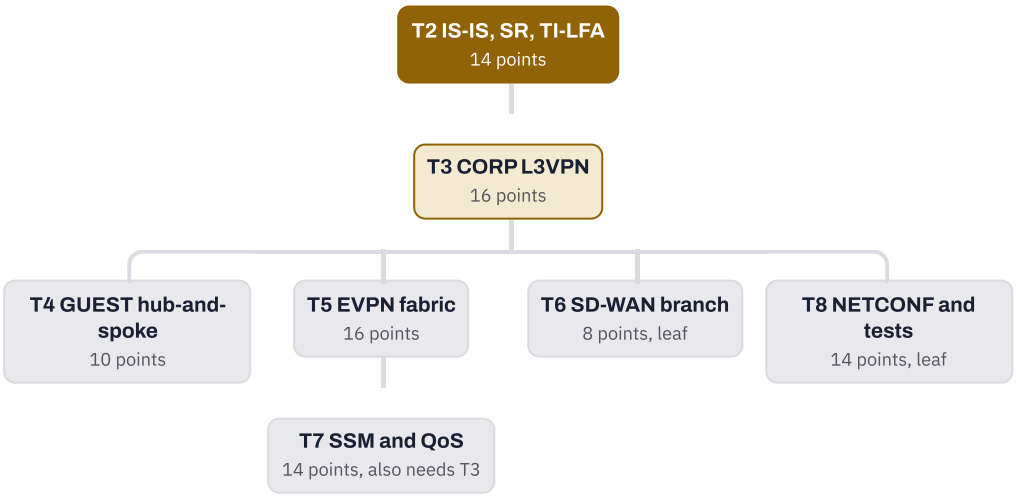
Tasks nothing depends on (T6, T8) can be deferred or dropped if time runs out.

REGR

Regression risk

Note which later tasks touch earlier configuration (QoS on PE-CE links, NETCONF on PEs).

The mock lab's dependency map



T2 is the root of almost everything and T7 ends the longest chain. T1 is independent, while T6 and T8 are leaves that can be deferred if time runs out.

TEMPLATE

Mock-lab scoring sheet

Score each task honestly against its verification statement, not against how much you configured.

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TASK	POINTS AVAILABLE	TIME BOX (MIN)	TIME USED (MIN)	STATUS (FULL / PARTIAL / NONE)	VERIFICATION EVIDENCE	LESSON LEARNED
T3 VPNv4 and PE-CE for CORP	*16*	*45*	*58*	*Full*	*Remote VRF routes; FIB shows 24210 over 16002*	*Typed wrong RT once; verify RTs before BGP*
T7 SSM and QoS	*14*	*40*	*35*	*Partial*	*(S,G) up; QoS test not run*	*Leave time for load tests*

SCENARIO · DECIDE · FORTY MINUTES LEFT, TWO TASKS OPEN

With 40 minutes left you have T6 (8 points, SD-WAN app policy, half done) and T8 (14 points, NETCONF change and verification script, not started). The final re-verification of earlier tasks has not been done. What is the best plan?

(A) Spend 15 minutes re-verifying T1 to T5 and T7, then use the remainder on T8, which is worth more and independent of T6.

(B) Finish T6 first because it is half done, then start T8 if time remains, skipping the final verification sweep.

(C) Start T8 immediately and use all 40 minutes on it, because it is the highest-value task still open.

(D) Use the full 40 minutes to re-verify everything twice, because accuracy matters more than new tasks.

Circle your choice. Answer and feedback: module answer key.

Mock lab self-review

☐ I read every task before configuring anything

☐ I drew a dependency map and set time boxes

☐ I verified each task against its stated verification before moving on

☐ I recorded assumptions for ambiguous requirements

☐ I did not change defaults or configuration outside task scope

☐ I ran a full re-verification sweep at the end

☐ I scored myself against evidence, not effort

☐ I wrote down three lessons for the next attempt

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Key takeaways

- Read the whole lab and draw the dependency map before configuring anything.
- Verify each task against its own verification statement before moving on.
- Protect points already earned with a sweep before chasing new ones.
- Score yourself against evidence, not effort.

KNOWLEDGE CHECK

16. In the mock, why is T2 (IS-IS with SR) treated as the highest-priority task despite not being the highest-scoring?

(A) It is the fastest task to configure, so finishing it early builds confidence for the rest.

(B) It is the only task that can be verified with show commands rather than with test traffic.

(C) Most other tasks depend on it, so an error there silently destroys points across the lab.

(D) It must be configured last, so it can be checked against every other task at the end.

17. The GUEST spoke VRF shows the hub route in the VPNv4 table but not in the VRF. What is the fastest correct diagnosis?

(A) The spoke VRF's import RT does not match the RT the hub exports, so the route is never imported.

(B) The hub PE uses a different RD, so the spoke PE cannot import its routes into the spoke VRF.

(C) The route reflector is filtering the route, so it reaches the global table but not the VRF.

(D) The transport label to the hub PE is missing, so BGP refuses to import the route into any VRF.

18. What is the purpose of a re-verification sweep at the end of a practical lab?

(A) To reformat configurations so the grader can read them more easily.

(B) To catch earlier tasks broken by later changes before the lab ends.

(C) To reload devices so that every configuration is saved and active.

(D) To add optional features that might earn partial credit on tasks.

19. A task's verification statement says 'spoke-to-spoke traffic traverses the hub'. Which evidence best proves it?

(A) The spoke VRF route table showing the hub route as the only route for that prefix.

(B) The hub PE configuration showing two VRFs with different import and export RTs.

(C) The BGP summary on the spoke PE showing the session to the reflector established.

(D) A traceroute between two spokes that shows the hub firewall hop in the path.

20. How should you score a task where you configured everything but the verification test fails?

(A) As fully achieved, because the configuration effort clearly shows that you understand the topic.

(B) As fully achieved if the failure is caused by a task completed by someone else.

(C) As not achieved or partial at best, because tasks are scored against the stated verification.

(D) As not attempted, because a failed verification means the work did not happen.

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · CODE · LESSON 5 · 35 MIN

Exam-Day Strategy and the ZCIE Capstone

Time, triage, verification discipline and the final project that proves expert practice

- Apply a time and triage strategy to a long practical exam.
- Handle ambiguous requirements and troubleshooting tickets without collateral damage.
- Plan the ZCIE capstone deliverables and how they will be assessed.

Expert practical exams are as much about judgement under time pressure as about technical depth. Candidates who fail often know the material but lose points to unread requirements, unverified work, rabbit holes and changes that break earlier tasks. This lesson turns the Method segment of the Mastery Ring into an exam-day routine and then introduces the **ZCIE capstone**, the project that closes the programme.

Exam-day timeline for a six-hour lab

DAY BEFORE
Prepare
Sleep, logistics, a familiar keyboard and environment

0:00
Read and map
Full read-through, dependency map, points per task, assumptions

0:25
Build
Foundations first; verify each task immediately

3:00
50% checkpoint
Progress against plan; re-prioritise

4:48
80% checkpoint
Defer or drop low-value leaf tasks

5:40
Sweep
Re-verify everything and save configurations

6:00
End

Marks for a 360-minute lab with 25 minutes of reading and a 20-minute sweep, matching the time budget below.

Time budget

MEASURE	FORMULA	WHAT IT TELLS YOU
Minutes per point	`(total time − read time − sweep time) ÷ total points`	(360 − 25 − 20) ÷ 100 = 3.15 minutes per point; a 14-point task gets about 44 minutes.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Rabbit-hole limit	`time box × 1.5`	When a task exceeds 150% of its box, park it, note where you are and move on.

Rules that protect points

READ

Read the whole task

Constraints often appear in the last sentence: 'without static routes', 'using only the provided loopbacks'.

SCOP

Stay in scope

Never change defaults, timers or configuration not requested; graders and later tasks may depend on them.

VERI

Verify immediately

Use the task's own verification statement as your test; save configuration after each success.

PARK

Park, do not sink

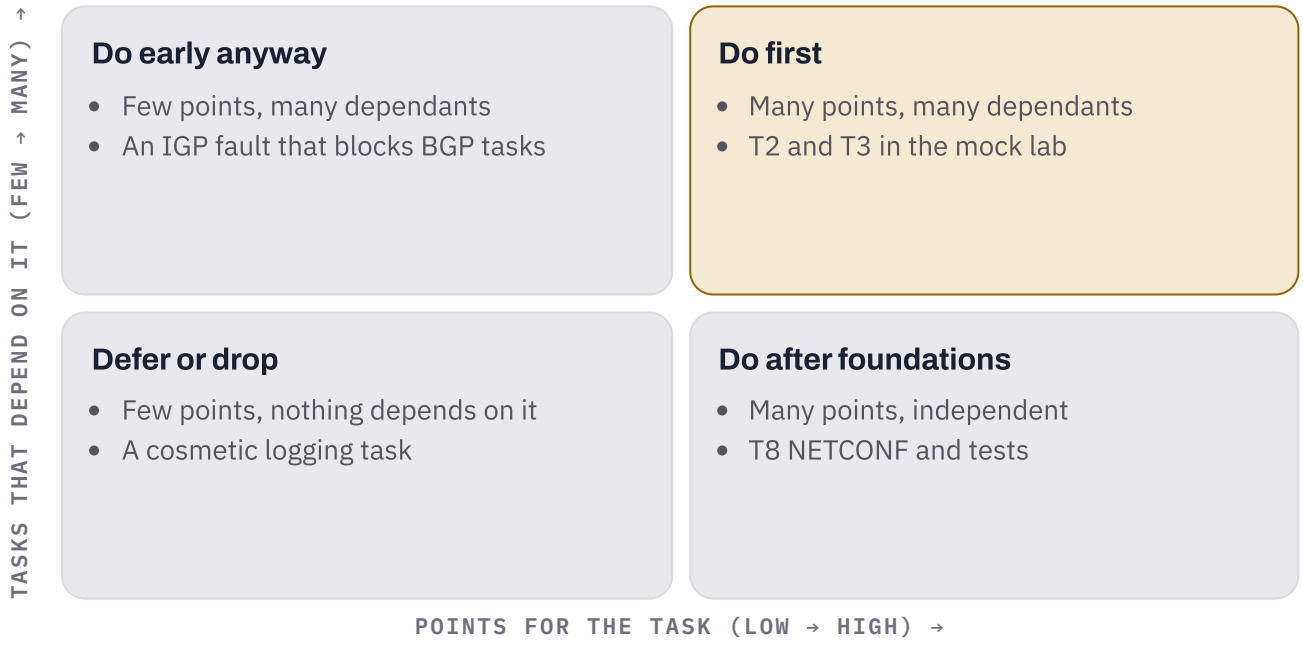
Use the rabbit-hole limit; return later with fresh eyes.

NOTE

Write assumptions

When wording is ambiguous, choose the minimal literal interpretation and record it.

Task triage under time pressure



Order by dependants first and points second: a cheap task that unlocks others belongs early, while an expensive independent task waits until foundations are proven.

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A personal verification sweep, one line per ring segment

Exam notes · IOS-style CLI

```
! Layer2: bundles, roots, blocked ports
show etherchannel summary | include SU
show spanning-tree root
! IGP: adjacencies and loopback reachability
show isis neighbors
show ip route 10.255.0.0 255.255.0.0 longer-prefixes
! BGP: sessions and prefix counts
show bgp all summary
! Label: labels for every PE loopback, VPN routes per VRF
show mpls forwarding-table | include 10.255.0.
show bgp vpnv4 unicast all summary
! Fabric: EVPN peers and VNIs
show nve peers
! Stream: PIM neighbours, RP mapping, QoS drops
show ip pim neighbor
show ip pim rp mapping
show policy-map interface | include Class-map|drops
```

Memorise a sweep like this and adapt it to the lab topology. Run it at the halfway checkpoint and again in the final minutes; differences between the two runs are the first place to look.

Troubleshooting tickets under exam conditions

1. **Reproduce** Confirm the symptom exactly as stated, from the stated source to the stated destination.
2. **Bound** Decide which ring segment the symptom lives in and which device holds the evidence.
3. **Narrow** Use the layered methods from this programme: underlay first, then labels, services, overlays, policies.
4. **Fix minimally** Change the smallest thing that restores the intended design; do not redesign.
5. **Prove and protect** Re-run the ticket's test and check that nothing else changed.

Exam-day habits

Do

- Plan time boxes and keep them
- Verify every task against its statement
- Save configurations regularly
- Keep a running list of parked items and assumptions

Avoid

- Configuring while still reading
- Fixing unrelated 'improvements' you notice
- Reloading devices without need
- Spending an hour on a five-point task

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SCENARIO · DECIDE · AN AMBIGUOUS REQUIREMENT

A task says: 'Ensure branch 2 prefers the MPLS path for all traffic to the data centre.' It does not say how. Branch 2 runs eBGP to its PE and also has an SD-WAN tunnel via the hub. What is the best approach?

(A) Shut down the SD-WAN tunnel on branch 2, because then MPLS is the only path and is preferred automatically.

(B) Use the least invasive standard tool on branch 2, such as local preference on routes from the MPLS PE, verify with a trace, and note the assumption.

(C) Change IS-IS metrics in the provider core, so the MPLS path toward the data centre looks shorter everywhere.

(D) Ask the proctor which method to use and wait for an answer before doing anything else on the task.

Circle your choice. Answer and feedback: module answer key.

The ZCIE capstone is an expert mock lab across the whole Mastery Ring. You build and document a lab of fifteen or more nodes for a fictional national carrier and its enterprise customers, then break and fix it under time pressure. It is assessed against a rubric with one criterion per ring segment, rising from trial and error, through structured and correct, to fast, evidence-based and proven. Volume of configuration earns nothing on its own; verified behaviour and clear reasoning do.

Capstone deliverables

D1

Topology and core design

Lab topology and addressing for 15+ nodes, and an IGP design with fast convergence (BFD, LFA or TI-LFA) proven by measurement.

D2

Routing and services

A BGP route-reflection and community policy framework, plus an MPLS L3VPN or SR-MPLS service for three customers.

D3

Fabric and streams

A VXLAN EVPN fabric with anycast gateway, a PIM-SM or SSM service with RP redundancy, and an end-to-end QoS policy with validation.

D4

Automation

An automated verification suite built with Python, Ansible and templating, run before and after every change.

D5

Pressure test and summary

A timed troubleshooting log of eight faults found and fixed, and an executive summary with lessons learned.

Exam-day kit

☐ Personal verification sweep per ring segment memorised
☐ Time budget calculated for the published total time and points
☐ Rabbit-hole rule agreed with yourself in advance
☐ Assumptions log ready (paper or notes file)
☐ Habit of saving configuration after each verified task
☐ Plan for the 50% and 80% checkpoints
☐ Rest, food and water planned for a long day

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REFLECT

Look back over all eight segments of the ZCIE Mastery Ring. Which segment would cost you the most points if the exam were tomorrow? Write a two-week plan that closes that gap, including one full mock lab under timed conditions.

Key takeaways

- Plan time boxes from the published duration and points, with checkpoints at 50% and 80%.
- Stay in scope, verify immediately and save after every verified task.
- For ambiguous wording choose the minimal literal interpretation and record the assumption.
- Fix tickets minimally and prove that nothing else changed.

KNOWLEDGE CHECK

21. With 360 minutes, 25 for reading, 20 for the final sweep and 100 points, how long should a 10-point task take?

(A) About 32 minutes, using roughly 3.15 minutes per point.

(B) About 36 minutes, using 3.6 minutes per point over the full exam time.

(C) About 45 minutes, using a fixed 1.5 multiplier on every single task.

(D) About 20 minutes, using two minutes per point for any core task.

22. During a task you notice an unrelated OSPF timer that you would normally tune. What should you do?

(A) Tune it, because experts are expected to improve every configuration they touch.

(B) Tune it only on one router, so that the change has a smaller effect if it is wrong.

(C) Leave it, because changes outside task scope can break other tasks or grading checks.

(D) Remove it, because non-default timers are always a sign of an intended fault.

23. A troubleshooting ticket is fixed by redesigning the routing between two sites. What is the risk?

(A) The redesign will be graded higher, because it shows more effort and skill than a minimal fix would.

(B) The redesign cannot be verified because ticket tests check only the configuration lines.

(C) The redesign will always be reverted automatically when the lab reaches its end time.

(D) The redesign may restore the symptom but violate the intended design and break other tasks.

24. Which capstone deliverable shows that you can restore the network under time pressure, not only build it?

(A) The timed troubleshooting log of eight faults found and fixed.

(B) The lab topology and addressing plan for more than fifteen nodes.

(C) The BGP route-reflection and community policy framework document.

(D) The executive summary with the lessons learned from the build.

25. At the 50% checkpoint you have earned fewer points than planned. What is the best response?

(A) Skip the final verification sweep, so the time can be spent on configuring more tasks.

(B) Re-prioritise remaining tasks by value and dependency, and drop or defer low-value leaf tasks.

(C) Work faster on every remaining task without verifying, then verify all at the very end.

(D) Return to the first task and redo it, because early mistakes are probably the cause.

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☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · CODE

Answer key

Knowledge checks

1. C Only confirmed commit recovers automatically when connectivity is lost. Locks prevent conflicts, validate catches errors but not lockouts, and a backup needs a working session to restore.

2. D On-change streaming is efficient for discrete state. Polling and high-frequency sampling add load, and get-config returns configuration, not operational state.

3. C 'ro' marks read-only operational data. Optional leaves carry a question mark, keys appear in brackets after the list name, and deprecation is shown differently.

4. A Consistency versus coverage is the usual trade-off. OpenConfig includes configuration, native models can stream, and both work over several protocols.

5. B Partial failure must be handled explicitly and safely. Blind retries, forced commits and leftover candidates create unpredictable state.

6. C Fail-fast rendering prevents commands such as 'neighbor remote-as' reaching devices. It neither skips blocks, encrypts data nor invents defaults.

7. A Assertions turn data into pass or fail. Printed output needs human review, late checks miss the rollback window, and accepted configuration can still be wrong.

8. B Decisions belong in data. Hard-coding them in templates hides intent, and running configs and tickets are not structured, authoritative sources.

9. D Unstructured text is brittle. Ansible is agentless, show commands do not change configuration, and assertions do evaluate output.

10. D Type-3 membership proves the flood list. Session counts do not prove ESI state, gateway MACs must be identical, and RDs are not advertised as type-5 routes.

11. A Policy evaluation needs a model of the network. Linting checks syntax, a canary touches devices, and snapshots show change, not correctness.

12. B Draining is gentle, observable and reversible. It does not remove the link from the topology instantly, does not disable counters, and is unrelated to LDP.

13. C Widespread triggering suggests a systemic issue that needs humans. Thresholds and sampling reduce sensitivity but do not cap actions, and dashboards do not act.

14. B AI output is treated like any other contribution. Direct device access, skipped checks and self-approval violate basic guardrails.

15. D Virtual labs run real protocol stacks and accept automation, but hardware ASIC features and scale differ. Canaries close that gap with limited risk.

16. C Dependency, not score, sets priority. Speed and confidence are side benefits; verification methods and ordering claims are wrong.

17. A The route exists in BGP, so distribution works; import depends only on RT match. RDs do not control import, and a missing transport label affects forwarding, not import.

18. B Later changes, such as QoS on PE-CE links, can break earlier work. Reformatting, reloads and unrequested features add risk without protecting points.

19. D The statement is about the data path, so a data-path test is the evidence. Route tables and configurations suggest it; they do not prove it.

20. C Results, not effort, earn points. Self-scoring should mirror that honestly; 'not attempted' misrepresents partial progress you may still salvage.

21. A (360 − 25 − 20) ÷ 100 = 3.15 minutes per point, so 10 points ≈ 32 minutes. Using full time ignores reading and sweep; the 1.5 multiplier is the rabbit-hole limit, not the budget.

22. C Scope discipline protects points. Non-default values may be intentional and other tasks may depend on them.

23. D Tickets expect the minimal fix that restores intended behaviour. Larger changes add risk without extra credit.

24. A The troubleshooting log is the evidence of finding and fixing faults against the clock. Topology, policy framework and summary document the build but do not test recovery under pressure.

25. B Re-planning by value and dependency recovers the most points. Skipping verification risks losing earned points, and redoing work without evidence wastes time.

Scenarios & case studies

> **Scenario · A RESTCONF call that removed too much**
> **C** is best. Correct. PATCH merges and the narrow URL limits the scope to one interface and one leaf, so nothing else can be replaced.

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Scenario · The emergency fix that disappeared
C is best. Correct. The source of truth must remain authoritative; drift detection catches out-of-band changes before the pipeline overwrites them.

Case study · A bank's automated change pipeline
Start with a **source of truth and verification tests**, because the biggest win is catching inconsistent data and proving outcomes: schema and policy checks (unique RDs, approved RT ranges, route-map naming), automated pre- and post-checks per change type, then rendering from templates to eliminate typos. Add **static analysis** to test route policy and ACL behaviour, and a **virtual lab** for the MPLS core and fabric templates. Deploy with confirmed commits, canaries and automatic rollback. Segregation of duties is enforced by the pipeline: authors cannot approve their own merge requests, production credentials live in a vault accessible only to the pipeline, and every stage's output is retained with the change record as evidence. The AI assistant should **draft** changes as merge requests into the same repository, with no device access; its output passes the same checks and human review, and its prompts and suggestions are logged. Measure change failure rate, mean change duration, rollback frequency, percentage of changes passing all automated checks first time and time to detect drift; a credible target is a failure rate under 3% within a year.

Scenario · Auto-remediation made it worse
D is best. Correct. Guardrails ensure automation never removes the last path and that a widespread problem escalates to humans instead of cascading.

Scenario · Forty minutes left, two tasks open
A is best. Correct. Protecting points already earned comes first; then pursue the higher-value independent task within the remaining time.

Scenario · An ambiguous requirement
B is best. Correct. A minimal, standard change that satisfies the literal requirement and is verified and documented protects points and avoids side effects.

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REFERENCE

Glossary

A

Action community A community a neighbour sets to request specific treatment, such as prepending or suppression. (BGP)

Active state A BGP state in which the router is trying to open a TCP connection to a peer that it cannot reach or that refuses the connection. (Method)

Add-path A BGP extension (RFC 7911) that advertises several paths per prefix with path identifiers. (BGP)

Adjacency The resolved layer 2 rewrite information for a next hop, built from ARP, neighbour discovery or label bindings. (Method)

Adjacency SID A local segment that forces forwarding over a specific IGP adjacency. (Label)

Administrative distance A per-source preference value used to choose between routes from different protocols. (IGP)

Administrative scoping Use of 239.0.0.0/8 groups with boundaries so streams stay inside a region or organisation. (Stream)

Advertise-map A route-map naming routes whose advertisement depends on a condition. (BGP)

AllDRouters The multicast group 224.0.0.6 used to send updates to the DR and BDR. (IGP)

Always-compare MED Comparing MED between paths from different neighbouring ASes. (BGP)

Anycast gateway Identical gateway IP and MAC on every leaf serving a subnet. (Fabric)

Anycast RP Several RPs sharing one address so unicast routing selects the nearest and provides failover. (Stream)

Anycast SID The same prefix SID advertised by several routers, so traffic goes to the nearest one. (Label)

Application-aware routing Per-application path selection based on measured loss, latency and jitter against SLA classes. (Fabric)

Area range ABR summarisation of an area's prefixes into one type 3 LSA. (IGP)

ARP ACL A static list of permitted MAC and IP pairs used by ARP inspection for hosts without DHCP leases. (Layer2)

ARP suppression Answering ARP requests locally on a leaf from EVPN-learned IP-to-MAC bindings. (Fabric)

AS_CONFED_SEQUENCE The AS-path segment listing the sub-ASes a route has crossed inside a confederation. (BGP)

AS_PATH The list of ASes a route has crossed, used for loop prevention and path length comparison. (BGP)

AS_SET An unordered AS-path segment created by aggregation, counted as one AS in length. (BGP)

ASPA Autonomous System Provider Authorisation: signed RPKI objects listing an AS's upstream providers, so routers can detect AS paths that violate valley-free routing. Still maturing; check current standard and product status. (BGP)

Assert PIM election that removes duplicate forwarders onto a shared segment. (Stream)

Assured Forwarding Four AF classes with three drop precedences each, for example AF41, AF42, AF43. (Stream)

Asymmetric IRB Model in which the ingress leaf routes and bridges into the destination VNI, and the egress leaf only bridges. (Fabric)

Attached bit Set by a level 1/2 router in its level 1 LSP when it has level 2 connectivity to other areas. (IGP)

Automated steering Placing BGP traffic into the SR policy whose colour and endpoint match the route's colour and next hop. (Label)

Automatic RT filtering Default PE behaviour of discarding VPN routes whose RTs match no local import policy. (Label)

B

Baseline snapshot A short capture of interface, protocol and routing state taken before any change, used to separate pre-existing faults from new ones. (Method)

Best-external Advertising an edge router's best external path to iBGP even when its overall best path is internal. (BGP)

BFD Bidirectional Forwarding Detection (RFC 5880): a lightweight hello protocol that detects forwarding failures in milliseconds. (IGP)

BGP labelled unicast BGP advertisement of IPv4 or IPv6 prefixes with an MPLS label, used for hierarchical and inter-AS LSPs. (Label)

BGP-LS BGP link-state: an address family that exports IGP topology, TE attributes and SIDs from the network to controllers and PCEs (current specification RFC 9552). (Label)

Binding table The snooping database of client MAC, IP, VLAN, port and lease. (Layer2)

Blast radius The set of devices, prefixes and tasks that a change can affect beyond its intended target. (Method)

Border leaf A leaf that connects tenant VRFs to external networks, firewalls or other sites. (Fabric)

Boundary port A port that connects to another region or to an RSTP or legacy STP bridge; all instances follow the CIST role on it. (Layer2)

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BPDU guard Error-disables an edge port that receives any BPDU. (Layer2)

Broadcast storm Uncontrolled multiplication of broadcast frames caused by a layer 2 loop. (Layer2)

BSR Bootstrap router: an elected router that floods the set of candidate RPs to all PIM routers. (Stream)

BUM traffic Broadcast, unknown-unicast and multicast frames that must be flooded within a segment. (Fabric)

C

C-tag The inner customer VLAN tag carried transparently through the provider network. (Layer2)

Canary deployment Applying a change to a small subset first and verifying before expanding. (Code)

Candidate datastore A scratch configuration that can be edited and validated before being committed to running. (Code)

Candidate RP A router that offers itself to the BSR as an RP for a range of groups. (Stream)

Carrier supporting carrier A service in which one provider carries another provider's labelled traffic inside a VRF. (Label)

CBWFQ Class-based weighted fair queuing: guaranteed bandwidth shares per class during congestion. (Stream)

Change record A short written entry for each change: what, where, why, proof and side effects. (Method)

CIST Common and internal spanning tree: the single tree that spans all regions and non-MST bridges. (Layer2)

CIST regional root The bridge in a region with the lowest external path cost to the CIST root. (Layer2)

Classification community A community attached at ingress to record a route's relationship, region or entry point. (BGP)

Client An iBGP peer of a reflector that receives reflected routes; it needs no special configuration. (BGP)

Closed-loop automation Automation that detects a condition from telemetry, acts and verifies the result. (Code)

Cluster ID The identifier of a reflection cluster, by default the reflector's router ID. (BGP)

CLUSTER_LIST An optional attribute listing the cluster IDs a route has been reflected through. (BGP)

Colour A numeric intent value attached to an SR policy and, as an extended community, to BGP routes. (Label)

Conditional debug A debug limited to a specific interface, neighbour or prefix to reduce load and noise. (Method)

Confederation An AS split into sub-ASes that exchange routes over eBGP-like sessions while appearing as one AS externally. (BGP)

Configuration digest A hash of the VLAN-to-instance table carried in MST BPDUs to detect mapping mismatches. (Layer2)

Confirmed commit A commit that rolls back automatically unless a confirming commit arrives before a timeout. (Code)

Containment The first response step that stops impact quickly, such as shutting one redundant link. (Layer2)

Control policy Policy that manipulates which routes and transport locators sites receive, shaping the overlay topology. (Fabric)

CSNP Complete sequence number PDU: a periodic list of all LSPs used to synchronise databases. (IGP)

D

Dependency map A sketch of which tasks depend on others, used to order work in a practical lab. (Method)

Designated router (DR) The PIM router on a segment that sends joins for receivers and registers for sources. (Stream)

Detect multiplier The number of missed BFD packets that declares a session down. (IGP)

Deterministic MED Grouping paths by neighbouring AS before comparing, so results do not depend on arrival order. (BGP)

DHCP snooping A switch function that permits DHCP server messages only on trusted ports and records client leases. (Layer2)

Direct internet access Sending internet-bound traffic out of a local branch transport rather than backhauling it. (Fabric)

DIS Designated intermediate system: the router that represents a broadcast segment with a pseudonode LSP. (IGP)

Discard route A route to a null interface installed with a summary to drop traffic for missing components. (IGP)

Divide and conquer Testing at the midpoint of a path or layer stack to halve the search space with each test. (Method)

Drift A difference between the source of truth and the running network. (Code)

DSCP Differentiated Services Code Point: the 6-bit field in the IP header used to mark per-hop behaviour. (Stream)

Dual-active A split-brain condition in which both multi-chassis peers act independently. (Layer2)

Dynamic ARP inspection Validation of ARP packets on untrusted ports against bindings or ARP ACLs. (Layer2)

E

ECN Explicit Congestion Notification: marking packets instead of dropping them to signal congestion. (Stream)

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Edge port A port declared to connect only to end hosts; it forwards immediately and does not generate topology changes. (Layer2)

EF Expedited Forwarding (DSCP 46): low-latency, low-loss behaviour for real-time traffic such as voice. (Stream)

Embedded RP IPv6 technique that encodes the RP address inside the multicast group address. (Stream)

End.DT4 SRv6 behaviour that decapsulates and performs an IPv4 lookup in a specific VRF table. (Label)

EVI EVPN instance: the set of Ethernet segments and bridge domains that make up one EVPN service. (Fabric)

Evidence ladder A ranking of proof from weak (configuration present) to strong (forwarding state and observed traffic). (Method)

Exact-route lookup A query that shows which load-sharing path the FIB selects for a given source and destination pair. (Method)

Explicit null Reserved label 0 (IPv4) or 2 (IPv6) that the egress receives, preserving TC bits for QoS until the last hop. (Label)

Explicit requirement An outcome stated directly in the task text, such as a preferred path or a single advertised prefix. (Method)

EXSTART An OSPF neighbour state in which the routers negotiate the master and slave roles for database exchange; being stuck here often indicates MTU mismatch. (Method)

Extended community A 64-bit attribute value with a type field, defined in RFC 4360, used for route targets and similar. (BGP)

External route A route injected into a link-state protocol from outside it, such as an OSPF type 5 LSA. (IGP)

Extranet Selective route sharing between VRFs of different customers or a shared-service VRF via RT policy. (Label)

F

Fault injection Deliberately introducing a failure or misconfiguration in a lab to practise detection and recovery. (Method)

Feedback loop A route redistributed out of its home domain and back in, where it can persist or loop. (IGP)

FIB Forwarding information base: the data-plane table derived from the routing table and used for actual packet forwarding. (Method)

Final sweep The reserved period at the end of a lab for re-verifying every task in order. (Method)

Flexible Algorithm IGP extension that computes extra constraint-based topologies, each with its own prefix SIDs. (Label)

Flowspec A BGP address family that distributes traffic-filtering rules (match on addresses, ports, protocol; actions such as drop, rate-limit or redirect), commonly used for DDoS mitigation. (BGP)

Forwarding address A field in external LSAs that, when non-zero, names the address traffic should be sent to instead of the ASBR. (IGP)

Forwarding equivalence class (FEC) The set of packets forwarded the same way and therefore given the same label, typically all traffic to one prefix. (Label)

G

gNMI gRPC network management interface providing Get, Set and Subscribe, widely used for streaming telemetry. (Code)

Graceful restart A mechanism that preserves forwarding while a neighbour's control plane restarts. (BGP)

Guardrail A control that limits what automation or AI assistance can do, such as preconditions, approvals and rate limits. (Code)

H

Hierarchical QoS Nested policies, typically a parent shaper with a child queuing policy. (Stream)

Hold time The negotiated time after which a silent BGP neighbour is declared down. (BGP)

Hot-potato routing Sending traffic to the nearest exit by IGP metric when other attributes are equal. (BGP)

Hub-and-spoke VPN A topology in which spokes import only hub routes and the hub imports spoke routes, forcing traffic through the hub. (Label)

Hypothesis-driven troubleshooting An approach in which each test is chosen to confirm or refute one stated cause. (Method)

I

IGMP / MLD Protocols hosts use to report multicast group membership to the local router, for IPv4 and IPv6 respectively. (Stream)

IGMP snooping A switch feature that forwards multicast only to ports where membership reports or routers were seen. (Stream)

Implicit null Reserved label 3, signalled by an egress router to request PHP; it never appears in a packet. (Label)

Implicit requirement A condition the task depends on without stating it, such as next-hop reachability or a matching RIB entry. (Method)

Ingress replication Head-end copying of BUM traffic as unicast to every remote VTEP in the flood list. (Fabric)

Inter-AS option A Back-to-back VRFs on the ASBRs, exchanging plain IP per customer across the boundary. (Label)

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Inter-AS option B ASBRs exchange VPN routes over eBGP and swap service labels, without holding VRFs. (Label)

Inter-AS option C Reflectors exchange VPN routes over multihop eBGP while ASBRs carry only labelled PE loopbacks. (Label)

IP source guard A per-port data-plane filter that permits only source addresses found in the binding table. (Layer2)

IRB Integrated routing and bridging: routing between subnets on the same device that bridges them, here on every leaf. (Fabric)

IST Internal spanning tree: instance 0, the portion of the CIST inside a region. (Layer2)

L

L3 VNI A VNI that identifies a tenant VRF for routed traffic between VTEPs. (Fabric)

Label-switched path (LSP) The unidirectional sequence of label operations that carries traffic from an ingress to an egress router. (Label)

Large community A 96-bit attribute value defined in RFC 8092 that fits 32-bit ASNs in the global administrator field. (BGP)

Layer 2 protocol tunnelling Carrying customer control frames such as spanning tree or LACP across a provider service instead of consuming them. (Layer2)

LDP transport address The address used for the LDP TCP session; it must be reachable and is normally the router ID loopback. (Label)

LDP-IGP synchronisation Mechanism that keeps the IGP from using a link for forwarding until LDP has converged on it. (Label)

LFA Loop-free alternate: a pre-computed backup neighbour that will not loop traffic back. (IGP)

LFIB The label forwarding table used for packets that arrive with a label; contains only bindings from the best next hop. (Label)

Liberal label retention Keeping bindings from every LDP neighbour, not just the current next hop, so reconvergence needs no new signalling. (Label)

Link-state database The synchronised collection of LSAs in an area from which every router computes its shortest-path tree. (IGP)

LLQ Low-latency queuing: a strict-priority queue with an implicit policer, added to weighted queuing. (Stream)

LOCAL_PREF A well-known attribute carried within an AS that expresses exit preference; higher wins. (BGP)

Loop guard Prevents a non-designated port from becoming designated when BPDUs stop arriving. (Layer2)

LS sequence number A counter in each LSA that increases with every new instance. (IGP)

LSA throttling Exponential back-off on originating new instances of the same LSA. (IGP)

LSP ping and trace MPLS echo tests (RFC 8029) that validate a label path end to end and hop by hop. (Label)

M

MAC flap A MAC address repeatedly learned on alternating ports, a strong indicator of a loop. (Layer2)

MAC mobility EVPN extended community with a sequence number that lets the newest location of a moved MAC win. (Fabric)

Mapping server A function that advertises SIDs on behalf of LDP-only prefixes to allow SR and LDP interworking. (Label)

Master down interval The time a VRRP backup waits without advertisements before declaring itself master. (Layer2)

Master port The boundary port on the CIST regional root that leads toward the CIST root. (Layer2)

Mastery Ring The eight-domain ZCIE framework: Method, Layer2, IGP, BGP, Label, Fabric, Stream and Code, treated as interdependent rather than sequential. (Method)

Max hops The hop count limit for BPDUs inside an MST region, replacing message age within the region. (Layer2)

MaxAge The LSA age (3600 s) at which an LSA is flushed from all databases. (IGP)

Maximum prefix A per-neighbour limit on accepted prefixes that warns or tears down the session when exceeded. (BGP)

maxLength The longest prefix length a ROA authorises; more specific announcements are Invalid. (BGP)

MED Multi-exit discriminator: a hint to a neighbouring AS about which entry point to prefer; lower wins. (BGP)

Micro-loop A transient loop caused by routers updating their forwarding tables at different times. (IGP)

Micro-SID A compressed SRv6 format that packs several short segments into one 128-bit address. (Label)

MRU Maximum receive unit for labelled packets on a hop, reported by LSP trace. (Label)

MSDP Multicast Source Discovery Protocol: shares active-source information between RPs over TCP 639. (Stream)

MST region Bridges with identical region name, revision number and VLAN-to-instance mapping. (Layer2)

MSTI A multiple spanning tree instance that exists only inside its region and carries a group of VLANs. (Layer2)

Multi-chassis link aggregation Bundling links from one device to two switches that present a single LACP partner identity. (Layer2)

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Multipath Installing several equal BGP paths for load sharing after the multipath comparison step. (BGP)

Mutual redistribution Redistributing routes in both directions between two routing domains. (IGP)

N

NET Network entity title: the area address, system ID and selector that identify an IS-IS router. (IGP)

NETCONF XML-based network configuration protocol over SSH with datastores, locks and transactions. (Code)

Network LSA Type 2 LSA originated by the DR describing a transit segment and its attached routers. (IGP)

Next-hop tracking Event-driven reaction to IGP changes that affect BGP next hops. (BGP)

NMDA Network Management Datastore Architecture: separates intended configuration from operational state. (Code)

NO_ADVERTISE A well-known community that prevents a route from being advertised to any peer. (BGP)

NO_EXPORT A well-known community that keeps a route inside the local AS or confederation. (BGP)

Non-exist-map A condition that allows advertisement only while the named route is absent. (BGP)

NSSA Not-so-stubby area: blocks type 5 LSAs but allows local external routes as type 7 LSAs. (IGP)

NSSA translator The ABR that converts type 7 LSAs into type 5 LSAs for the rest of the domain. (IGP)

O

Object tracking Monitoring an interface, route or other condition and adjusting priority when it changes. (Layer2)

On-change subscription A telemetry mode that sends updates only when a value changes. (Code)

On-demand next hop (ODN) Automatic creation of an SR policy when a coloured route arrives for an endpoint without one. (Label)

Only-to-Customer An attribute defined with BGP roles (RFC 9234) that marks routes which may only be sent to customers. (BGP)

Opaque LSA Types 9, 10 and 11, used to carry application-specific information such as traffic engineering. (IGP)

OpenConfig Operator-led, vendor-neutral YANG models for configuration and telemetry. (Code)

Option 82 The DHCP relay agent information option that identifies the switch and port a request came from. (Layer2)

ORIGIN An attribute indicating how a route entered BGP: IGP, EGP or incomplete. (BGP)

ORIGINATOR_ID An optional attribute holding the router ID of the route's originator inside the AS, used to drop looping routes. (BGP)

Orphan port A port attached to only one multi-chassis peer. (Layer2)

Overload bit Tells other routers not to use this router for transit traffic. (IGP)

P

Packet walk A hop-by-hop analysis of one specific flow that asks the same ingress, lookup, resolve, rewrite, egress and return questions at every device. (Method)

Path hiding Loss of alternative paths because a reflector advertises only its own best path. (BGP)

Path MTU blackhole Silent loss of full-size packets on a path whose MTU is smaller than expected, with no usable feedback. (BGP)

PCE Path computation element: a controller that computes paths for headends, often over PCEP. (Label)

Peer keepalive A separate heartbeat path used to tell a dead peer from a failed peer link. (Layer2)

Peer link The inter-switch link that synchronises state and carries traffic between multi-chassis peers. (Layer2)

Penultimate hop popping (PHP) Removal of the top label by the router before the egress, triggered when the egress advertises implicit null. (Label)

Point budget The time allocated to a task in proportion to its score. (Method)

Policer A rate enforcer that drops or re-marks traffic exceeding a token bucket, without buffering. (Stream)

PQ node A node reachable from the source without the failed link and able to reach the destination without it; the target of remote LFA tunnels. (IGP)

Preemption delay The time a higher-priority router waits before reclaiming the master role. (Layer2)

Prefix prioritisation Installing important prefixes such as loopbacks before others during convergence. (IGP)

Prefix SID index The offset added to a node's SRGB base to form the label for a prefix segment. (Label)

Prefix-independent convergence A forwarding design in which prefixes share path lists, so one change repairs all of them. (BGP)

Prohibition constraint A limit on how a task may be solved, for example no static routes or no changes on a named device. (Method)

Propagate bit The flag in a type 7 LSA that permits translation into a type 5 LSA. (IGP)

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Proposal and agreement The rapid spanning tree handshake that lets a designated port forward immediately on a point-to-point link. (Layer2)

Pseudonode The virtual node created by the DIS to represent a broadcast segment in the topology. (IGP)

Q

QinQ Common name for IEEE 802.1ad provider bridging, which adds an outer service tag to customer frames. (Layer2)

Querier The router or switch that sends general queries on a segment; with IGMP the lowest IP address wins. (Stream)

R

Recursive lookup Resolving a next hop that is not directly connected by looking it up again in the routing table until a connected interface is found. (Method)

Redistribution point A router that redistributes between two routing domains. (IGP)

Register message Unicast encapsulation of multicast packets from the source DR to the RP until native forwarding begins. (Stream)

Rendezvous point (RP) The root of the shared tree where sources and receivers first meet in PIM-SM. (Stream)

RESTCONF HTTP-based protocol that exposes YANG data as resources, usually with JSON. (Code)

Return path The route the reply takes back to the source address; it can differ from the forward path and fail independently. (Method)

RIB-failure A BGP path that is valid and best but not installed because the routing table holds a route from a source with lower administrative distance. (Method)

ROA Route origin authorisation: a signed statement that an AS may originate a prefix up to a maximum length. (BGP)

Root guard Blocks a port that receives a superior BPDU, protecting the root placement. (Layer2)

Root-inconsistent The blocking state applied by root guard while superior BPDUs are received. (Layer2)

Route leak Propagation of routes beyond their intended scope, such as from one provider to another. (BGP)

Route origin validation Classifying routes as Valid, Invalid or NotFound against validated ROA data (RFC 6811). (BGP)

Route reflector An iBGP speaker allowed to re-advertise iBGP-learned routes to its clients and non-clients according to fixed rules. (BGP)

Route tag A numeric label attached to a route, used to recognise its origin during later redistribution. (IGP)

Route target constraint An MP-BGP address family in which PEs advertise the RTs they import, so reflectors send them only relevant VPN routes. (Label)

Router LSA Type 1 LSA describing a router's links in an area, with ABR, ASBR and virtual-link flags. (IGP)

Router MAC EVPN extended community carrying the egress VTEP's MAC, used as inner destination MAC for routed traffic. (Fabric)

RSVP-TE Signalling protocol that establishes traffic-engineered LSPs with per-hop state and optional bandwidth reservation. (Label)

S

S-tag The outer service VLAN tag added by the provider edge. (Layer2)

Scrubbing Removing communities that should not leave the network or that a neighbour should not have set. (BGP)

Seamless MPLS A hierarchical design that builds end-to-end LSPs across separate IGP domains using BGP labelled unicast. (Label)

Seed metric The metric assigned to routes when they are redistributed into a protocol. (IGP)

Segment An instruction in a source-routed path: a label in SR-MPLS or an IPv6 address in SRv6. (Label)

Segment routing header (SRH) An IPv6 routing extension header that carries the SRv6 segment list. (Label)

Session protection A targeted LDP hello that preserves a session and its bindings across a direct link failure. (Label)

Shaper A rate enforcer that buffers excess traffic and releases it at a target rate. (Stream)

Shared tree The (*,G) distribution tree rooted at the RP. (Stream)

Shortest-path tree The (S,G) distribution tree rooted at the source's first-hop router. (Stream)

SLA class A set of loss, latency and jitter thresholds that a path must meet for an application. (Fabric)

Snapshot A saved state of all node configurations that can be restored to reset a scenario quickly. (Method)

Source of truth The authoritative, structured record of intended network state from which configurations are generated. (Code)

Source-active message MSDP announcement that a source is actively sending to a group via a given RP. (Stream)

SPF throttling Exponential back-off that delays SPF runs during repeated changes. (IGP)

SR policy A source-routed path identified by headend, colour and endpoint, with one or more candidate paths. (Label)

SRGB Segment Routing Global Block: the label range a node reserves for global segments such as prefix SIDs. (Label)

SRLB Segment Routing Local Block: a label range for locally significant segments such as adjacency SIDs. (Label)

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SRv6 locator The routable IPv6 prefix of a node from which its SRv6 SIDs are allocated. (Label)

SSE / SASE Cloud-delivered security edge services, combined with SD-WAN in a secure access service edge architecture. (Fabric)

SSM Source-specific multicast: receivers join (S,G) channels directly; no RP is used. (Stream)

Standard community A 32-bit BGP attribute value defined in RFC 1997, written as two 16-bit numbers. (BGP)

Static network analysis Offline modelling of configurations to predict routing, reachability and policy outcomes. (Code)

Storm control Rate limiting of broadcast, multicast or unknown unicast traffic per port. (Layer2)

Stub area An area that receives no type 4 or type 5 LSAs and uses a default route from the ABR. (IGP)

Suboptimal routing Forwarding along a longer path because a redistributed copy wins on distance. (IGP)

Summary LSA Type 3 or 4 LSA originated by an ABR to describe inter-area prefixes or ASBR reachability. (IGP)

Sunk-cost trap Continuing an unproductive effort because of the time already invested rather than the expected return. (Method)

Symmetric IRB Model in which both the ingress and egress leaf route, using a per-VRF L3 VNI between them. (Fabric)

Sync Blocking all non-edge designated ports before agreeing to a proposal, so no temporary loop can form. (Layer2)

T

Task interaction map A note of which tasks share devices or protocols, used to re-verify earlier work after later changes. (Method)

TCP-AO TCP Authentication Option (RFC 5925), the successor to the TCP MD5 signature option for BGP sessions. (BGP)

TI-LFA Topology-independent LFA: a segment-routing-based backup that follows the post-convergence path. (IGP)

Time-boxing Allocating a fixed time budget to a task and parking it with notes when the budget runs out. (Method)

TLV Type-length-value: the record format that carries all IS-IS information and makes the protocol extensible. (IGP)

Topology as code Describing nodes, links and startup configuration in version-controlled text files so a lab can be rebuilt identically. (Method)

Topology change A notification that triggers MAC flushing; in rapid spanning tree it is caused by a non-edge port moving to forwarding. (Layer2)

Topology sheet A single annotated drawing of devices, addressing, protocol boundaries and redistribution points. (Method)

Totally stubby area A stub area that also receives no type 3 LSAs except a default route. (IGP)

TPID Tag protocol identifier, the EtherType value that marks a VLAN tag (0x8100 or 0x88A8). (Layer2)

Transit area The area whose intra-area path a virtual link uses. (IGP)

Transport locator An SD-WAN tunnel endpoint on one transport, defined by addresses, colour and encapsulation. (Fabric)

Trust boundary The point in the network where markings start to be accepted rather than rewritten. (Stream)

Trusted port A port exempt from snooping and inspection checks, normally an uplink toward infrastructure. (Layer2)

TTL security Generalised TTL Security Mechanism (RFC 5082): accepting only packets with a TTL that proves a nearby sender. (BGP)

Type-2 route EVPN MAC/IP advertisement carrying a host MAC and optional IP with L2 and L3 VNI labels. (Fabric)

Type-3 route EVPN inclusive multicast route announcing VTEP membership in a VNI for BUM replication. (Fabric)

Type-5 route EVPN IP prefix route used for subnets, external routes and summaries without a MAC. (Fabric)

U

Unidirectional link A link that passes traffic in only one direction, which can defeat spanning tree's loop prevention. (Layer2)

Up/down bit Marks a prefix leaked from level 2 into level 1 so it is never advertised back into level 2. (IGP)

uRPF Unicast reverse path forwarding: a check that drops packets whose source address is not reachable back through the arrival interface. (Method)

V

Valley-free export The rule that routes from peers and transit providers are exported only to customers. (BGP)

Verification evidence The specific command output that proves a requirement is met, defined before configuring. (Method)

Virtual lab A software-based replica of part of the network used to test changes before production. (Code)

Virtual link A logical area 0 adjacency between two ABRs across a non-stub transit area. (IGP)

VNI VXLAN network identifier: a 24-bit segment ID that maps to a bridge domain (L2 VNI) or a VRF (L3 VNI). (Fabric)

VRRP Virtual Router Redundancy Protocol, a standard first-hop redundancy protocol defined in RFC 5798. (Layer2)

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VTEP VXLAN tunnel endpoint: the device that encapsulates and decapsulates VXLAN, identified by a loopback address. (Fabric)

W

Weight A router-local preference value used by some implementations before any protocol attribute. (BGP)

Wide metrics 32-bit link and 32-bit prefix metrics carried in extended TLVs 22 and 135. (IGP)

WRED Weighted random early detection: probabilistic early drops weighted by drop precedence. (Stream)

Y

YANG Data modelling language (RFC 7950) describing configuration and operational data as a tree. (Code)

Z

Zero-touch provisioning Automatic onboarding of a new device using a hardware identity and a cloud discovery service. (Fabric)

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CAPSTONE

Capstone: Expert mock lab across the Mastery Ring

Build and document a full expert lab for a fictional national carrier and enterprise: an IS-IS/OSPF core with BGP route reflection, an MPLS L3VPN or SR-MPLS backbone, an EVPN data-centre fabric, multicast IPTV, an end-to-end QoS policy and automated verification. Then break and fix it under time pressure.

Deliverables

☐ Lab topology and addressing for 15+ nodes

☐ IGP design with fast convergence (BFD, LFA/TI-LFA)

☐ BGP route-reflection and community policy framework

☐ MPLS L3VPN or SR-MPLS service for three customers

☐ VXLAN EVPN fabric with anycast gateway

☐ Multicast (PIM-SM/SSM) service with RP redundancy

☐ End-to-end QoS policy and validation

☐ Automated verification suite (Python/Ansible/templating)

☐ Timed troubleshooting log: 8 faults found and fixed

☐ Executive summary and lessons learned

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Methodology	○ Trial and error	○ Structured	○ Fast, evidence-based, documented
Layer 2	○ Unstable	○ Stable and protected	○ Optimised with forensics ready
IGP	○ Converges slowly	○ Correct design	○ Sub-second convergence proven
BGP	○ Peering only	○ Policy framework	○ Scalable, secure, deterministic
MPLS/SR	○ Partial	○ Working VPNs	○ Optimised paths and TE
Fabric	○ Not built	○ Working EVPN	○ Multihomed, verified, documented
Multicast & QoS	○ Missing	○ Working	○ Validated under load
Automation	○ Manual	○ Scripted config	○ Automated verification and rollback

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CAPSTONE

Planning pages

Sketch your design, timeline or architecture here.

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EXAM PREPARATION

Practice exam · 60 questions

The official online exam draws 45 questions at random from a separate, secret bank that tests the same skills, with a 90-minute limit and a 85% pass mark. Practise with these questions first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Method — Expert Methodology

1. A task reads: "R7 must reach 10.80.0.0/16 from its Loopback0 through R3, and the path must survive the loss of the R3–R4 link." You configured a cost change and a ping from R7 succeeds. What is the best proof that the task is complete?

   (A) A trace from Loopback0 via R3, repeated with the R3–R4 link shut

   (B) A ping from R7 to 10.80.0.1 with a repeat count of 1000 and zero loss

   (C) The routing table on R7 showing 10.80.0.0/16 with R3 as the next hop

   (D) The running configuration on R3 showing the new cost on the right link

2. `show ip route 10.20.0.0` on R1 lists two equal-cost next hops, R2 and R3. The task requires traffic from Loopback0 to 10.20.0.10 to use R2. `show ip cef exact-route 10.0.0.1 10.20.0.10` returns `adj out of Gi0/1, addr 10.1.13.3`. What is the correct conclusion?

   (A) The task is met because R2 is one of the two installed equal-cost paths

   (B) The task is not met because the forwarding hash for that flow selects R3

   (C) The task is met because CEF load-shares every flow evenly across both paths

   (D) The output is inconclusive until a debug of packet switching is enabled

3. An iBGP session from R5 to 10.0.0.9 stays in Active. `show ip route 10.0.0.9` returns `% Subnet not in table`, and R5's OSPF database has no router LSA from 10.0.0.9. Where should you look next?

   (A) At the BGP update-source configuration on both R5 and R9

   (B) At the BGP authentication passwords configured on both of the peers

   (C) At the OSPF adjacencies of R9, because its loopback is not reachable

   (D) At an access list that may be blocking TCP port 179 on R5

4. `debug ip ospf adj` on R2 shows `Nbr 10.0.0.3 has larger interface MTU` and the neighbour is stuck in EXSTART. The task says "do not change OSPF settings". What is the best fix?

   (A) Configure the router to ignore the MTU in database descriptions

   (B) Change the OSPF network type on both ends to point-to-point

   (C) Clear the OSPF process on R2 and R3 to restart the exchange

   (D) Align the interface MTU on both routers so that they match

5. During a practical lab you have 70 minutes left. Open tasks: a 3-point IGP fault that blocks two BGP tasks worth 6 points in total, and an independent 4-point QoS task. What should you do first?

   (A) Fix the IGP fault first; its dependency weight is 9 points

   (B) Do the 4-point QoS task, because it is the largest single task left

   (C) Do the BGP tasks first, as they are worth more combined than the IGP

   (D) Spend the time on a full verification sweep of the completed tasks

6. A traceroute sourced from R1's Loopback0 replies from hops 1 and 2 and then times out at every later hop. All routers on the path have a route to the destination. What is the most likely cause?

   (A) The destination network has an MTU smaller than the trace probes

   (B) The trace's maximum TTL value is too low for the path length

   (C) The third hop has no route back to R1's loopback address

   (D) Hop 2 is load-sharing the probes across two separate paths

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7. A colleague proposes tuning BGP timers to stop sessions that reset a few minutes after establishing. The sessions show thousands of TCP retransmissions and `max data segment is 1460 bytes`. What should you do first?

   (A) Lower the keepalive and hold timers so failures are detected sooner
   (B) Test the path with a sourced don't-fragment ping at full size
   (C) Raise the hold timer so that the sessions survive the slow transfer
   (D) Replace the route reflector that serves the affected routers

Layer2 — Layer 2 at Scale

8. `show spanning-tree mst configuration` on a new switch shows `Name [DC-C0RE]` while every other switch uses `DC-CORE`. Its uplinks show the Bound type and one uplink blocks. VLAN 210, allowed only on that uplink, is down. What is the root cause?

   (A) The switch's instance 2 priority is lower than the core's priority for that instance
   (B) VLAN 210 is not mapped to any instance on the switch and falls back to instance 0
   (C) The uplinks are configured as edge ports and therefore ignore received BPDUs
   (D) It is in its own region, so all instances follow the CIST role at the boundary

9. `show spanning-tree inconsistentports` on a distribution switch lists `Te1/2/1 Loop Inconsistent` on a fibre uplink that was previously an alternate port. What is the most likely cause?

   (A) A downstream switch began sending BPDUs with a superior bridge priority
   (B) A host connected to the port while it was configured as an edge port
   (C) The port's VLAN allowed list no longer matches the instance mapping
   (D) BPDUs stopped arriving, typically because of a unidirectional fault

10. After enabling ARP inspection on VLAN 40, `show ip arp inspection statistics vlan 40` shows all drops under DHCP Drops, and logs show invalid ARP from 10.1.40.20, a statically addressed printer. What is the best fix?

    (A) Configure the printer's access port as trusted for ARP inspection to stop the drops
    (B) Add an ARP ACL or static binding for the printer's MAC and IP address pair
    (C) Disable the IP validation check of ARP inspection on VLAN 40 for all ports
    (D) Raise the ARP inspection rate limit on the printer's port to a higher value

11. Clients on an access switch stopped receiving DHCP leases after DHCP snooping was enabled. The uplink is trusted. Packet captures on the distribution relay show requests arriving with option 82 and a gateway address of 0.0.0.0, and no relayed requests leaving. What is the most likely cause?

    (A) The relay discards option 82 requests with a zero gateway address
    (B) The DHCP snooping rate limit on access ports is dropping the requests
    (C) IP source guard on access ports is blocking clients with no address
    (D) ARP inspection is dropping the DHCP traffic on the trusted uplink

12. `show vrrp brief` on DSW1 shows `Vl110 10 100 ... Master`, and on DSW2 `Vl110 10 100 ... Backup`. DSW1's configured priority is 110 with `track 30 decrement 10`, and its uplink has failed. Traffic still uses DSW1. Why?

    (A) Preemption is disabled by default in VRRP, so the backup never takes over
    (B) Object tracking only lowers priority after the preemption delay timer expires
    (C) The decrement only creates a tie, and a tie does not cause preemption
    (D) The backup cannot become master while DSW1 is still sending advertisements

13. A multi-chassis pair's peer link and keepalive both ran over the same line card. The card failed, and a downstream switch began seeing MAC flaps and duplicate frames. What state did the pair enter?

    (A) Dual-active, with both peers forwarding independently under one LACP identity
    (B) Secondary suspended, with all member ports on the secondary peer disabled
    (C) Primary isolated, with the secondary taking over all forwarding duties
    (D) Normal operation, with the downstream switch failing its LACP hashing

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14. A provider core carries QinQ customers. The core trunk's native VLAN is 500, which is also customer ACME's S-VLAN. ACME reports that traffic arrives in the wrong VLAN at the far site. What is the cause?

    (A) The core MTU is too small for the added service tag, so frames are dropped

    (B) The TPID of the service tag is 0x8100 rather than 0x88A8 on the core trunk

    (C) Layer 2 protocol tunnelling is rewriting the destination MAC of data frames

    (D) The outer tag is stripped as native, so the customer tag is misread

15. During a storm, `show logging` shows several unrelated hosts flapping between Te1/1/1 and Te1/1/2, and Te1/1/2 receives 11 million packets per second, mostly broadcast. What is the best next action?

    (A) Reboot the core switch to rebuild the MAC table and spanning tree state

    (B) Shut the link toward the switch behind Te1/1/2 and watch the counters

    (C) Enable a debug of spanning tree events to find the looping switch

    (D) Raise the storm-control level on Te1/1/2 to protect legitimate traffic

IGP — IGP Mastery

16. `show ip ospf database external 172.16.50.0` shows the LSA with `Forward Address: 10.7.7.9`. `show ip route 10.7.7.9` returns `% Subnet not in table`. No router installs 172.16.50.0/24. What is the fix?

    (A) Change the external metric type of the LSA from type 2 to type 1

    (B) Convert the area containing the ASBR into a not-so-stubby area

    (C) Make the forwarding address reachable through internal OSPF

    (D) Clear the OSPF process on the ASBR to reoriginate the LSA

17. A router LSA from R2 shows `LS Seq Number: 80000A3C` and `LS age: 37`. R2 has been up for 20 days. What does this indicate?

    (A) R2 has been refreshing its LSA normally every thirty minutes

    (B) R2's links have changed far more often than normal refreshing would explain

    (C) R2's LSA was flushed recently and then reached MaxAge again

    (D) R2 originates its LSA in a different area from the reader

18. R3 has an inbound distribute-list that removes 10.50.0.0/16 from its routing table. R4, behind R3 in the same area, cannot reach 10.50.0.0/16 and its traces stop at R3. What is the best design fix?

    (A) Add the same distribute-list on R4 so that both routers agree on reachability

    (B) Increase the cost of the R3 to R4 link so R4 chooses a different exit path

    (C) Configure a static default route on R3 toward the backbone for the prefix

    (D) Filter the type 3 LSA for the prefix at the area border router instead

19. `show isis database detail R3.00-00 level-1` shows `ATT/P/OL 1/0/0` and one prefix listed as `IP-Interarea 10.0.0.9/32`. What do these show?

    (A) R3 has level 2 connectivity, and 10.0.0.9/32 was leaked from level 2 with the up/down bit set

    (B) R3 is overloaded, and 10.0.0.9/32 is an external route redistributed from another protocol

    (C) R3 is the DIS for the segment, and 10.0.0.9/32 is the loopback of its pseudonode

    (D) R3 uses narrow metrics, and 10.0.0.9/32 is carried in the legacy reachability TLV

20. `show bfd neighbors detail` on R1 shows: local desired TX 50 ms, required RX 50 ms, multiplier 3; remote desired TX 300 ms, required RX 300 ms, multiplier 3. What is R1's detection time?

    (A) 150 ms, because the local interval and multiplier are used for detection

    (B) 450 ms, because the two configured intervals are averaged together

    (C) 900 ms, because the remote side's 300 ms interval governs detection

    (D) 300 ms, because detection equals the slower of the two intervals

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21. A ring of eight routers runs per-prefix LFA. Coverage reports show about half of the destinations unprotected, and sub-50 ms recovery is required for all of them. What is the best action?

    (A) Reduce the BFD interval so failures are detected before traffic is lost
    (B) Set the initial SPF delay to zero so new paths are computed immediately
    (C) Set every ring link to an equal cost so that more equal-cost paths appear
    (D) Deploy remote LFA or TI-LFA for the destinations basic LFA cannot cover

22. R2 shows 10.20.5.0/24 as `Known via "ospf 1", distance 110, type extern 2, Tag 115`, while the prefix also exists natively in IS-IS. Tags are denied correctly on both redistribution routers. What causes the suboptimal path?

    (A) The tag value 115 makes R2 prefer the OSPF route by policy over IS-IS
    (B) OSPF external distance 110 beats the native IS-IS distance of 115
    (C) The OSPF external route has a lower metric than the IS-IS route
    (D) R2 is redistributing the prefix back into IS-IS from OSPF

23. After a merger, a data-centre prefix was removed from IS-IS but stayed reachable for hours in both OSPF and IS-IS, while traffic looped between the two redistribution routers. What would have prevented it?

    (A) Raising the OSPF external distance on one of the two redistribution routers only
    (B) Setting a higher seed metric for the routes that are redistributed into OSPF
    (C) Tags by source domain, with the other domain's tag denied at every point
    (D) Enabling BFD on the links between the two redistribution routers

BGP — BGP Mastery

24. A route reflector receives a route whose CLUSTER_LIST contains its own cluster ID. Two reflectors serve the same clients with the same cluster ID. What is the consequence for the reflectors?

    (A) Each drops the other's reflected copies, so clients must peer with both
    (B) Each reflector accepts the other's copies and stores two paths per prefix
    (C) Both reflectors stop reflecting routes learned from shared clients
    (D) The reflectors replace each other's ORIGINATOR_ID with their own router ID

25. `show ip bgp 192.0.2.0/24` shows three iBGP paths with LOCAL_PREF 100 and equal AS-path length: via 10.0.0.7 (AS 64510, MED 50, IGP 20), via 10.0.0.8 (AS 64520, MED 10, IGP 40) and via 10.0.0.9 (AS 64510, MED 0, IGP 50). Deterministic MED is enabled. Which path is best?

    (A) Via 10.0.0.9, because it has the lowest MED of all three paths
    (B) Via 10.0.0.8, because it beats the AS 64510 winner on IGP metric
    (C) Via 10.0.0.7, because it has the lowest IGP metric of all three
    (D) No single best, because MEDs from different ASes are incomparable

26. A customer tags its prefix with your documented do-not-announce-to-peers community. At the ingress router the route shows the community; at the egress router it shows no communities at all, and the route reaches a peer. What is the root cause?

    (A) The egress router's export policy matches on the large-community form
    (B) The customer must also attach NO_EXPORT for the signal to be honoured
    (C) Reflectors remove all communities that were set by external customers
    (D) A session between ingress and egress does not send communities

27. You send MED 10 on link A and MED 100 on link B to the same provider, yet inbound traffic prefers link B. The provider confirms both MEDs are received. What is the most effective fix?

    (A) Use the provider's community to lower its local preference on link B
    (B) Lower the MED on link A to zero so that the difference becomes larger
    (C) Advertise more-specific prefixes over link B than over link A
    (D) Remove the MED from link B so it is treated as the worst value

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28. A backup provider session uses `advertise-map RM-ADVERTISE non-exist-map RM-ISP1-ALIVE`, where RM-ISP1-ALIVE matches 0.0.0.0/0. Both providers send a default route. The primary fails, but the backup announcement never appears. Why?

    (A) Non-exist-maps only check routes that were learned through iBGP

    (B) The advertise-map must match the default route rather than the address block

    (C) The default from the backup provider keeps the condition route present

    (D) Conditional advertisement requires add-path on the backup session

29. eBGP sessions with BFD at 50 ms × 3 over a provider layer 2 service drop briefly during the provider's maintenance, which involves protection switching of about 200 ms. What is the best fix?

    (A) Enable graceful restart on both sides of every affected session

    (B) Raise BFD detection time above the provider's switching time

    (C) Remove BFD and lower the BGP hold time to three seconds

    (D) Disable next-hop tracking on the routers at both ends of the session

30. A ROA authorises AS 64496 to originate 198.51.100.0/22 with maxLength 22. A route for 198.51.100.0/24 arrives with origin AS 64496. Your policy drops Invalid. What happens?

    (A) The route is accepted as Valid, because the origin AS matches the ROA

    (B) The route is accepted as NotFound, because no ROA lists the /24

    (C) The route is accepted, because covered routes are always Valid

    (D) The route is dropped, as it exceeds the authorised maximum length

Label — Label — MPLS and Segment Routing

31. An LSP trace from PE3 to PE1's loopback 10.255.1.1/32 shows 'L' at P1 and 'U' at P2. P2's configuration contains: `ip access-list standard LDP-FEC permit 10.255.0.0 0.0.0.255` and `mpls ldp advertise-labels for LDP-FEC`. What is the root cause and best fix?

    (A) P2 has no LDP session toward P1, so labels learned from PE1 cannot be passed upstream; restart LDP on P2 to rebuild the session.

    (B) PE1 advertises implicit null for its loopback, so P2 pops the label too early; configure explicit null on PE1 to keep the label.

    (C) The LDP label filter covers only 10.255.0.0/24, so no label is advertised for 10.255.1.1; widen the ACL to every PE loopback range.

    (D) The IS-IS metric toward PE1 changed, so P2 uses a path without LDP; restore the old metric so traffic returns to the labelled path.

32. A customer site is dual-homed to PE1 and PE2. Both PEs use RD 65000:50 for the VRF. When the PE1-CE link fails, remote PEs take about 40 seconds to move traffic to PE2. Reflectors show only one path for the site prefixes. What change most directly improves convergence?

    (A) Give each PE a unique RD for the VRF so the reflectors keep and reflect both paths, enabling fast failover at remote PEs.

    (B) Lower the BGP keepalive and hold timers on the VPNv4 sessions between the PEs and the route reflectors in the core.

    (C) Enable LDP session protection on the core so that the transport labels survive the PE-CE link failure without relearning.

    (D) Configure as-override toward the CE on both PEs so the site accepts routes that pass through the other PE as well.

33. PE1's forwarding table shows local label 16004 but outgoing label 900004 toward P2 for the SR prefix with index 4. All traffic to the prefix is delivered correctly. What is the correct interpretation?

    (A) P2 has an index collision for prefix SID 4, so PE1 uses a fallback label; renumber the index on the destination router.

    (B) P2's SRGB starts at 900000, so PE1 correctly sends P2's label for index 4; aligned SRGBs would simplify operations.

    (C) PE1 has fallen back to an LDP label for the prefix, so SR is not in use; enable SR preference on PE1 to fix the label.

    (D) PE1 is applying a TI-LFA repair label, so the primary path has failed; check for a down link between PE1 and P2 now.

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34. Inside VRF ORANGE, `ping vrf ORANGE 172.16.30.1 size 1500 df-bit` fails while size 1488 succeeds. Traffic crosses a core using a transport label, a service label and one additional TE label. What is the root cause?

    (A) The CE router has an IP MTU of 1488 on its PE-facing interface, which was configured to match the provider's link.

    (B) The PE fragments labelled packets at 1488 bytes because the DF bit is ignored for traffic carried in a VRF table.

    (C) The service label is 12 bytes long for VPN routes, so the core must reserve 12 bytes for every VPN packet sent.

    (D) A core link has an MPLS MTU of 1500, which leaves room for only 1488 bytes of IP once three labels are added.

35. Customer routes from OSPF site A arrive at OSPF site B as type-5 external routes instead of inter-area routes, so a backup static default at site B wins. PE1 runs `router ospf 10 vrf BLUE` and PE2 runs `router ospf 20 vrf BLUE`. What is the most likely cause?

    (A) The PEs use different route distinguishers for VRF BLUE, so the routes are marked as external during import.

    (B) The down bit is not being set on PE1, so PE2 treats the routes as having originated outside the VPN backbone.

    (C) A sham-link is missing between the PEs, so OSPF can only advertise VPN routes into the site as external routes.

    (D) The PEs derive different OSPF domain IDs from their process IDs, so routes are treated as external.

36. On PE1, `show segment-routing traffic-eng policy color 100` shows the only candidate path invalid with the reason 'segment-list first SID 16099 unresolved'. Coloured VPN routes are following the IGP path. What is the best fix?

    (A) Correct the explicit segment list so it references a node SID that exists, or add a dynamic candidate path as fallback.

    (B) Increase the SRGB on PE1 so that label 16099 falls inside the range and can be resolved by the headend router.

    (C) Remove the colour community from the VPN routes, because coloured routes cannot use an explicit segment list.

    (D) Enable LDP on all core links, because explicit segment lists need LDP labels to resolve their first segment.

37. In a hub-and-spoke VPN, traceroutes show spoke-to-spoke traffic going directly between spoke PEs instead of through the hub firewall. Spoke VRFs import RTs 65000:202 (hub) and 65000:201 (spoke) and export 65000:201. What is the best fix?

    (A) Change the spoke RDs to a single shared value so all spoke routes collapse into one path pointing at the hub PE.

    (B) Add a static default route in each spoke VRF pointing to the hub PE's loopback address in the global table.

    (C) Remove the 65000:201 import from the spoke VRFs so spokes learn only hub routes and must route via the hub.

    (D) Disable route reflection for the spoke PEs so they no longer receive each other's routes from the reflector.

38. In an inter-AS option C design, PE1 in AS 65000 has VPNv4 routes from PE9 in AS 65100 with next hop 10.200.0.9, but `show ip cef vrf RED` shows the route unusable with no label for 10.200.0.9. What should be checked first?

    (A) Whether the route reflectors in the two ASes are exchanging VPNv4 routes over a multihop eBGP session with each other.

    (B) Whether 10.200.0.9/32 is advertised with a label via BGP labelled unicast across the ASBRs into AS 65000.

    (C) Whether the ASBRs have the VRF RED configured with matching route targets for the customer's routes.

    (D) Whether the PE9 loopback is redistributed into the IS-IS of AS 65000 as a plain route without any label.

Fabric — Fabric — Overlays, VXLAN EVPN and SD-WAN

39. Leaves use eBGP with one AS per leaf and `route-target import auto` for VNI 10010. Leaf1 (AS 65101) shows type-2 routes from Leaf4 (AS 65104) in the BGP table but does not install them in the MAC table. What is the most likely cause?

    (A) Leaf1 and Leaf4 use different route distinguishers for VNI 10010, so the routes are treated as separate entries.

    (B) The spines are rewriting the next hop to themselves, so Leaf1 cannot build a VXLAN tunnel toward Leaf4's VTEP.

    (C) ARP suppression is enabled on Leaf1, so it ignores type-2 routes that carry both a MAC and an IP address together.

    (D) Auto-derived RTs include each leaf's own AS, so Leaf4 exports 65104:10010 while Leaf1 imports only 65101:10010.

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40. In an eBGP EVPN fabric, every leaf shows its EVPN sessions to the spines established, but no leaf receives EVPN routes from any other leaf. The spines have no VRFs or VNIs configured. What is the most likely cause?

(A) The leaves are missing an underlay route to the spine loopbacks, so the EVPN updates are dropped in transit.

(B) The spines discard EVPN routes through route-target filtering, because they import no RTs and retain none.

(C) The spines have ARP suppression enabled, so they cannot forward type-2 routes that contain host IP addresses.

(D) The leaves use the same L3 VNI for every tenant VRF, so the spines merge all routes and then discard them.

41. After a new fabric goes live, pings and small transactions between VMs on different leaves succeed, but VM live migrations and large file copies fail. Underlay interfaces are configured with MTU 1500. What is the root cause?

(A) The VXLAN UDP source port changes per flow, which causes reordering that breaks large transfers on the underlay.

(B) The anycast gateway MAC differs between leaves, so long-lived flows break when they move between the leaves.

(C) The underlay cannot carry 1500-byte tenant frames plus 50 bytes of VXLAN overhead, so large packets are dropped.

(D) The type-3 routes are missing for the VNI, so only unknown unicast fails while small packets are still delivered.

42. In a symmetric IRB fabric, hosts in the same subnet on different leaves communicate, but routed traffic to hosts behind Leaf6 fails. Type-2 routes from Leaf6 show 'Received label 10020' only and no Router MAC community. What is the best fix?

(A) Configure the tenant L3 VNI on Leaf6 and bind it to the NVE interface so routes carry label 2 and router MAC.

(B) Enable ARP suppression on Leaf6 so that it advertises the IP part of each type-2 route to the other leaves in the fabric.

(C) Configure a unique anycast gateway MAC on Leaf6 so that remote leaves can address routed traffic to it directly.

(D) Advertise every host on Leaf6 as a type-5 route so the remote leaves can reach them without any L3 VNI at all.

43. A legacy switch connects to Leaf2 and Leaf5 with two separate, unbundled trunks. Hosts behind it are repeatedly flagged as duplicate MACs and lose connectivity. What is the best fix?

(A) Disable MAC mobility on Leaf2 and Leaf5 so the leaves stop counting moves and keep advertising the host MACs.

(B) Increase the duplicate-detection threshold so that more MAC moves are tolerated before the advertisement is frozen.

(C) Assign the two trunks to different VLANs so that each leaf learns a different subset of the hosts behind the switch.

(D) Bundle the two links into one LACP port channel configured as an all-active Ethernet segment with a shared ESI on both leaves.

44. Two border leaves advertise a default route as type 5 into the tenant VRF. When Border1's WAN link fails, half of all outbound flows black-hole. Border1 originates the default with an 'always' option. What is the best fix?

(A) Originate the default only while it is learned from the external peer, so Border1 withdraws it when the WAN link fails.

(B) Lower the BGP hold timers between the leaves and the spines so the failure is detected faster by every leaf in the fabric.

(C) Configure Border1 and Border2 with the same VTEP address so remote leaves cannot distinguish the two border paths.

(D) Add a static default route on each leaf pointing at Border2 with a higher administrative distance than the BGP default.

45. An SD-WAN application-aware routing policy steers traffic matching DSCP EF onto MPLS. Phones mark voice EF, but captures at the edge's LAN interface show voice arriving with DSCP 0, and voice is using the internet path. What is the root cause?

(A) The MPLS transport currently fails the VOICE SLA class, so the edge correctly falls back to the internet transport.

(B) The branch access switch does not trust the phones and re-marks their traffic, so the edge policy never matches EF.

(C) The edge copies DSCP to the tunnel header, which hides the original marking from the application-aware policy.

(D) The controller has not distributed the MPLS transport locator, so the edge cannot build tunnels on MPLS at all.

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46. During a data-centre migration, the fabric's anycast gateway was enabled for VLAN 140 while the legacy FHRP gateway remained active. Firewall logs show asymmetric-flow drops for VLAN 140 hosts. What should have been done?

(A) Give the anycast gateway a different IP address from the legacy gateway, so hosts can use both gateways in parallel.

(B) Enable ARP suppression for the VLAN, so hosts only ever learn the fabric gateway's MAC and ignore the legacy one.

(C) Disable the legacy FHRP for the VLAN in the same change that enables the anycast gateway, so one gateway is active.

(D) Extend additional VLANs to the legacy core, so the firewall sees traffic from both gateways on the same interfaces.

Stream — Stream — Multicast and QoS

47. Receivers behind R7 have (*,G) state for 239.30.1.1 pointing at RP 10.255.0.100, but the first-hop router of the source shows RP 10.255.0.200 for the same group and registers there. No traffic reaches the receivers. What is the root cause?

(A) The source uses a TTL of 1, so the first-hop router cannot register the packets with any rendezvous point.

(B) Routers disagree on the RP for the group, so sources register with one RP while receivers join another.

(C) The receivers are using IGMPv2, so their (*,G) joins cannot reach an RP that is configured with BSR.

(D) The RP at 10.255.0.100 fails the RPF check for the source, so it drops all register messages it receives.

48. A new IPTV service uses SSM groups in 232.0.0.0/8. The last-hop router shows IGMP membership for the groups but no (S,G) state, and set-top boxes report IGMPv2. What is the best fix?

(A) Configure a static RP for 232.0.0.0/8 so that the last-hop router can build a (*,G) shared tree for the SSM groups.

(B) Enable MSDP between the head-end routers so that the last-hop router can learn which sources are active.

(C) Disable IGMP snooping on the access switches so that the IGMPv2 reports reach the last-hop router unchanged.

(D) Use IGMPv3 on the boxes, or SSM mapping on the last-hop router to turn group-only reports into (S,G) joins.

49. Branch receivers can join 239.192.10.1 (regional training video) but not 239.1.1.1 (company-wide broadcast). Both groups use the same RP. The branch WAN interface has `ip multicast boundary MCAST-SCOPE`, which permits 239.192.0.0/16 only. What is the best fix?

(A) Add the company-wide group range to the boundary ACL, which is blocking 239.1.1.1 at the WAN edge.

(B) Move 239.1.1.1 to a different RP, because the shared RP cannot serve two groups from different scopes at once.

(C) Convert the regional group to SSM, because ASM groups in 239.192.0.0/16 take precedence over other ranges.

(D) Raise the source TTL for 239.1.1.1, because company-wide groups need a larger TTL to cross the WAN boundary.

50. `show policy-map interface` on a WAN edge shows the VOICE priority class with offered rate 24 Mbit/s against a 20 Mbit/s priority limit and many exceed drops. Call volume has not changed, but a desktop video client was rolled out last week. What is the most likely cause and fix?

(A) The voice codec changed to a higher bit rate; raise the priority limit to 40% of the link to accommodate it.

(B) The shaper is too slow for voice; remove the parent shaper so voice can use the full physical interface rate.

(C) The video client marks its traffic EF and is trusted; re-mark it at the access trust boundary to its proper video class.

(D) The WRED thresholds are too low in the priority class; disable random-detect so voice is no longer dropped.

51. Voice at a branch shows 40 ms of jitter whenever backups run. The WAN policy is a parent shaper at 50 Mbit/s with a child LLQ policy. The shaper shows `bc 6250000` (Tc = 125 ms). What is the best change?

(A) Reduce the shaper's committed burst so Tc is 10 ms or less, releasing traffic in short, smooth intervals.

(B) Move voice out of the priority queue into a bandwidth class, so it is no longer released in bursts by the shaper.

(C) Replace the parent shaper with a policer at 50 Mbit/s, because policers never add any delay to voice traffic.

(D) Increase the child policy's queue limits, so backup traffic no longer fills the queue that voice shares with it.

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52. A business application gets only 30 Mbit/s through a 100 Mbit/s ingress policer configured with a 1,500-byte committed burst, even though the link is otherwise idle. Packet captures show regular TCP retransmissions. What is the best fix?

(A) Lower the policer's rate to 30 Mbit/s so it matches the throughput the application can actually reach in practice.

(B) Apply WRED to the policer, so drops become random and TCP flows back off more gradually than they do with tail drops.

(C) Mark the application traffic EF so that it bypasses the policer and is forwarded through the priority queue instead.

(D) Increase the policer's committed burst to absorb normal TCP bursts, or shape upstream to the contract rate.

53. A VPN customer reports that DSCP values change after crossing the provider: AF21 traffic arrives as DF whenever their traffic exceeds its contracted class rate. The provider re-marks excess traffic to TC 0 at ingress. Which provider change prevents this?

(A) Use explicit null on the egress PEs, so the customer DSCP is restored from the explicit null label at the last hop.

(B) Use the pipe or short-pipe DiffServ model, so provider TC changes are not copied back into customer DSCP.

(C) Use per-VRF label allocation, so the egress PE performs an IP lookup and resets DSCP to the original value.

(D) Use penultimate hop popping, so the label carrying the re-marked TC is removed before reaching the egress PE.

Code — Code — Expert Automation and Lab Mastery

54. A pipeline job fails on PE4 with the NETCONF error `lock-denied` and an error-info session-id of 4127. Another automation job is scheduled for the same device. What is the best response?

(A) Kill session 4127 immediately, because a NETCONF lock always means a stuck session that is blocking changes.

(B) Retry without locking, editing the running datastore directly so the job completes inside the change window.

(C) Treat it as a concurrency conflict: let the holding session finish, and enforce per-device job serialisation in the pipeline.

(D) Commit the candidate anyway with a forced commit, because the lock only protects against accidental edits.

55. Rendered configurations for three PEs contain the line `neighbor remote-as 65010` with two spaces. Devices rejected the line during deployment. Which pipeline improvement prevents this class of error earliest?

(A) Add a canary stage so that the rejected line is detected on fewer devices before the full deployment begins.

(B) Use strict undefined rendering and validate intent against a schema, so missing fields fail the build.

(C) Add a post-deployment check that counts BGP neighbours, so the missing peer is reported after the change.

(D) Add a human review of the rendered configuration diff, so engineers can spot blank values before approval.

56. A change passed schema checks and static analysis but failed on the production canary because a feature behaves differently on the router's software release. Which additional pipeline stage would most likely have caught it?

(A) A virtual lab test on the same software release as production, running the verification suite on real stacks.

(B) A stricter YAML lint stage that rejects any intent data not formatted to the team's indentation and style rules.

(C) A snapshot test that compares the rendered configurations with the previous approved version in the repository.

(D) A second static analysis run with a different snapshot of the network taken from the previous night's backups.

57. A closed-loop rule drains a link when its error rate exceeds a threshold and restores it when the rate falls. On a marginal optic, the link has been drained and restored 26 times today. What is the best improvement?

(A) Lower the threshold so the link is drained earlier, which keeps traffic off the marginal optic for longer each time.

(B) Remove the verification step after draining, so the rule completes faster and generates fewer events per action.

(C) Increase the telemetry sample interval, so the rule sees fewer error spikes and triggers less often overall.

(D) Add hysteresis and hold-downs before restoring, cap actions per device per day, and escalate at the cap.

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58. The pipeline's dry-run diff for PE5 shows it will remove 40 lines that exist on the device but not in the source of truth, including a static route added during last week's incident. What is the best action?

(A) Deploy as planned, because the source of truth is authoritative and anything outside it is unauthorised by definition.

(B) Switch the pipeline to merge-only mode for PE5, so the extra lines stay in place and the deployment can continue.

(C) Pause PE5, investigate the drift, and either add the needed lines to the source of truth or confirm removal.

(D) Copy the device's running configuration into the source of truth wholesale, so the diff for PE5 becomes empty.

59. An AI assistant drafted a VRF change that included a route target belonging to a different customer. A reviewer caught it by chance. Which control most directly prevents this from reaching production in future?

(A) A rule that only senior engineers may review changes drafted by the AI assistant before they reach the pipeline.

(B) A requirement that the AI assistant explains its reasoning in the merge request description for every change.

(C) A pipeline policy check that every RT in a change belongs to the range allocated to that customer in the source of truth.

(D) A canary deployment that applies AI-drafted changes to one PE first and watches for customer complaints.

60. A NETCONF deployment script commits with `<confirmed/>` and a 300-second timeout, runs post-checks that pass, and exits. Five minutes later every change is reverted. What is the root cause?

(A) The device's candidate datastore is locked by another session, so the committed change is overwritten automatically.

(B) The script never sends the confirming commit after post-checks pass, so devices roll back at the timeout.

(C) The post-checks run against the candidate datastore, so the devices treat the change as untested and discard it.

(D) The confirm timeout is too short for the post-checks to complete, so the device rolls back before checks finish.

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EXAM PREPARATION

Practice exam answers & explanations

1. A The task names a source, a path and a failure condition. Only a sourced trace in both the normal and failed state proves all three; the other options prove reachability or intent only.

2. B The exact-route lookup shows the forwarding decision for that specific flow, which uses R3. Being one of the installed paths is not the same as carrying the traffic.

3. C No route and no router LSA means the IGP has no knowledge of R9, so BGP cannot even open TCP. The other checks matter only once the peer address is reachable.

4. D Ignoring the MTU check is an OSPF setting and hides a real mismatch that can drop large updates. Aligning the MTU fixes the cause without touching OSPF.

5. A Dependency weight (3 + 6) exceeds the 4 independent points, and the BGP tasks cannot be proven until the IGP fault is fixed.

6. C Complete silence from hop 3 onward with intact forward routing indicates the return path to the source is missing at hop 3.

7. B Retransmissions of full-size segments with resets on the hold timer are an MTU signature; the DF-bit test decides it in seconds.

8. D A name mismatch creates a separate region. At a region boundary all instances share the CIST port role, so the blocked uplink blocks VLAN 210.

9. D Loop guard blocks a non-designated port when BPDUs stop arriving, which is the classic symptom of one-way fibre failure.

10. B The printer has no DHCP binding. An explicit entry validates it; trusting the port or disabling checks weakens protection for everyone.

11. A The relay sees option 82 inserted by a non-relay switch and treats the request as untrusted. Disable insertion on the switch or trust option 82 on the relay.

12. C A backup preempts only with a strictly higher priority. The decrement must take the master below 100. VRRP preempts by default, so that option is wrong.

13. A With both peer link and keepalive lost, neither peer can tell whether the other is alive, so both act alone with the same system ID.

14. D Native VLAN traffic leaves untagged, so the customer's C-tag becomes the outermost tag and is misread. Tag the native VLAN or avoid it as an S-VLAN.

15. B A targeted, reversible shutdown contains the loop and confirms its location. Rebooting destroys evidence and debugging a saturated CPU makes things worse.

16. C A non-zero forwarding address must resolve through an internal OSPF route; otherwise the LSA is ignored everywhere.

17. B 20 days of normal refreshing is about 960 instances; over 2,600 means repeated changes. The low age shows a recent new instance.

18. D The distribute-list changes only R3's table, while R4 still computes its path through R3. Filtering at the ABR removes the prefix consistently from the database.

19. A ATT = 1 means attached to level 2; the inter-area marking shows a leaked prefix with the up/down bit, which is never re-advertised into level 2.

20. C Detection = remote multiplier × max(local required RX, remote desired TX) = 3 × 300 ms.

21. D Rings defeat basic LFA. Tunnelling to a PQ node or encoding the post-convergence path closes the coverage gap; faster detection or SPF does not create backups.

22. B Protocols are compared by administrative distance. Raising OSPF external distance above 115 lets the native IS-IS route win.

23. C The redistributed copies kept each other alive. Tag-based filtering on every redistribution point stops routes returning to their home domain.

24. A A shared cluster ID makes partner reflections look like loops. That saves memory but makes each client's dual peering essential.

25. B Grouping by AS picks 10.0.0.9 (MED 0) in AS 64510. Against the AS 64520 path MED is not compared, so IGP metric decides: 40 beats 50.

26. D Communities present at ingress and absent at egress were stripped in transit, typically by a session without community sending.

27. A Local preference in the provider's AS decides before MED. Provider communities act at that earlier step; a missing MED is treated as best by default, not worst.

28. C The condition must match a signal only the primary sends. With both sending a default, the condition route never disappears.

29. B Detection must be slower than the protection of the layer below; otherwise BFD reports failures the transport would recover from.

30. D The route is covered but exceeds maxLength, so it is Invalid and dropped. Origin match alone is not enough.

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31. C The filter permits only 10.255.0.x, so P2 never advertises (or uses) a label for 10.255.1.1 and forwards unlabelled. A down session would affect all prefixes, PHP is normal behaviour that appears as implicit null at the penultimate hop rather than unlabelled output mid-path, and nothing in the output points to a metric change.

32. A With identical RDs the reflector selects one best VPN path and hides the other, so remote PEs must wait for a withdrawal and new best path. Unique RDs expose both paths. BGP timers do not detect a PE-CE link failure, the core transport is unaffected, and as-override addresses AS-path loops.

33. B The outgoing label equals the next hop's SRGB base plus index, so different SRGBs yield different labels per hop and are valid. A collision would cause conflict errors, LDP labels would not follow the SRGB arithmetic, and TI-LFA repair would push additional labels on a backup interface.

34. D Three labels add 12 bytes: 1488 + 12 = 1500, exactly the core MPLS MTU. If the CE limited the size, the ping from the PE would not be affected that way; labelled packets are not fragmented when DF is set; and each label is 4 bytes.

35. D When domain IDs differ, VPN routes are redistributed as externals. Setting the same domain ID (or process ID) restores inter-area routes. RDs do not influence OSPF route type, the down bit prevents loops, and sham-links are for intra-area preference, not for avoiding externals.

36. A No node advertises the SID behind 16099, so the explicit path is invalid and steering falls back. Fix the list, ideally with a dynamic fallback path. 16099 is already inside a 16000-23999 SRGB; removing colours abandons the intent; SR does not need LDP.

37. C Importing the spoke RT gives spokes each other's specific routes, which beat the hub path. Removing it enforces the topology. RDs do not filter imports, a default route loses to more specific spoke routes, and disabling reflection breaks the whole VPN.

38. B Option C needs a labelled path to remote PE loopbacks; the VPN routes already arrived, so the multihop session works. ASBRs hold no VRFs in option C, and a plain unlabelled IGP route still leaves the next hop without a label.

39. D Auto RT derivation uses the local AS; with per-leaf ASes the values differ and nothing matches. Use explicit RTs or an import option that ignores the AS. Different RDs are normal, a next-hop rewrite would break forwarding rather than import, and ARP suppression uses type-2 routes.

40. B Without VRFs a spine's default RT filtering drops every EVPN route; it must retain all route targets. Established multihop sessions prove underlay reachability, ARP suppression is a leaf feature, and L3 VNI choices do not affect spine propagation.

41. C VXLAN adds 50 bytes; full-size tenant frames exceed a 1500-byte underlay. Per-flow ports keep packets of one flow on one path, gateway MAC problems would break routing for all sizes, and missing type-3 routes break BUM rather than large packets.

42. A Without an L3 VNI the leaf cannot participate in symmetric routing: its routes lack label 2 and the router MAC. ARP suppression does not add labels, anycast gateway MACs must be identical, and type-5 routes also require the L3 VNI and router MAC.

43. D Two unbundled links create a Layer 2 loop or constant MAC moves between leaves; an ES with LACP makes them one logical attachment with DF election. Disabling mobility or raising thresholds hides the loop, and splitting VLANs breaks the hosts' design.

44. A The 'always' default keeps attracting traffic after Border1 lost its exit. Conditional origination ties advertisement to real reachability. Faster timers do not help because the EVPN sessions stay up, a shared VTEP does not fix the advertisement, and a floating static never wins while BGP still offers the route.

45. B The packets arrive at the edge already re-marked to 0, so the EF match fails before any SLA check. SLA fallback would show EF traffic arriving correctly, DSCP copy affects the outer header only, and a missing locator would affect all traffic on MPLS.

46. C Two active gateways split return paths across firewalls or interfaces, causing asymmetric drops. One gateway per VLAN, switched in a single change, avoids this. Different gateway IPs require host changes, ARP suppression does not stop the legacy gateway answering, and extending VLANs increases the problem.

47. B Sources and receivers meet only if they agree on the RP. The output shows two different RPs for one group. A TTL of 1 would prevent registering at all, IGMP version does not affect RP choice, and register messages are unicast and not subject to the multicast RPF check.

48. D SSM needs source-specific joins. IGMPv2 cannot carry the source, so either upgrade or map. SSM ranges do not use an RP, MSDP is for ASM between RPs, and the reports already reach the router.

49. A The boundary ACL defines which groups may cross the interface; 239.1.1.1 is not permitted. RPs can serve any number of groups, scope ranges have no precedence over each other, and TTL is not the filtering mechanism here.

50. C New EF traffic appeared with the video rollout, so the priority class is overloaded by traffic that does not belong there. Raising the limit risks starving other classes, removing the shaper exposes the provider policer, and priority classes do not use WRED.

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51. A A long Tc lets the shaper send large bursts then stall, adding jitter for voice behind it. A short Tc smooths output. A bandwidth class gives voice worse treatment, a policer loses class protection, and voice does not share a queue with backups.

52. D A burst of one packet drops normal TCP bursts and triggers repeated backoff. A realistic Bc, or smoothing with a shaper upstream, restores throughput. Lowering the rate formalises the problem, WRED does not apply to policers, and EF marking does not bypass a policer.

53. B In the uniform model, TC changes are propagated into the IP header. Pipe and short-pipe preserve customer DSCP. Explicit null keeps TC visible but does not restore DSCP, label allocation mode does not affect marking, and PHP does not undo a copy.

54. C lock-denied means another session legitimately holds the lock; serialising jobs per device prevents collisions. Killing sessions or bypassing locks risks corrupting another change, and NETCONF has no forced commit that overrides locks.

55. B The blank came from a missing variable; strict rendering and schema checks stop it at build time. Canaries and post-checks detect it later, and human review is valuable but less reliable than an automated fail-fast check.

56. A Behavioural differences in software are found only by running that software. Linting and snapshots check form, and static analysis models behaviour generically rather than release-specifically.

57. D Flapping automation needs hysteresis, limits and escalation; a marginal optic needs a human and a replacement. A lower threshold increases triggers, removing verification removes safety, and slower sampling hides data without solving the oscillation.

58. C Drift must be reconciled deliberately: some lines may be required fixes, others stale. Blind deployment re-creates outages, merge-only leaves drift permanently, and wholesale import pollutes the source of truth.

59. C An automated policy check against authoritative allocations catches the error every time. Senior reviewers and explanations help but rely on attention, and a canary would expose the wrong customer's routes in production.

60. B A confirmed commit must be followed by a confirming commit; otherwise rollback is the designed behaviour. Locks do not overwrite committed changes, post-checks do not influence commit state, and here the checks finished well before the timeout.

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IT operations & support

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ZITSS

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ZCIE

Certified Internetwork Expert

CERTIFICATE OF ACHIEVEMENT

This certifies that

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has successfully completed all modules of the ZyeemZia™ Certified Internetwork Expert track and passed the ZCIE certification exam

- Expert troubleshooting
- IGP & BGP mastery
- MPLS & segment routing
- EVPN fabrics & SD-WAN
- Multicast, QoS & automation

ISSUED

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CREDENTIAL ID

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NOTES

My notes

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